



US 20220054876A1

(19) **United States**(12) **Patent Application Publication**  
**Conboy**(10) **Pub. No.: US 2022/0054876 A1**(43) **Pub. Date: Feb. 24, 2022**(54) **ENVIRONMENTALLY-CLEAN  
BIODEGRADABLE WATER-BASED  
CONCENTRATES FOR PRODUCING FIRE  
INHIBITING AND FIRE EXTINGUISHING  
LIQUIDS AND FOAMS FOR FIGHTING  
CLASS A AND CLASS B FIRES**(71) Applicant: **M-Fire Holdings LLC**, Carlsbad, CA  
(US)(72) Inventor: **Stephen Conboy**, Carlsbad, CA (US)(73) Assignee: **M-Fire Holdings LLC**, Carlsbad, CA  
(US)(21) Appl. No.: **17/233,461**(22) Filed: **Apr. 17, 2021****Related U.S. Application Data**

(63) Continuation-in-part of application No. 16/805,811, filed on Mar. 1, 2020, Continuation-in-part of application No. 16/449,389, filed on Jun. 22, 2019, now abandoned, Continuation-in-part of application No. 15/829,944, filed on Dec. 3, 2017, now Pat. No. 10,919,178, Continuation-in-part of application No. 16/914,067, filed on Jun. 26, 2020, Continuation-in-

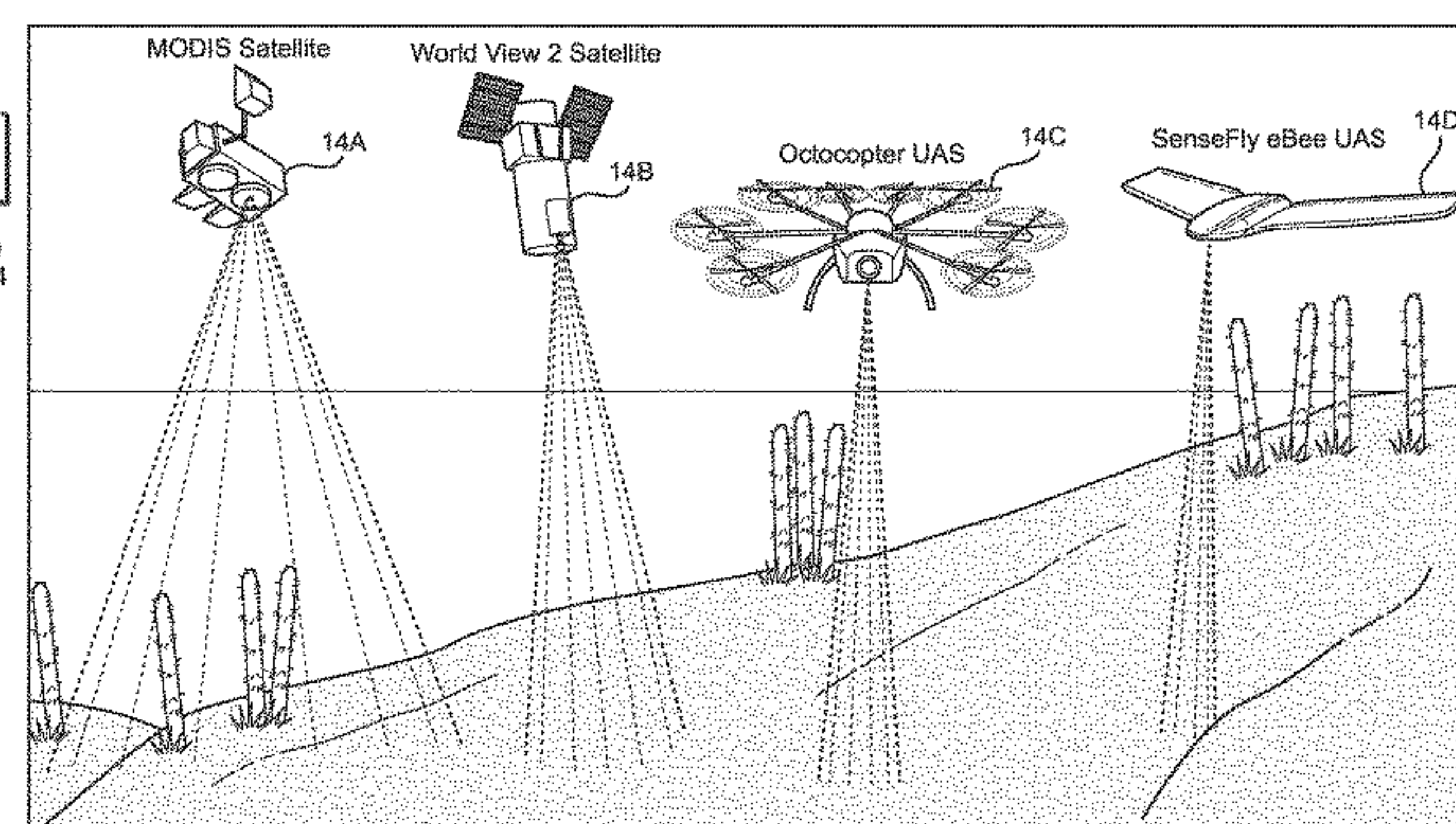
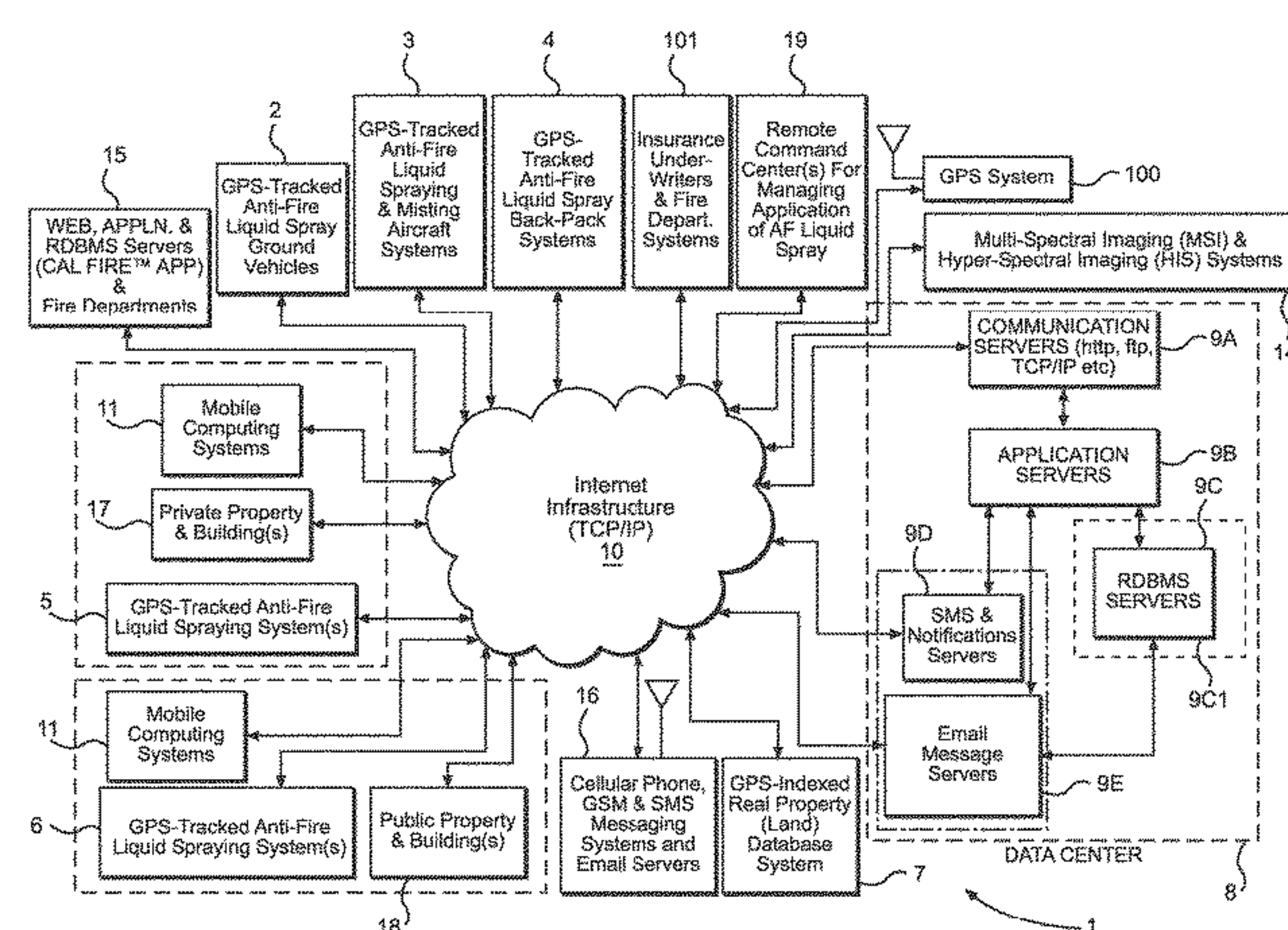
part of application No. 16/029,861, filed on Jul. 9, 2018, Continuation-in-part of application No. 17/167,084, filed on Feb. 4, 2021.

**Publication Classification**(51) **Int. Cl.****A62D 1/02** (2006.01)**A62D 1/00** (2006.01)(52) **U.S. Cl.**CPC ..... **A62D 1/0078** (2013.01); **A62C 3/0242**  
(2013.01); **A62D 1/0042** (2013.01)

(57)

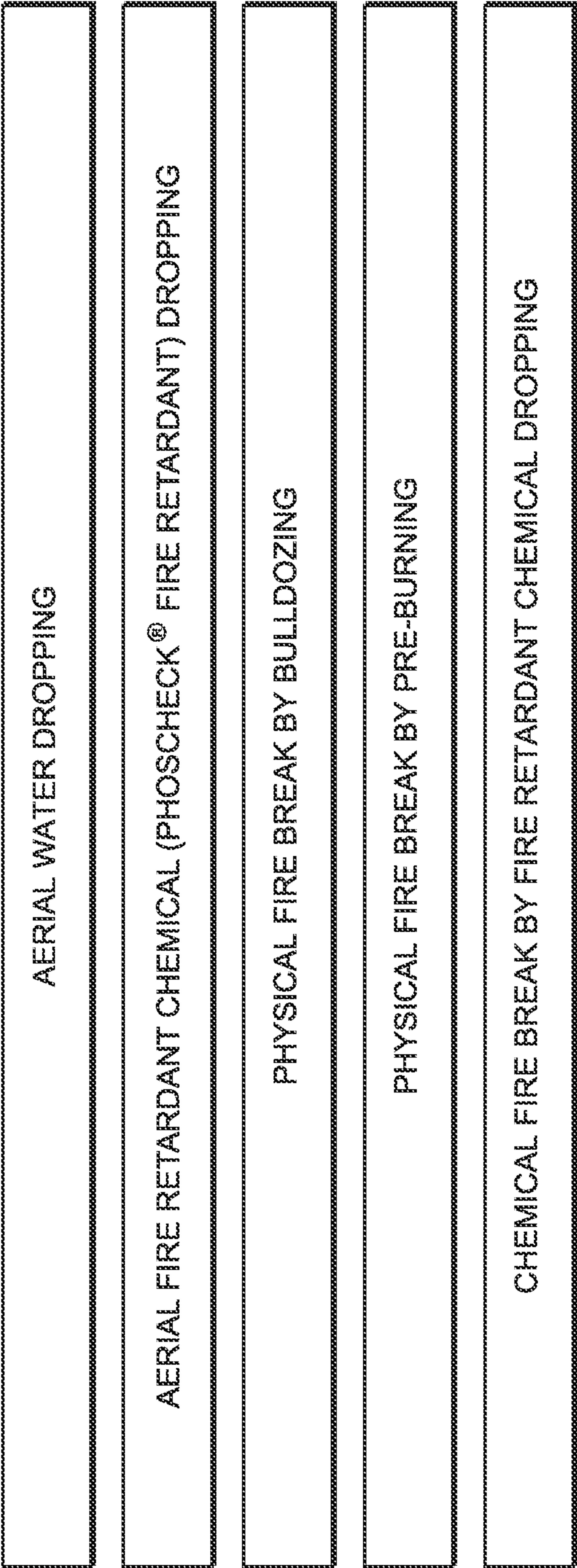
**ABSTRACT**

Environmentally-clean biodegradable water-based biochemical concentrates for producing fire-inhibiting and fire-extinguishing liquids and foams for fighting Class A and/or B fires. The concentrates are formulated using generally safe food-grade chemicals, for foaming agents, surfactants and dispersants, and fire inhibiting extinguishing agents, without the use of phosphates and ammonia compounds. Automated proportioning and mixing devices and systems can be used to mix the biochemical concentrates with proportioned quantities of pressurized water so as to produce fire inhibiting and extinguishing water streams, as well as finished firefighting foam materials, to both inhibit and extinguish fire involving Class A and/or B fuels.



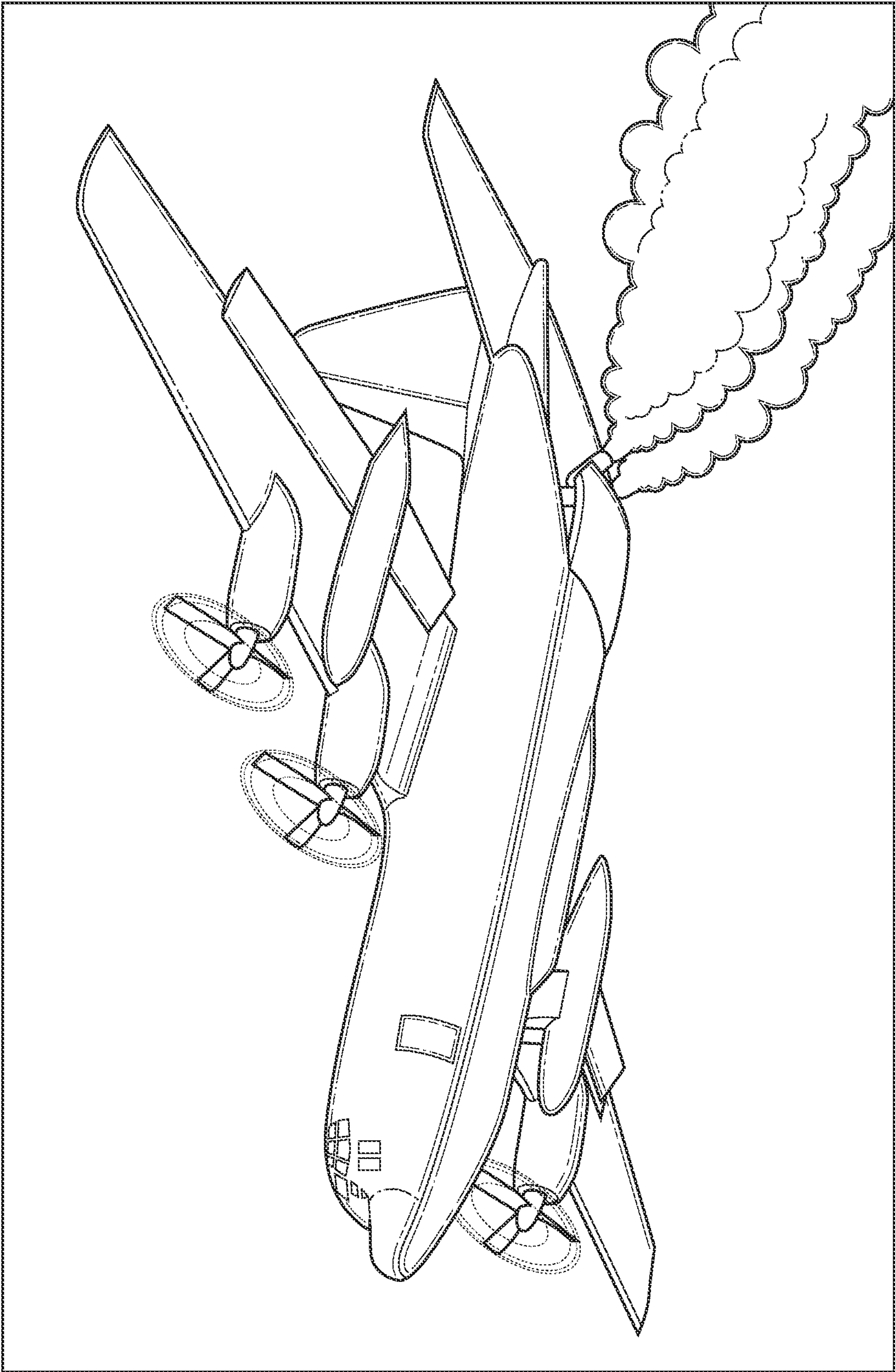


CONVENTIONAL METHODS FOR FIGHTING AND DEFENDING AGAINST WILD FIRES



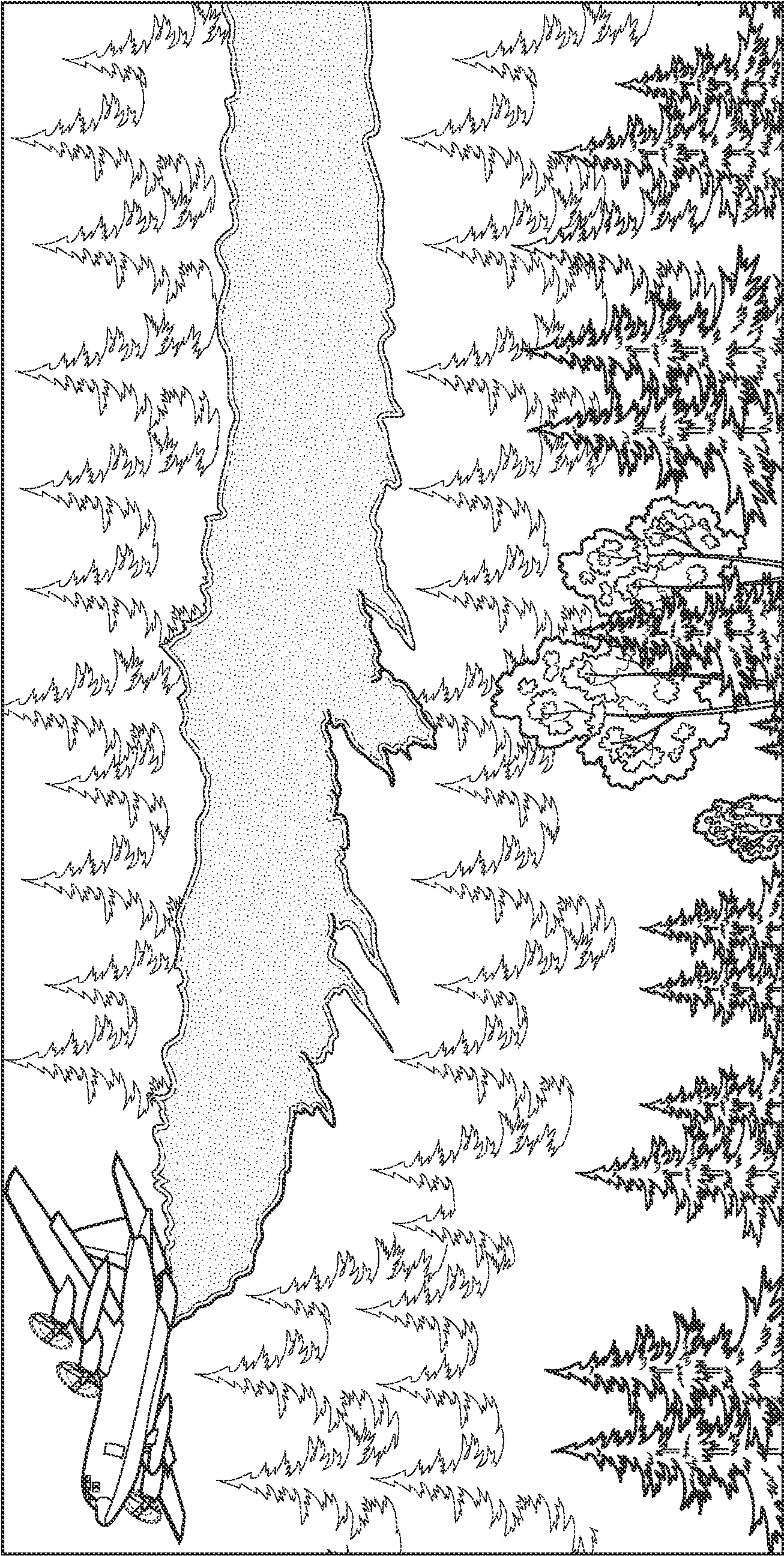
**FIG. 1**  
(PRIOR ART)





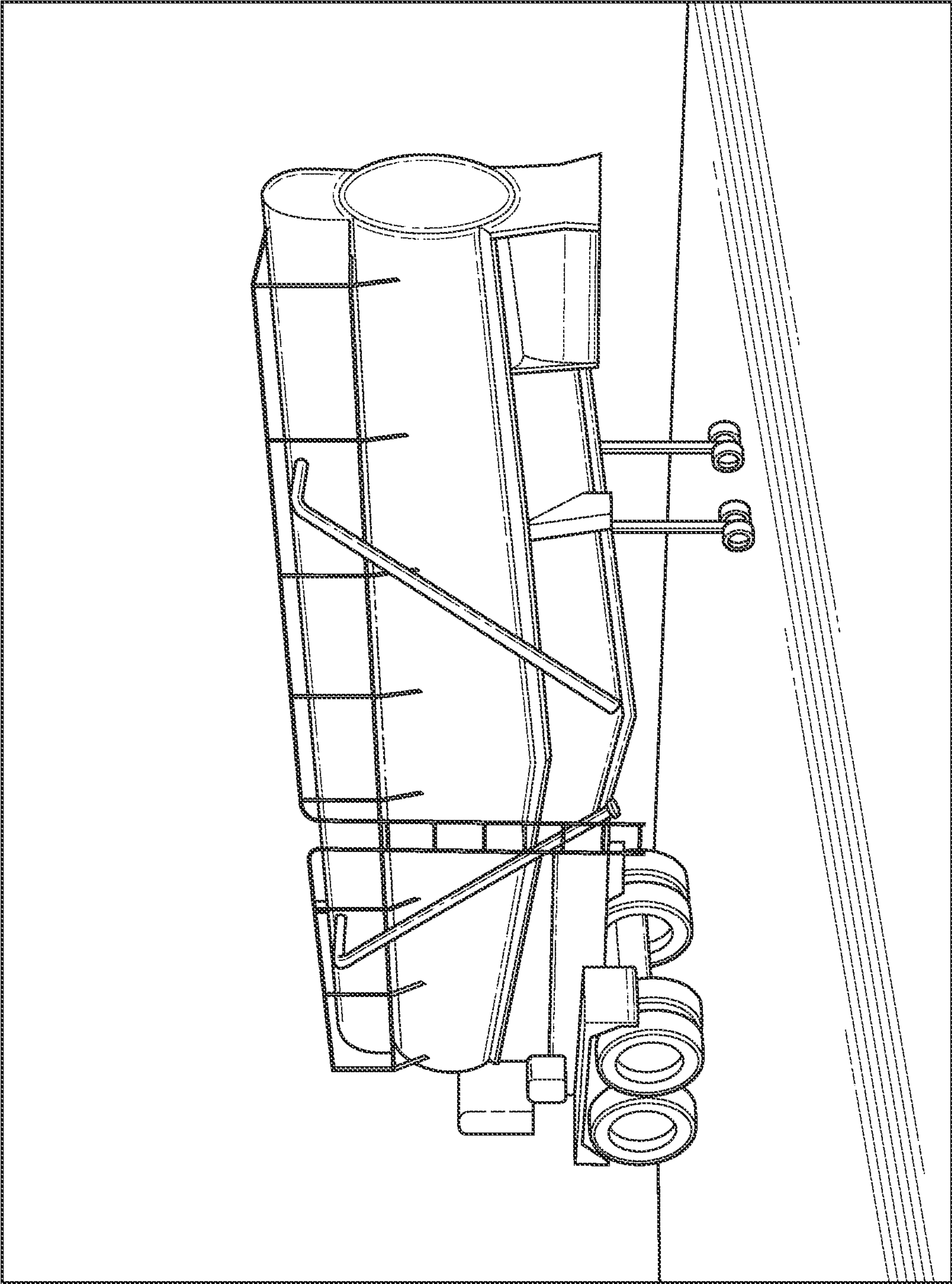
**FIG. 2A**  
(PRIOR ART)





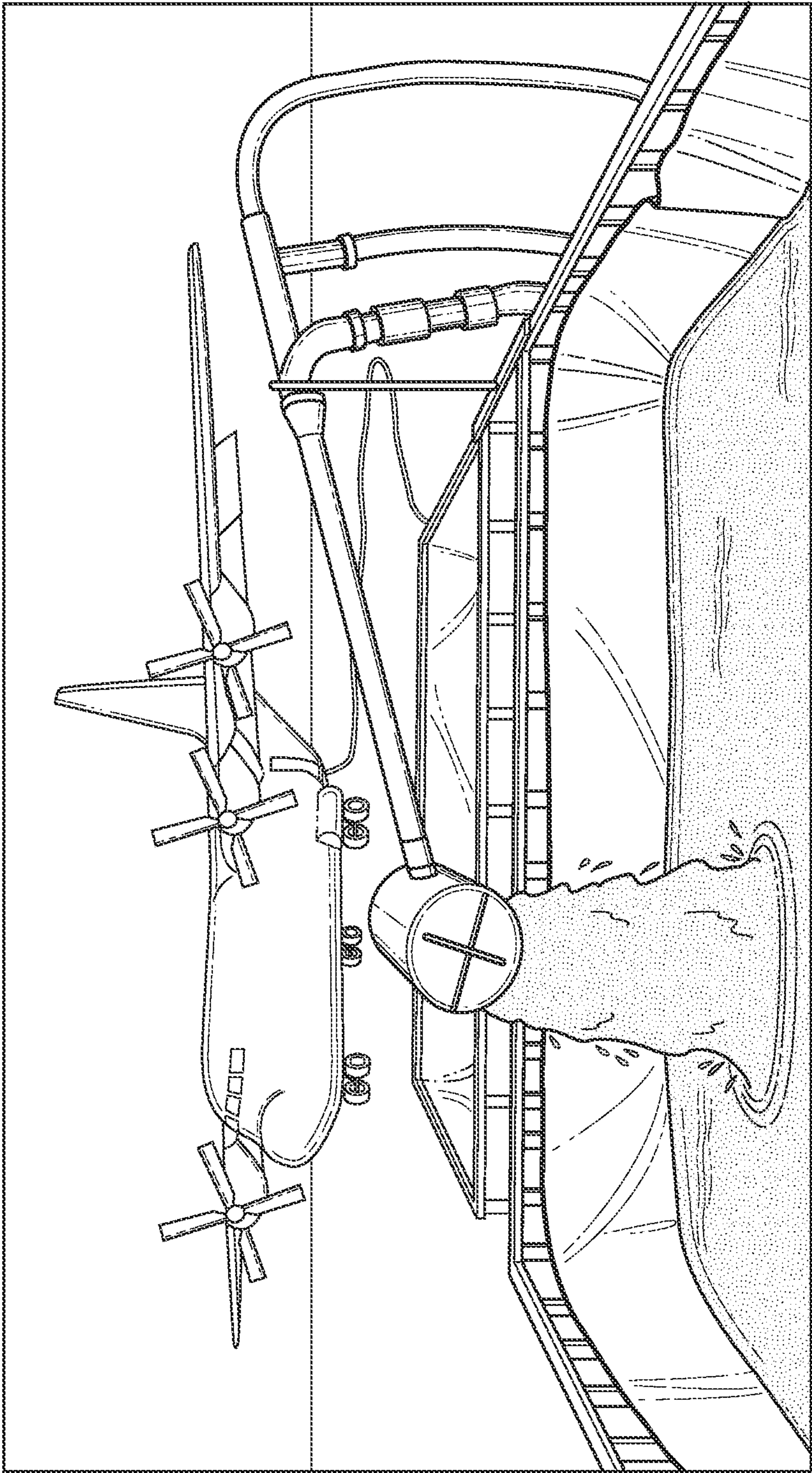
**FIG. 2B1**  
**(PRIOR ART)**





**FIG. 2B2**  
(PRIOR ART)



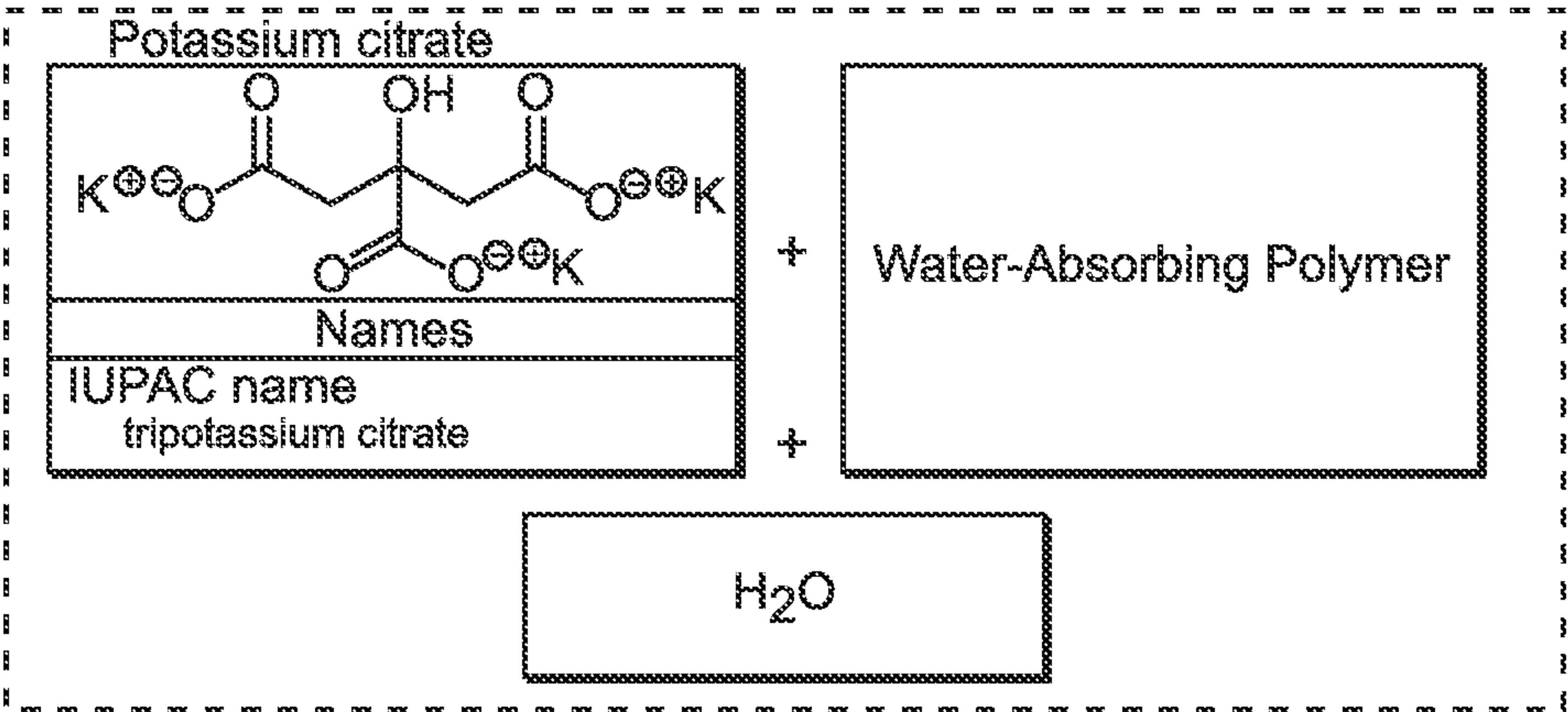


**FIG. 2B3**  
**(PRIOR ART)**









(PRIOR ART)

FIG. 3A  
(PRIOR ART)

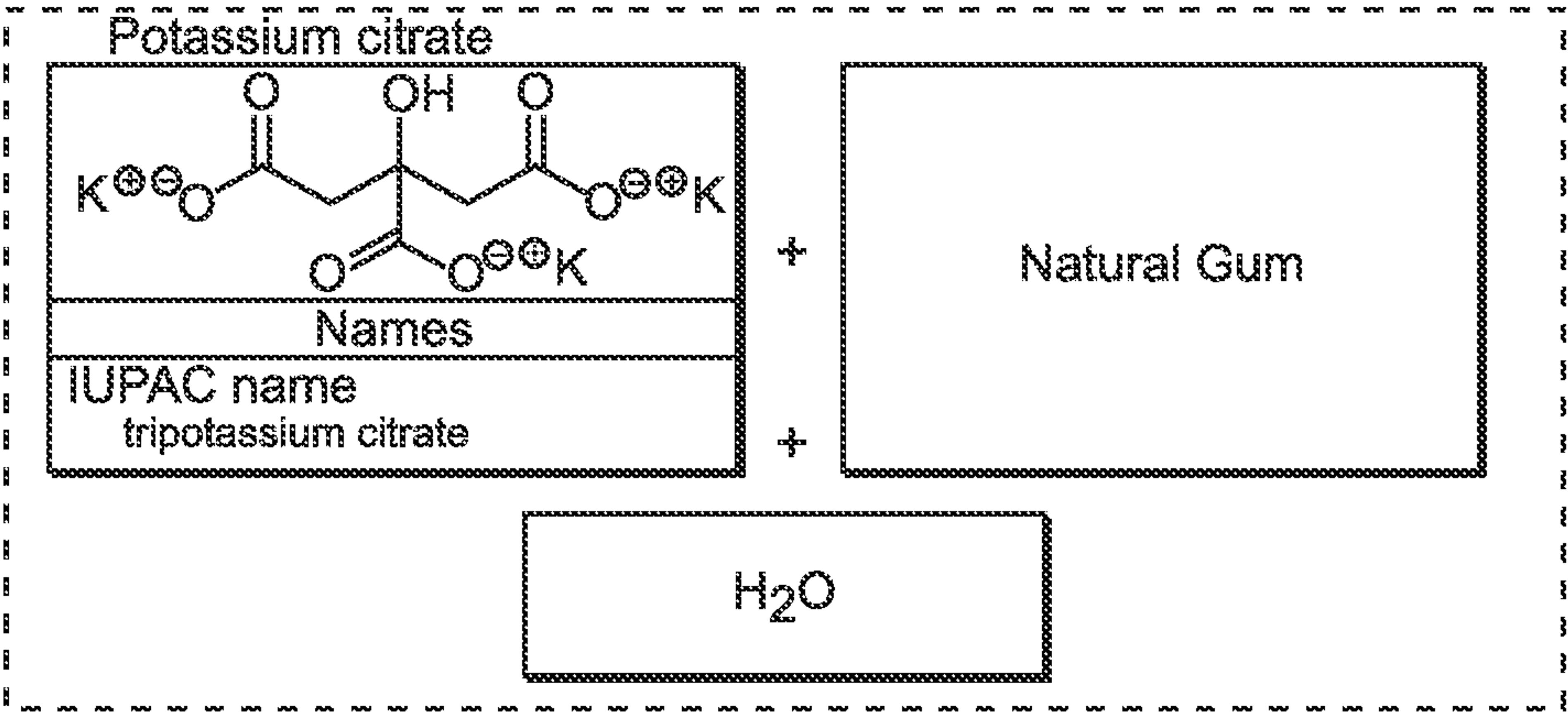
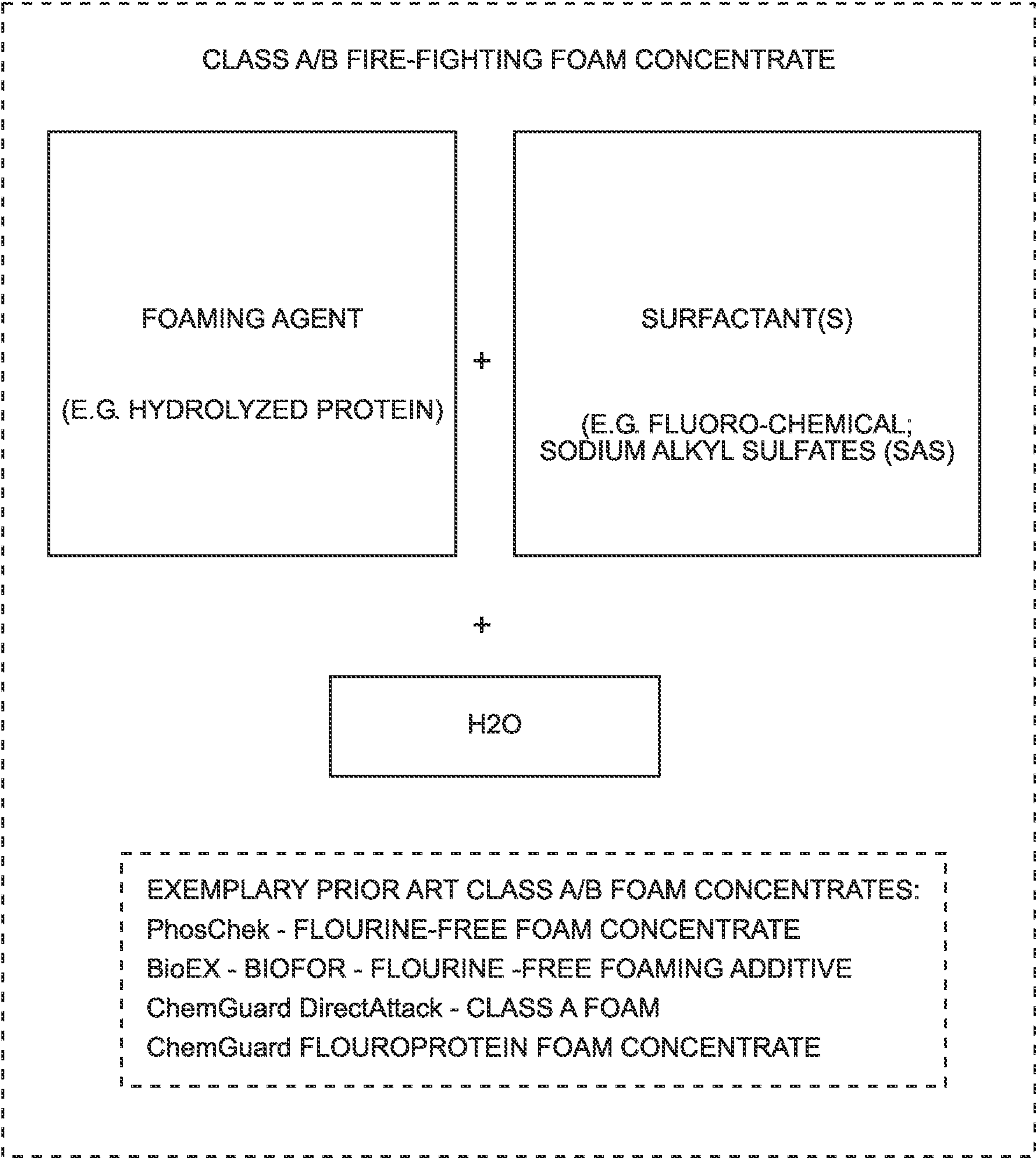


FIG. 3B  
(PRIOR ART)





**FIG. 3C**  
**(PRIOR ART)**



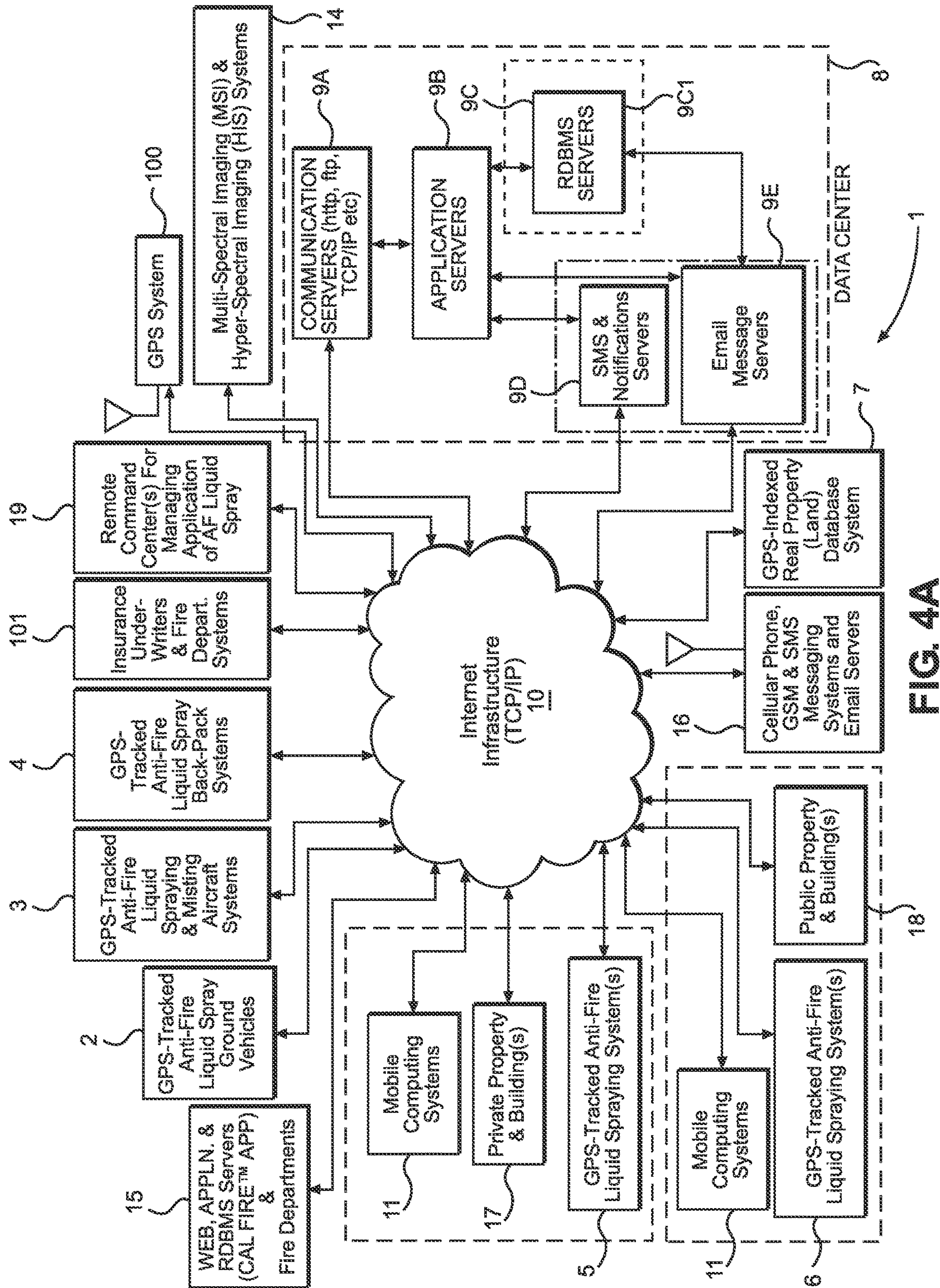
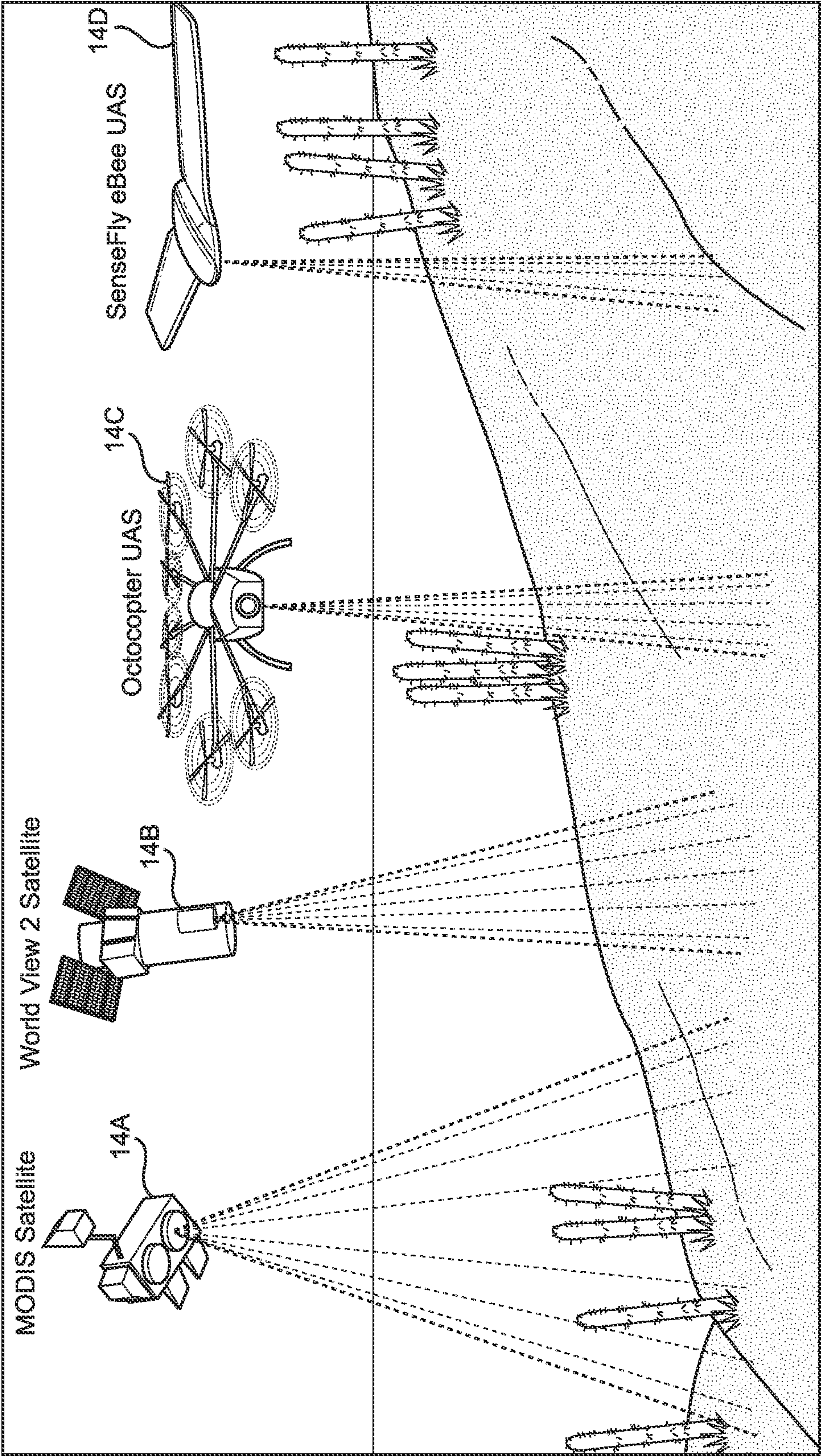


FIG. 4A





**FIG. 4B**



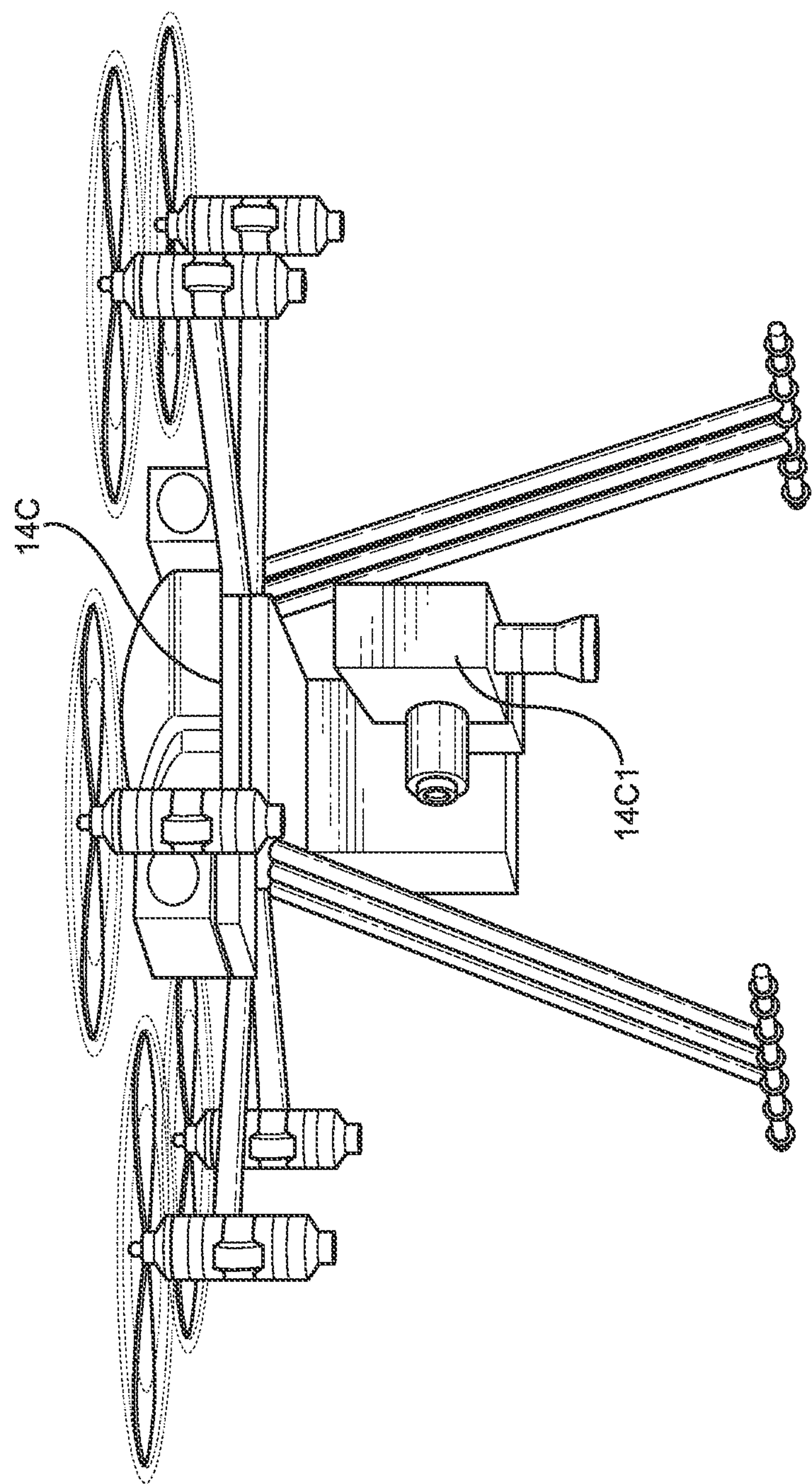


FIG. 4C



MOBILE COMPUTING DEVICES DEPLOYED ON THE SYSTEM NETWORK OF THE PRESENT INVENTION

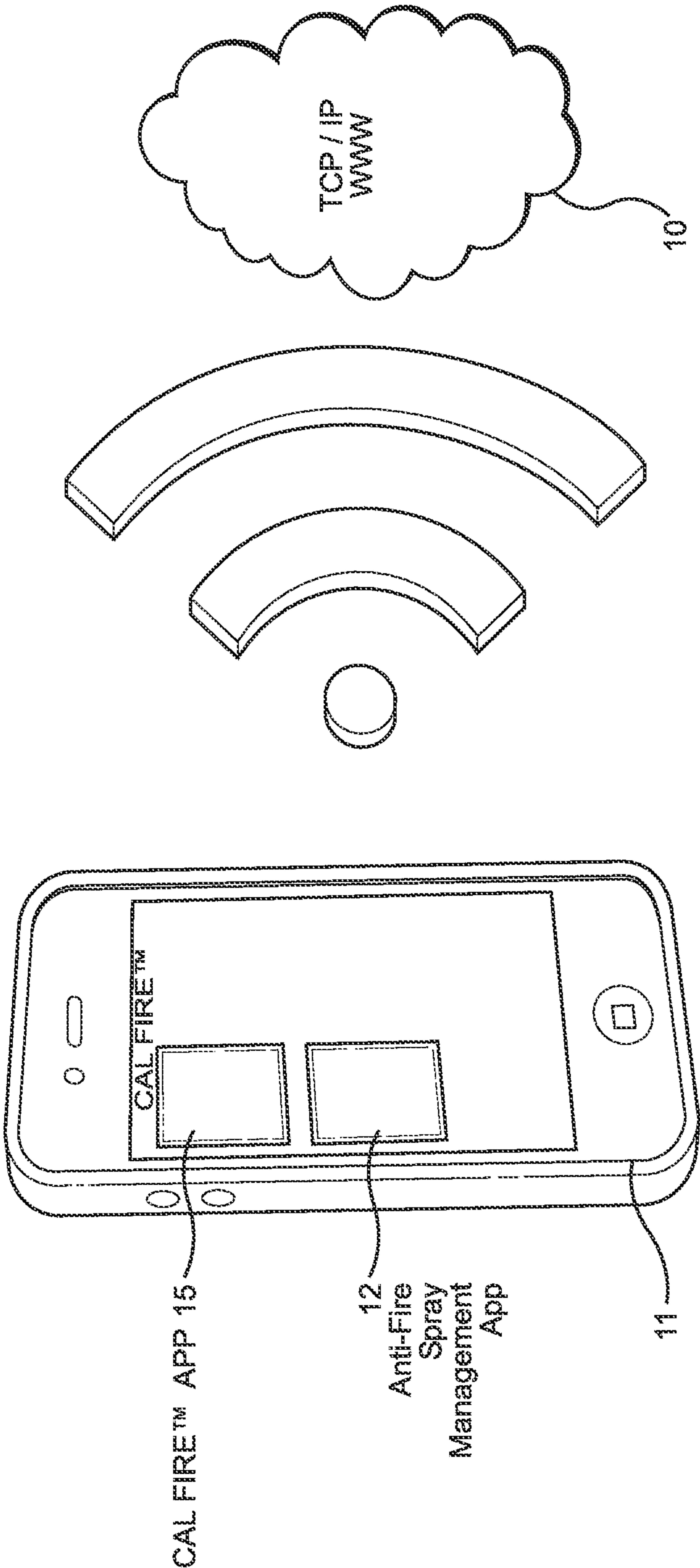
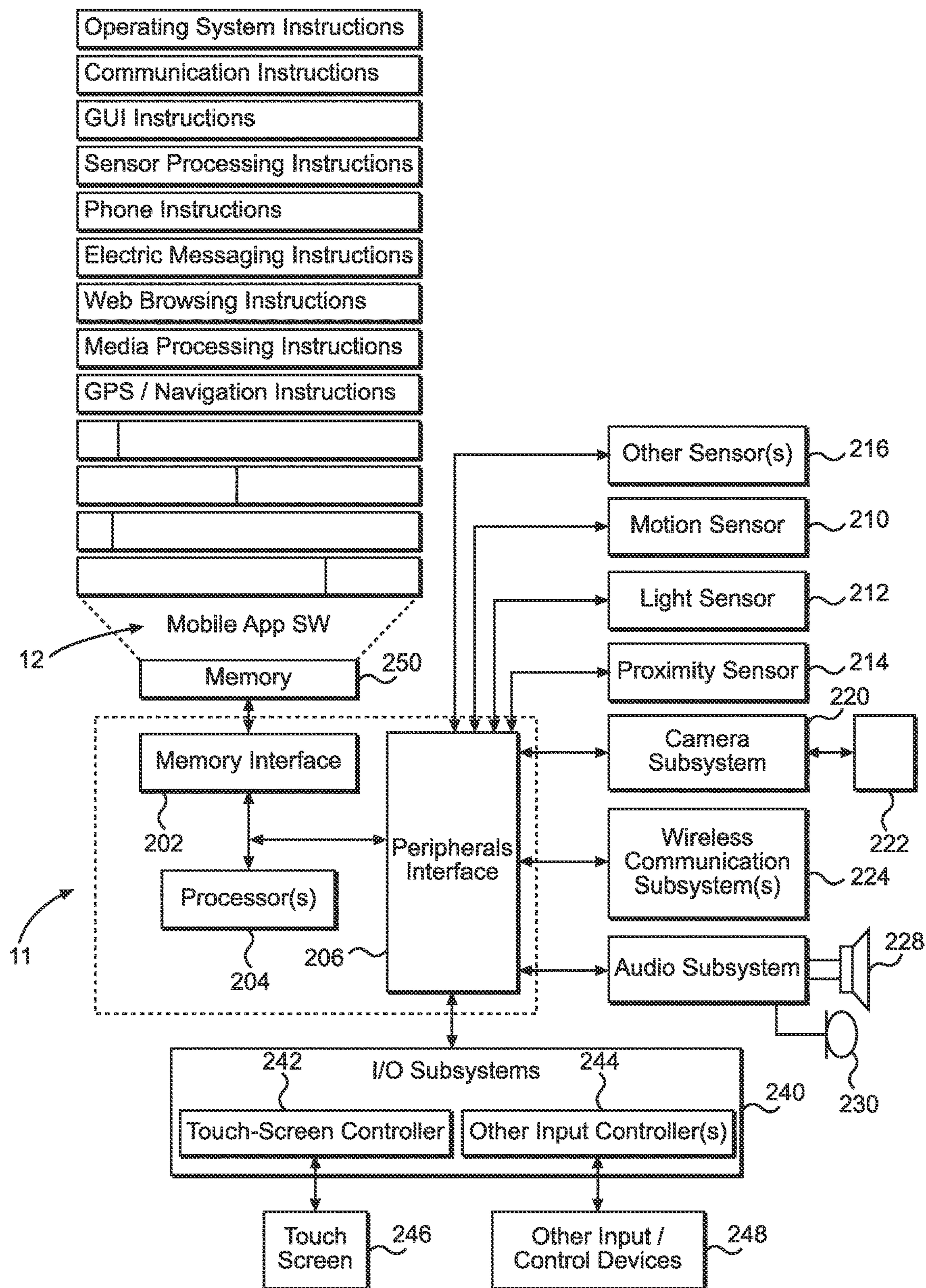


FIG. 5A



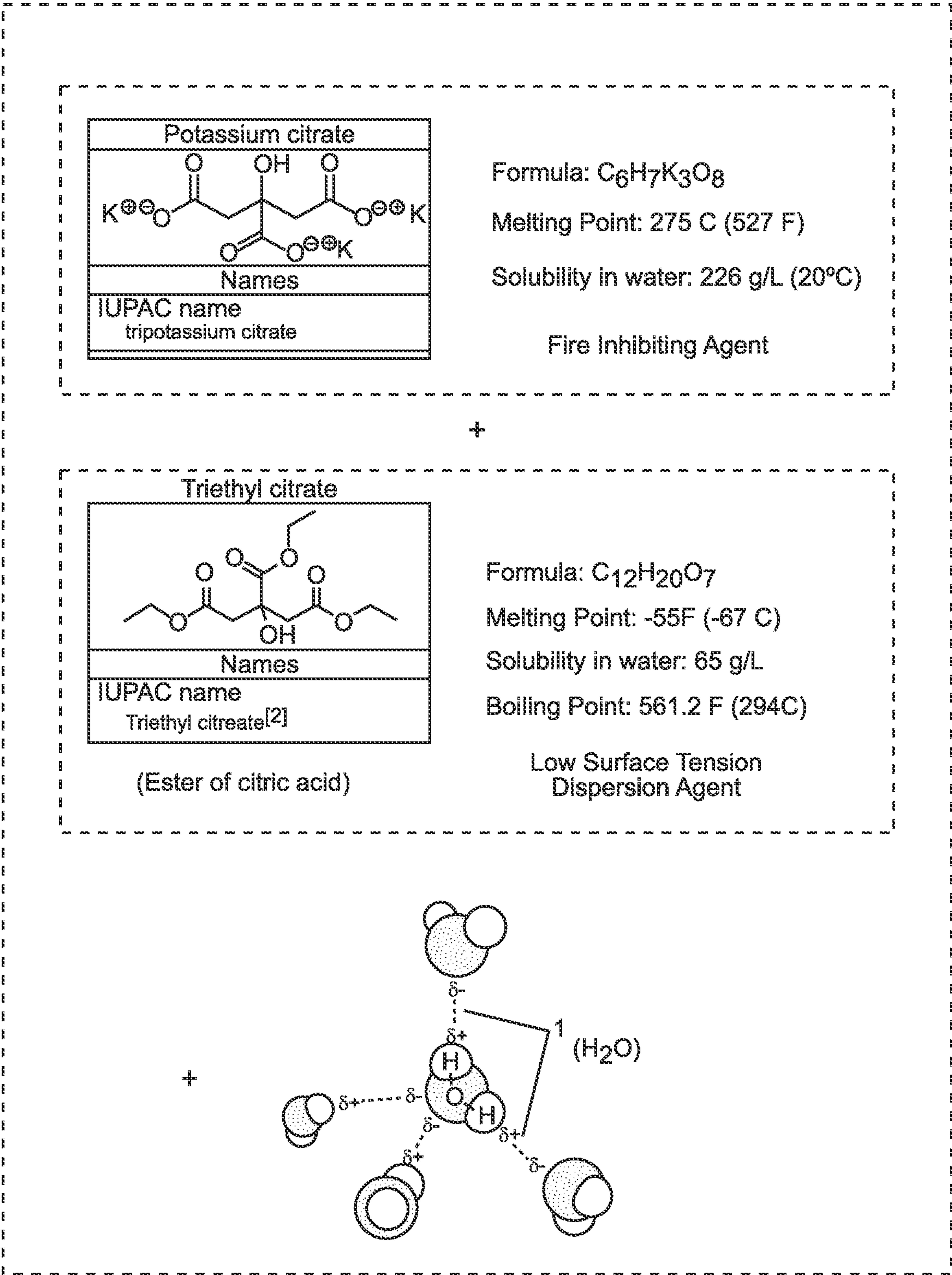
# GENERAL SYSTEM ARCHITECTURE OF MOBILE CLIENT SYSTEM



**FIG. 5B**



WILDFIRE INHIBITOR BIOCHEMICAL COMPOSITION FOR  
PROACTIVELY FIRE-PROTECTING COMBUSTABLE SURFACES





WILDFIRE INHIBITOR BIOCHEMICAL COMPOSITION KIT FOR USE IN  
PROACTIVELY FIRE-PROTECTING COMBUSTABLE SURFACES

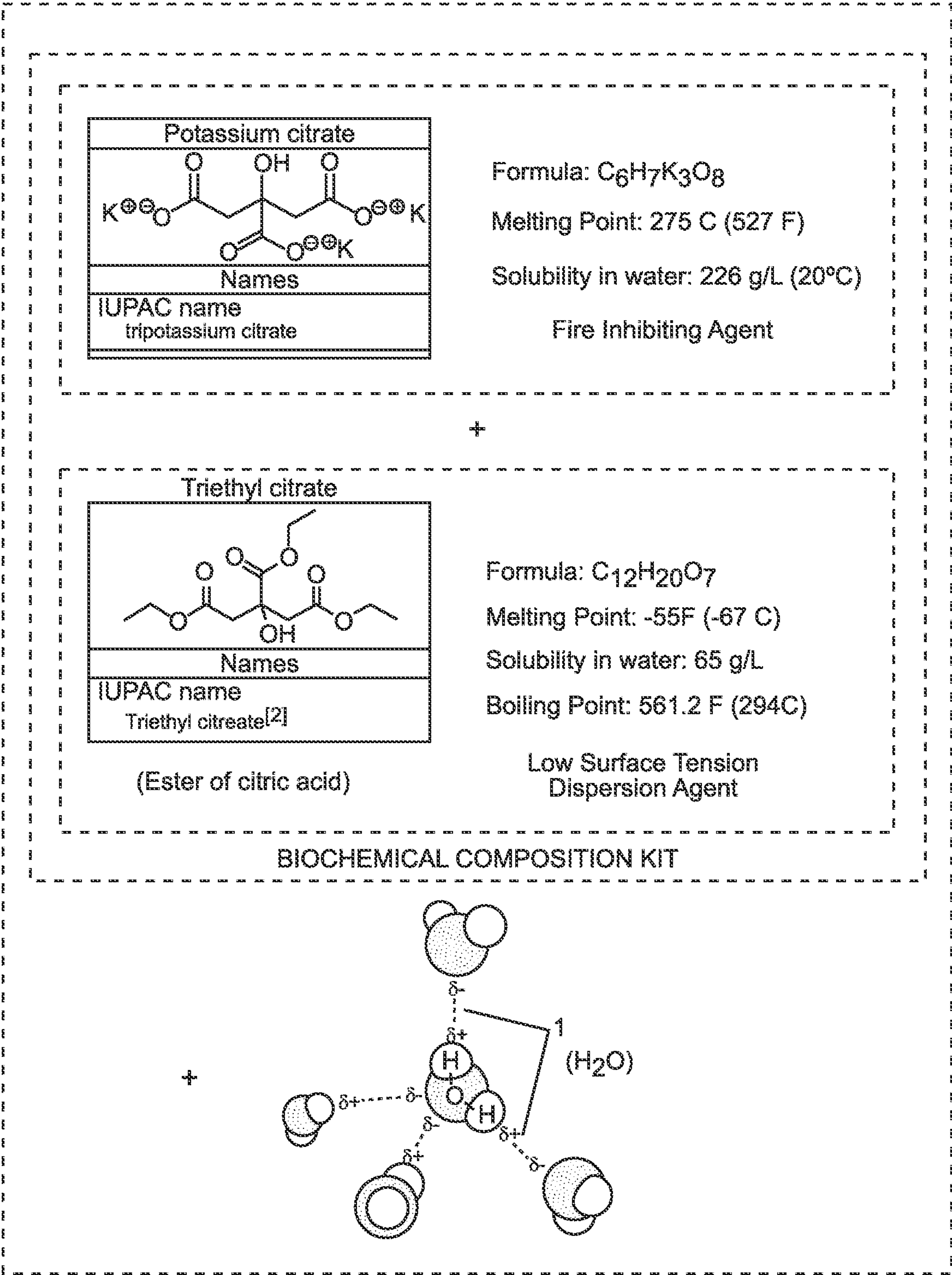


FIG. 6A2



WILDFIRE INHIBITOR BIOCHEMICAL COMPOSITION FOR PROACTIVELY  
PROTECTING WOOD PRODUCTS

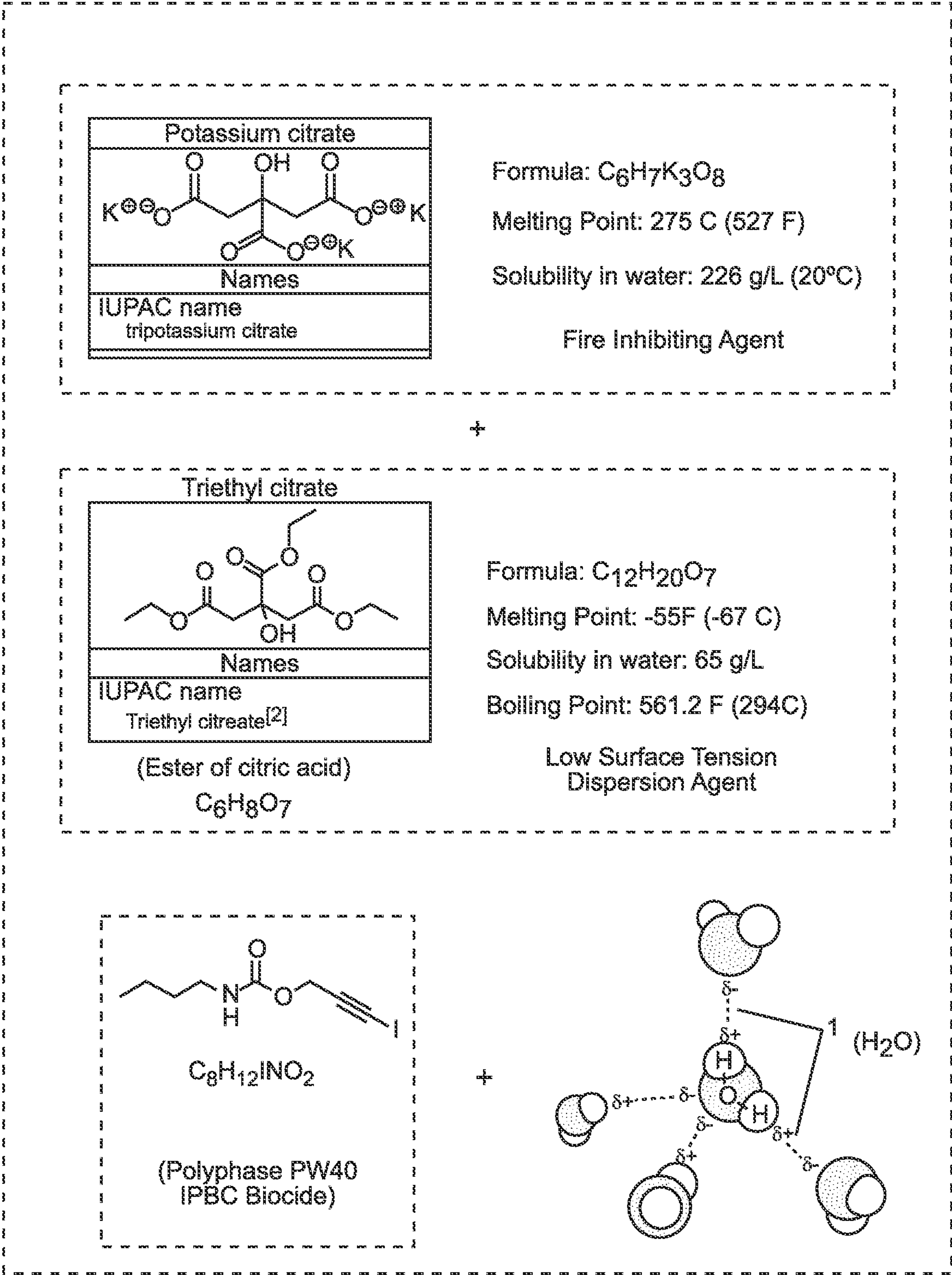


FIG. 6B1



**FIRE INHIBITOR BIOCHEMICAL COMPOSITION KIT FOR PROACTIVELY  
PROTECTING WOOD PRODUCTS**

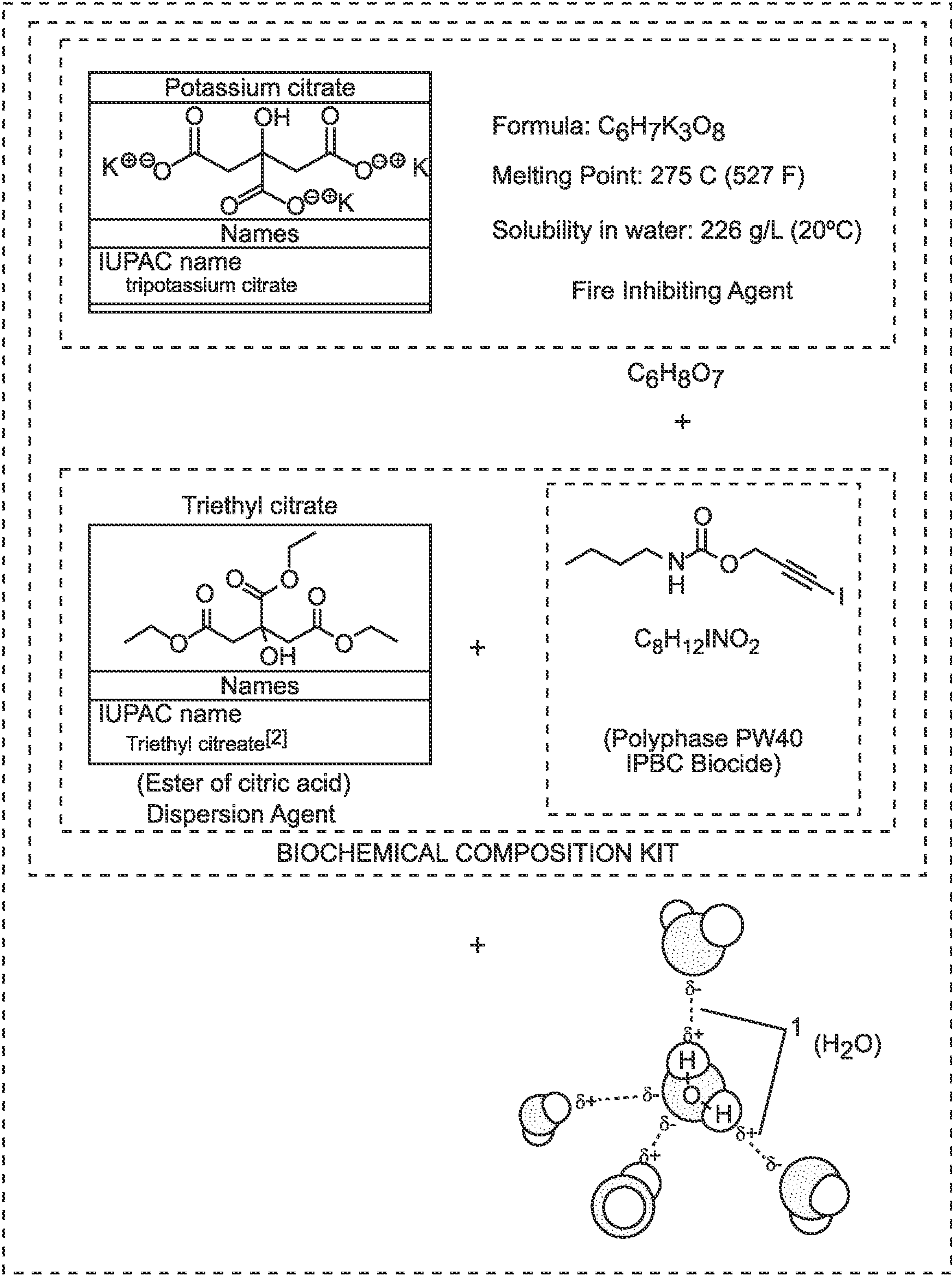


FIG. 6B2



FIRE INHIBITOR BIOCHEMICAL COMPOSITION FOR PROACTIVELY  
PROTECTING WOOD PRODUCTS

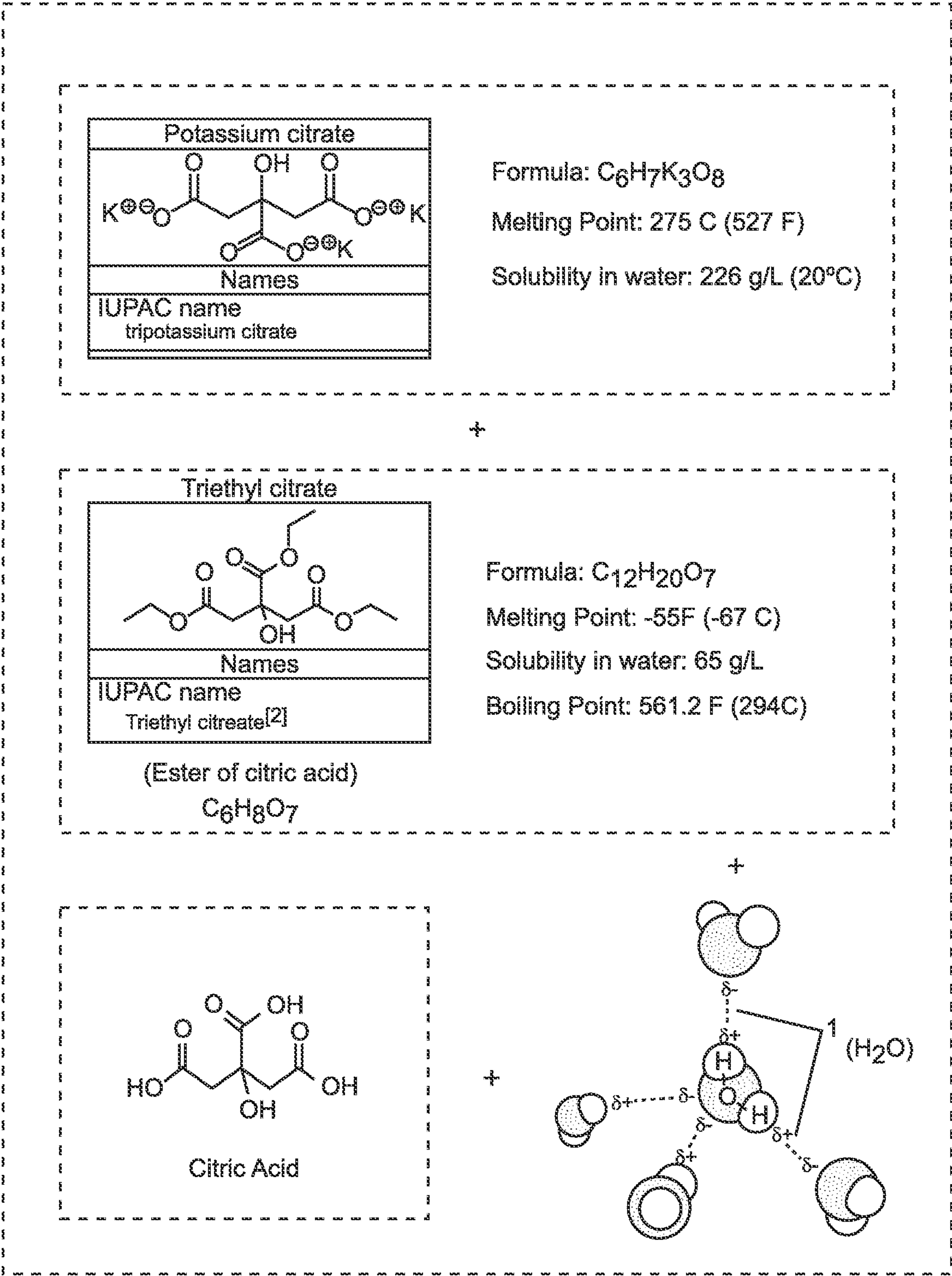


FIG. 6C1



**FIRE INHIBITOR BIOCHEMICAL COMPOSITION KIT FOR PROACTIVELY  
PROTECTING WOOD PRODUCTS**

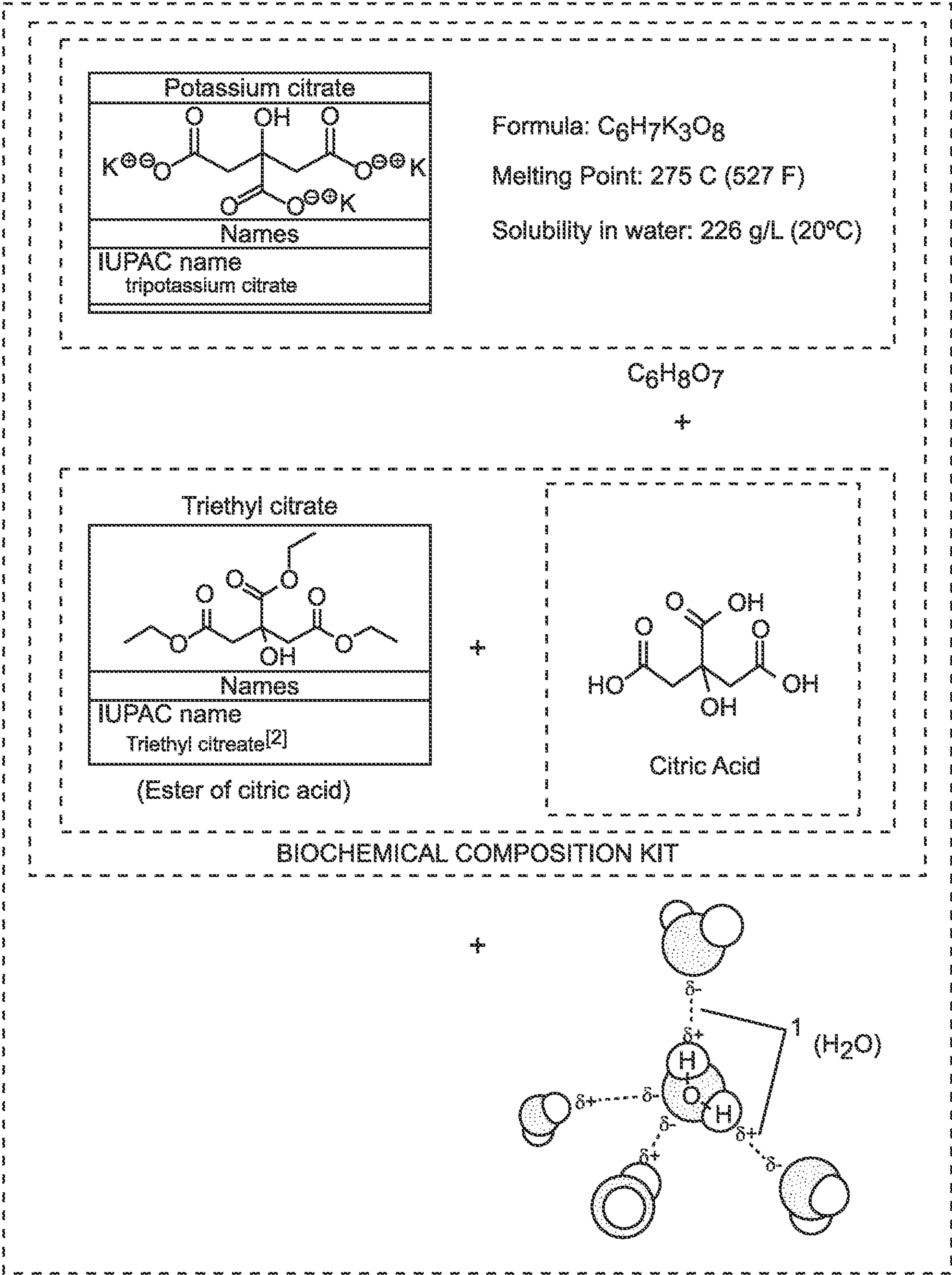
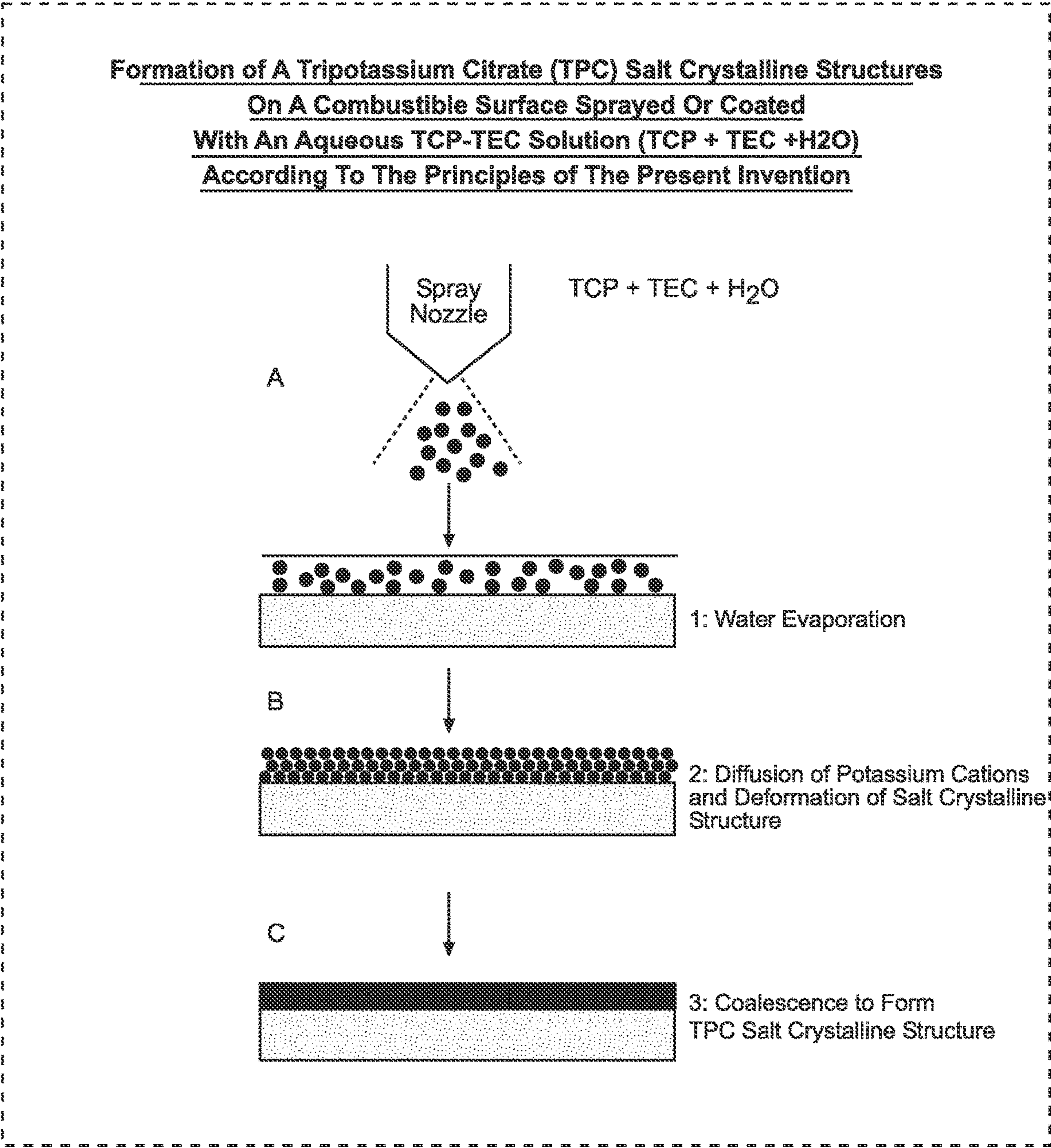


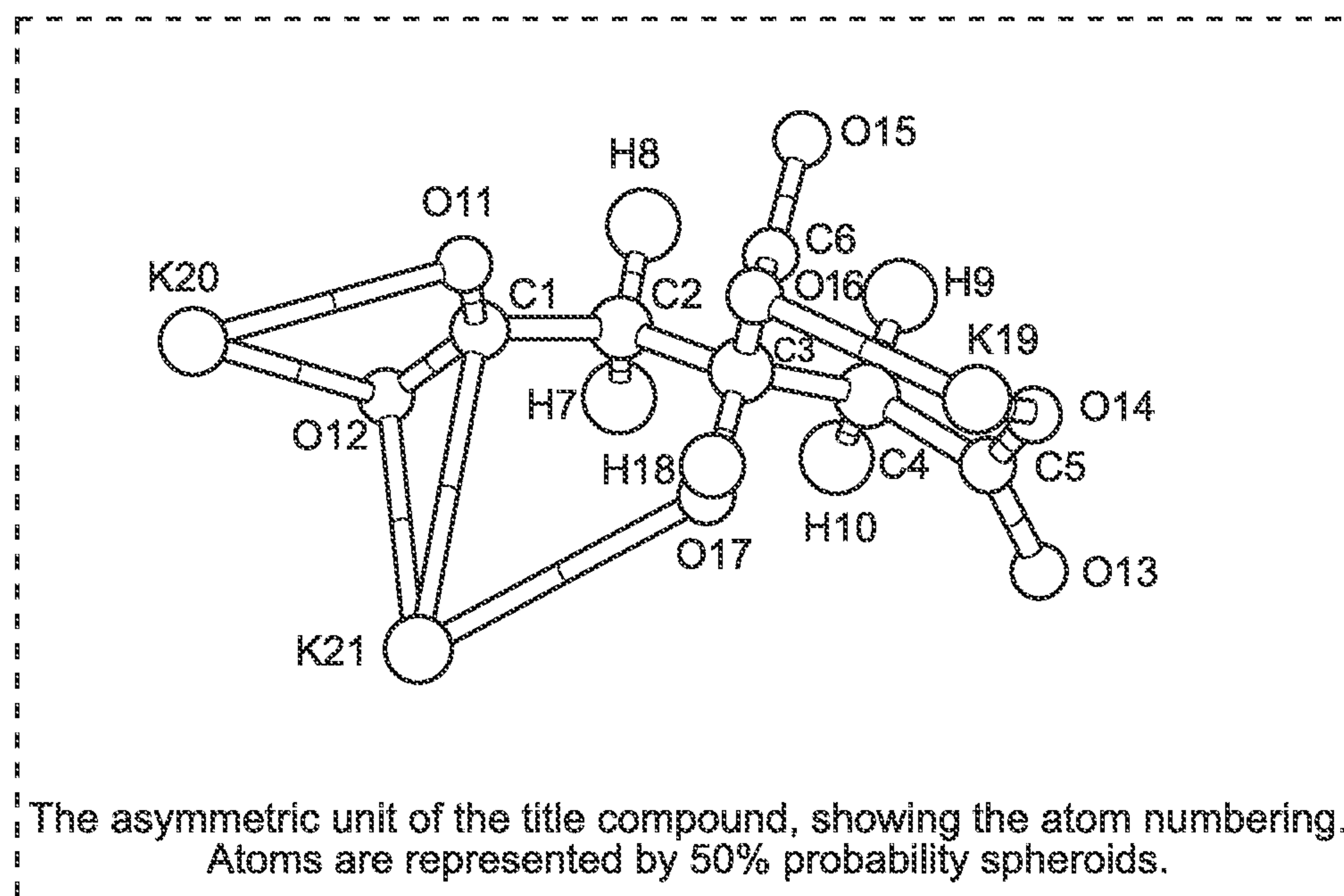
FIG. 6C2



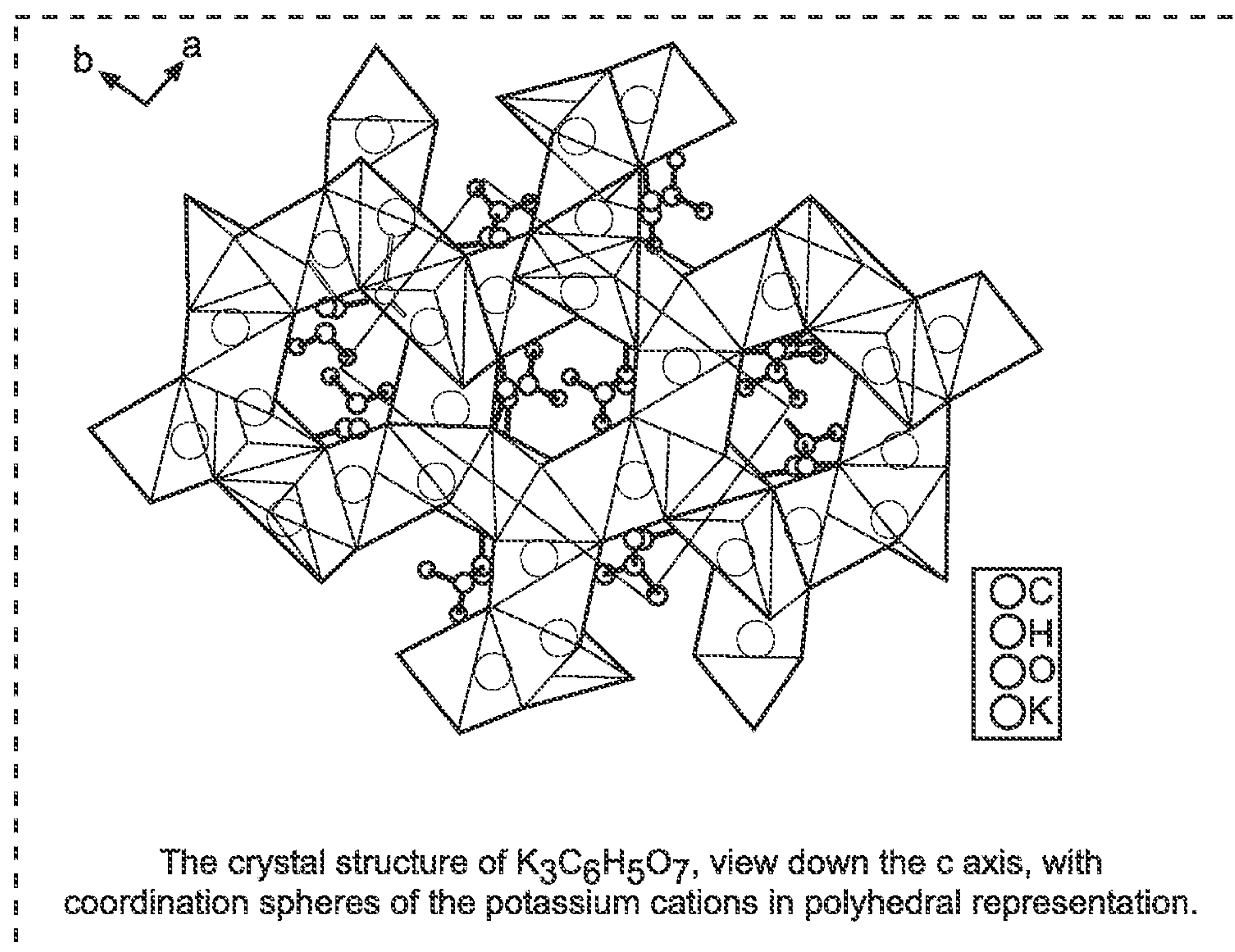


**FIG. 7A**





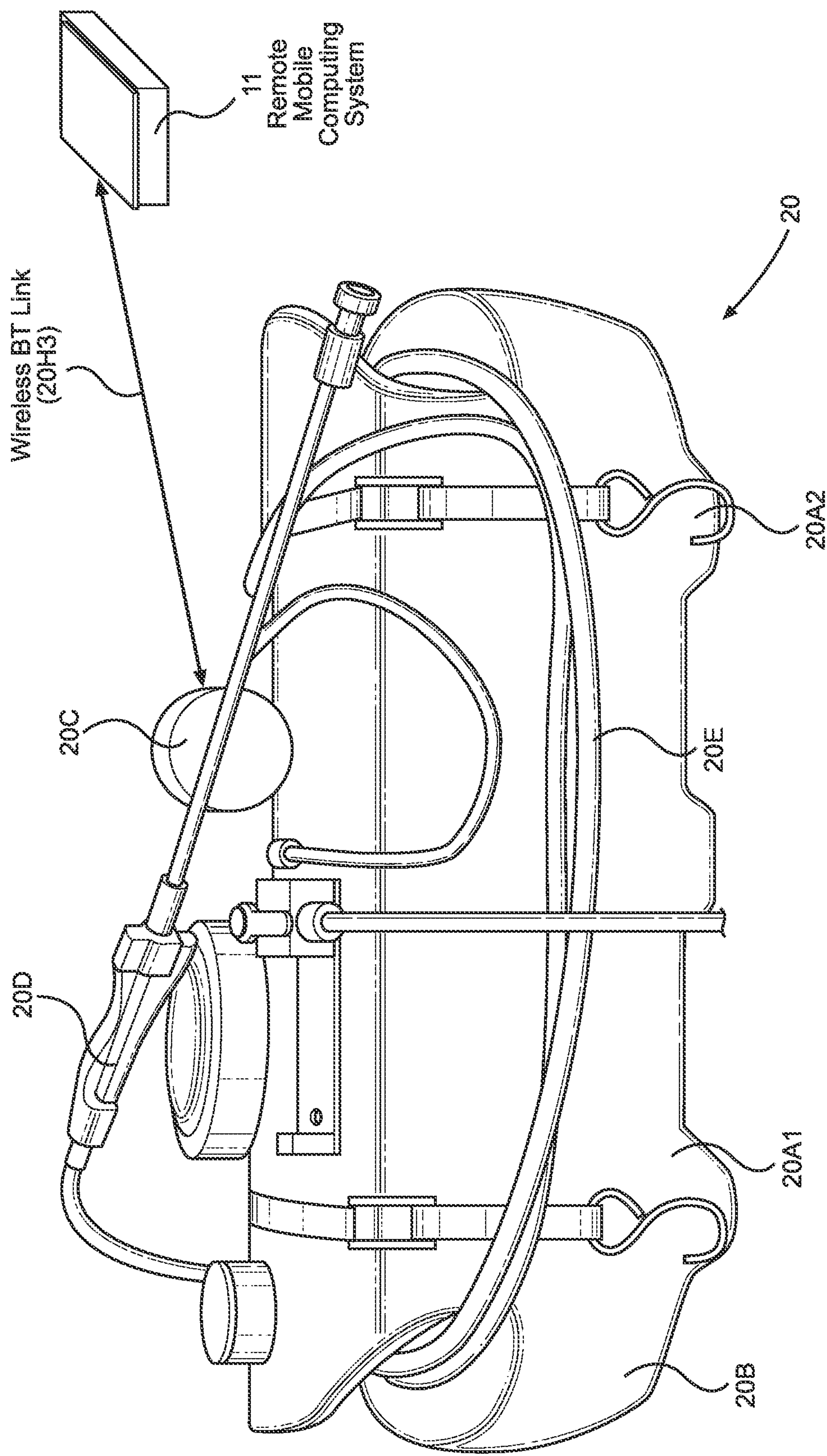
**FIG. 7B**



**FIG. 7C**



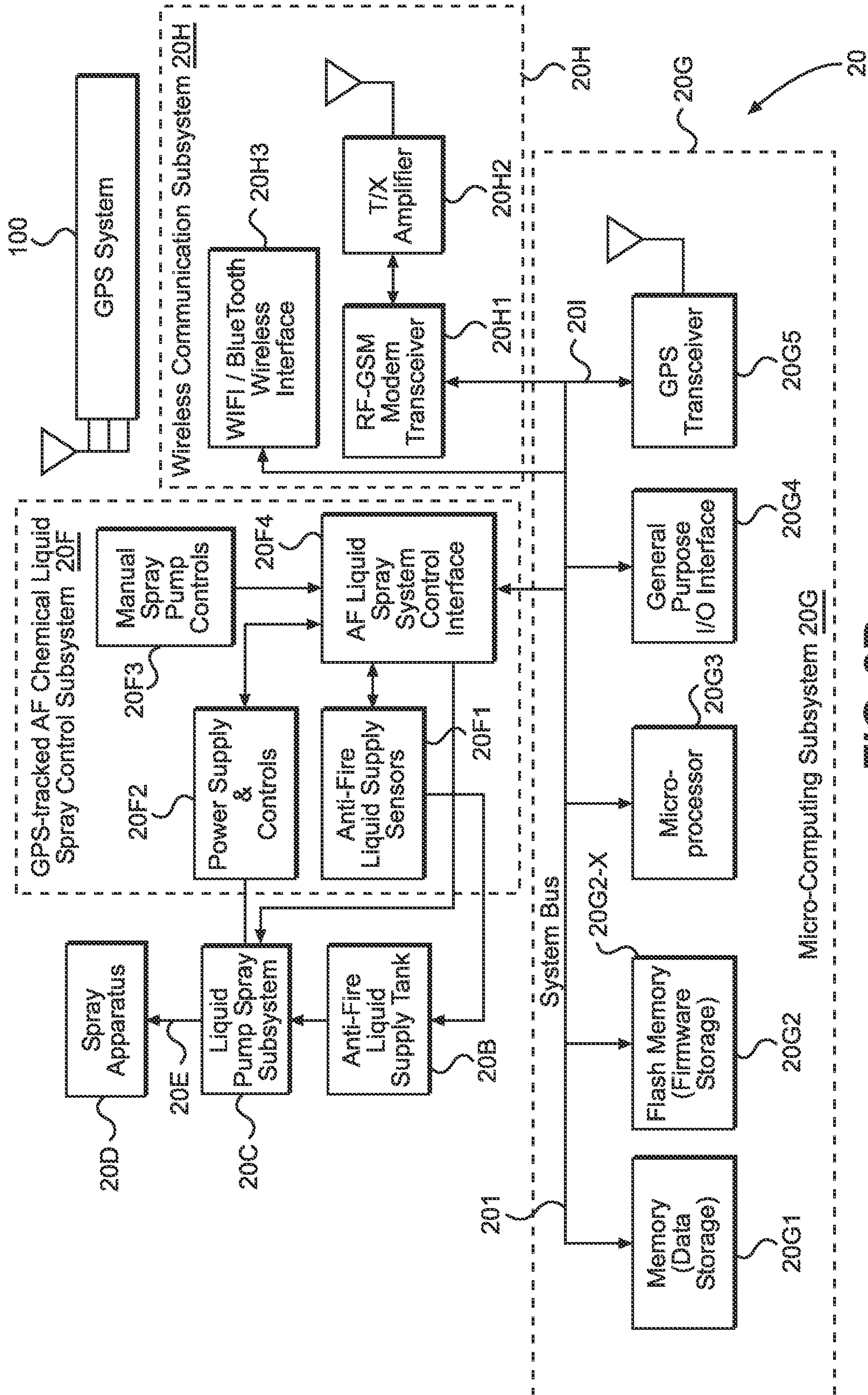
GPS-TRACKED MOBILE ANTI-FIRE (AF) CHEMICAL LIQUID SPRAY SYSTEM



**FIG. 8A**



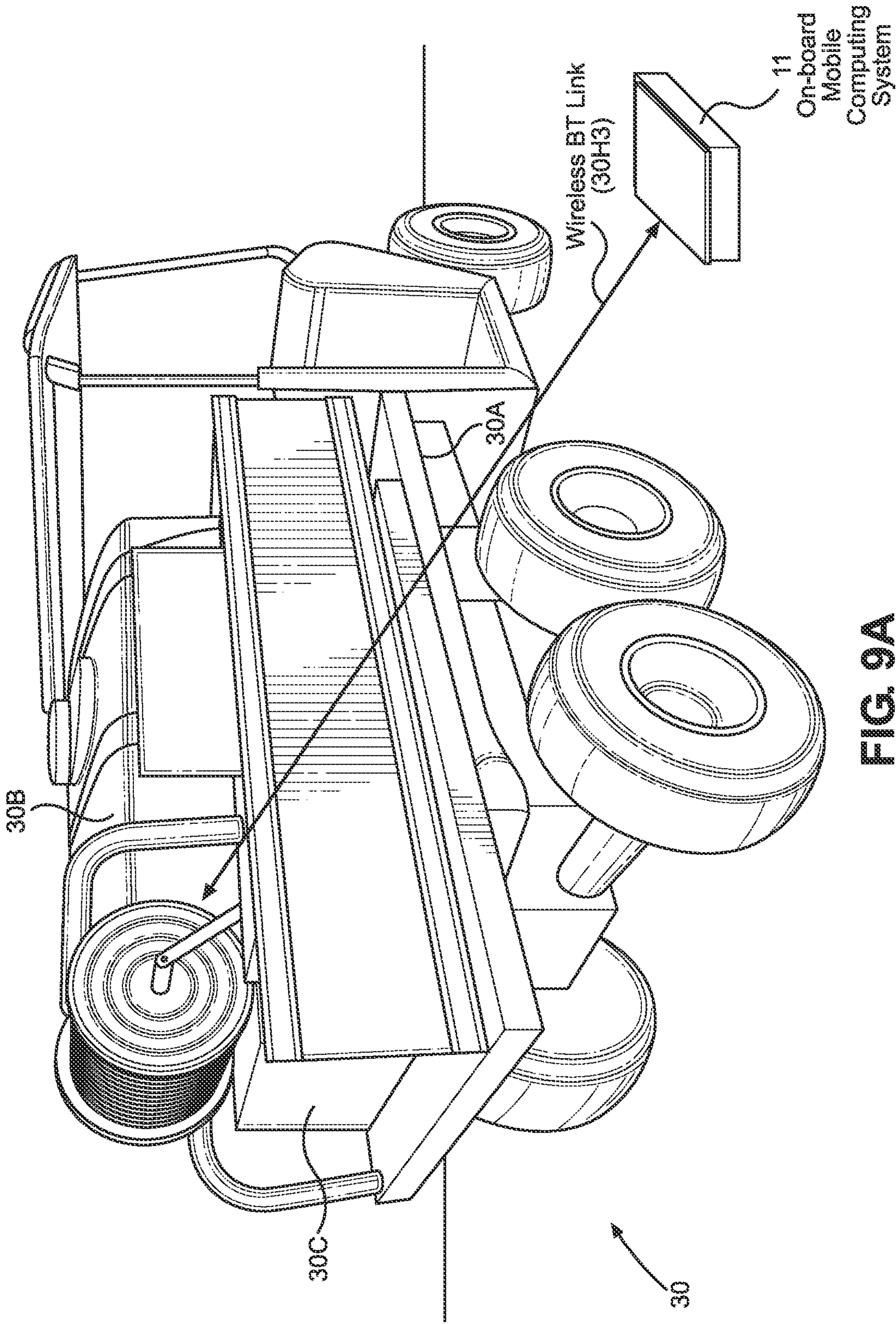
GPS-TRACKED MOBILE ANTI-FIRE (AF) CHEMICAL LIQUID SPRAY SYSTEM



**FIG. 8B**



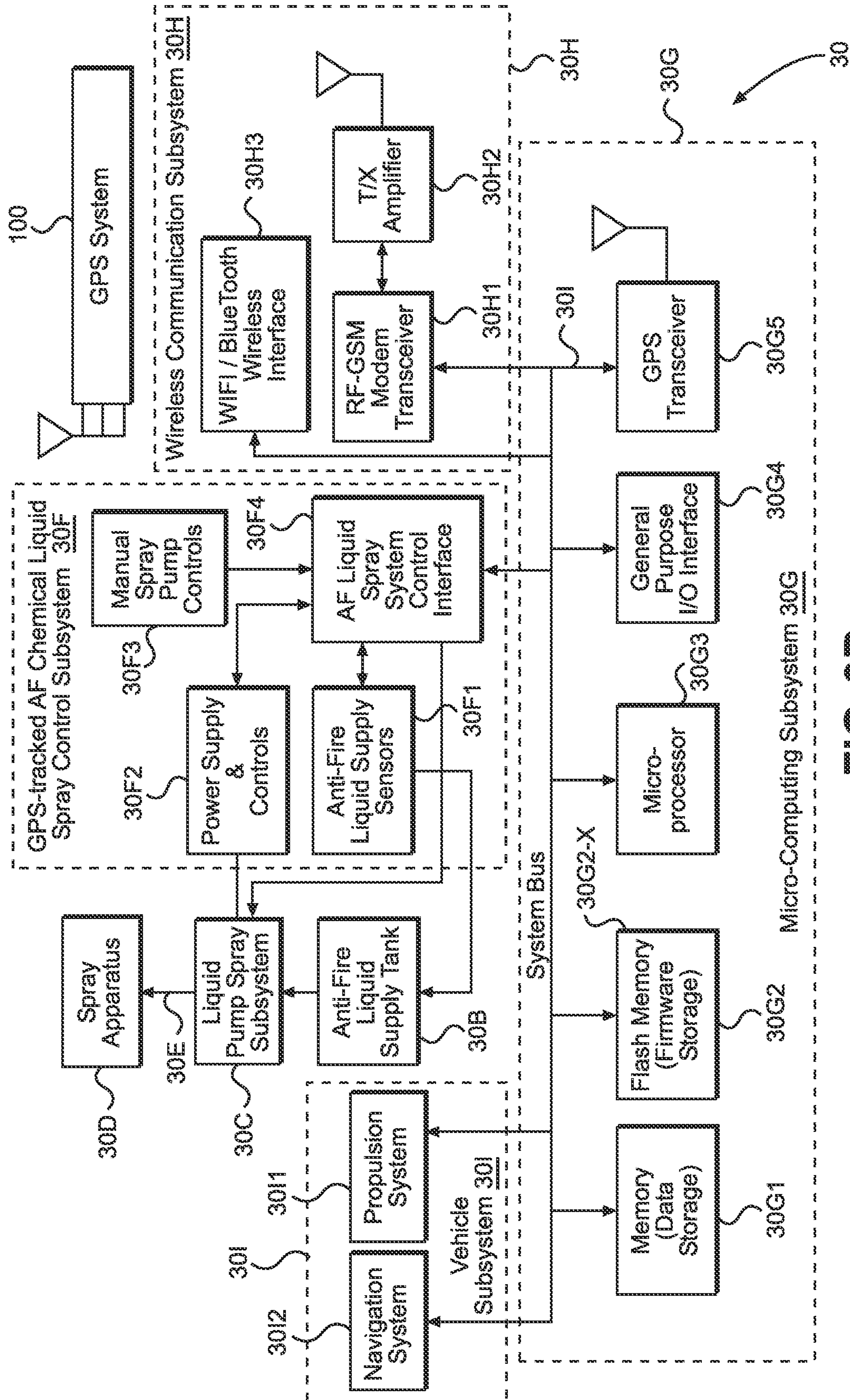
GPS-TRACKED MANNED OR AUTONOMOUS VEHICLE FOR SPRAYING ANTI-FIRE (AF) CHEMICAL LIQUID  
ON GROUND SURFACES TO CREATE FIRE BREAK REGIONS IN HIGH RISK AREAS



**FIG. 9A**



GPS-TRACKED MANNED OR AUTONOMOUS VEHICLE FOR SPRAYING ANTI-FIRE (AF) CHEMICAL LIQUID ON GROUND SURFACES TO CREATE FIRE BREAK REGIONS ON HIGH RISK AREAS



**FIG. 9B**



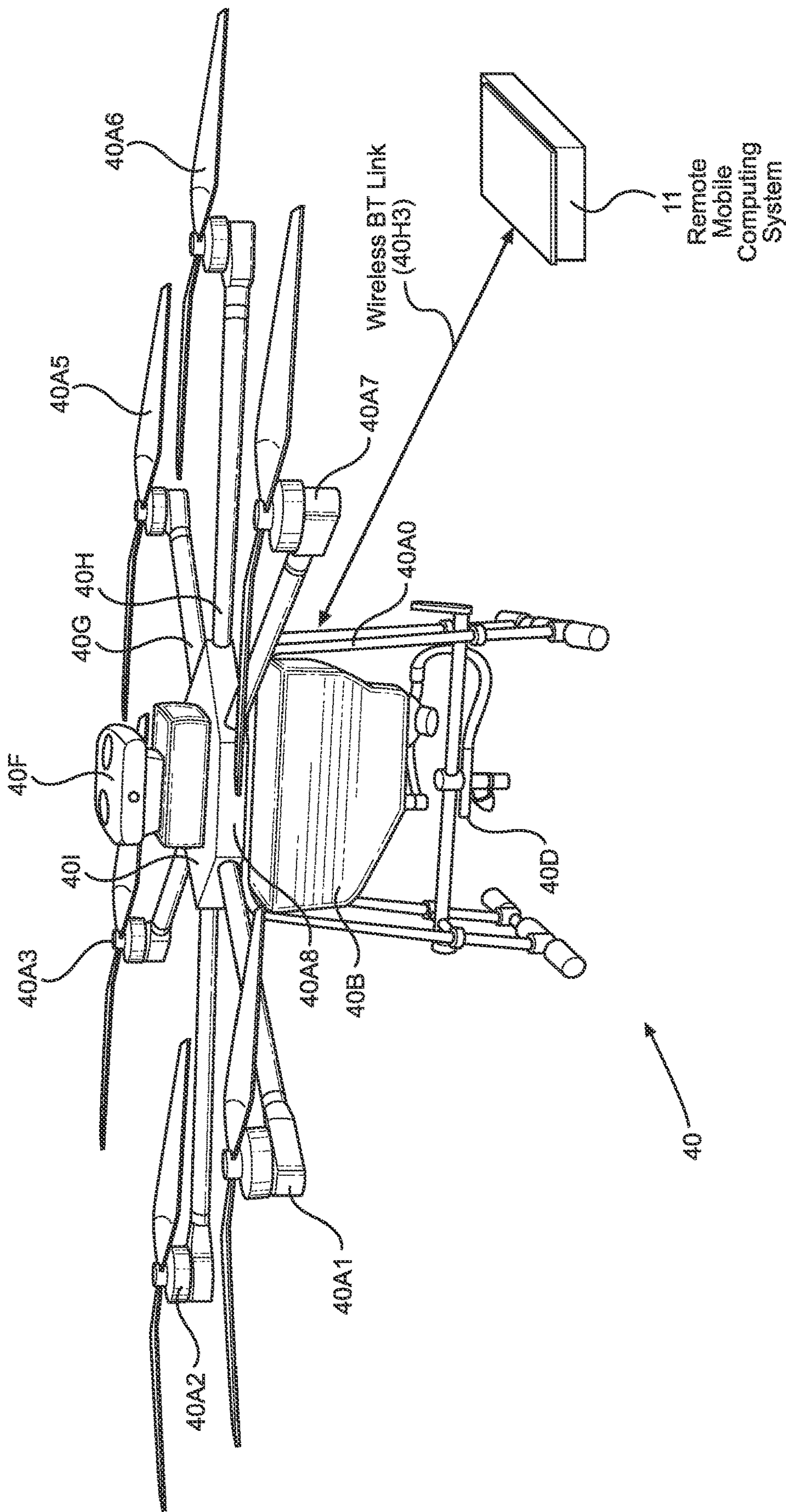


FIG. 10A



AUTONOMOUS (DRONE) AIRCRAFT SYSTEM FOR SPRAYING  
HOUSES, BUILDINGS AND GROUND SURFACES WITH ANTI-FIRE (AF) CHEMICAL LIQUID

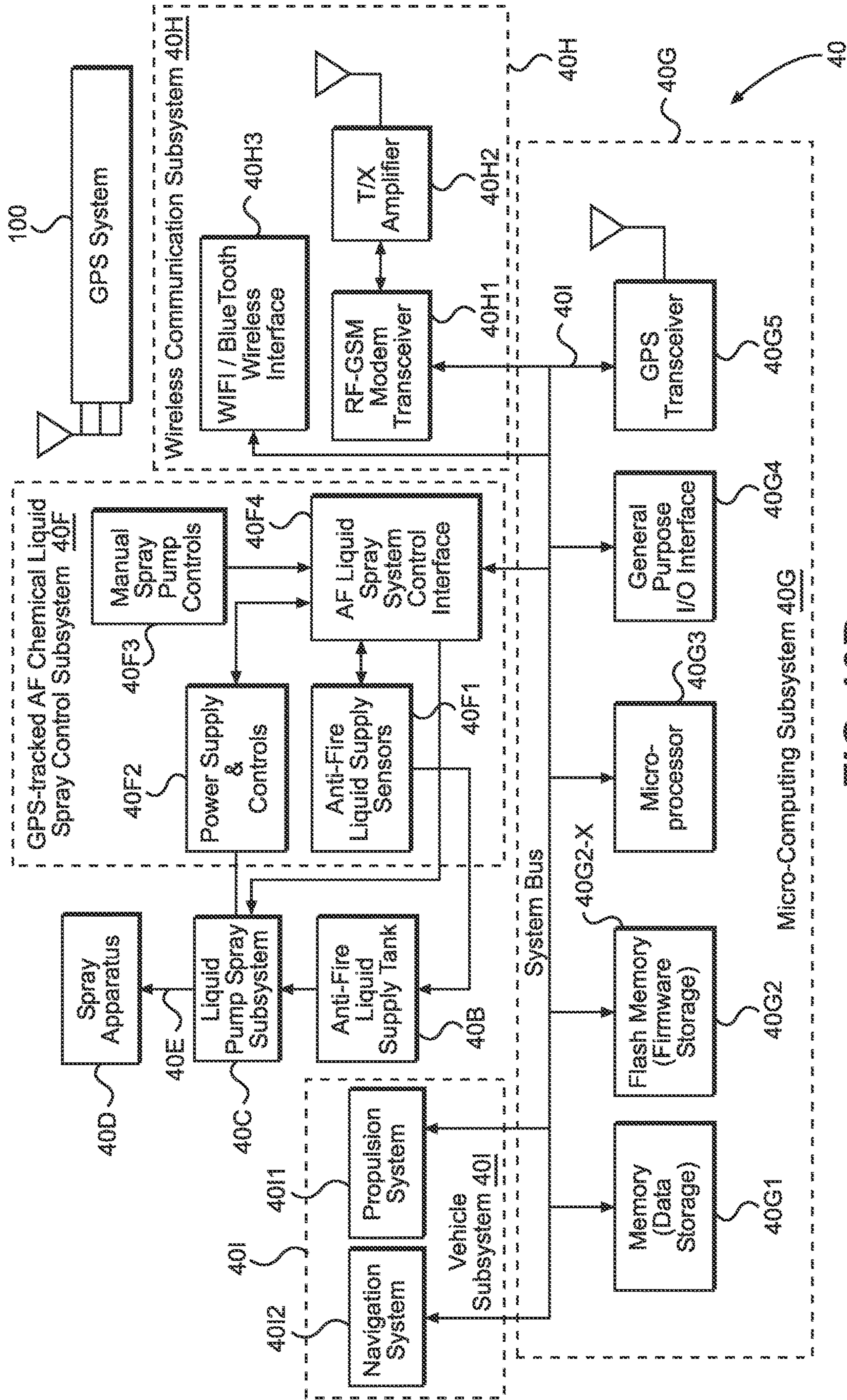


FIG. 10B



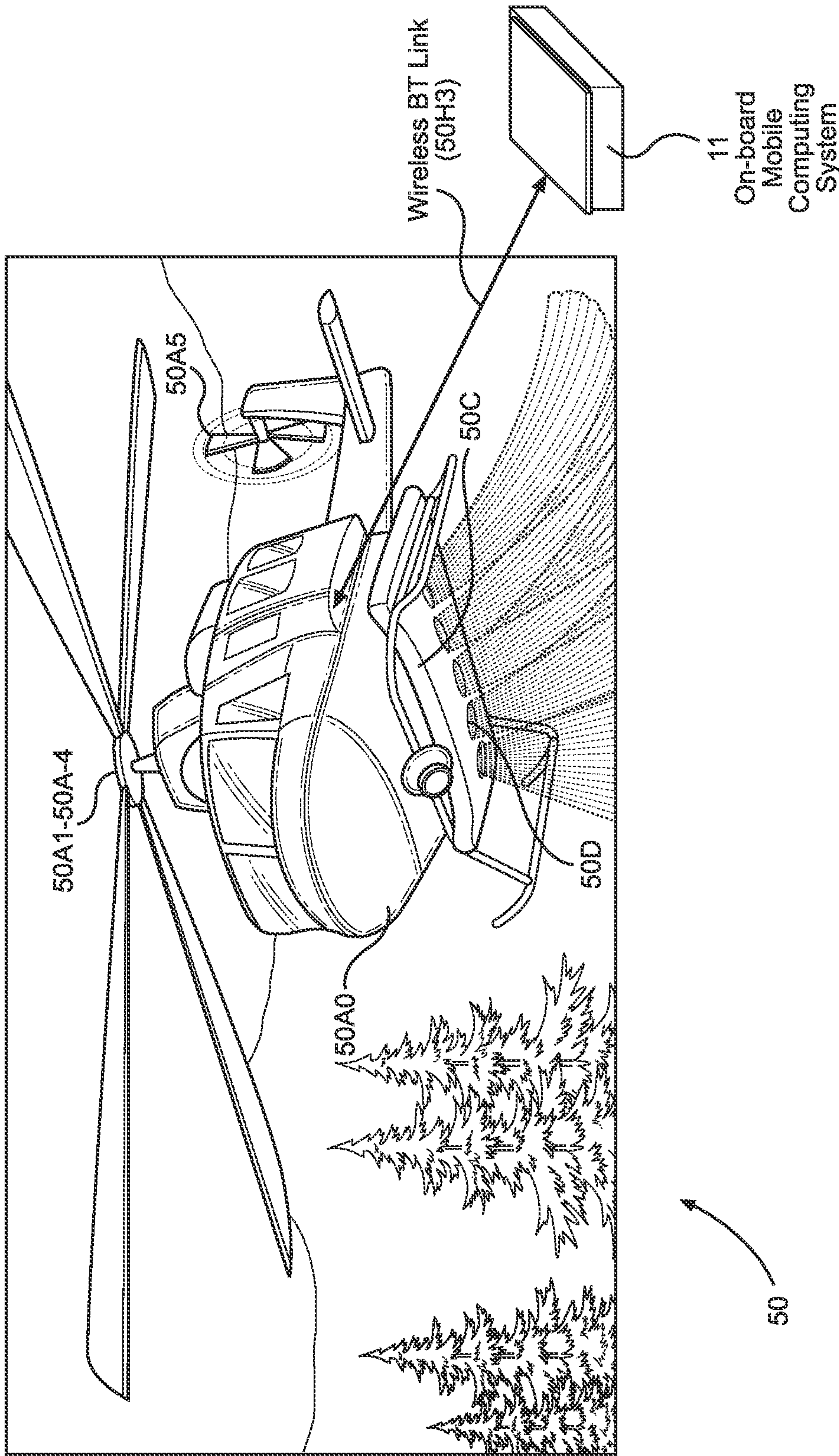


FIG. 11A



GPS-TRACKED (HELICOPTER) AIRCRAFT SYSTEM FOR SPRAYING FORESTS  
GROUND SURFACES WITH ANTI-FIRE (AF) CHEMICAL LIQUID

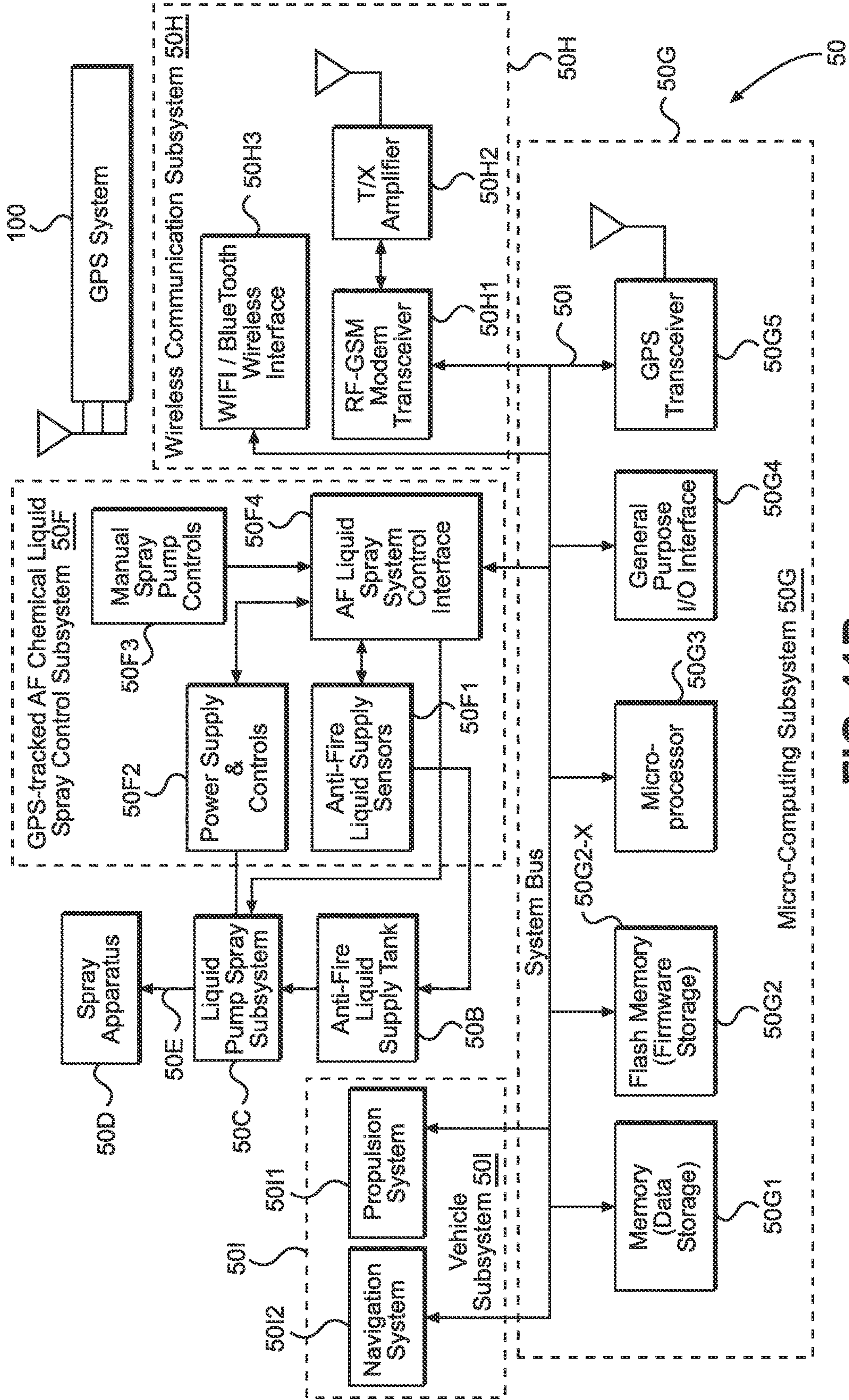
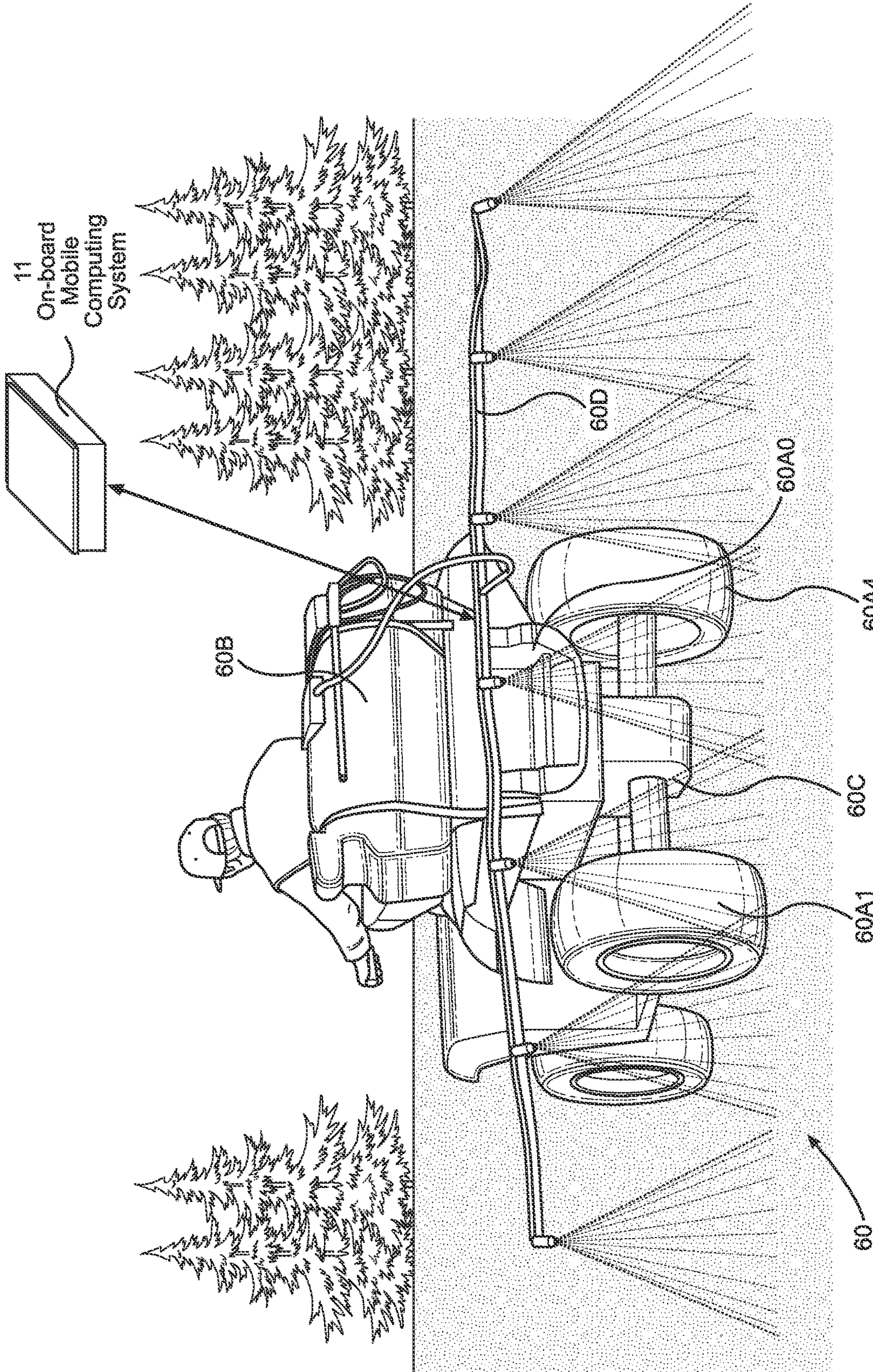


FIG. 11B



**GPS-TRACKED ALL TERRAIN VEHICLE (ATV) SYSTEM FOR SPRAYING  
GROUND SURFACES WITH ANTI-FIRE (AF) CHEMICAL LIQUID**



**FIG. 12A**



GPS-TRACKED ALL TERRAIN VEHICLE (ATV) SYSTEM FOR SPRAYING  
GROUND SURFACES WITH ANTI-FIRE (AF) CHEMICAL LIQUID

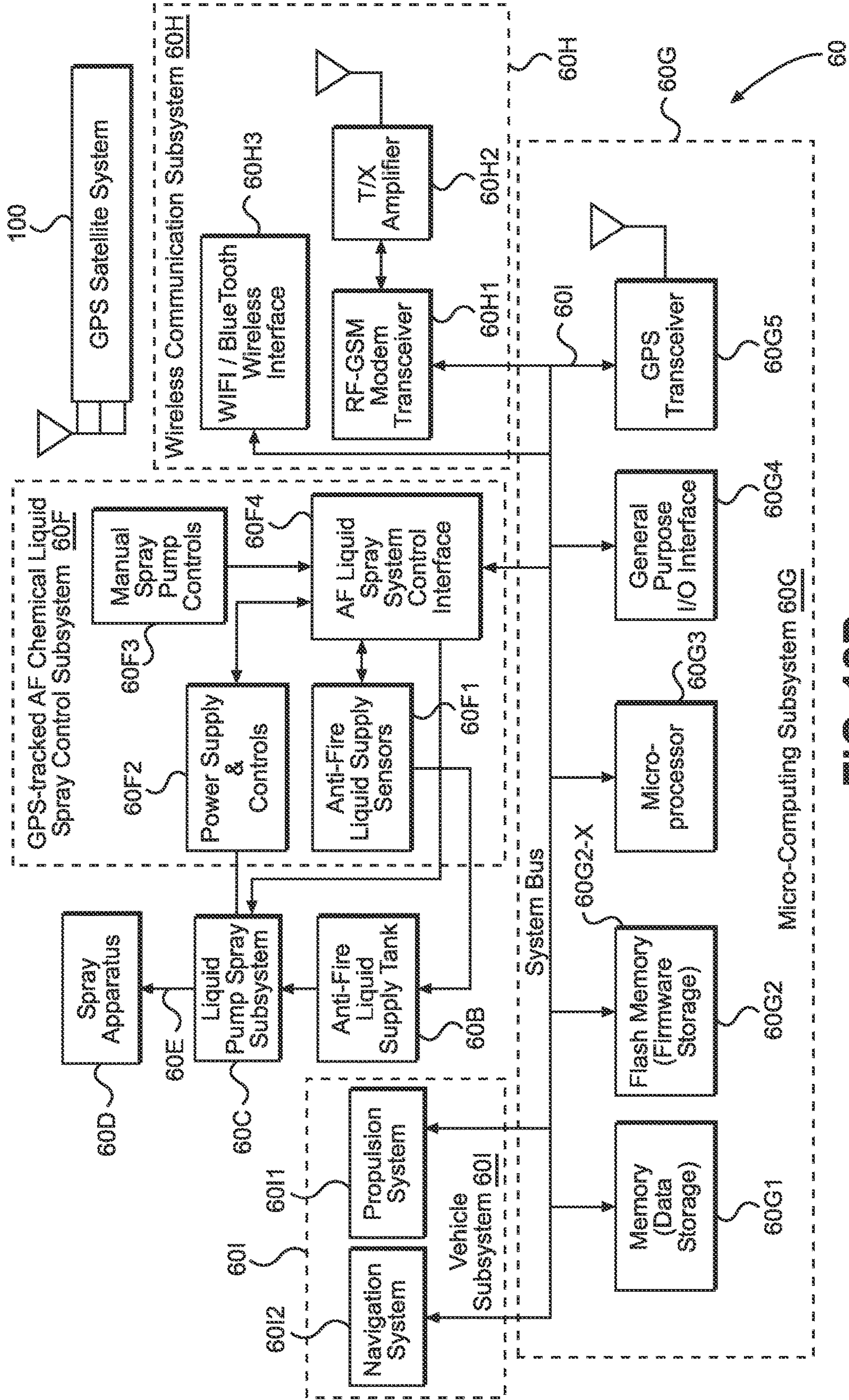
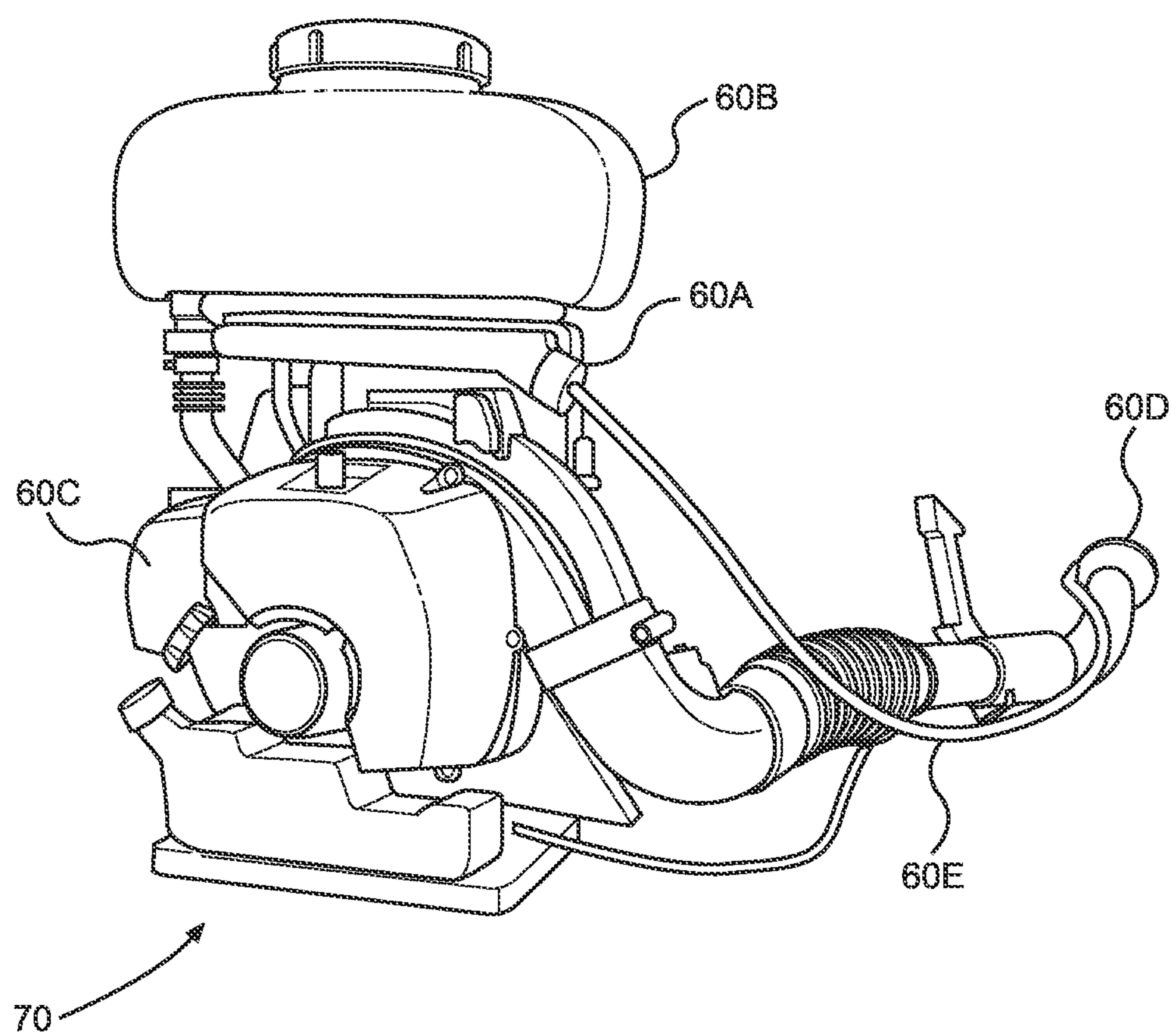


FIG. 12B



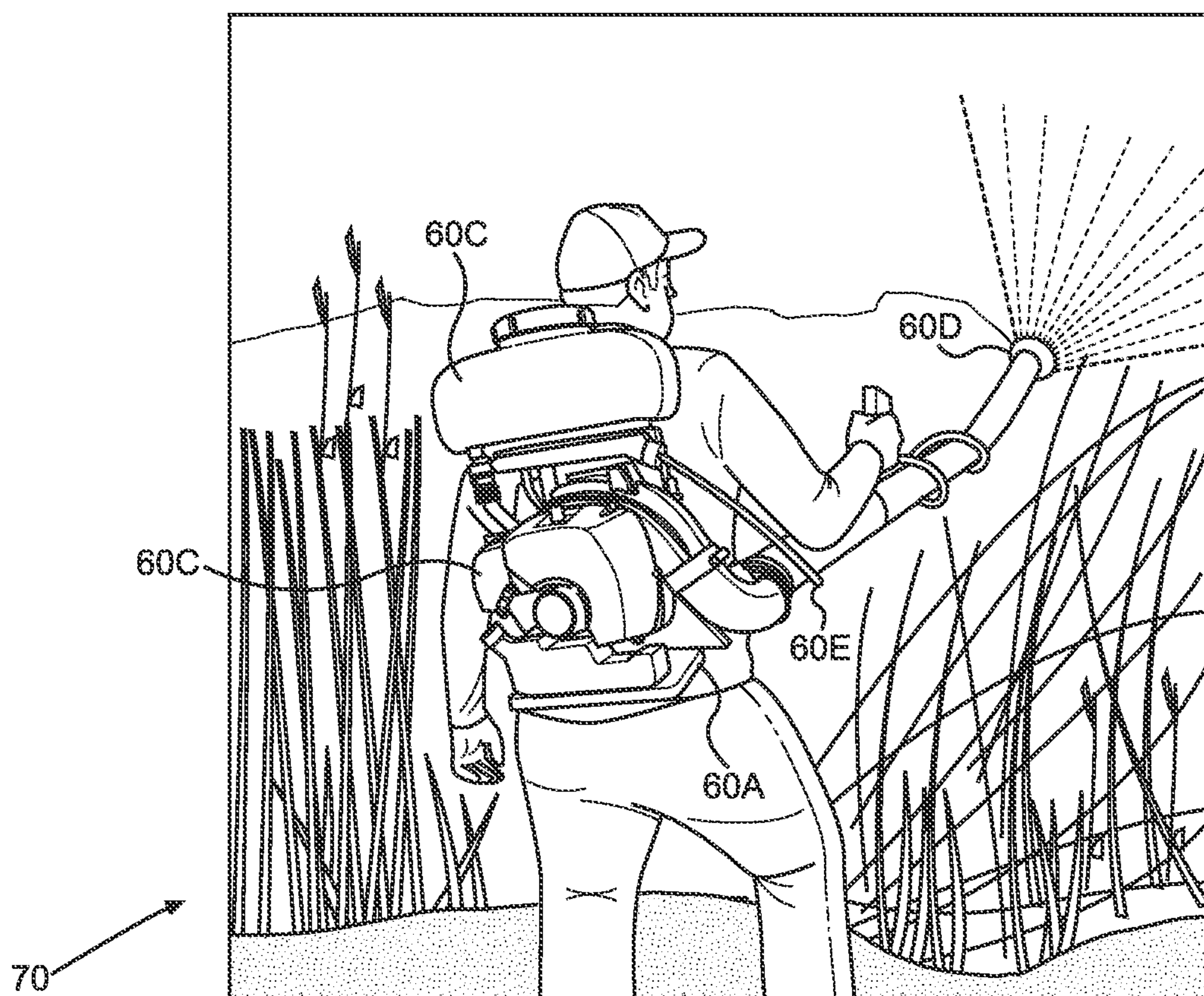
GPS-TRACKING BACKPACK-MOUNTED ATOMIZING SPRAY CANNONS



**FIG. 13A**



GPS-TRACKING BACKPACK-MOUNTED ATOMIZING SPRAY CANNONS



**FIG. 13B**



GPS-TRACKED BACKPACK-MOUNTED ATOMIZING SPRAY CANNONS FOR SPRAYING  
GROUND SURFACES WITH ANTI-FIRE (AF) CHEMICAL LIQUID

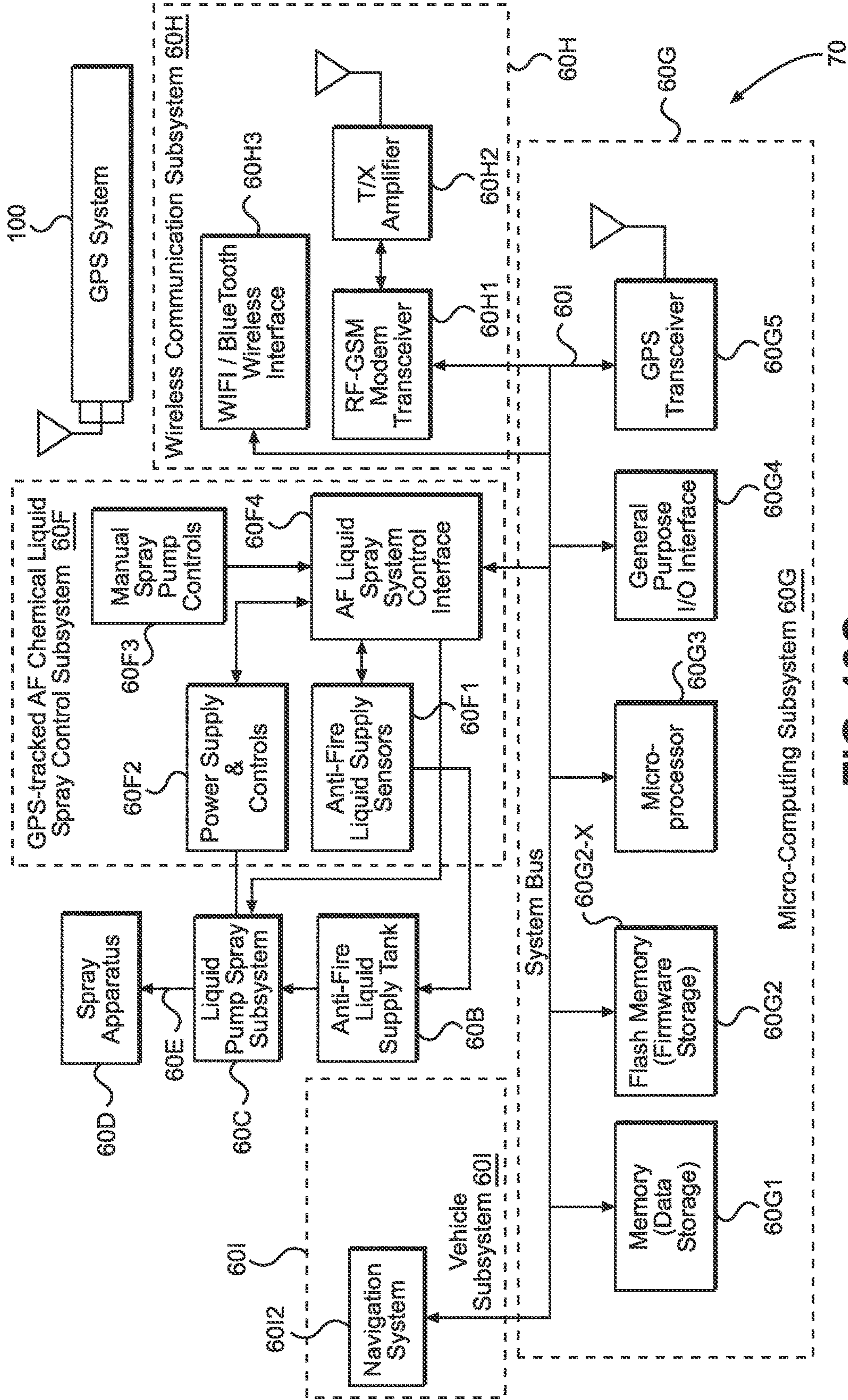


FIG. 13C



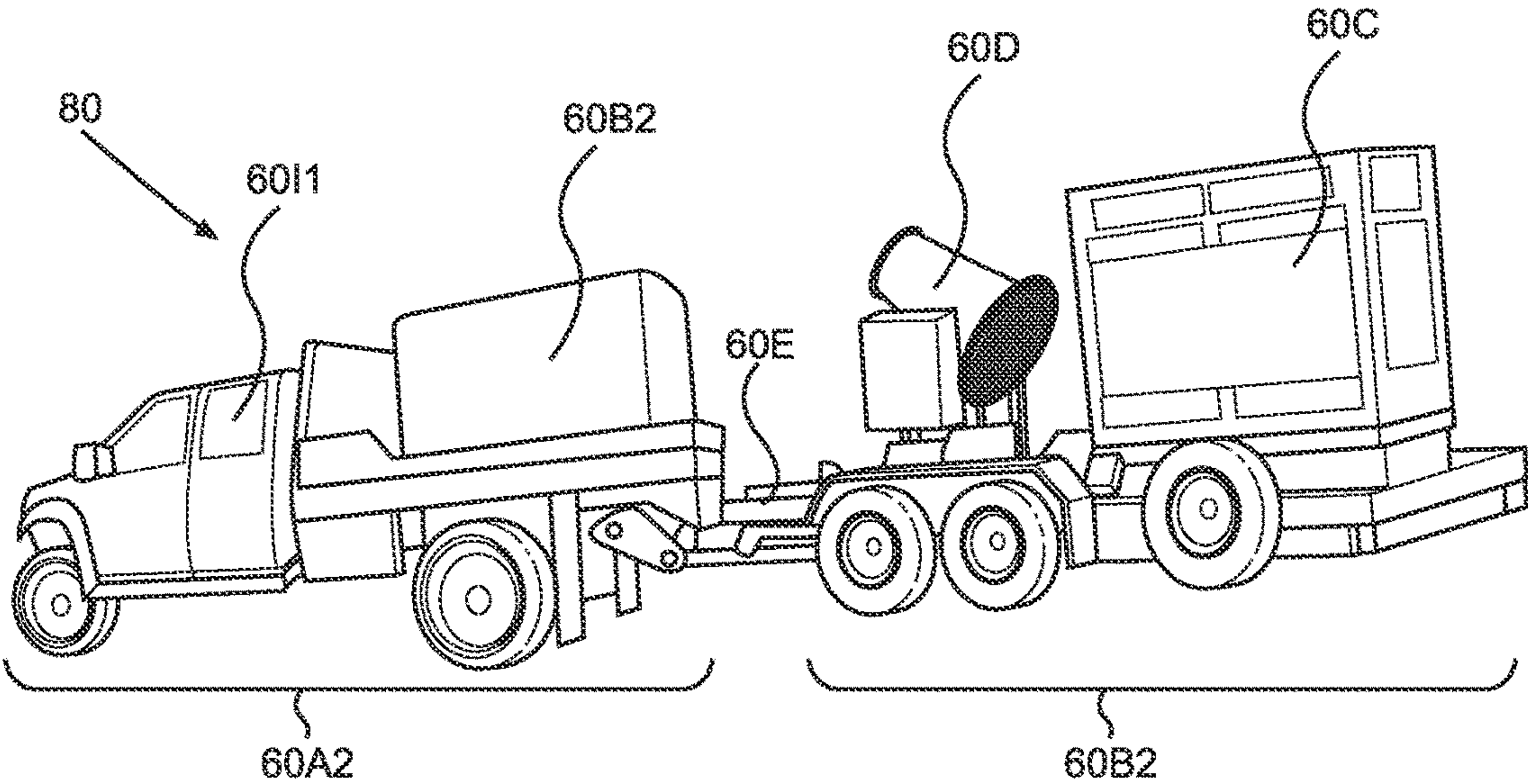


FIG. 14A

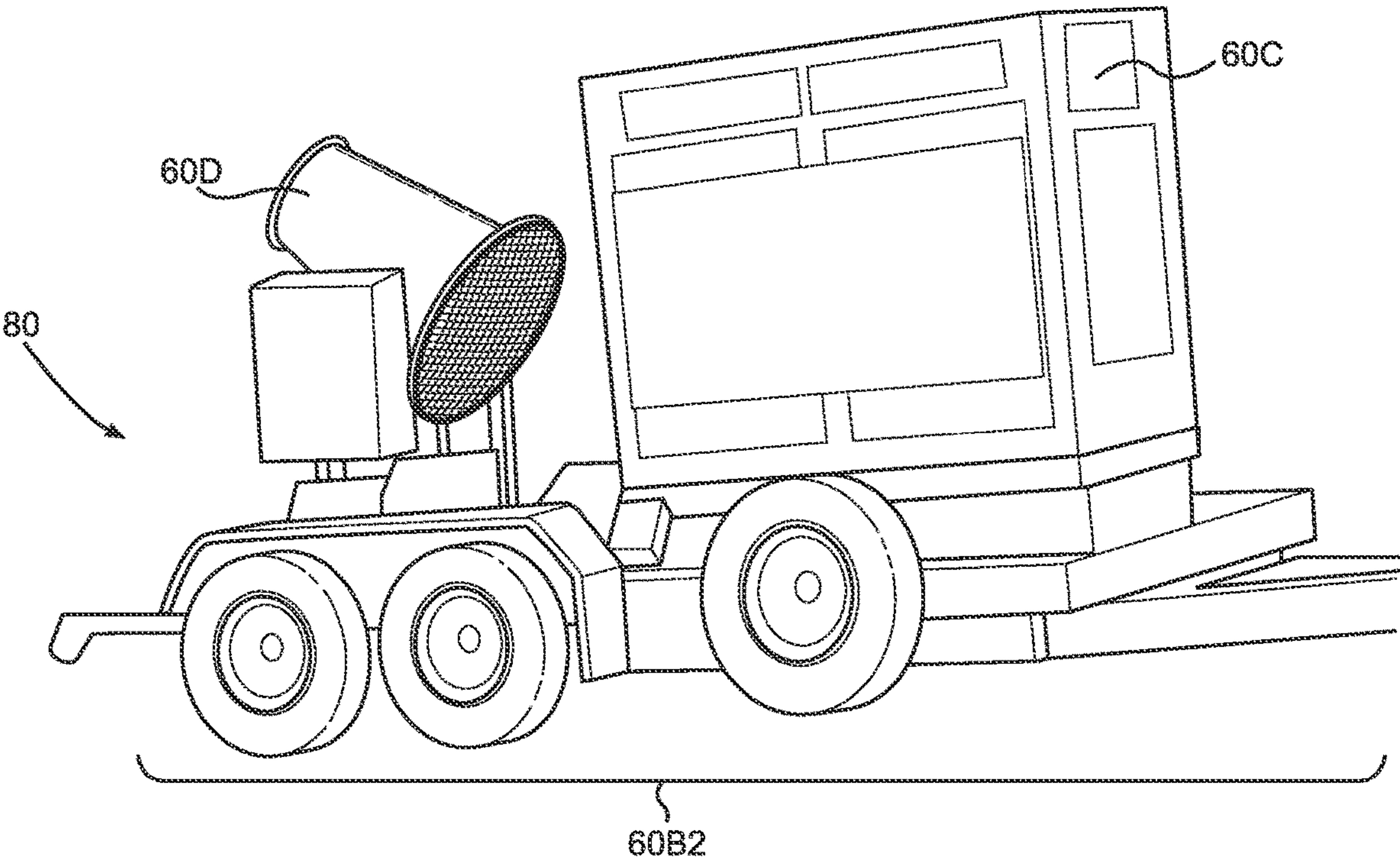
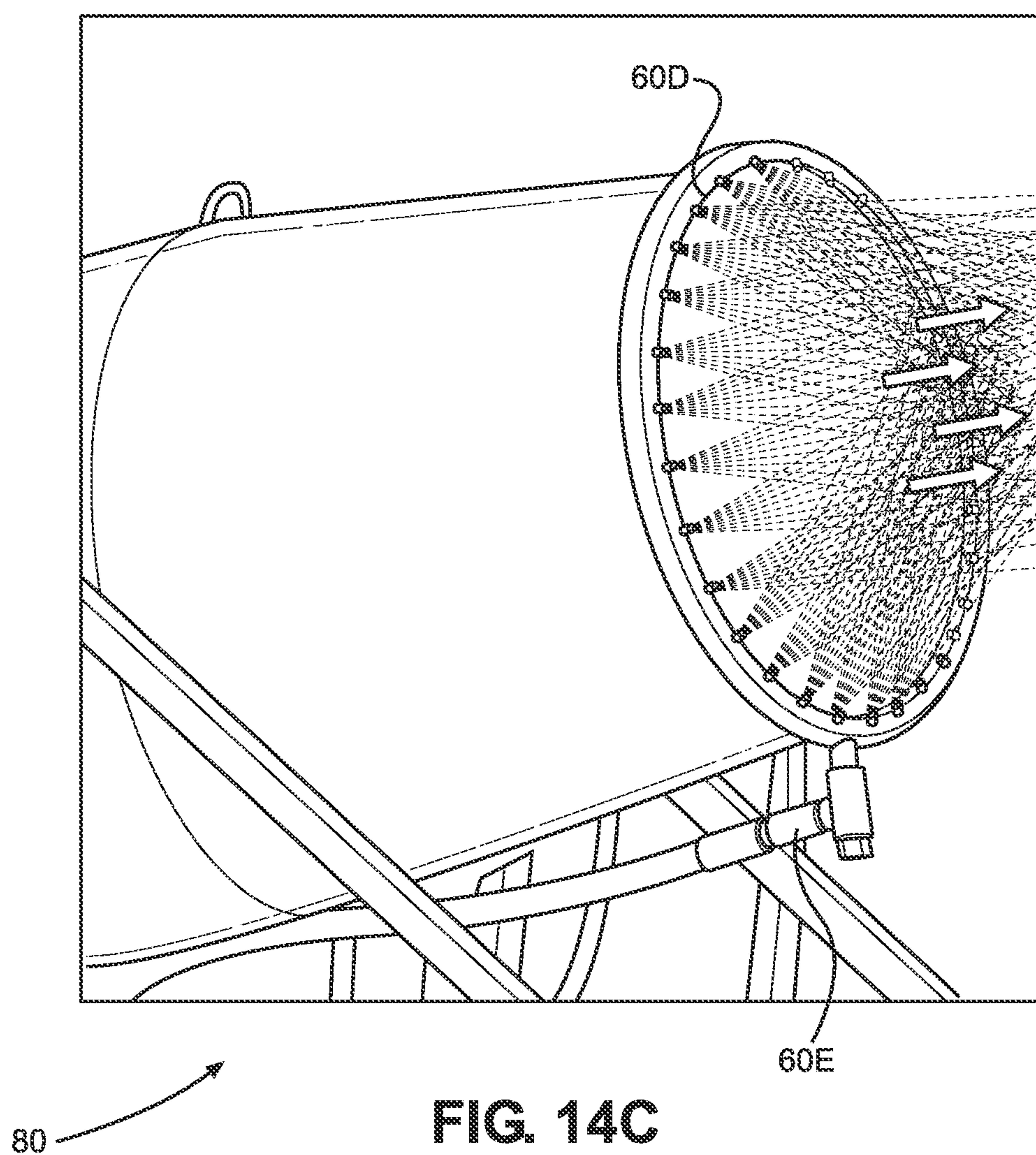


FIG. 14B



ATOMIZING LIQUID SPRAY MICRODROPLETS ENTRAINED IN A FORCEFUL AIR  
STREAM PROJECTED OUT OF THE TURBINE-FAN DRIVEN CANNON STRUCTURE





GPS-TRACKED ALL TERRAIN VEHICLE (ATV) SYSTEM FOR SPRAYING  
GROUND SURFACES WITH LIQUID ANTI-FIRE (AF) CHEMICAL LIQUID

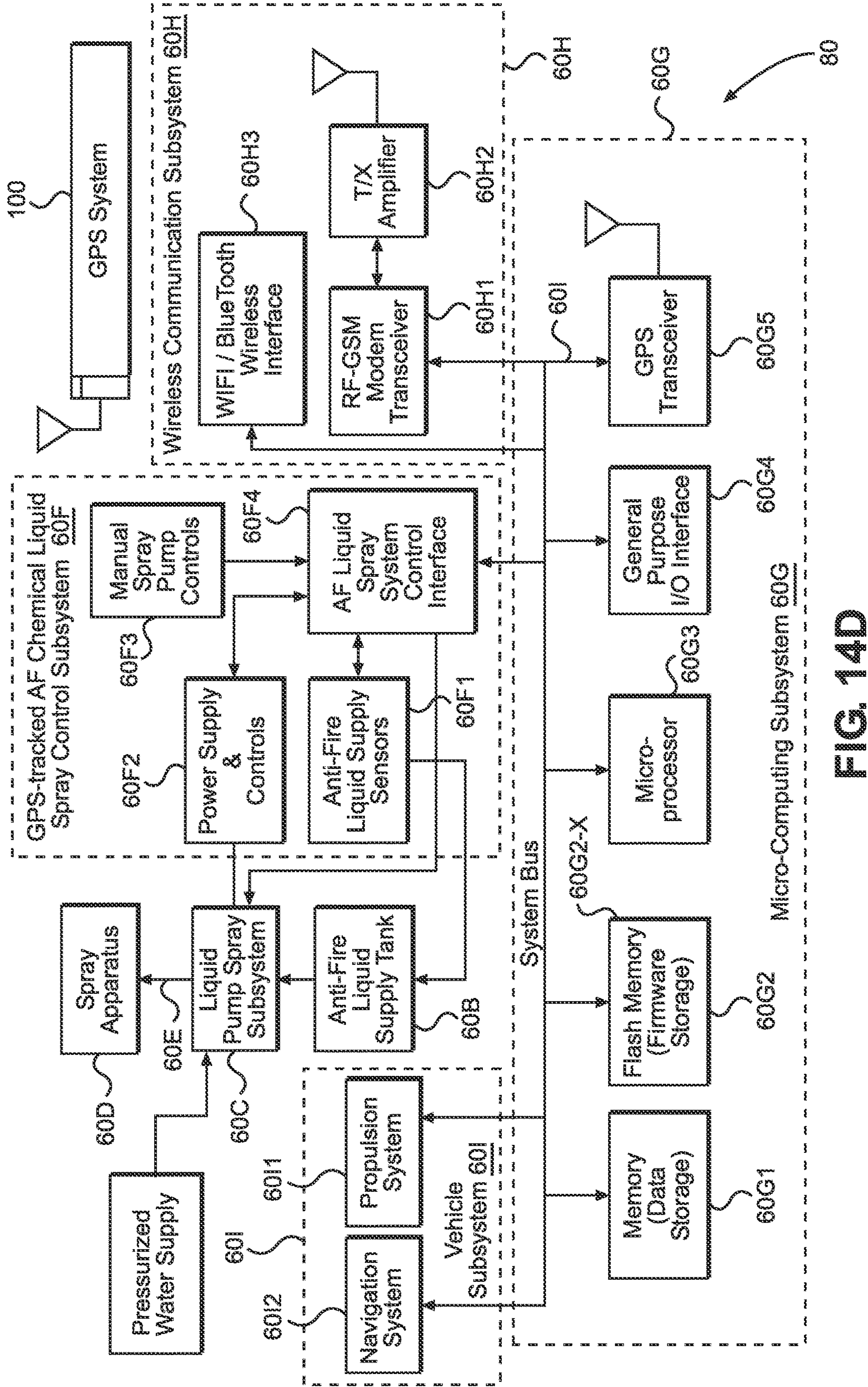
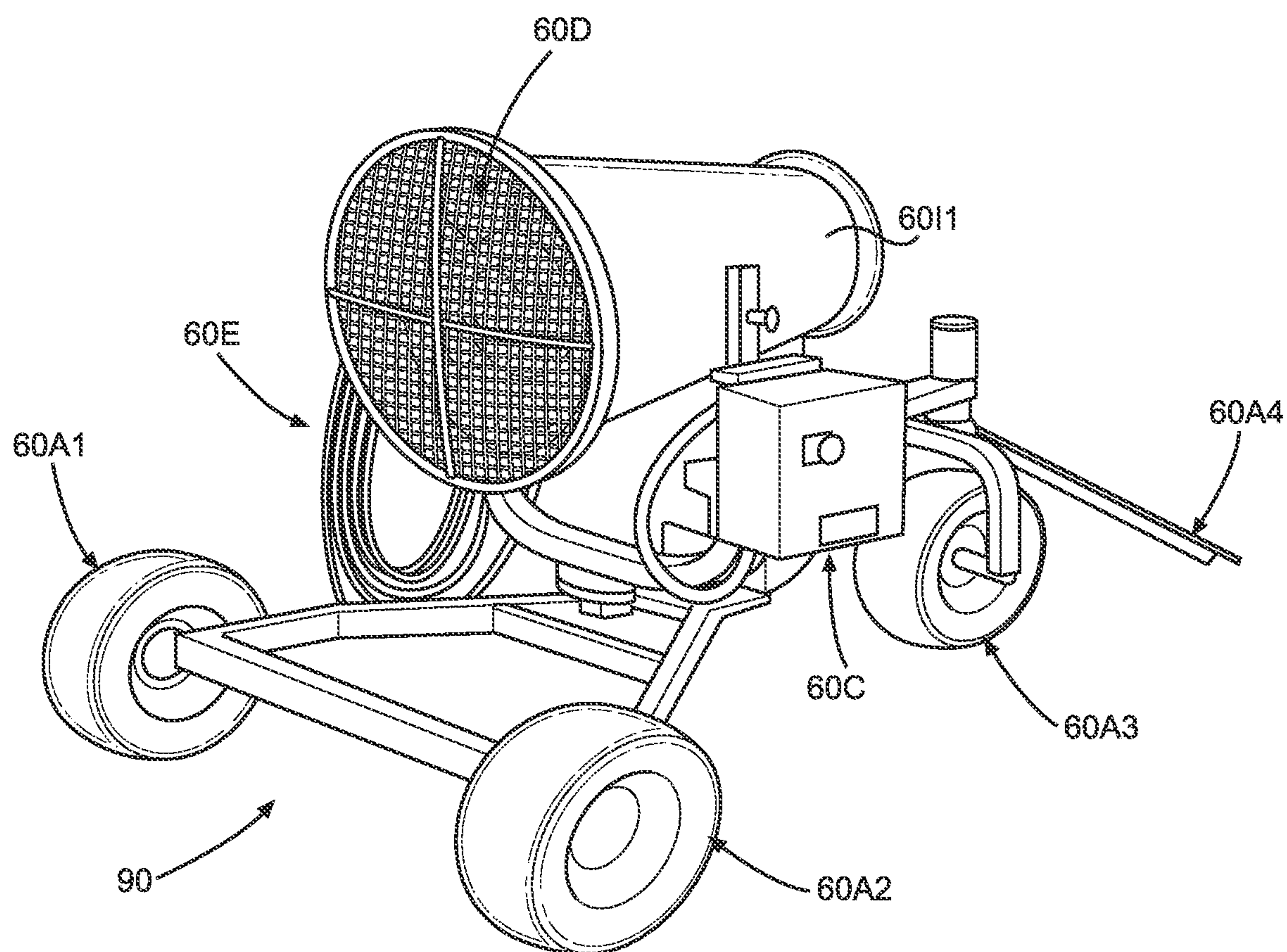


FIG. 14D



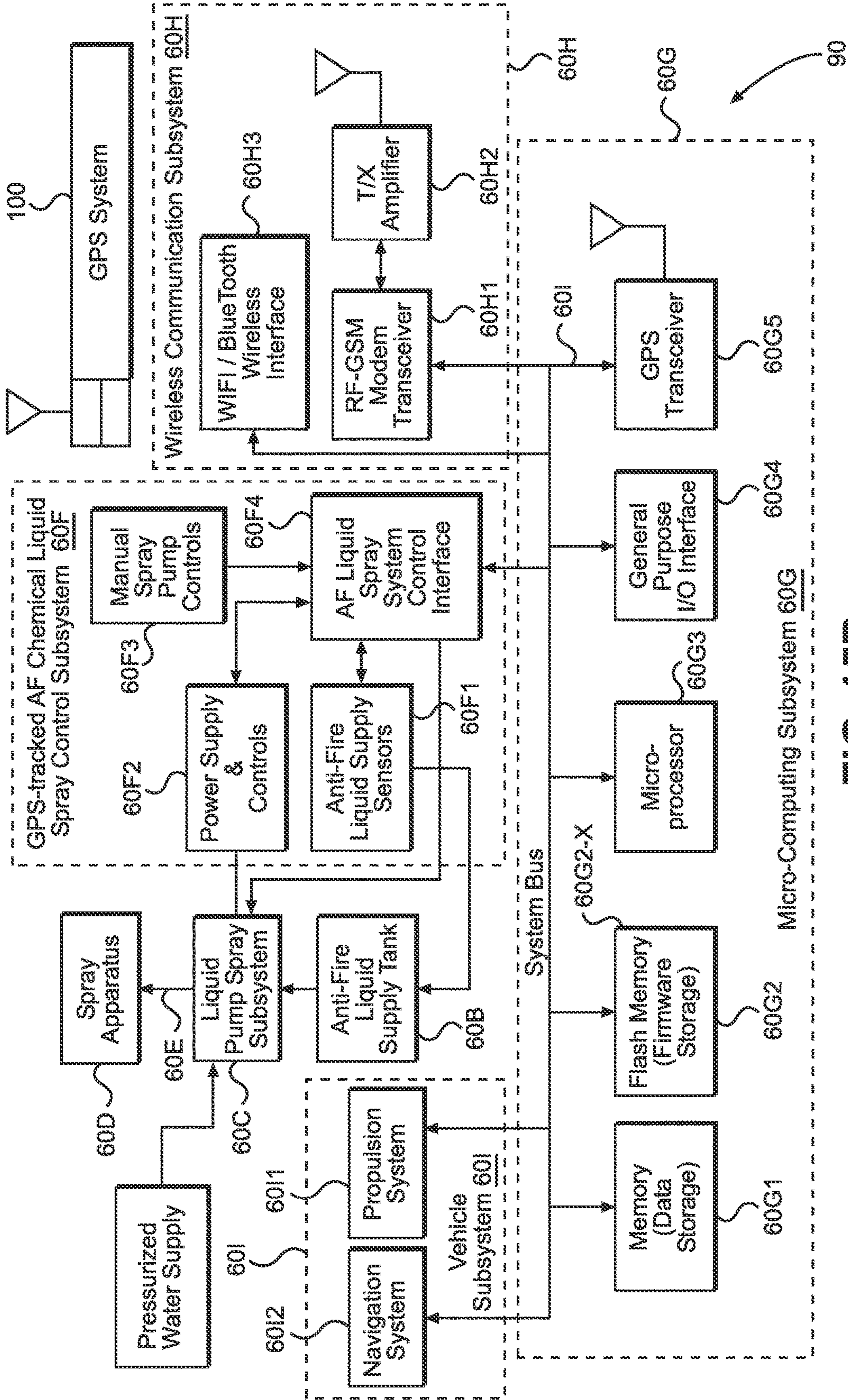
ELECTRIC TURBINE FAN DRIVEN CANNON SPRAYING SYSTEM



**FIG. 15A**



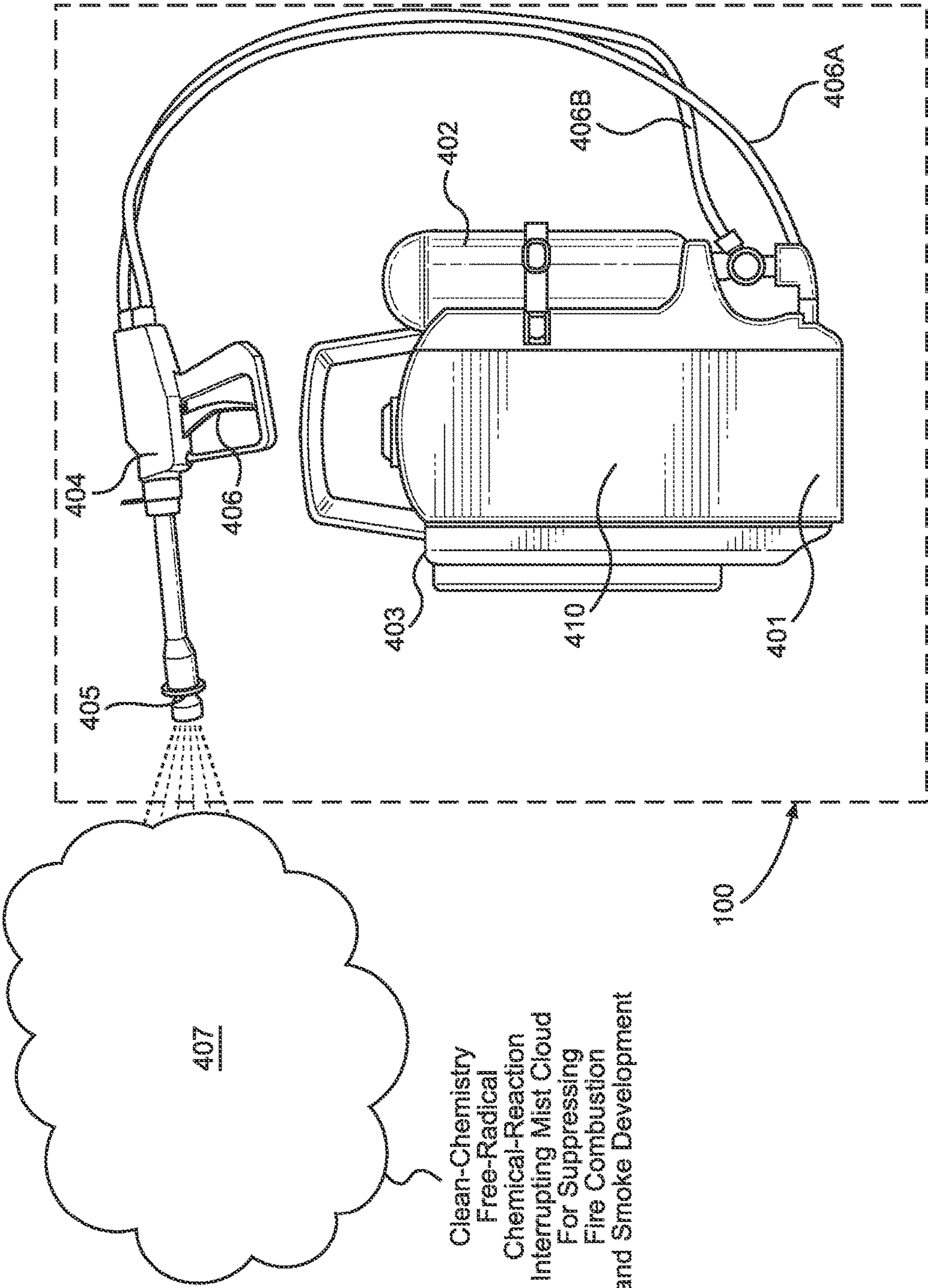
**GPS-TRACKED ALL TERRAIN VEHICLE (ATV) SYSTEM FOR SPRAYING  
GROUND SURFACES WITH LIQUID ANTI-FIRE (AF) CHEMICAL LIQUID**



**FIG. 15B**



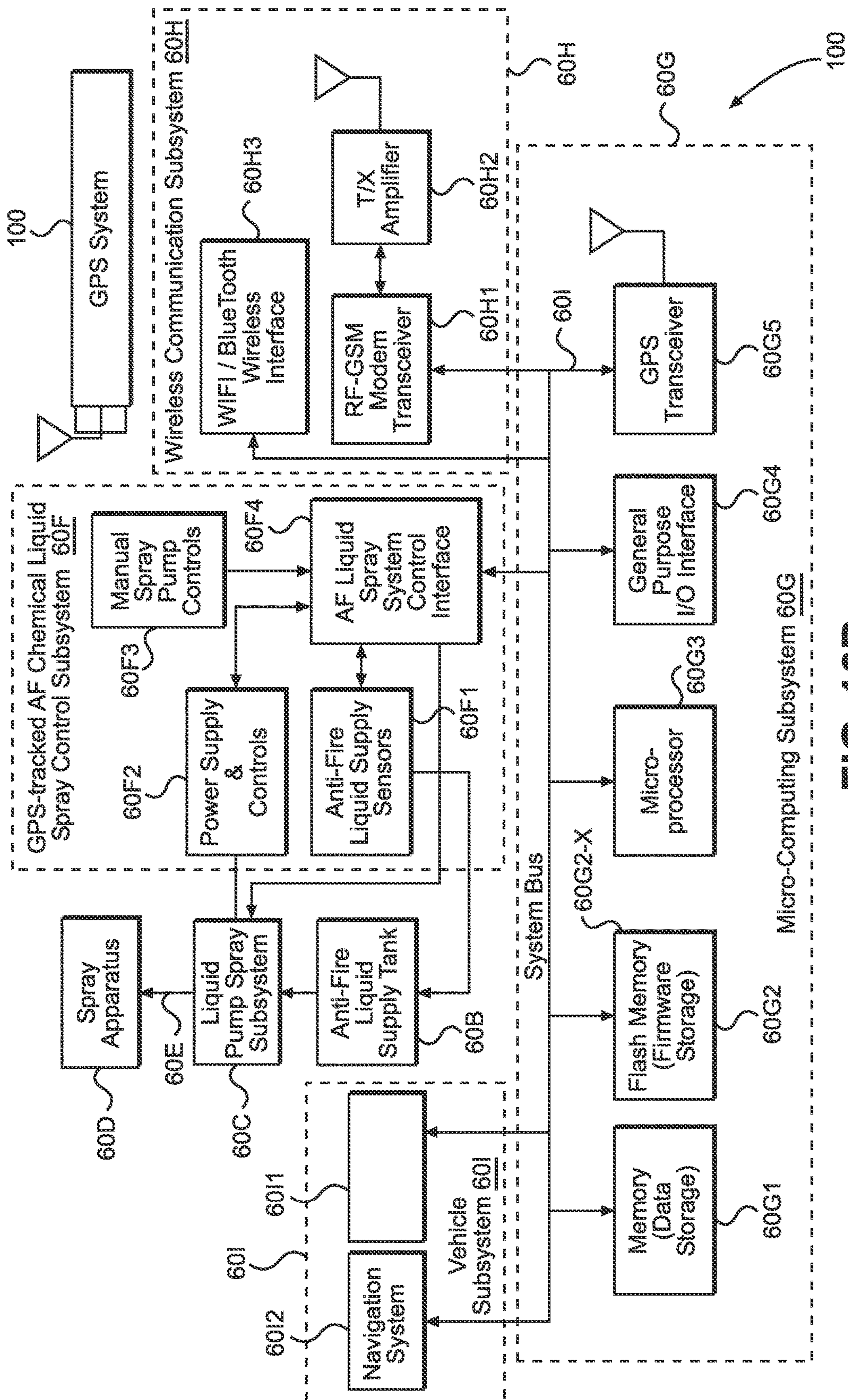
PORTABLE FIRE EXTINGUISHER FOR PRODUCING CLEAN-CHEMISTRY FREE-RADICAL  
CHEMICAL-REACTION INTERRUPTING MISTING CLOUD FOR FIRE SUPPRESSION



**FIG. 16A**

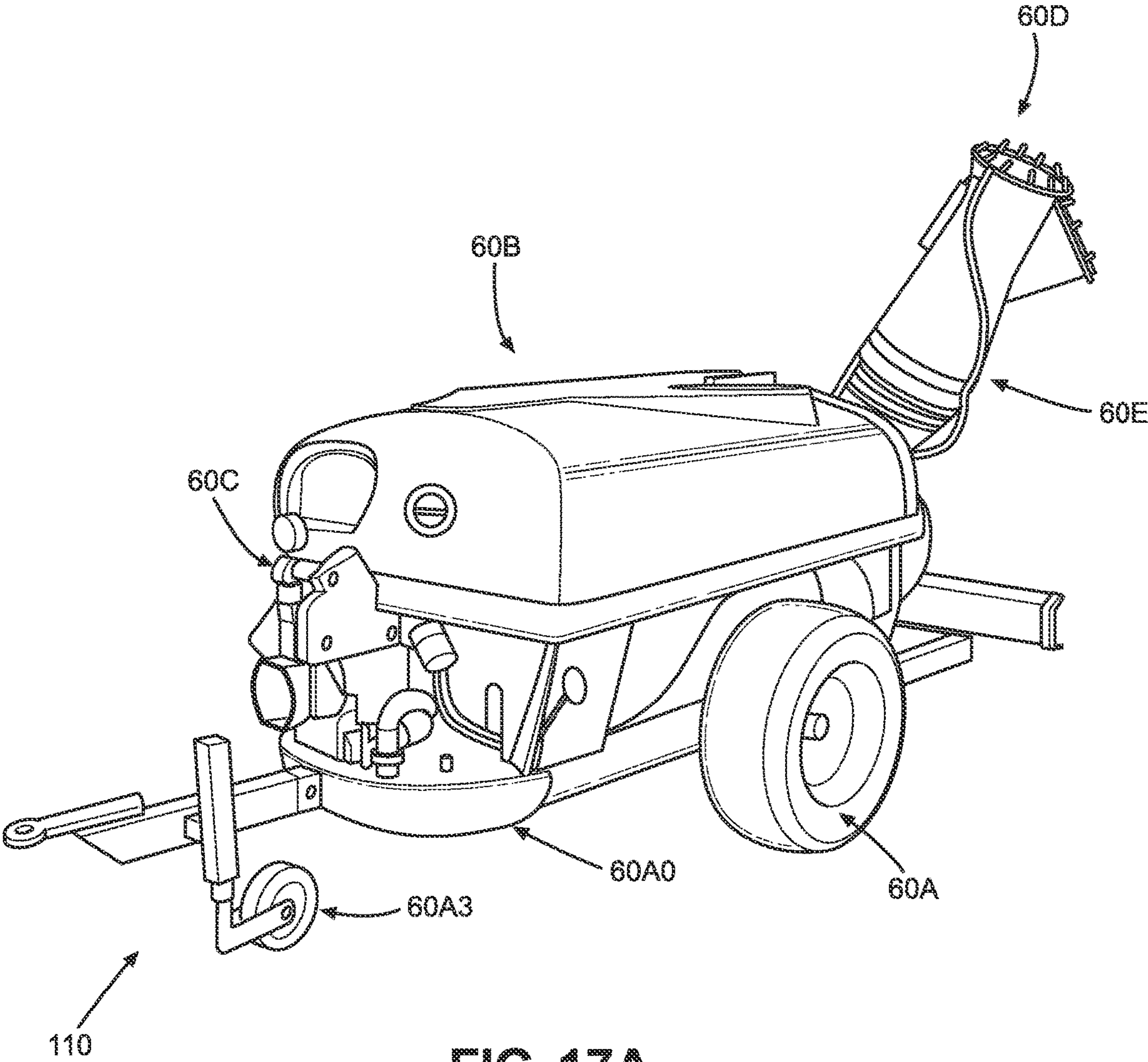


# GPS-TRACKED FIRE EXTINGUISHING SYSTEM FOR SPRAYING SURFACES WITH ANTI-FIRE (AF) CHEMICAL LIQUID



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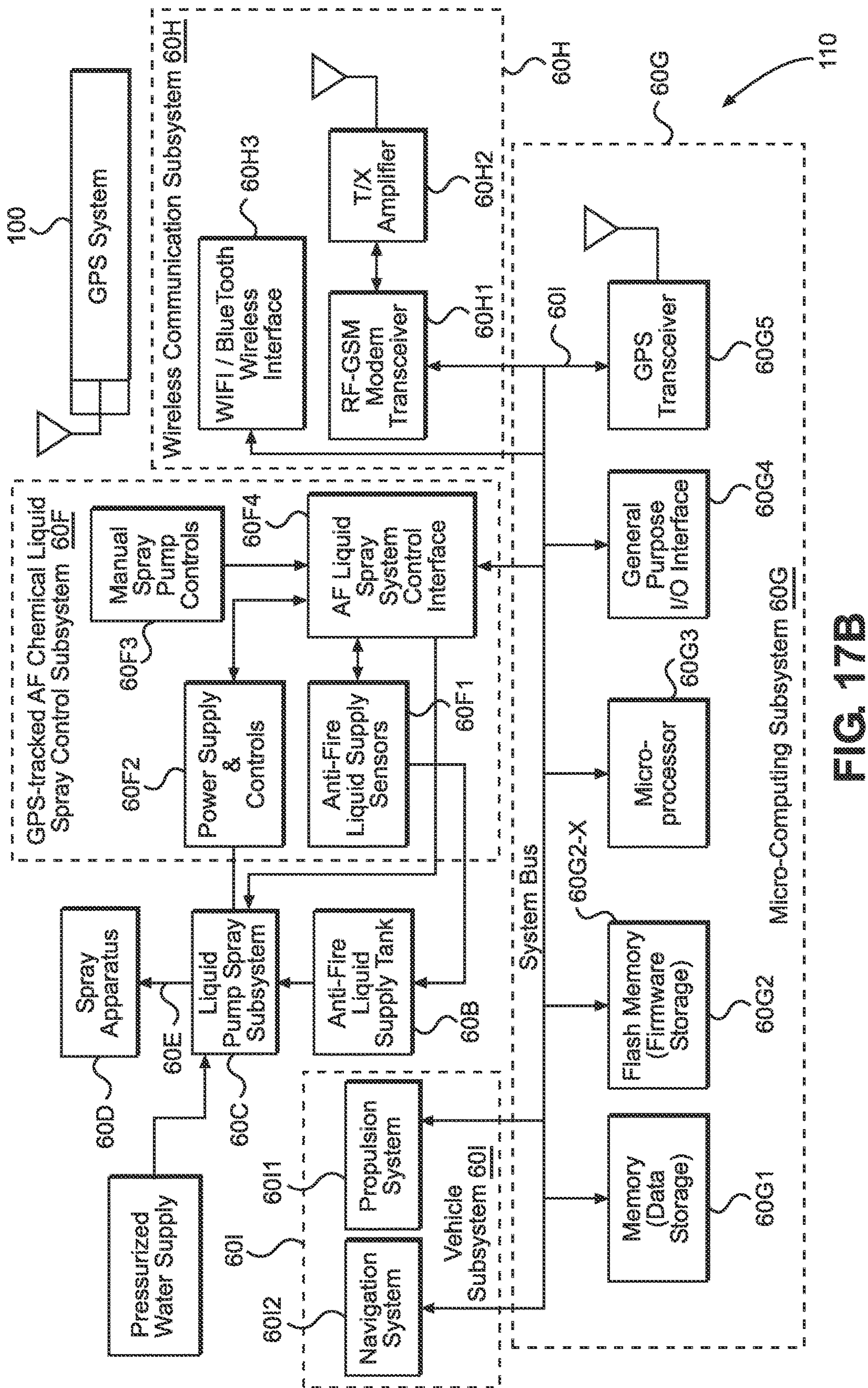




**FIG. 17A**



**GPS-TRACKED REMOTELY-CONTROLLABLE SPRAY ATOMIZING CANNON SYSTEM  
FOR SPRAYING ANTI-FIRE (AF) CHEMICAL LIQUID**



A vertical sequence of six images showing the progression of a handwritten letter 'L' from a simple dot to a fully formed character.



SCHEMA FOR NETWORK DATABASE SUPPORTED BY SYSTEM NETWORK OF THE PRESENT INVENTION

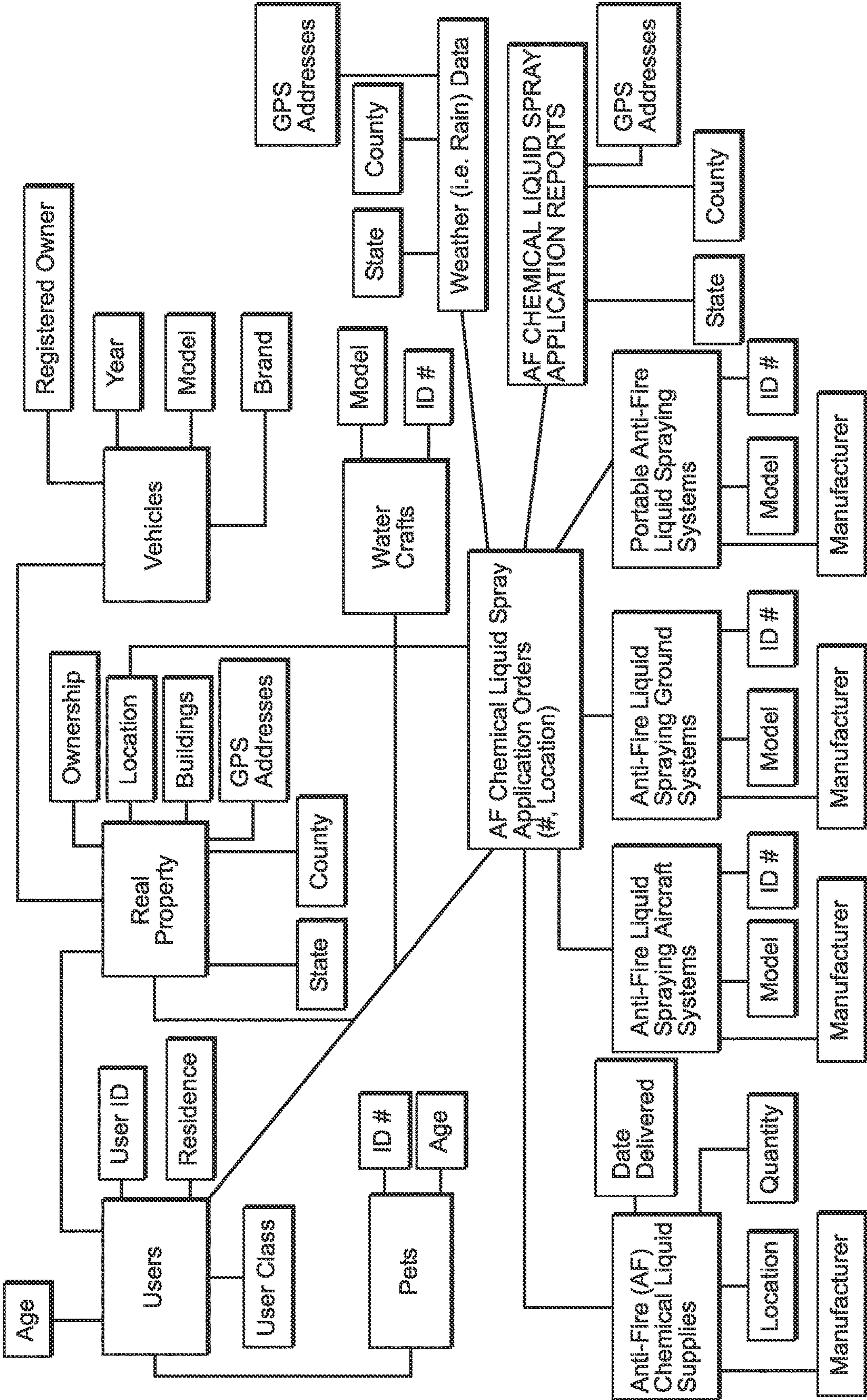
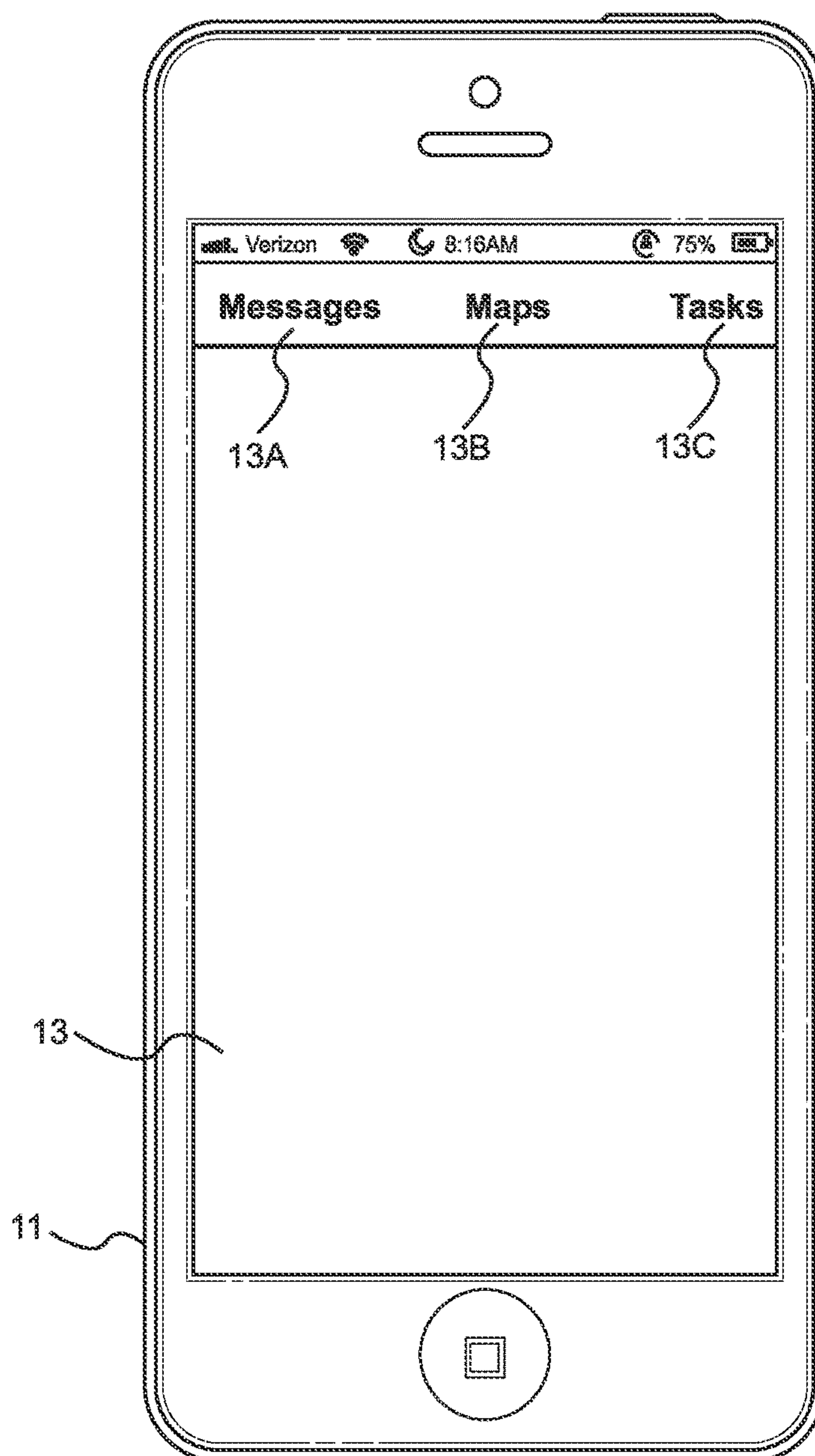


FIG. 18



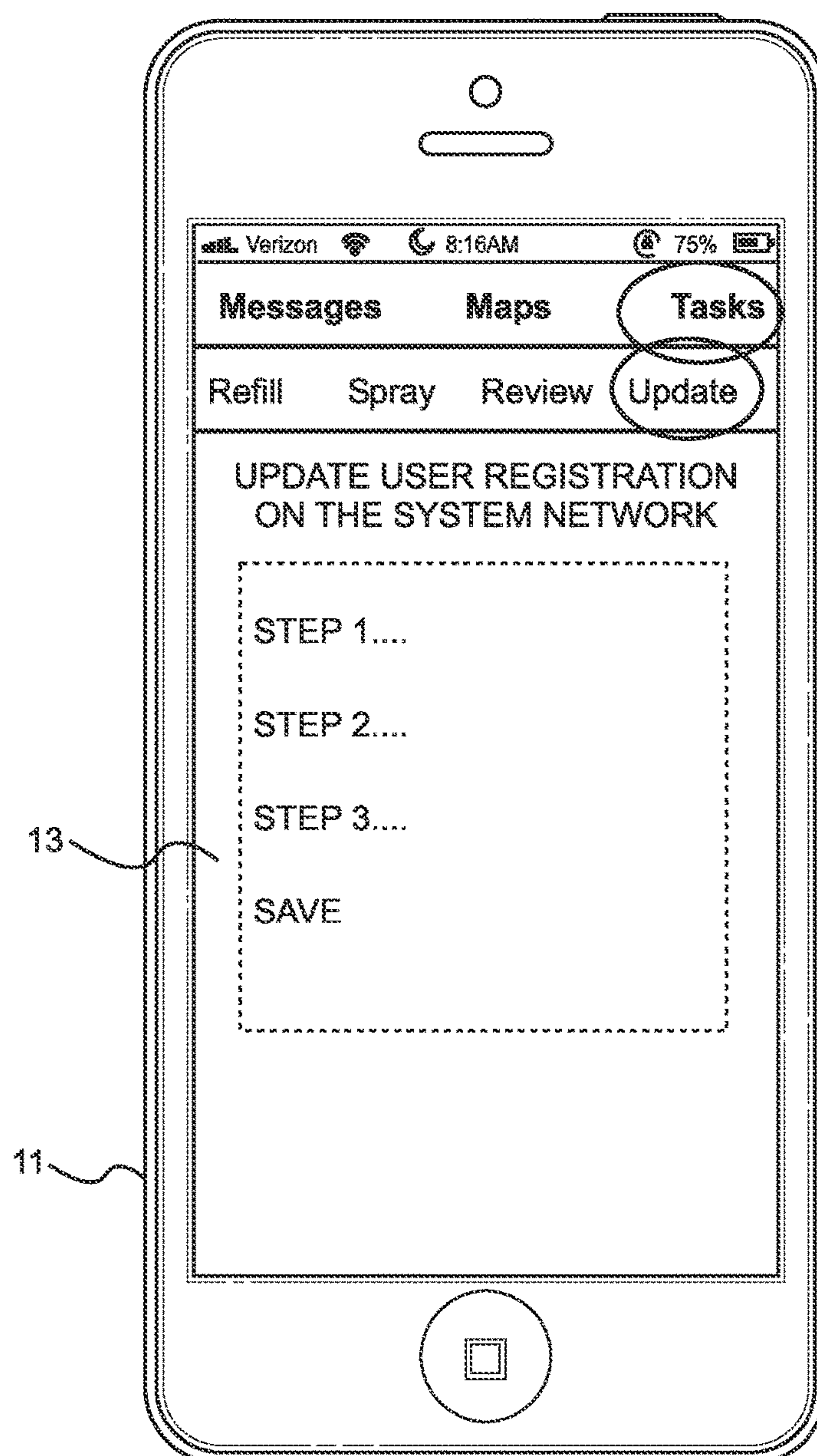
GRAPHICAL USER INTERFACE SUPPORTED BY MOBILE APPLICATION FOR USE BY  
REGISTERED USERS (E.G. PROPERTY PARCEL OWNERS, CONTRACTORS AND/OR  
AGENTS, RESIDENTS, OFFICES, ET AL)



**FIG. 19**



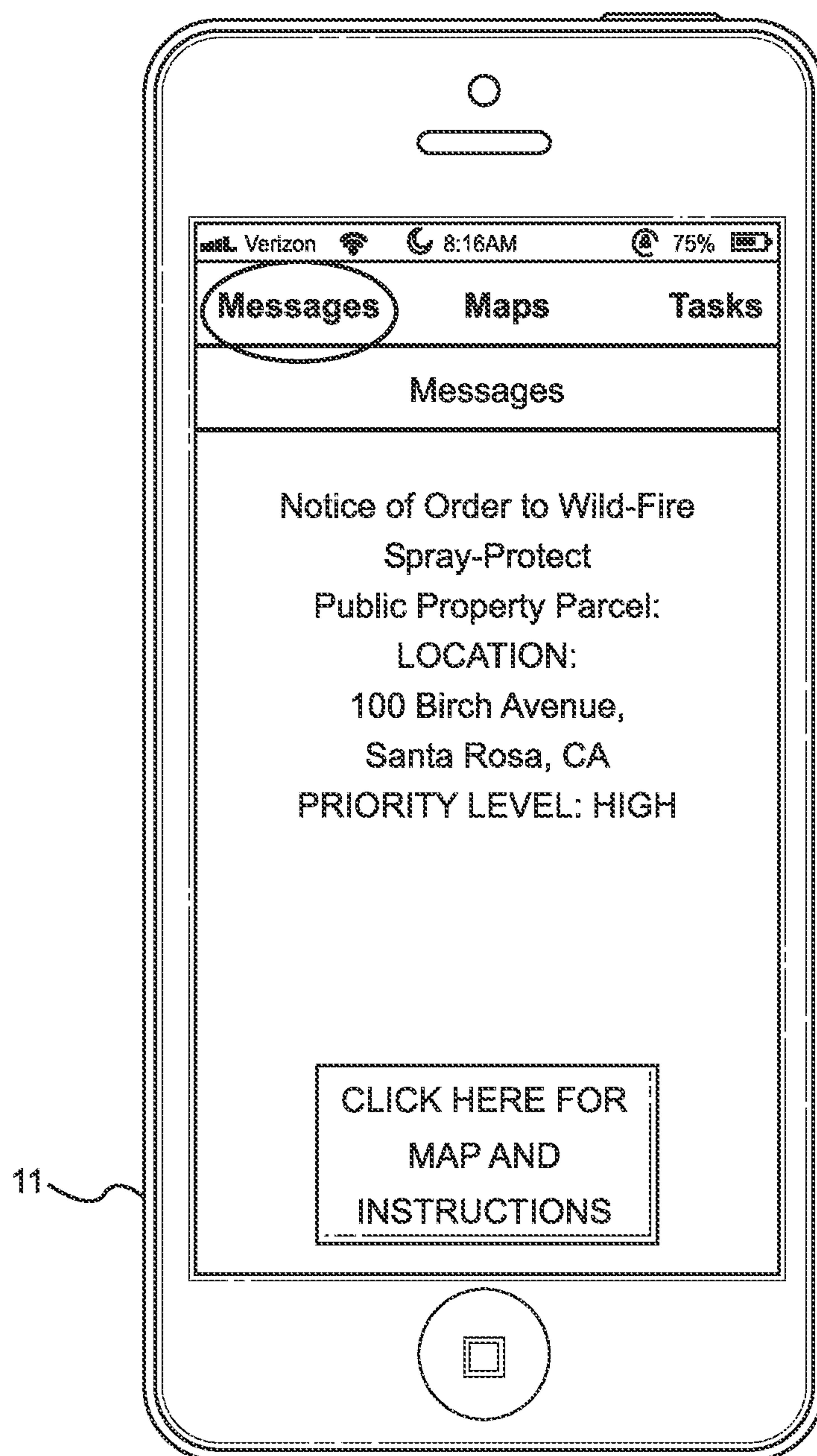
GRAPHICAL INTERFACE SUPPORTED BY THE MOBILE APPLICATION SHOWING A  
USER UPDATING THE REGISTRATION PROFILE AS A TASK ON THE SYSTEM NETWORK



**FIG. 19A**



GRAPHICAL USER INTERFACE OF THE MOBILE APPLICATION SHOWING A USER  
RECEIVING A MESSAGE REQUEST FROM THE COMMAND CENTER TO SPRAY  
GPS-SPECIFIED PRIVATE PROPERTY PARCEL(S) WITH  
CLEAN ANTI-FIRE (AF) CHEMICAL LIQUID



**FIG. 19B**



GRAPHICAL USER INTERFACE SUPPORTED BY THE MOBILE APPLICATION SHOWING  
A USER RECEIVING A REQUEST / NOTICE OF ORDER TO WILD-FIRE SPRAY-PROTECT  
GPS-SPECIFIED PUBLIC PROPERTY PARCEL(S) WITH CLEAN AF LIQUID TO CREATE  
GPS-SPECIFIED PUBLIC FIREBREAK

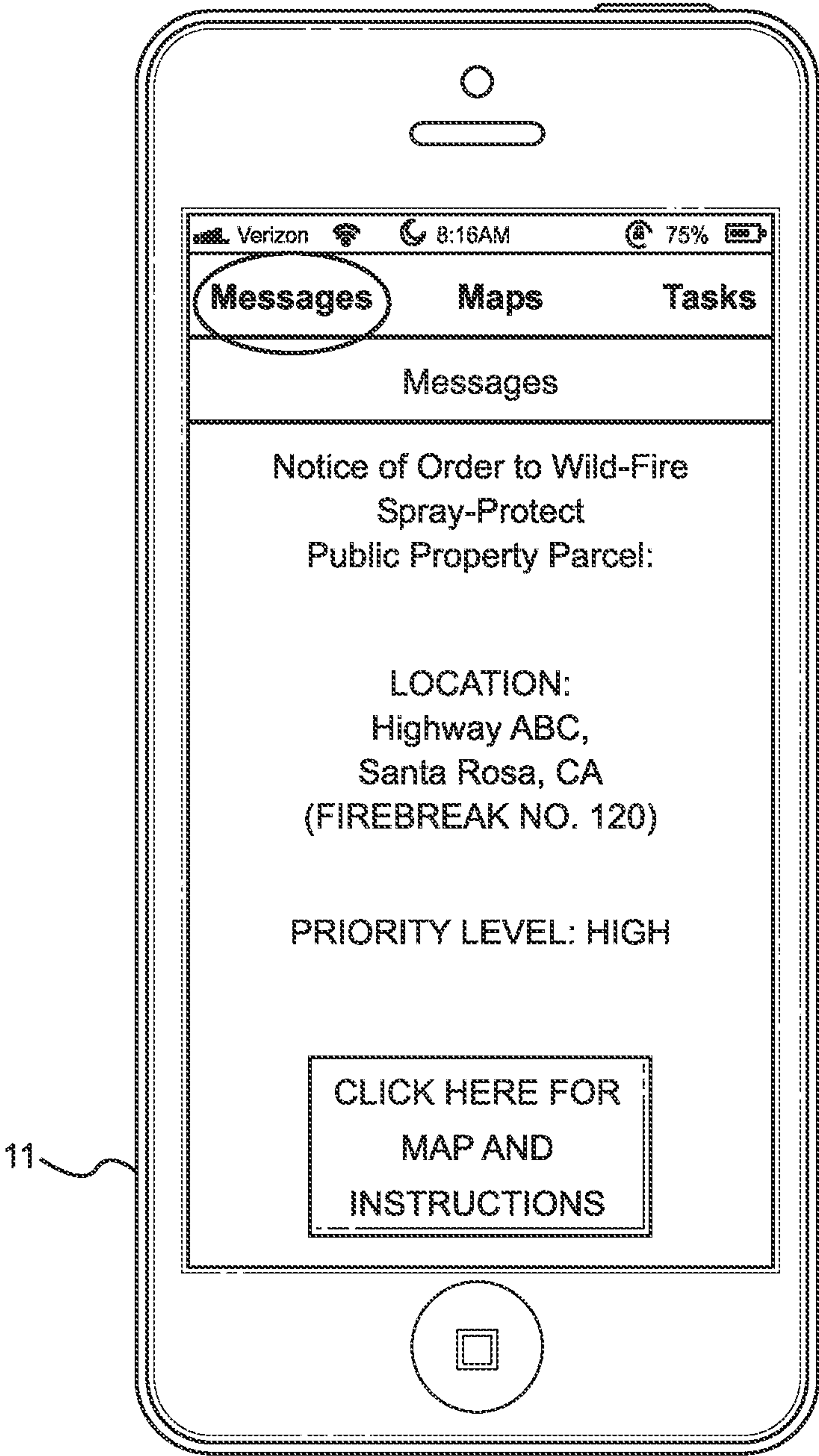


FIG. 19C



GRAPHICAL USER INTERFACE SUPPORTED BY THE MOBILE APPLICATION  
SHOWING A USER REQUESTING A REFILL OF CLEAN ANTI-FIRE (AF) CHEMICAL  
LIQUID FOR SUPPLY TO GPS-SPECIFIED SPRAY EQUIPMENT

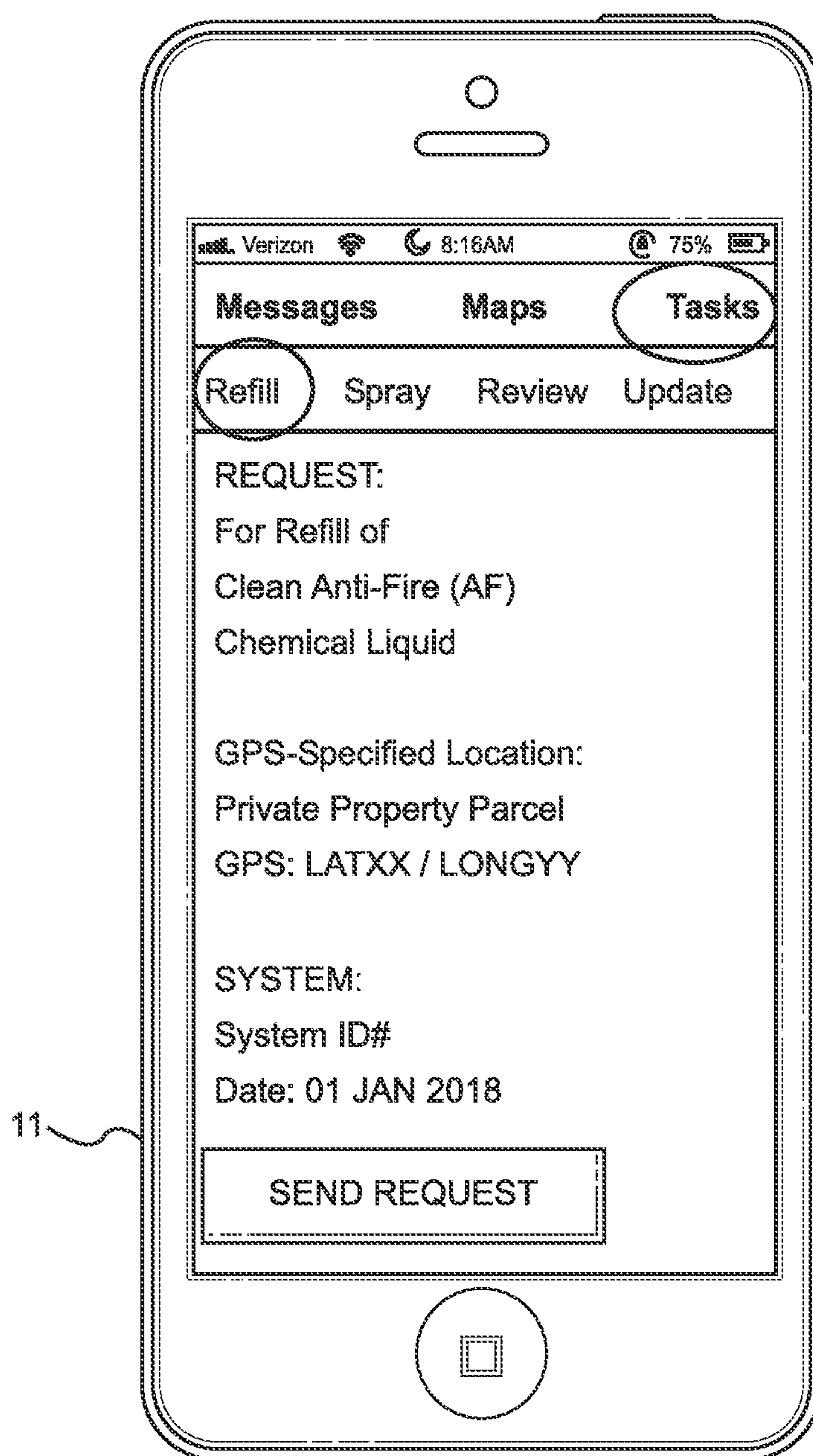
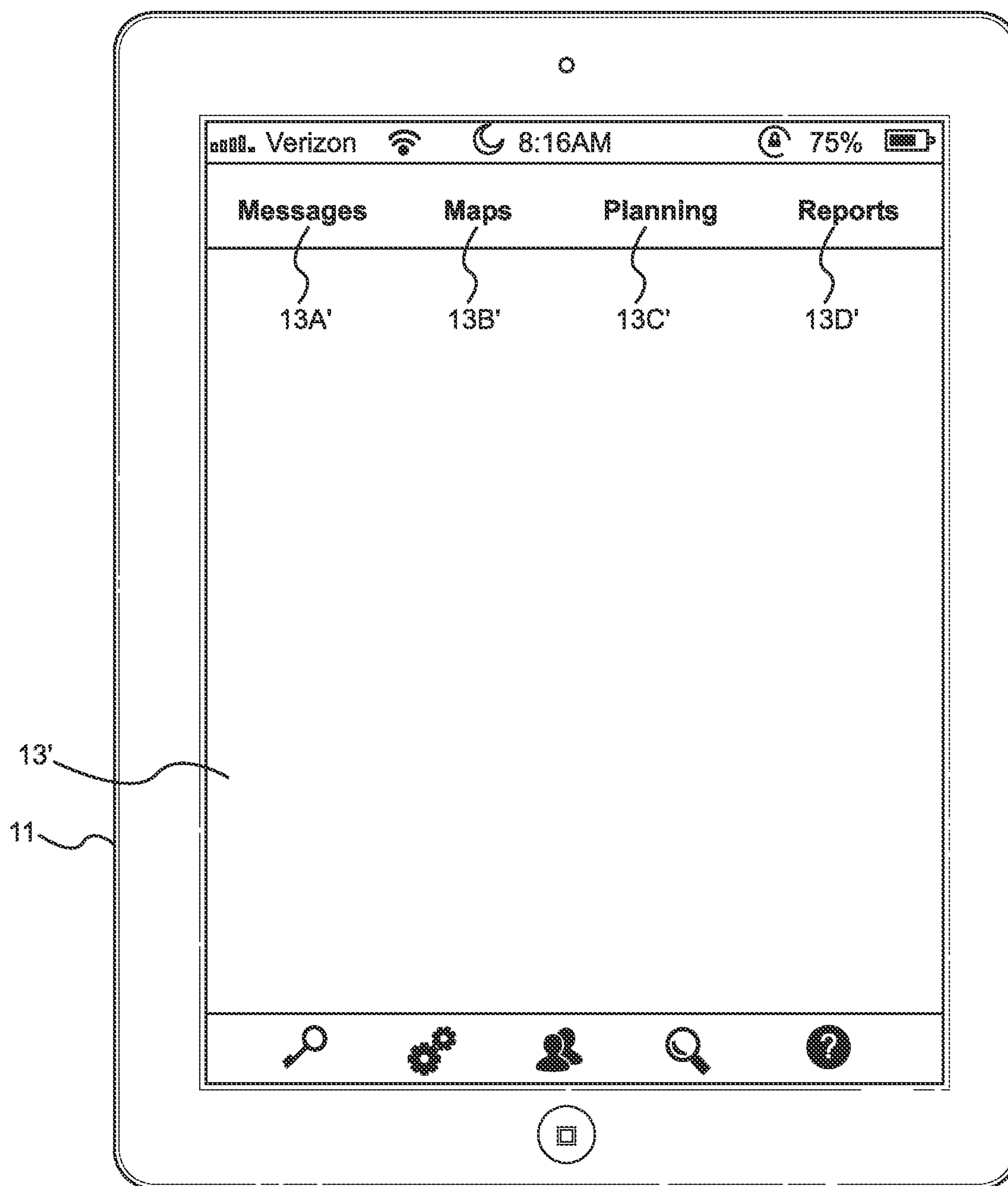


FIG. 19D



GRAPHICAL USER INTERFACE SUPPORTED BY MOBILE APPLICATION FOR USE  
BY COMMAND CENTER ADMINISTRATORS TO ISSUE WILD-FIRE PROTECTION  
ORDERS, PLAN WILD-FIRE PROTECTION TASKS, GENERATE REPORTS, AND  
SEND AND RECEIVE MESSAGES





GRAPHICAL USER INTERFACE SUPPORTED BY THE MOBILE APPLICATION FOR USE BY  
COMMAND CENTER ADMINISTRATORS TO ISSUE WILD-FIRE PROTECTION ORDERS

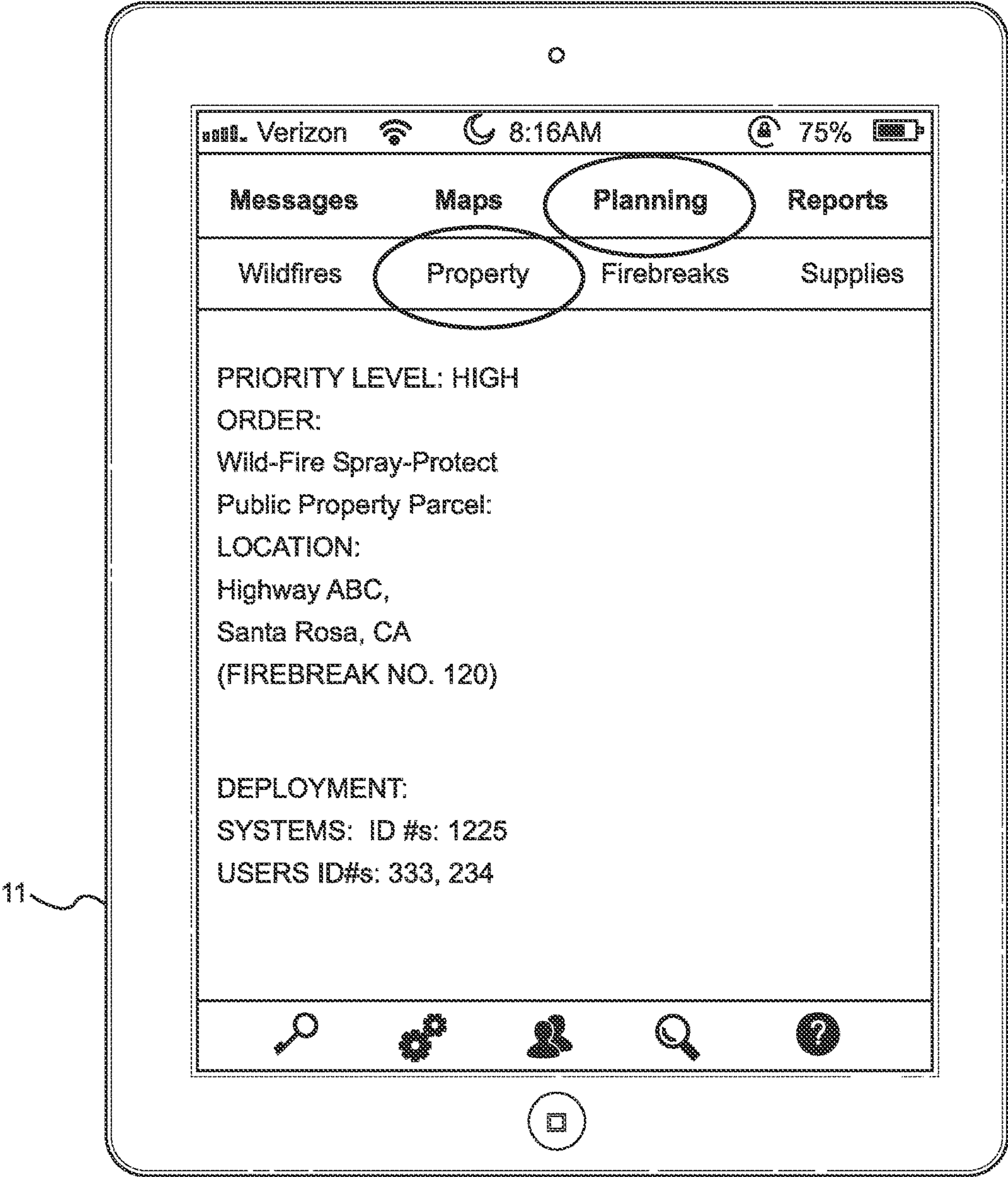


FIG. 20A



GRAPHICAL USER INTERFACE SUPPORTED BY MOBILE APPLICATION FOR USE BY COMMAND CENTER ADMINISTRATORS TO ISSUE WILD-FIRE PROTECTION ORDERS INVOLVING THE CREATION AND MAINTENANCE OF CLEAN AF-BASED CHEMICAL FIREBREAK

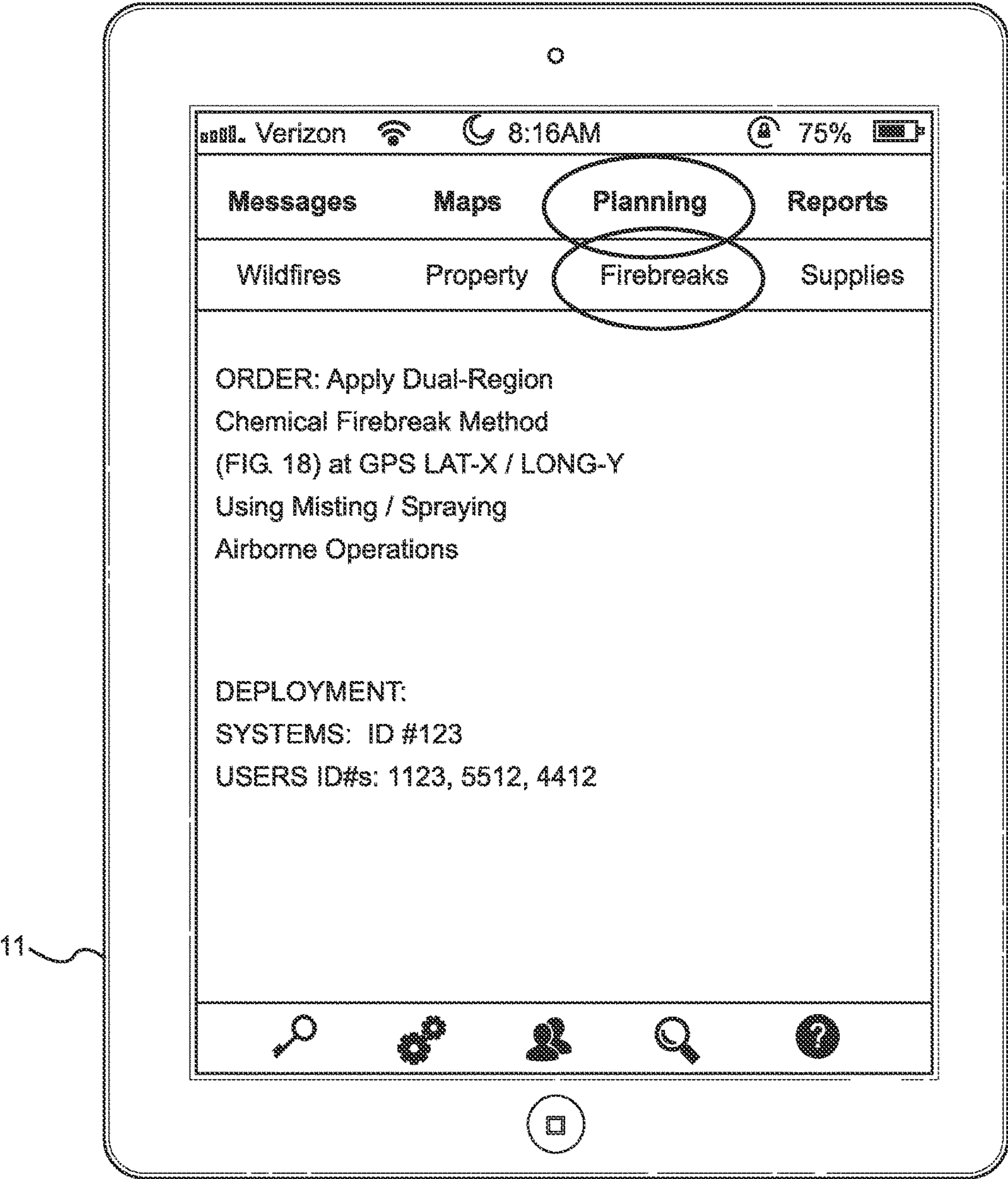


FIG. 20B



GRAPHICAL USER INTERFACE SUPPORTED BY MOBILE APPLICATION FOR USE BY COMMAND CENTER ADMINISTRATORS TO ORDER THE CREATION OF GPS-SPECIFIED CLEAN AF-BASEDCHEMICAL FIREBREAKS ON ONE OR MORE PUBLIC / PRIVATE PROPERTY PARCELS

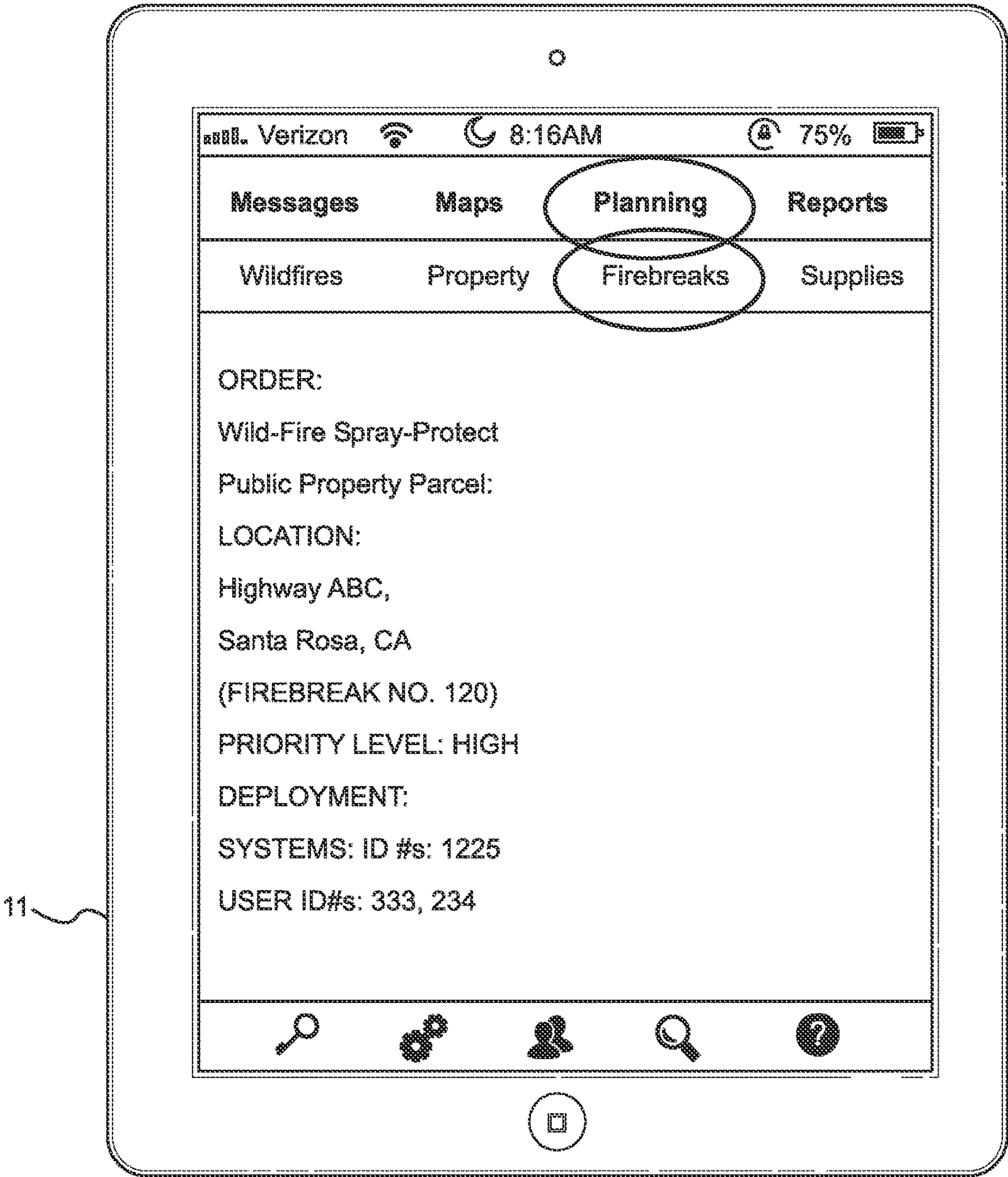


FIG. 20C



GRAPHICAL USER INTERFACE FOR MOBILE APPLICATION USED BY COMMAND CENTER ADMINISTRATORS TO RECEIVE MESSAGES FROM USERS INCLUDING PROPERTY OWNERS AND CONTRACTORS REQUESTING REFILLS FOR CLEAN ANTI-FIRE (AF) CHEMICAL LIQUID

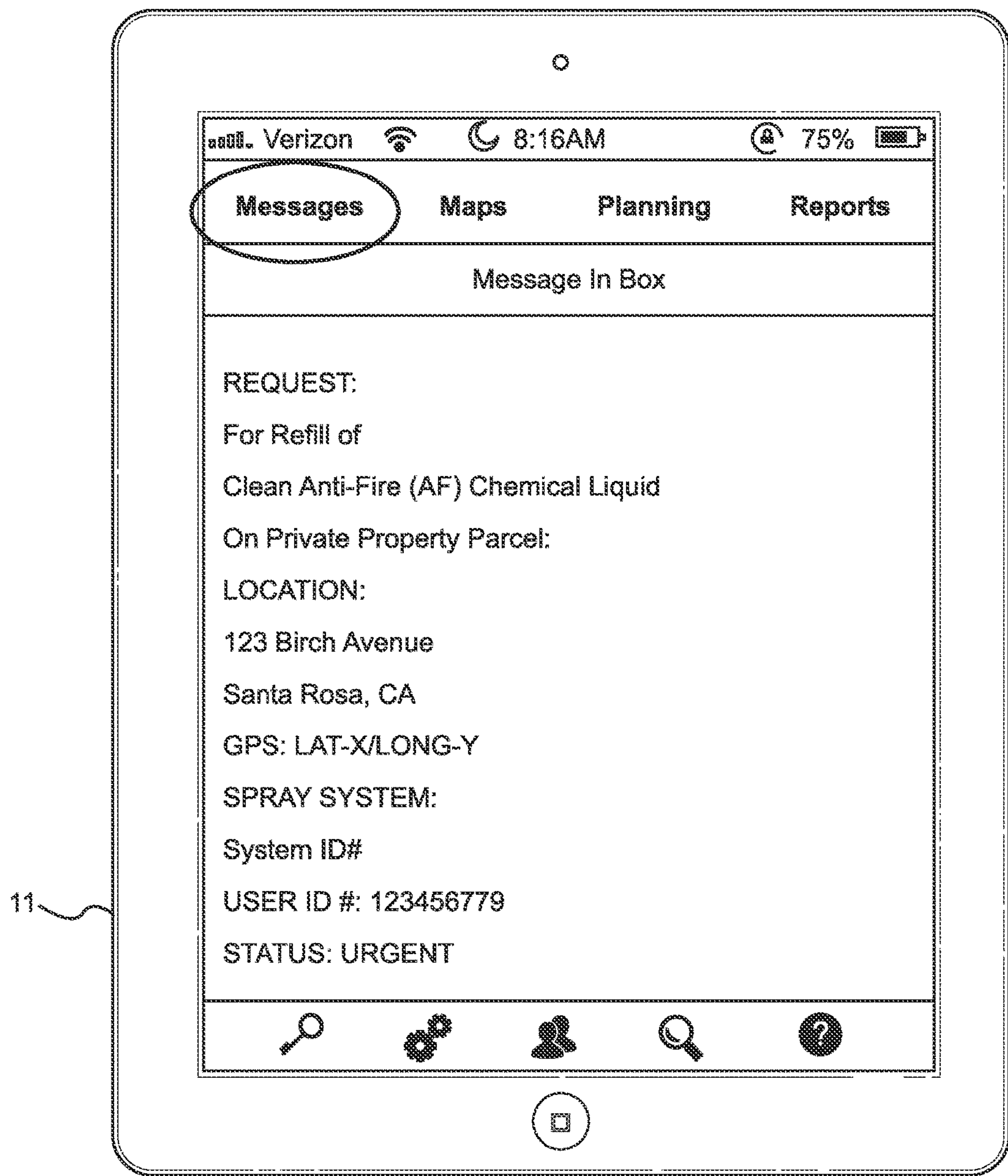


FIG. 20D



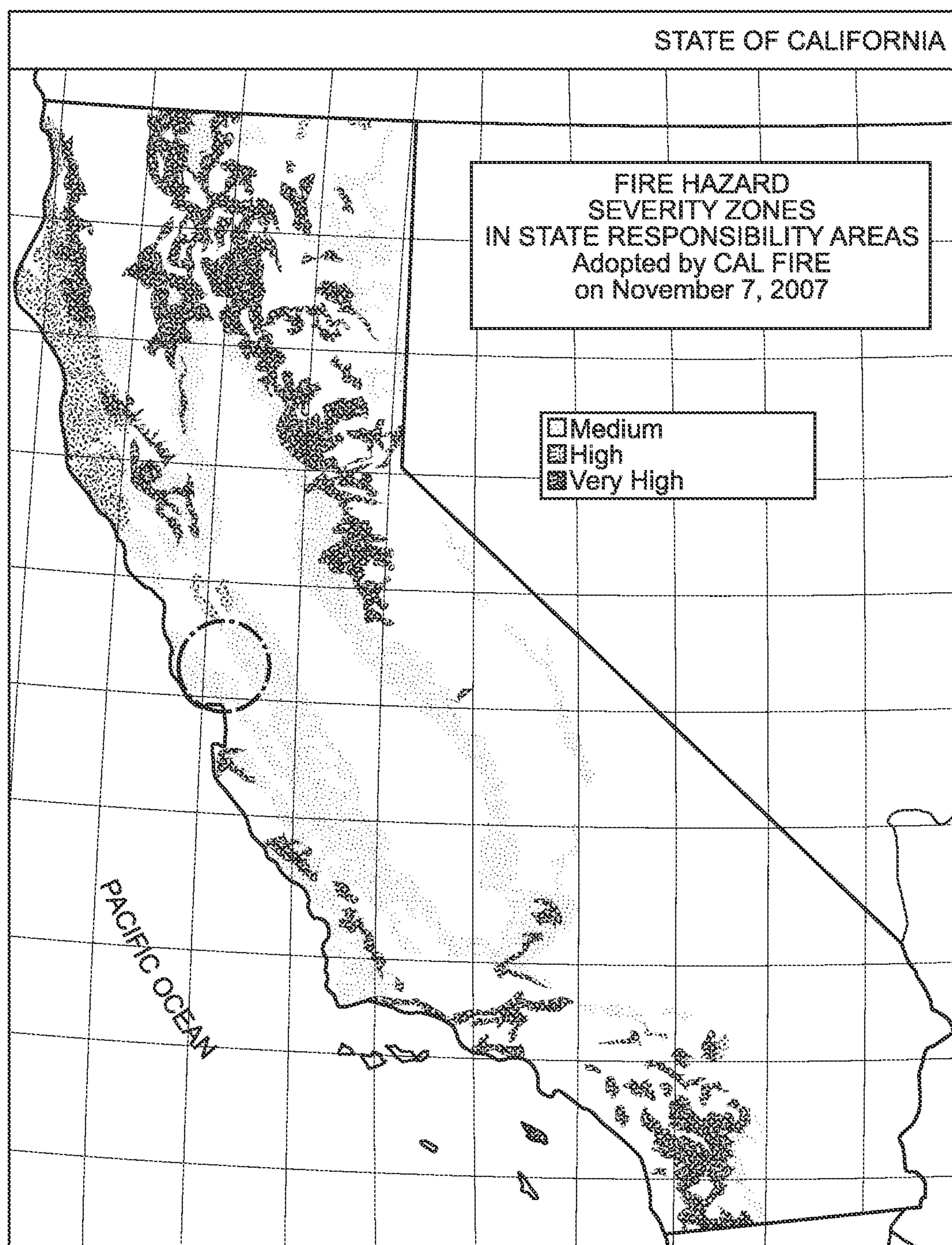


FIG. 21



EXEMPLARY ANTI-FIRE SPRAY PROTECTION MAP GENERATED BY THE SYSTEM NETWORK  
SHOWING HOUSES AND STRUCTURES SPRAYED AND NOT-SPRAYED WITH ANTI-FIRE (AF) LIQUID  
AS OF RECORD DATE 15 DECEMBER 2017

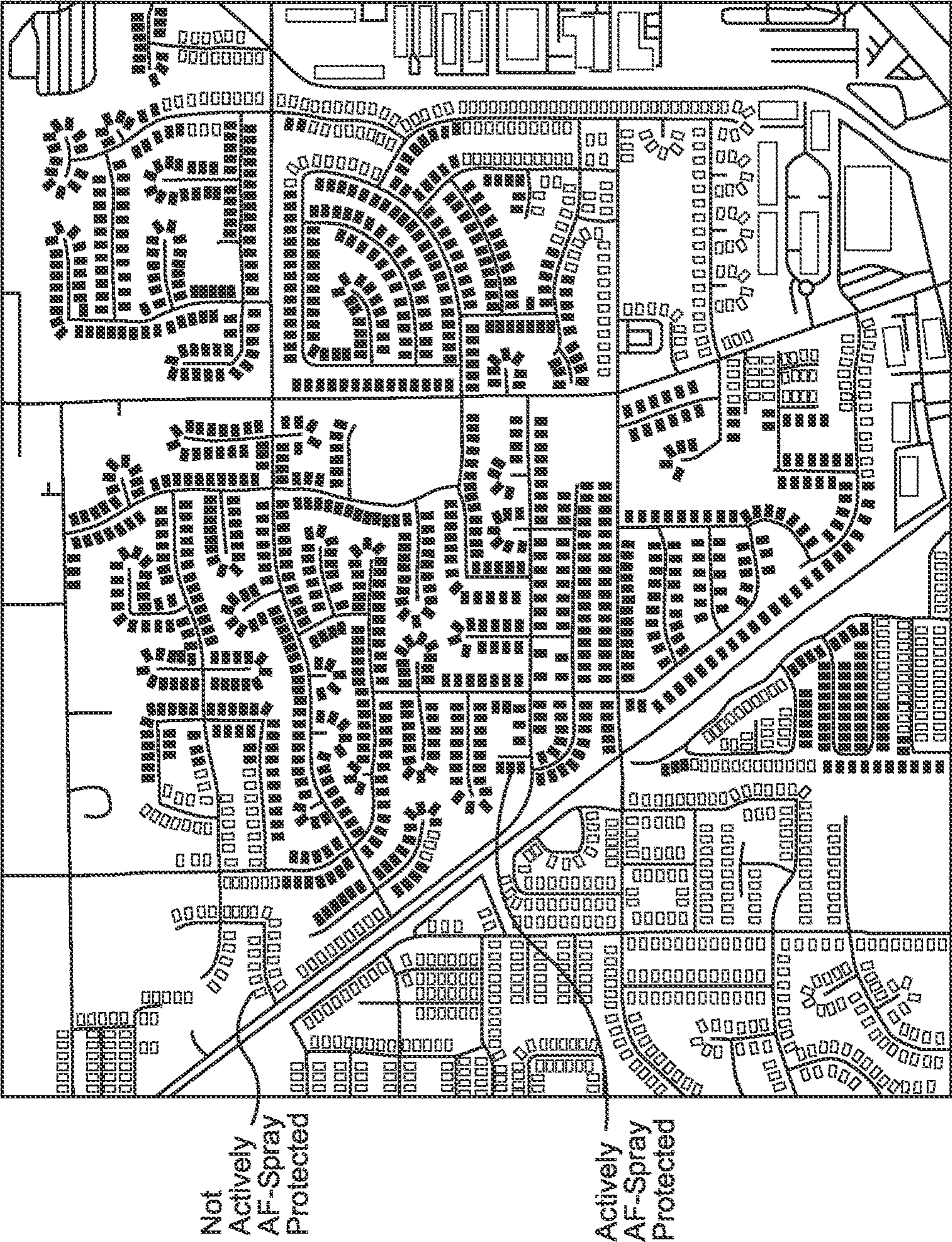


FIG. 22



EXEMPLARY ANTI-FIRE (AF) SPRAY PROTECTION TASK REPORT

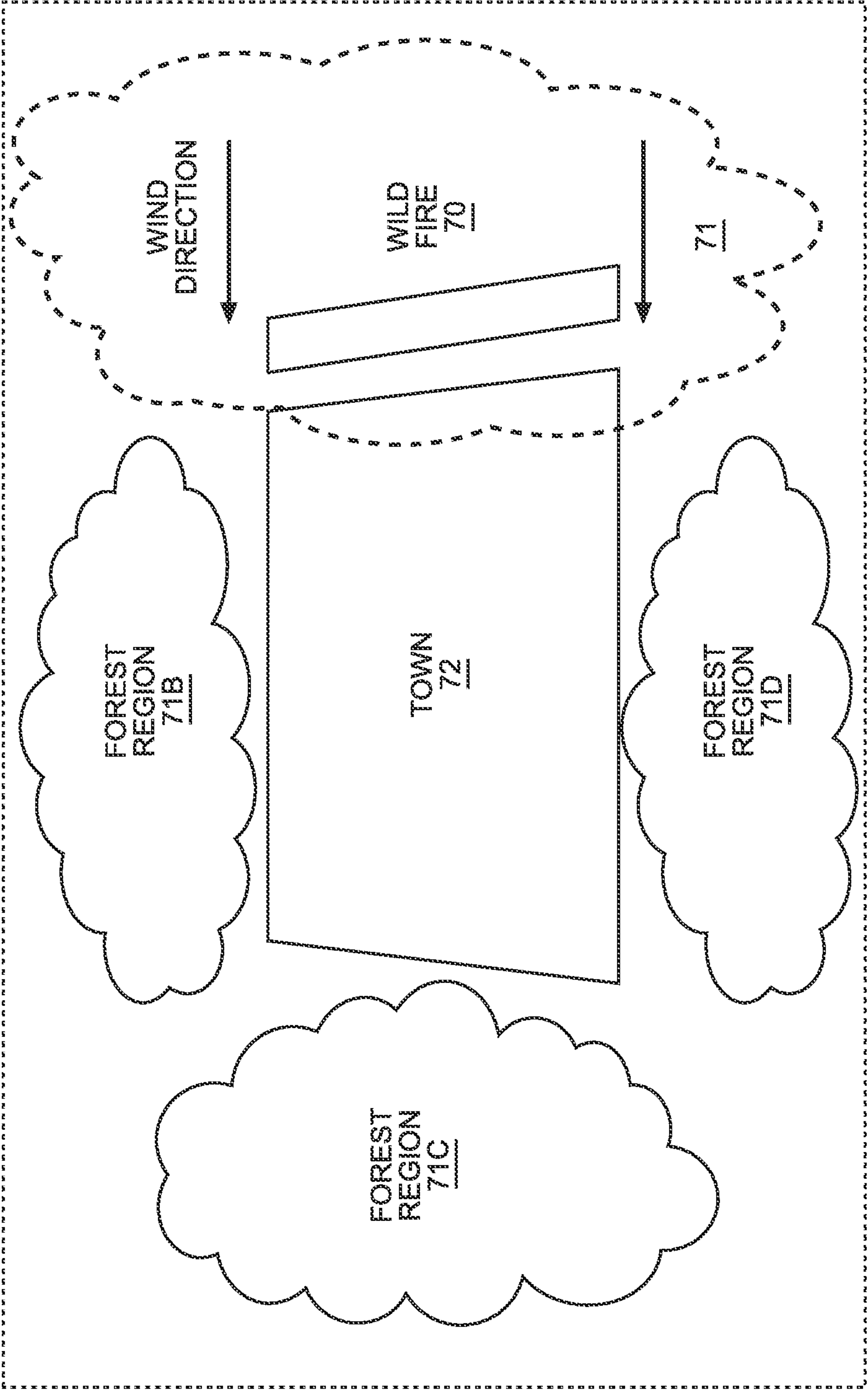
GENERATED BY THE SYSTEM FOR COUNTY, STATE ON 15 DECEMBER 2017

| AF Liquid Spray Order # | GPS<br>Addresses | Street Addresses,<br>Town, County | Building<br>Structure (Y/N) | Last Appln.<br>Date | Spray<br>Equip. | SMS | Supervisor<br>ID |
|-------------------------|------------------|-----------------------------------|-----------------------------|---------------------|-----------------|-----|------------------|
|                         |                  |                                   |                             |                     |                 |     |                  |
|                         |                  |                                   |                             |                     |                 |     |                  |
|                         |                  |                                   |                             |                     |                 |     |                  |
|                         |                  |                                   |                             |                     |                 |     |                  |
|                         |                  |                                   |                             |                     |                 |     |                  |
|                         |                  |                                   |                             |                     |                 |     |                  |
|                         |                  |                                   |                             |                     |                 |     |                  |
|                         |                  |                                   |                             |                     |                 |     |                  |

FIG. 23



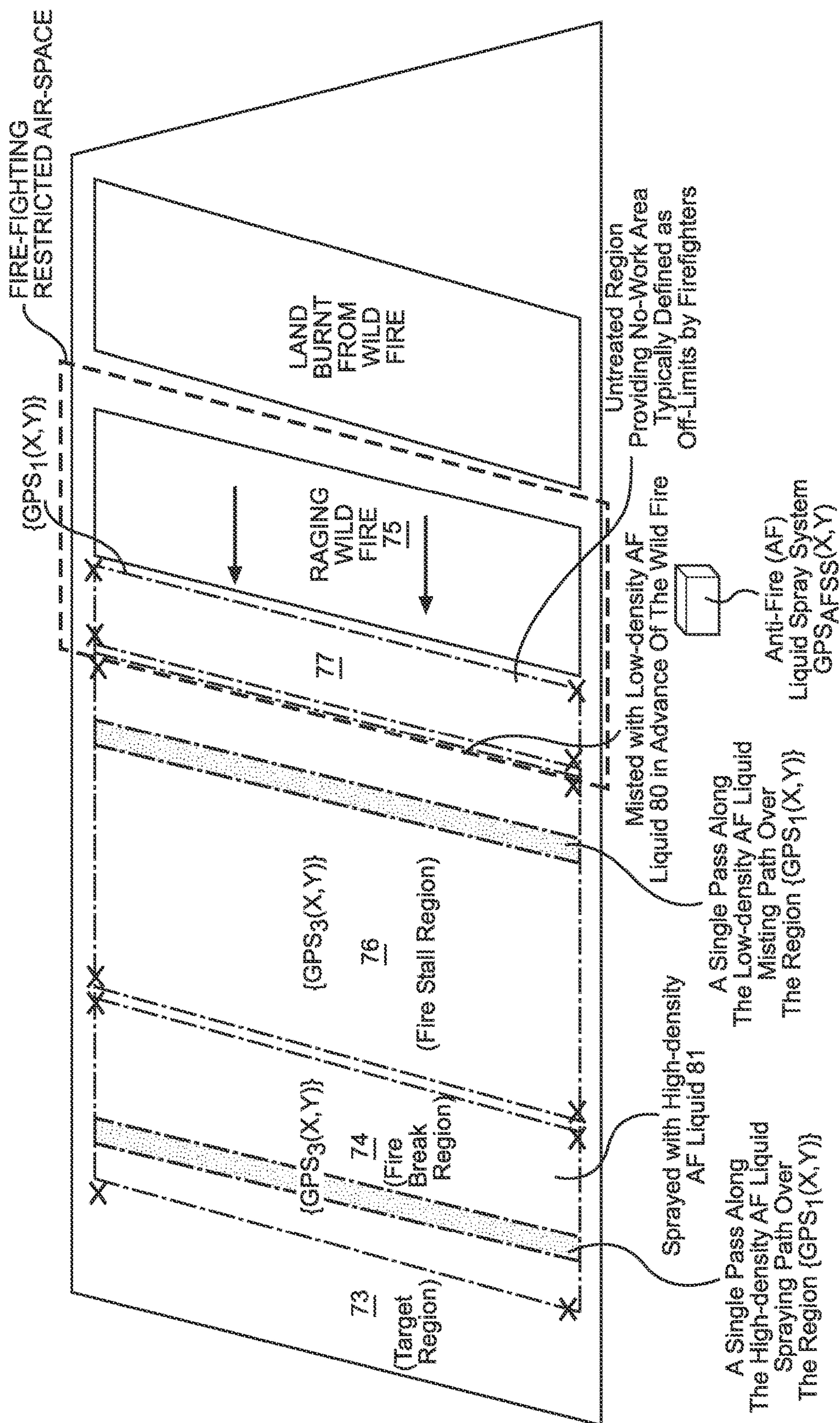
PLAN VIEW OF A WILD FIRE APPROACHING A NEIGHBORING TOWN  
IN THE DIRECTION OF PREVAILING WINDS



**FIG. 24**



METHOD OF SUPPRESSING A WILD FIRE RAGING ACROSS A REGION OF LAND BY FORMING A MULTI-STAGE ANTI-FIRE CHEMICAL FIRE-BREAK SYSTEM USING THE REMOTELY MANAGED GPS-CONTROLLED APPLICATION OF ANTI-FIRE (AF) LIQUID MIST AND SPRAY STREAMS FROM GROUND AND AIR BASED GPS-TRACKED ANTI-FIRE (AF) LIQUID SPRAY VEHICLES



26



**METHOD OF SUPPRESSING A WILD FIRE RAGING TOWARDS A TARGET REGION OF LAND IN A DIRECTION DETERMINED BY PREVAILING WINDS AND OTHER ENVIRONMENTAL AND WEATHER FACTORS**

A: PRIOR TO THE WILD FIRE REACHING THE SPECIFIED TARGET REGION OF LAND, APPLYING A LOW-DENSITY ANTI-FIRE (AF) LIQUID MIST IN ADVANCE OF THE WILD FIRE SO AS TO FORM A FIRE STALL REGION WHILE PROVIDING A NON-TREATED REGION OF SUFFICIENT SIZE BETWEEN THE FRONT OF THE WILD FIRE APPROACHING THE TARGET REGION OF LAND AND THE FIRE STALL REGION, WHEREIN SAID NON-TREATED REGION IS DEFINED BY A FIRST SET OF GPS COORDINATES {GPS1(X,Y)} MAPPED OUT USING GLOBAL POSITIONING SYSTEM (GPS) METHODS, WHEREIN SAID FIRE STALL REGION IS DEFINED BY A SECOND SET OF GPS COORDINATES {GPS2(X,Y)} MAPPED OUT USING GLOBAL POSITIONING SYSTEM (GPS) METHODS, AND WHEREIN SAID FIRE STALL REGION IS FORMED BY A FIRST GPS-GUIDED AIRCRAFT FLYING OVER SAID FIRE STALL REGION DURING MULTIPLE PASSES AND APPLYING SAID LOW-DENSITY AF LIQUID MIST OVER SAID FIRE STALL REGION;

B: APPLYING A HIGH-DENSITY ANTI-FIRE (AF) LIQUID SPRAY IN ADVANCE OF THE WILD FIRE TO FORM A FIRE BREAK REGION BEYOND AND CONTIGUOUS WITH SAID FIRE STALL REGION, WHEREIN SAID FIRE BREAK REGION IS DEFINED BY A THIRD SET OF GPS COORDINATES {GPS3(X,Y)} MAPPED OUT USING GLOBAL POSITIONING SYSTEM (GPS) METHODS, AND WHEREIN SAID FIRE BREAK REGION IS FORMED BY A SECOND GPS-GUIDED AIRCRAFT FLYING OVER SAID FIRE BREAK REGION DURING MULTIPLE PASSES AND APPLYING SAID HIGH-DENSITY AF LIQUID SPRAY OVER SAID FIRE BREAK REGION; AND

A

**FIG. 26A**



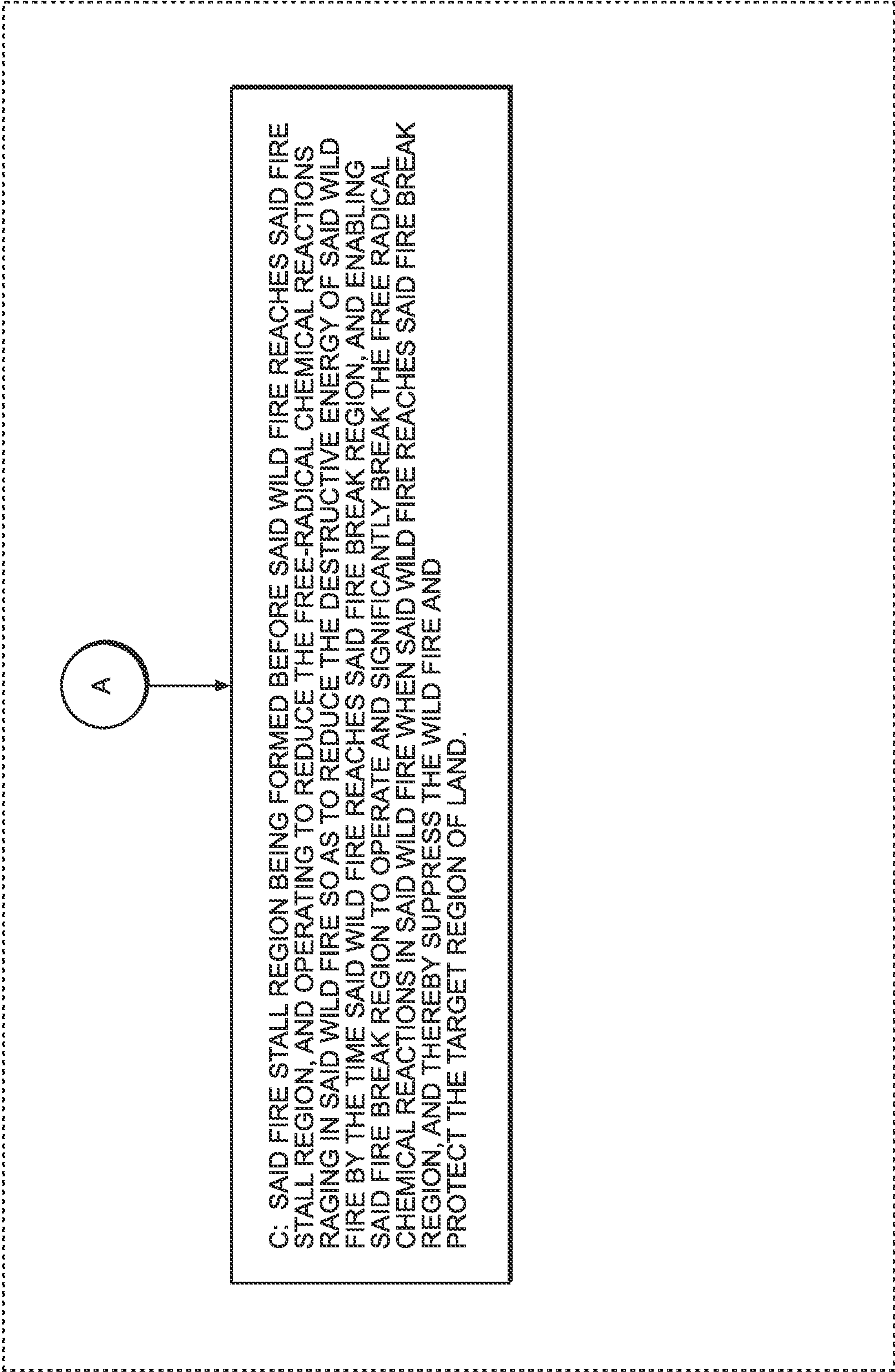
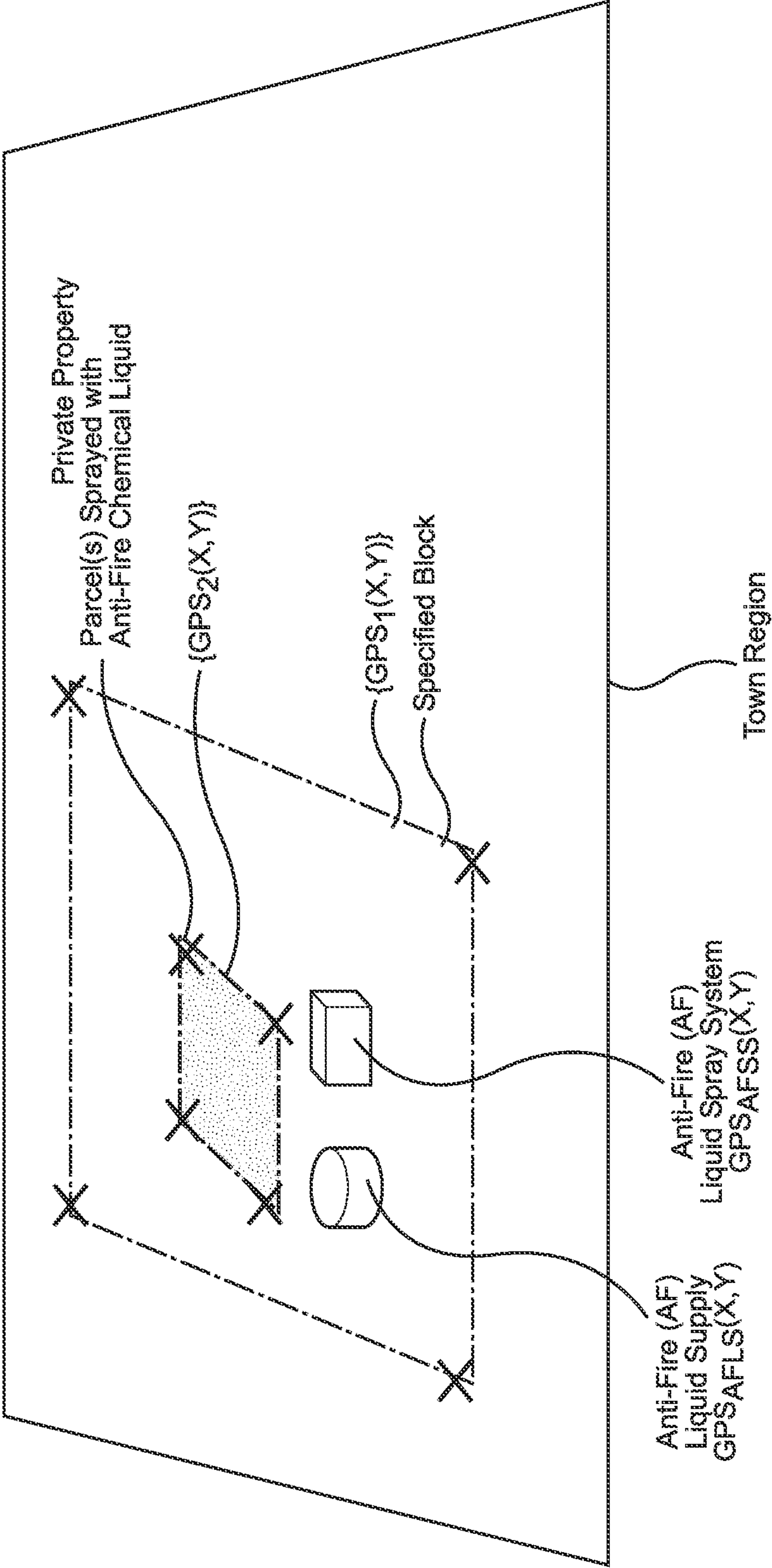


FIG. 26B

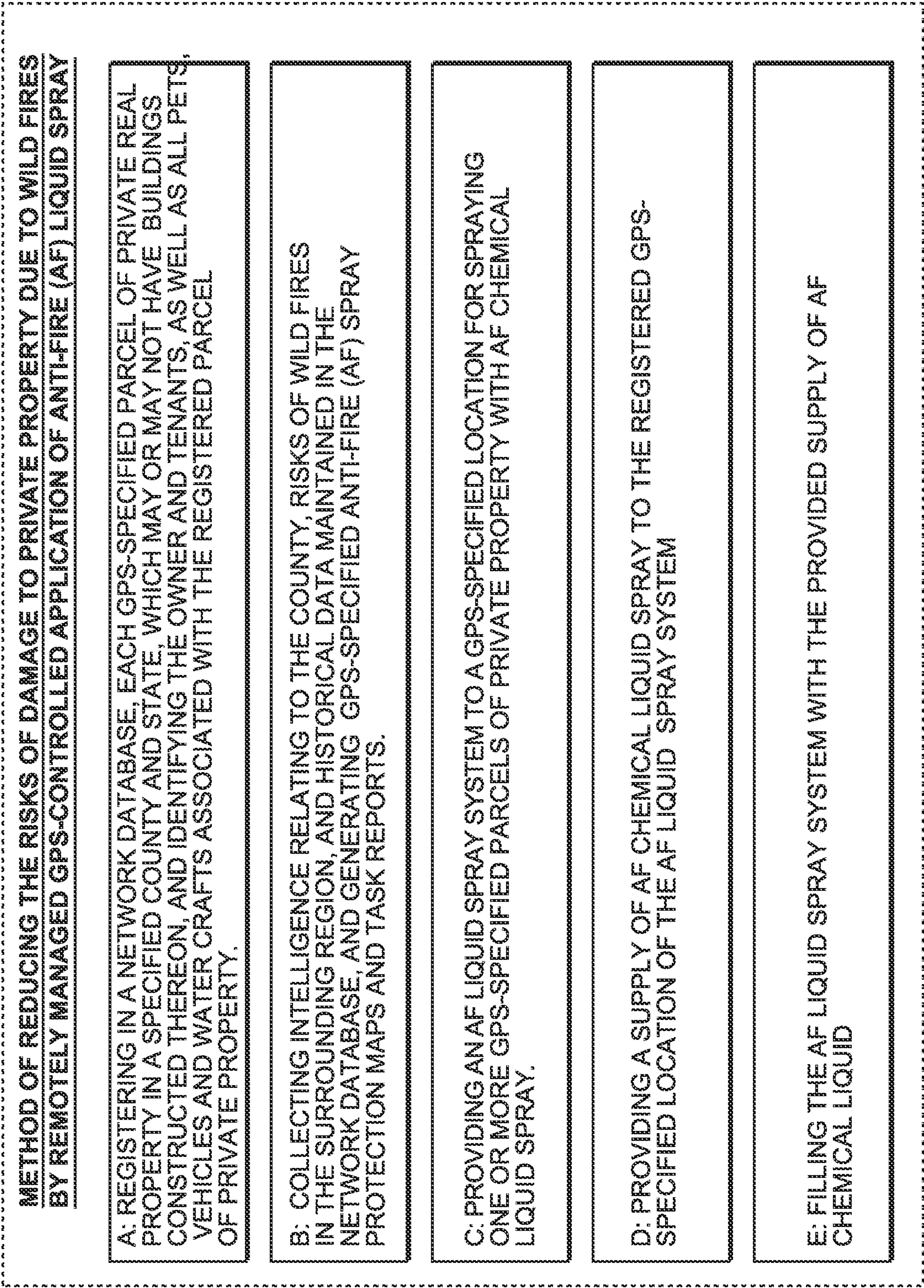


**METHOD OF REDUCING THE RISKS OF DAMAGE TO PRIVATE PROPERTY DUE TO WILD FIRES  
BY REMOTELY MANAGED GPS-CONTROLLED APPLICATION OF ANTI-FIRE (AF) LIQUID SPRAY  
FROM GROUND AND AIR BASED GPS-TRACKED ANTI-FIRE (AF) LIQUID SPRAY VEHICLES**



**FIG. 27**





**FIG. 28A**



F: BASED ON THE GPS-SPECIFIED ANTI-FIRE (AF) SPRAY PROTECTION MAPS AND TASK REPORTS THE SYSTEM ISSUING ORDERS TO THE PRIVATE PROPERTY OWNER OVER AN ELECTRONIC MESSAGING SYSTEM, TO APPLY AF CHEMICAL LIQUID SPRAY ON THE PRIVATE PROPERTY USING THE AF LIQUID SPRAY SYSTEM PRIOR TO THE PRESENCE OF WILD FIRE IN THE REGION.

G: THE PRIVATE PROPERTY OWNER EXECUTING THE ORDER AND APPLYING AF CHEMICAL LIQUID SPRAY ON THE PRIVATE PROPERTY USING THE AF LIQUID SPRAY SYSTEM, AND THE SYSTEM REMOTELY MONITORING THE CONSUMPTION AND APPLICATION OF AF CHEMICAL LIQUID AT THE PRIVATE PROPERTY ON A GIVEN DATE AND DATE AND AUTOMATICALLY RECORDING THE TRANSACTION IN THE NETWORK DATABASE.

H: THE SYSTEM UPDATING RECORDS IN THE NETWORK DATABASE ASSOCIATED WITH EACH APPLICATION OF AF CHEMICAL LIQUID SPRAY ON A REGISTERED PARCEL OF PRIVATE PROPERTY.

I: THE SYSTEM SCHEDULING THE NEXT APPLICATION OF AF CHEMICAL LIQUID SPRAY ON THE PARCELS OF PRIVATE PROPERTY, FACTORING WEATHER CONDITIONS, WEAR AND TIME.

J: THE SYSTEM ISSUING ANOTHER ORDER TO THE GPS-SPECIFIED PARCELS OF PRIVATE PROPERTY TO RE-APPLY AF CHEMICAL LIQUID SPRAY THEREON TO MAINTAIN ACTIVE FIRE PROTECTION.

**FIG. 28B**



K: THE PROPERTY OWNER EXECUTING THE ORDER TO REAPPLY AF CHEMICAL LIQUID SPRAY ON THE PARCELS OF PRIVATE PROPERTY USING THE AF LIQUID SPRAY SYSTEM, AND THE SYSTEM REMOTELY MONITORING THE APPLICATION OF AF CHEMICAL LIQUID AT THE PRIVATE PROPERTY ON A GIVEN TIME AND DATE, AND RECORDING THIS TRANSACTION IN THE NETWORK DATABASE.

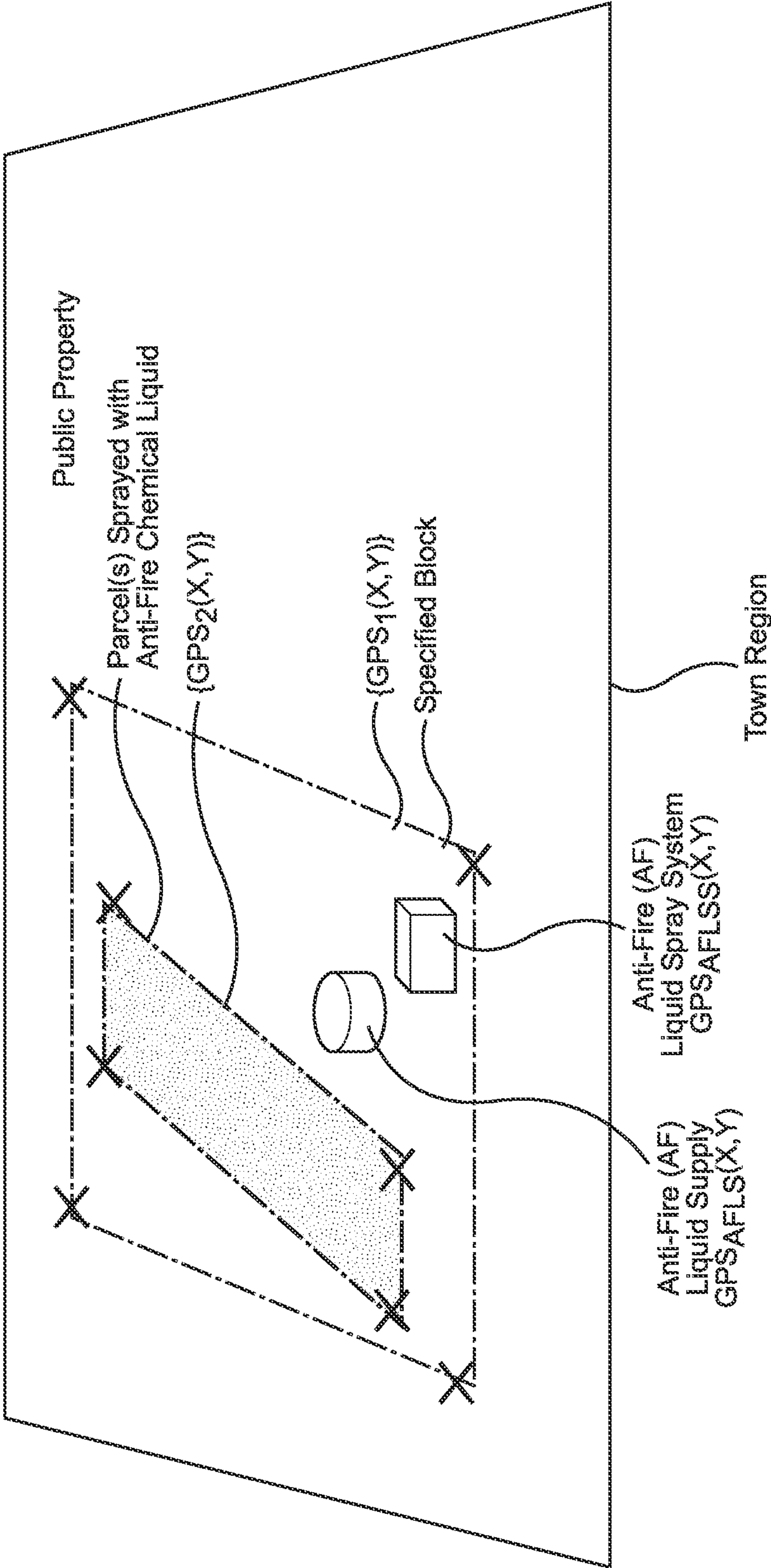
L: THE SYSTEM UPDATING RECORDS ON AF CHEMICAL LIQUID SPRAY APPLICATION IN THE NETWORK DATABASE, CONCERNING REAPPLICATION OF AF CHEMICAL LIQUID ON THE PARCELS OF PRIVATE PROPERTY.

M: THE SYSTEM SCHEDULING THE NEXT APPLICATION OF AF LIQUID SPRAY ON THE PARCELS OF PRIVATE PROPERTY, FACTORING WEATHER CONDITIONS AND THE PASSAGE OF TIME, AND UPDATING THE NETWORK DATABASE.

FIG. 28C



METHOD OF REDUCING THE RISKS OF DAMAGE TO PUBLIC PROPERTY DUE TO WILD FIRES  
BY GPS-CONTROLLED APPLICATION OF ANTI-FIRE (AF) LIQUID SPRAY  
FROM GROUND AND AIR BASED GPS-TRACKED ANTI-FIRE (AF) LIQUID SPRAY VEHICLES



**FIG. 29**



**METHOD OF REDUCING THE RISKS OF DAMAGE TO PUBLIC PROPERTY DUE TO WILD FIRES**  
**BY GPS-CONTROLLED APPLICATION OF ANTI-FIRE (AF) LIQUID SPRAY**

A: REGISTERING WITH A NETWORK DATABASE, EACH GPS-SPECIFIED PARCEL OF PUBLIC REAL PROPERTY IN A SPECIFIED COUNTY AND STATE, WHICH MAY OR MAY NOT HAVE BUILDINGS CONSTRUCTED THEREON, AND IDENTIFYING THE OWNER AND TENANTS, ALL PETS, VEHICLES AND WATERCRAFTS ASSOCIATED WITH THE REGISTERED PARCEL OF PUBLIC PROPERTY.

B: COLLECTING INTELLIGENCE RELATING TO THE COUNTY, RISKS OF WILD FIRES IN THE SURROUNDING REGION, AND HISTORICAL DATA MAINTAINED IN THE NETWORK DATABASE, AND GENERATING ANTI-FIRE (AF) SPRAY PROTECTION MAPS AND TASK REPORTS FOR EXECUTION

C: PROVIDING AN AF LIQUID SPRAY SYSTEM TO A GPS-SPECIFIED LOCATION FOR SPRAYING ONE OR MORE GPS-SPECIFIED PARCELS OF PUBLIC PROPERTY WITH AF CHEMICAL LIQUID SPRAY.

D: PROVIDING A SUPPLY OF AF LIQUID SPRAY TO THE GPS-SPECIFIED LOCATION OF THE AF LIQUID SPRAY SYSTEM

E: FILLING THE AF LIQUID SPRAY SYSTEM WITH THE PROVIDED SUPPLY OF AF LIQUID

**FIG. 30A**



F: BASED ON GPS-SPECIFIED ANTI-FIRE (AF) SPRAY PROTECTION MAPS AND TASK REPORTS, THE SYSTEM ISSUING ORDERS TO THE PUBLIC PROPERTY OWNER, OR ITS CONTRACTOR, TO APPLY AF LIQUID SPRAY ON THE PROPERTY PARCELS USING THE AF LIQUID SPRAY SYSTEM.

G: THE PUBLIC PROPERTY OWNER EXECUTING THE ORDER AND APPLYING AF CHEMICAL LIQUID SPRAY ON THE PUBLIC PROPERTY USING THE AF LIQUID SPRAY SYSTEM, AND THE SYSTEM REMOTELY MONITORING THE CONSUMPTION AND APPLICATION OF AF CHEMICAL LIQUID AT THE PUBLIC PROPERTY ON A GIVEN DATE AND TIME, AND AUTOMATICALLY RECORDING THE TRANSACTION IN THE NETWORK DATABASE.

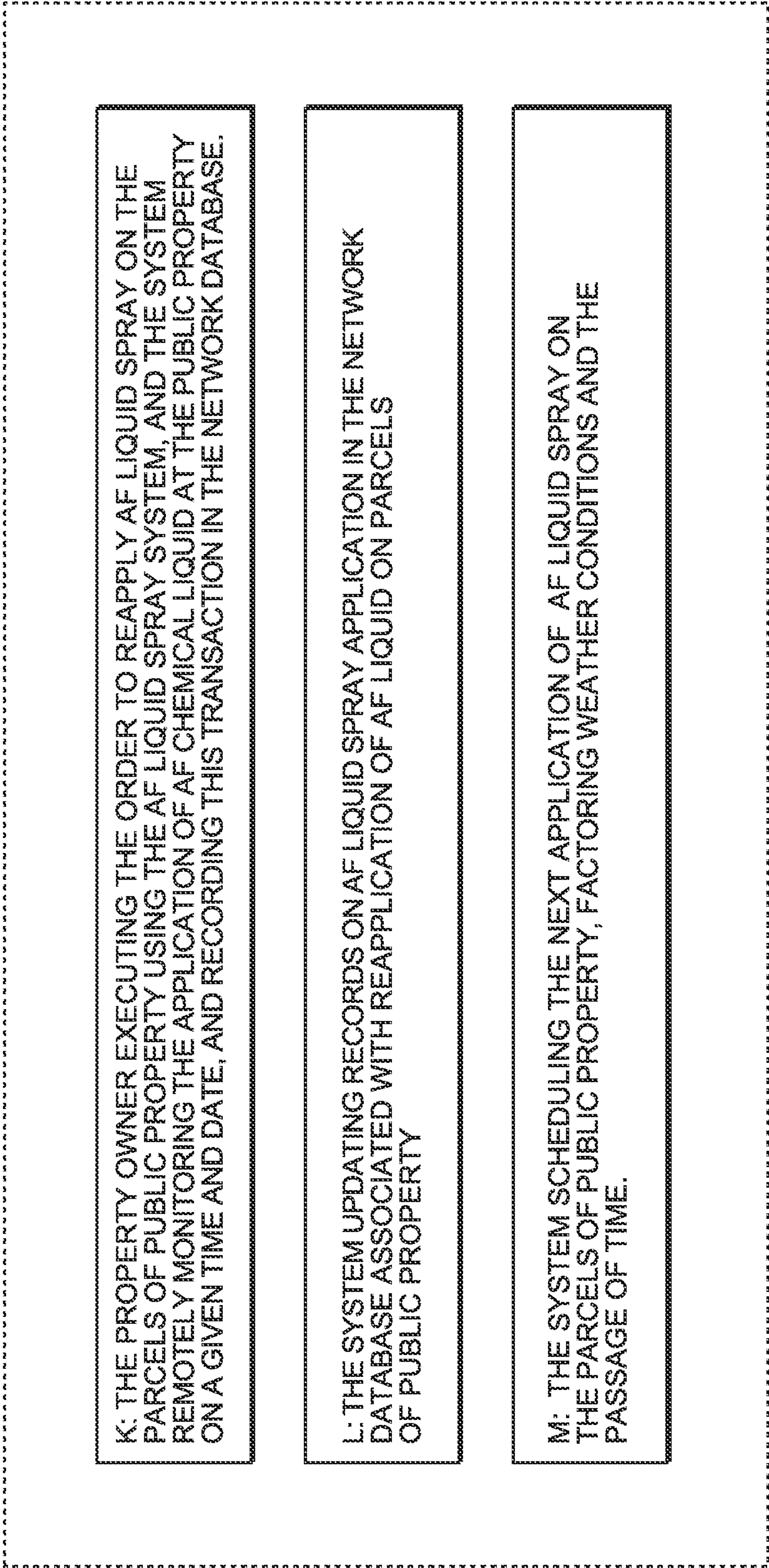
H: THE SYSTEM UPDATING RECORDS IN THE NETWORK DATABASE ASSOCIATED WITH EACH APPLICATION OF AF CHEMICAL LIQUID SPRAY ON GPS-SPECIFIED PARCELS OF PUBLIC PROPERTY.

I: THE SYSTEM SCHEDULING THE NEXT APPLICATION OF AF LIQUID SPRAY ON THE PARCELS OF PUBLIC PROPERTY, FACTORING WEATHER CONDITIONS AND THE PASSAGE OF TIME.

J: THE SYSTEM ISSUING ANOTHER ORDER TO THE GPS-SPECIFIED PARCELS OF PUBLIC PROPERTY TO RE-APPLY AF LIQUID SPRAY THEREON TO MAINTAIN ACTIVE FIRE PROTECTION.

**FIG. 30B**





**FIG. 30C**



METHOD OF REDUCING THE RISKS OF DAMAGE TO REAL PROPERTY PARCELS  
DUE TO WILD FIRES BY GPS-CONTROLLED MANAGED APPLICATION OF ANTI-FIRE (AF) LIQUID  
SPRAY FROM GROUND AND AIR BASED GPS-TRACKED ANTI-FIRE (AF) LIQUID SPRAY VEHICLES

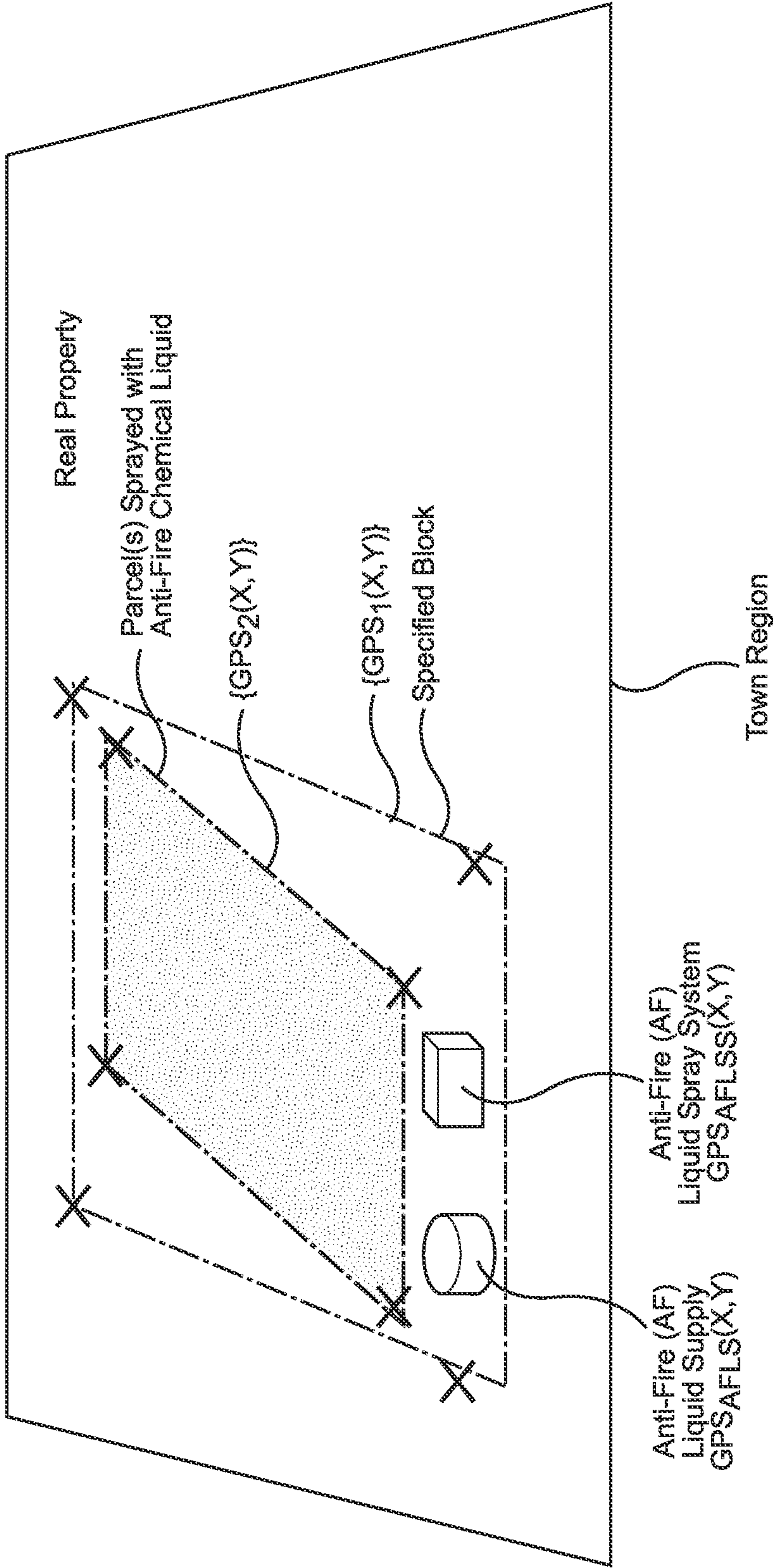


FIG. 31



**METHOD OF REMOTELY MANAGING THE GPS-CONTROLLED APPLICATION OF ANTI-FIRE (AF) LIQUID SPRAY TO GROUND COVER AND BUILDINGS SO AS TO REDUCE THE RISKS OF DAMAGE DUE TO WILD FIRES**

A: REGISTERING WITH A NETWORK DATABASE, EACH GPS-SPECIFIED PARCEL OF REAL PROPERTY IN A SPECIFIED COUNTY AND STATE, WHICH MAY OR MAY NOT HAVE BUILDINGS CONSTRUCTED THEREON, AND IDENTIFYING THE OWNER AND TENANTS, AS WELL AS ALL PETS, VEHICLES AND WATERCRAFTS ASSOCIATED WITH THE GPS-SPECIFIED PARCELS OF REAL PROPERTY.

B: COLLECTING INTELLIGENCE RELATING TO THE COUNTY, RISKS OF WILD FIRES IN THE SURROUNDING REGION, AND HISTORICAL DATA MAINTAINED IN THE NETWORK DATABASE, AND GENERATING GPS-SPECIFIED ANTI-FIRE (AF) SPRAY PROTECTION MAPS AND TASK REPORTS.

C: PROVIDING AN AF LIQUID SPRAY SYSTEM TO A GPS-SPECIFIED LOCATION FOR SPRAYING GPS-SPECIFIED PARCELS OF REAL PROPERTY WITH AF LIQUID SPRAY.

D: PROVIDING A SUPPLY OF AF CHEMICAL LIQUID SPRAY TO THE GPS-SPECIFIED LOCATION OF THE AF LIQUID SPRAY SYSTEM

E: FILLING THE AF LIQUID SPRAY SYSTEM WITH THE PROVIDED SUPPLY OF AF LIQUID

**FIG. 32A**



F: PRIOR TO THE ARRIVAL OF A WILD FIRE TO THE REGION, AND BASED ON THE GPS-SPECIFIED ANTI-FIRE (AF) SPRAY PROTECTION MAPS AND TASK REPORTS, THE SYSTEM ISSUING A REQUEST TO THE PROPERTY OWNER, OR REGISTERED CONTRACTOR, TO APPLY AF CHEMICAL LIQUID SPRAY ON THE GPS-SPECIFIED PARCELS OF PROPERTY USING THE AF LIQUID SPRAY SYSTEM.

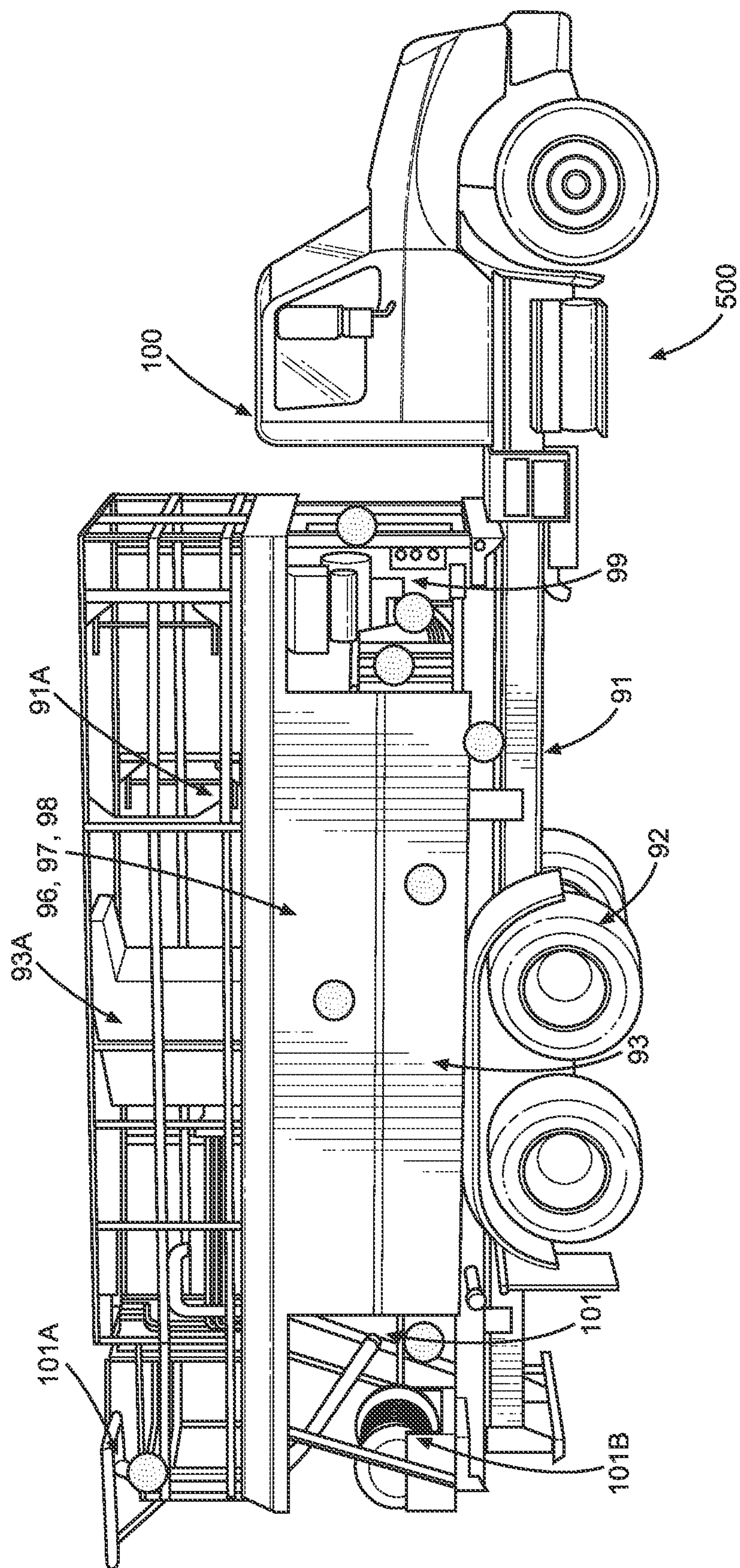
G: IN RESPONSE TO THE ISSUED REQUEST, THE PROPERTY OWNER OR CONTRACTOR THEREOF APPLYING AF LIQUID SPRAY ON THE REAL PROPERTY USING THE AF LIQUID SPRAY SYSTEM, AND THE SYSTEM REMOTELY MONITORING THE CONSUMPTION AND APPLICATION OF THE AF CHEMICAL LIQUID ON THE PROPERTY ON A GIVEN DATE AND TIME, AND AUTOMATICALLY RECORDING THE TRANSACTION IN THE NETWORK DATABASE.

H: THE SYSTEM UPDATING RECORDS IN THE NETWORK DATABASE ASSOCIATED WITH EACH APPLICATION OF AF CHEMICAL LIQUID SPRAY ON A GPS-SPECIFIED PARCEL OF REAL PROPERTY.

**FIG. 32B**



**MOBILE VEHICLE FOR MIXING AND SPRAYING CLEAN FIRE AND SMOKE INHIBITING SLURRY  
ON GROUND SURFACES TO CREATE FIRE BREAKS AND SMOTHER AMBERS AND ASH, AND IN OTHER APPLICATION**



# 3364



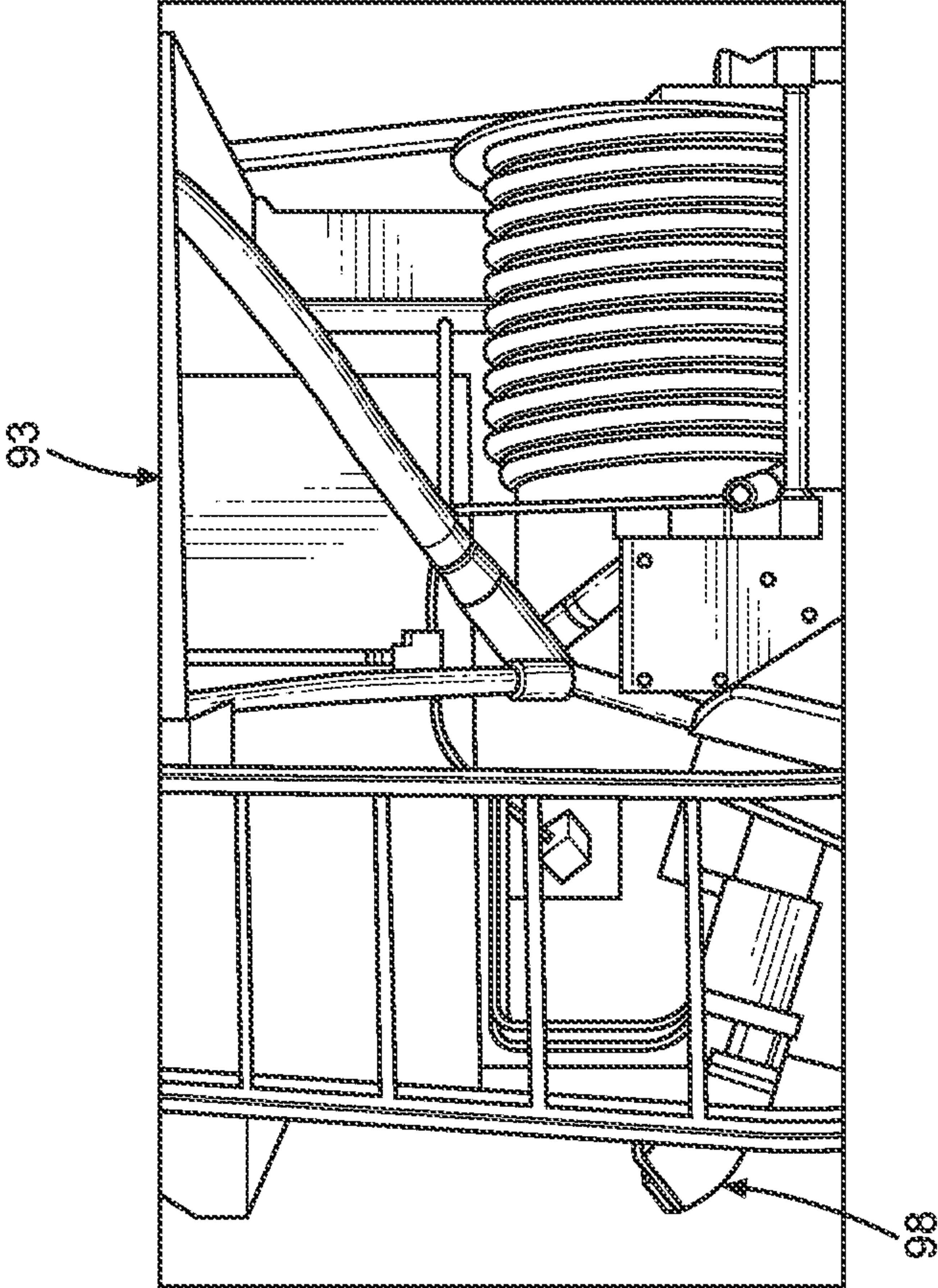


FIG. 33B

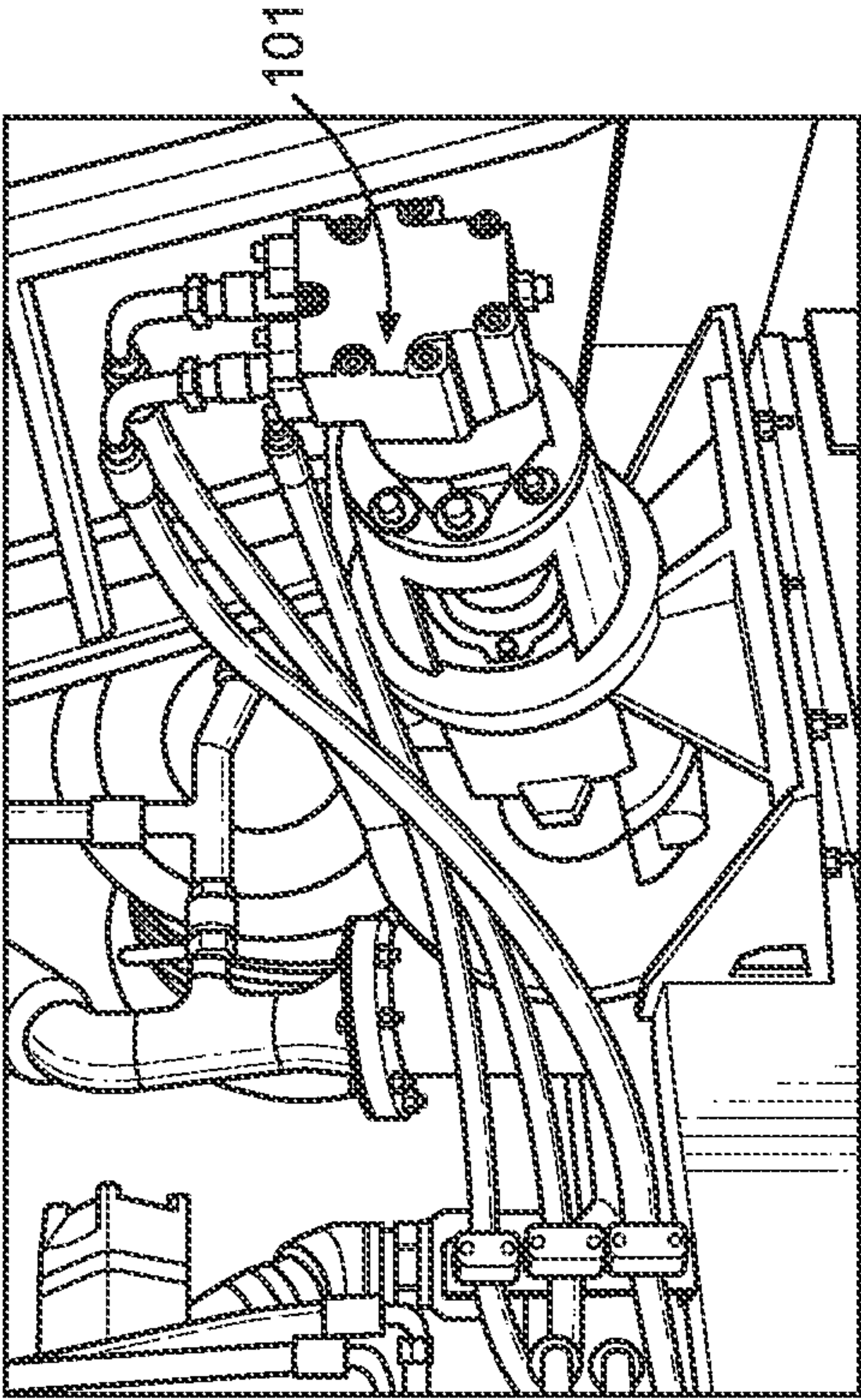


FIG. 33C



APPARATUS FOR APPLYING FIRE AND SMOKE INHIBITING COMPOSITIONS ON GROUND SURFACES BEFORE THE  
INCIDENCE OF WILD-FIRES, AND ALSO THEREAFTER, UPON SMOLDERING AMBERS AND ASHES TO REDUCE  
SMOKE AND SUPPRESS FIRE RE-IGNITION

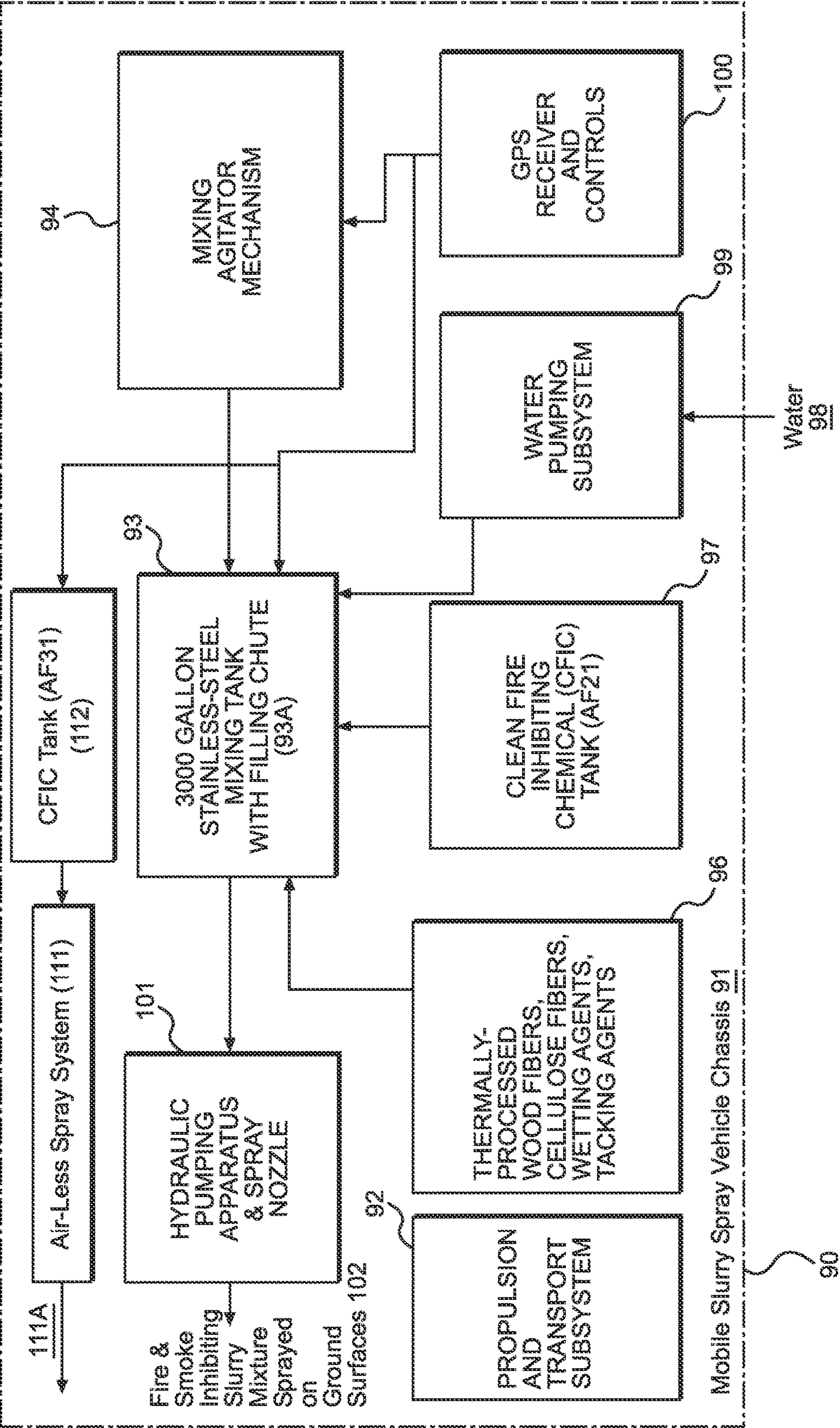


FIG. 34



METHOD OF METHOD OF APPLYING FIRE AND SMOKE INHIBITING SLURRY COMPOSITIONS ON GROUND SURFACES BEFORE THE INCIDENCE OF WILD-FIRES, AND ALSO THEREAFTER, UPON SMOLDERING AMBERS AND ASHES TO REDUCE SMOKE AND SUPPRESS FIRE RE-IGNITION

A : MEASURING AND STAKING OUT AREA USING GPS COORDINATES TO ENSURE PROPER APPLICATION

B : BLENDING PROCESSED WOOD FIBERS, CELLULOSE FIBER, WETTING AGENTS, TACKING AGENTS, AND CLEANFIRE INHIBITING CHEMICALS (CFIC), WITH A SUPPLY OF WATER TO MAKE UP A FIRE AND SMOKE INHIBITING SLURRY COMPOSITION ACCORDING TO THE PRESENT INVENTION

C : MIXING THE FIRE AND SMOKE INHIBITING SLURRY MIXTURE IN A MIXING TANK ON A MOBILE VEHICLE SUPPORTING HYDRAULIC SPRAY EQUIPMENT

D : SPRAYING THE MIXED FIRE AND SMOKE INHIBITING SLURRY MIXTURE ON THE SPECIFIC GROUND SURFACE USING HYDRAULIC SPRAY EQUIPMENT SUPPORTED ON THE MOBILE SPRAY VEHICLE

E : SPRAYING CFIC LIQUID MIST OVER THE THE APPLIED FIRE AND SMOKE INHIBITING SLURRY COATING ON THE GROUND SURFACE TO PROVIDE EROSION CONTROL, FIRE PROTECTION AND SMOKE REDUCTION IN THE PRESENT OF A WILDFIRE

**FIG. 35**



A BASE HYDRAULIC MULCH LOADING CHART FOR MAKING  
THE FIRE AND SMOKE INHIBITING SLURRY MIXTURE OF THE PRESENT INVENTION

| Water (gal) | Mulch Fiber (50-lb bales) | Mulch Fiber (lb) | 1,500  |         | 2,000  |         | 2,500  |         |
|-------------|---------------------------|------------------|--------|---------|--------|---------|--------|---------|
|             |                           |                  | Sq ft  | lb/acre | Sq ft  | lb/acre | Sq ft  | lb/acre |
| 100         | 1                         | 50               | 1,452  | 0.033   | 1,089  | 0.025   | 871    | 0.020   |
| 300         | 3                         | 150              | 4,356  | 0.099   | 3,267  | 0.075   | 2,614  | 0.060   |
| 500         | 5                         | 250              | 7,260  | 0.166   | 5,445  | 0.125   | 4,356  | 0.100   |
| 700         | 7                         | 350              | 10,164 | 0.233   | 7,623  | 0.175   | 6,098  | 0.140   |
| 900         | 9                         | 450              | 13,068 | 0.299   | 9,801  | 0.225   | 7,841  | 0.180   |
| 1,100       | 11                        | 550              | 15,972 | 0.366   | 11,979 | 0.275   | 9,583  | 0.220   |
| 1,300       | 13                        | 650              | 18,876 | 0.433   | 14,157 | 0.325   | 11,326 | 0.260   |
| 1,500       | 15                        | 750              | 21,780 | 0.499   | 16,335 | 0.375   | 13,068 | 0.300   |
| 1,700       | 17                        | 850              | 24,684 | 0.566   | 18,513 | 0.425   | 14,810 | 0.340   |
| 1,900       | 19                        | 950              | 27,588 | 0.633   | 20,691 | 0.475   | 16,553 | 0.380   |
| 2,100       | 21                        | 1,050            | 30,492 | 0.699   | 22,869 | 0.525   | 18,295 | 0.420   |
| 2,300       | 23                        | 1,150            | 33,396 | 0.766   | 25,047 | 0.575   | 20,038 | 0.460   |
| 2,500       | 25                        | 1,250            | 36,300 | 0.833   | 27,225 | 0.625   | 21,780 | 0.500   |
| 2,700       | 27                        | 1,350            | 39,204 | 0.899   | 29,403 | 0.675   | 23,522 | 0.540   |
| 2,900       | 29                        | 1,450            | 42,108 | 0.966   | 31,581 | 0.725   | 25,265 | 0.580   |

FIG. 36



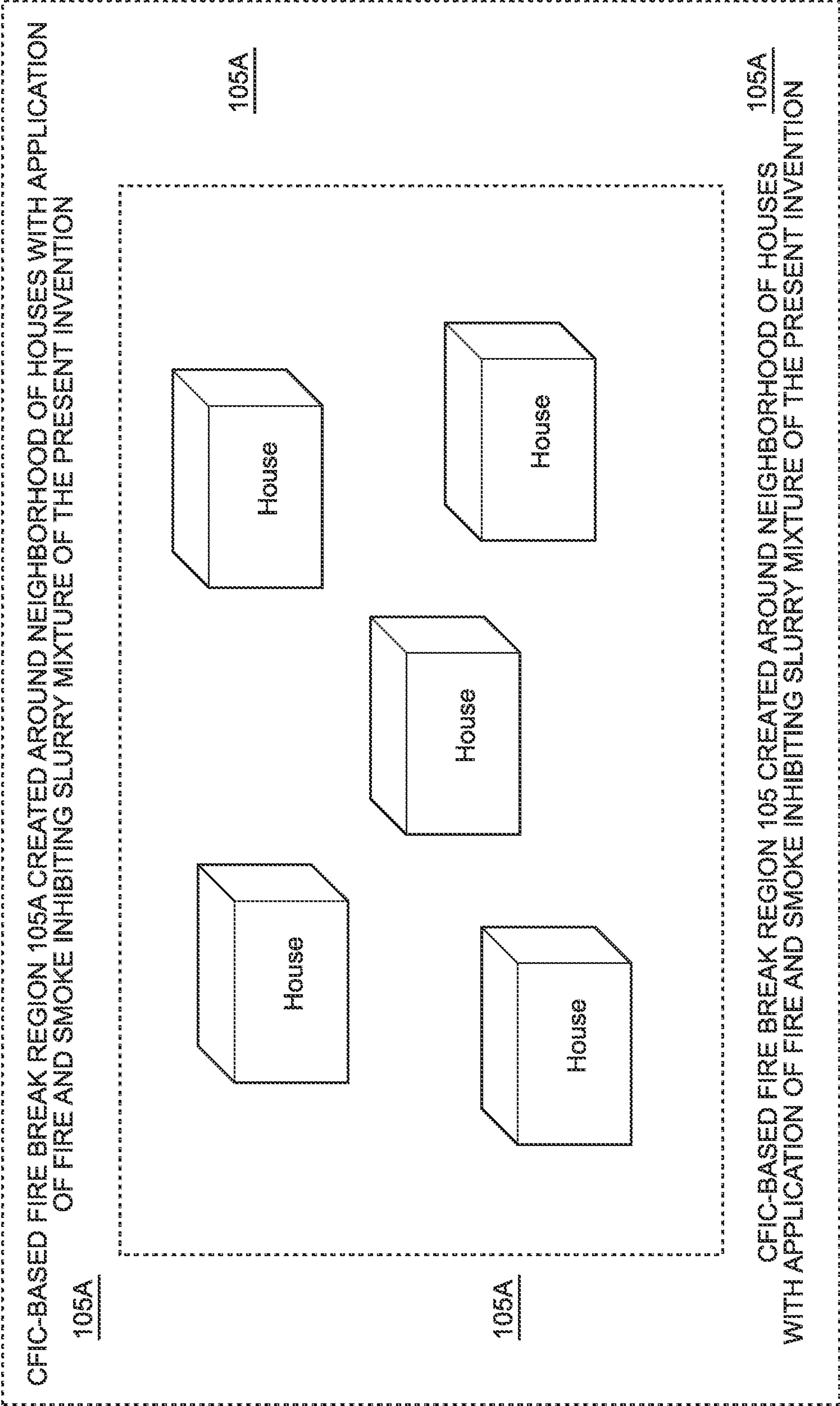
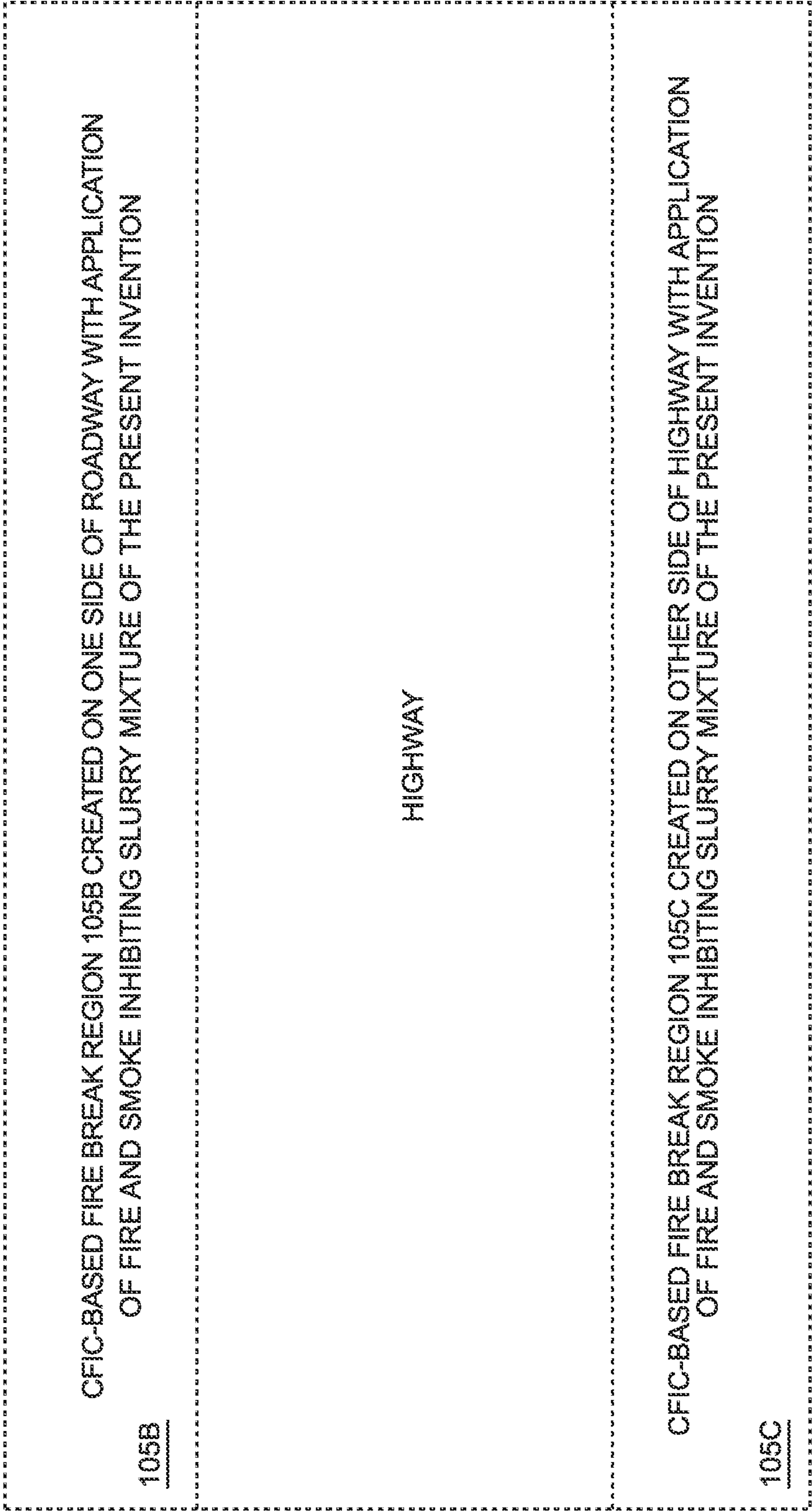


FIG. 37





**FIG. 38**



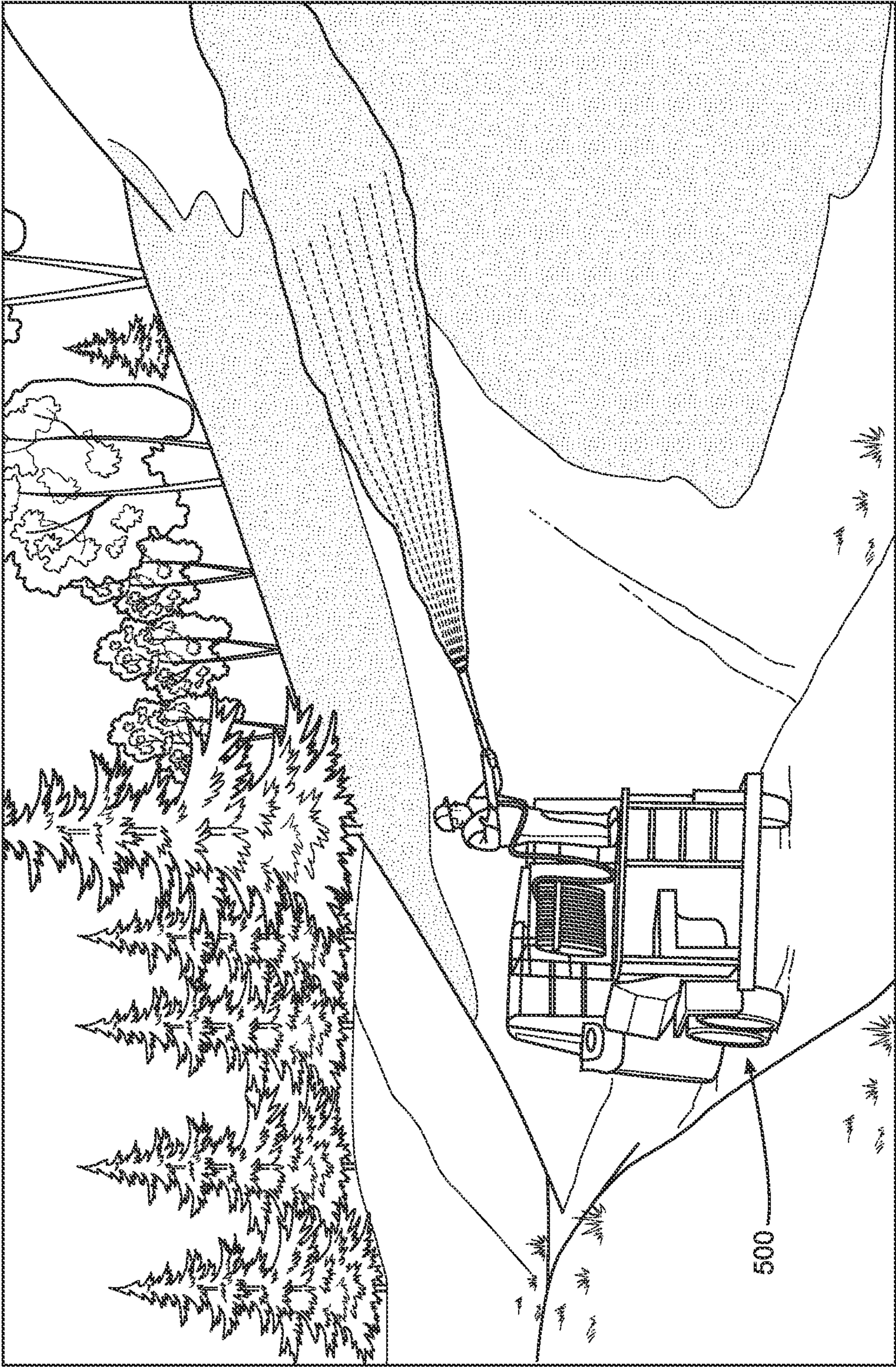


FIG. 39



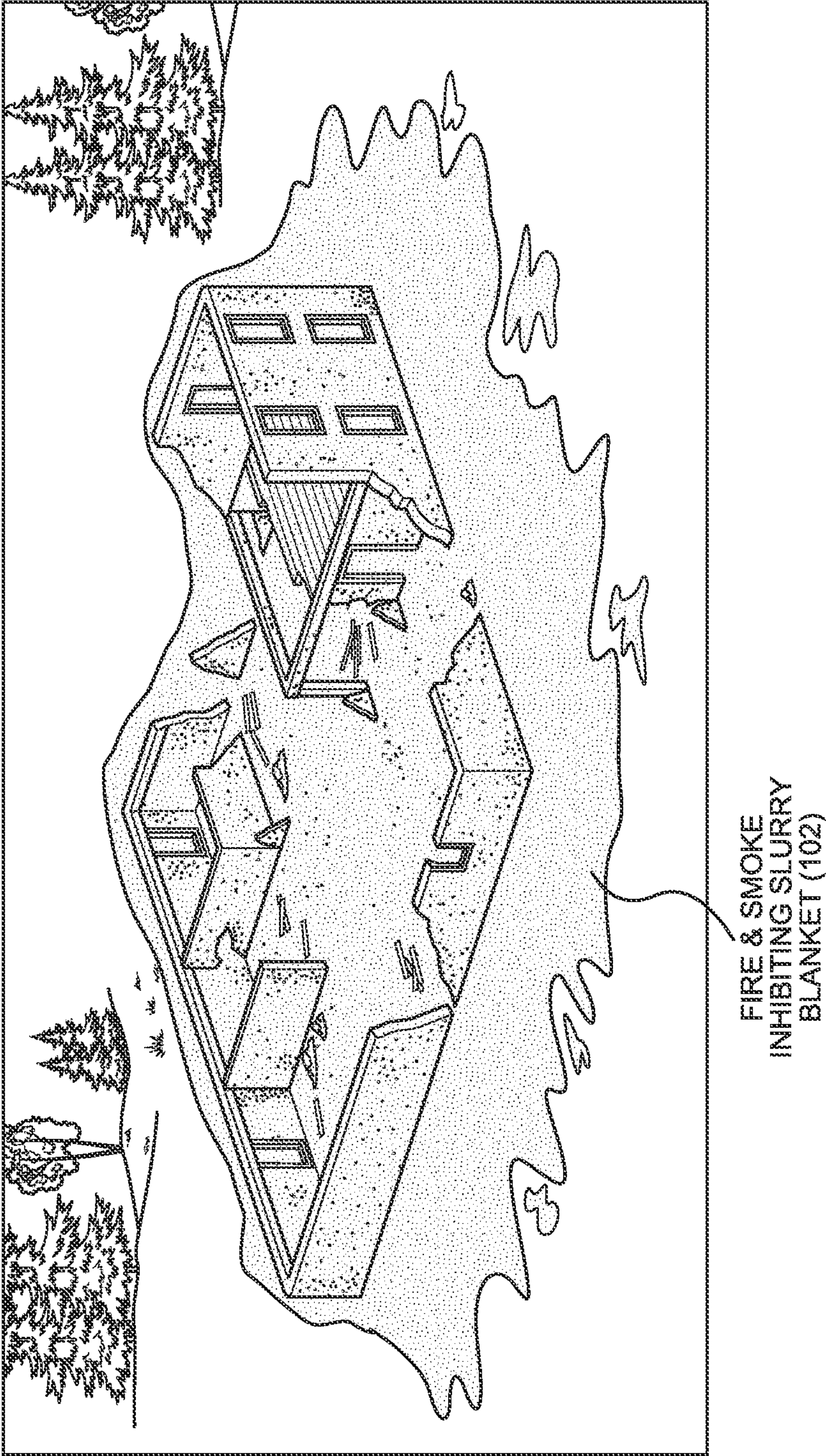


FIG. 40



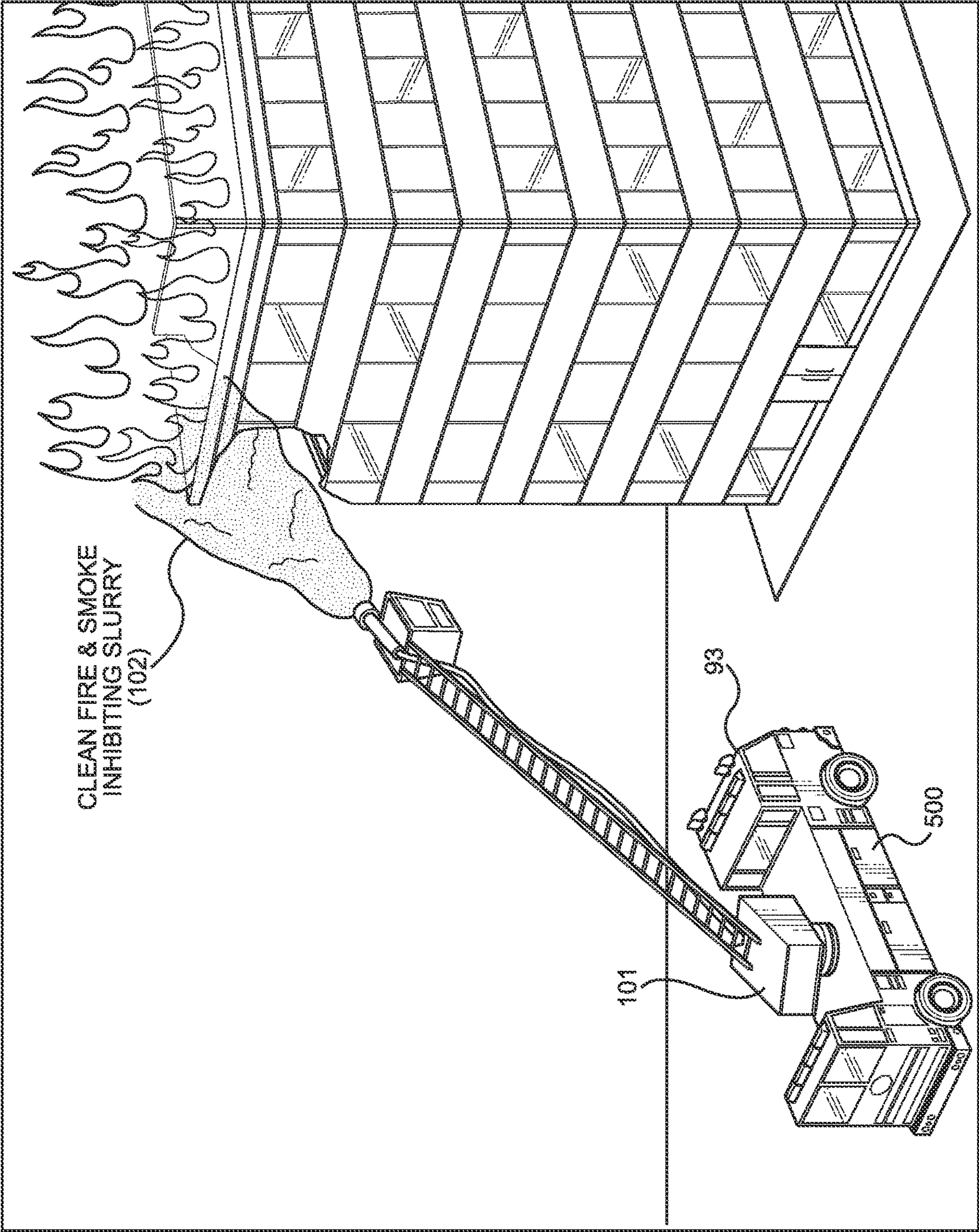
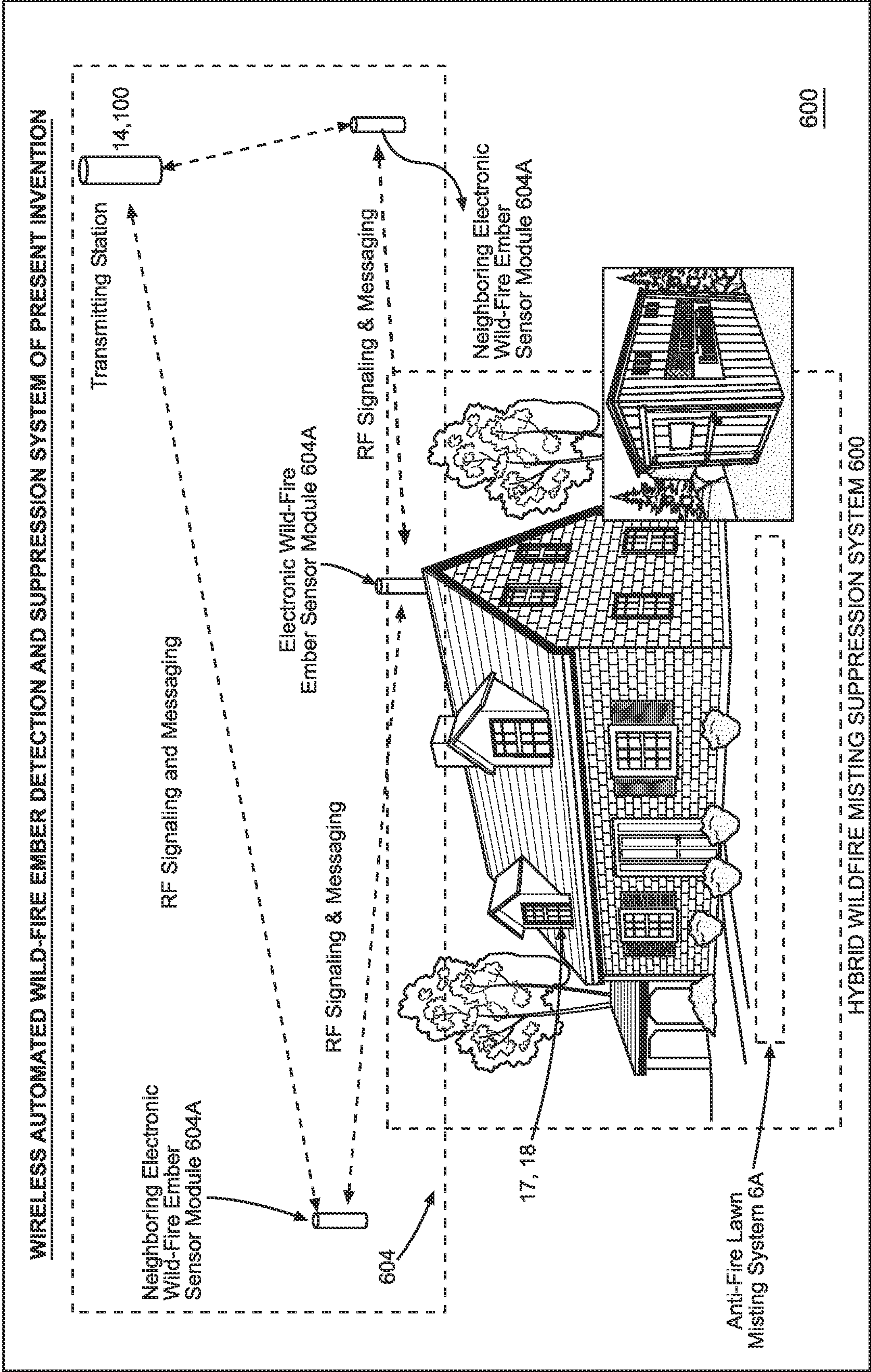


FIG. 41





**FIG. 42A**



WIRELESS PASSIVE INFRARED (PIR) THERMAL IMAGING FIRE-OUTBREAK AND ARSON-ATTACK SENSOR

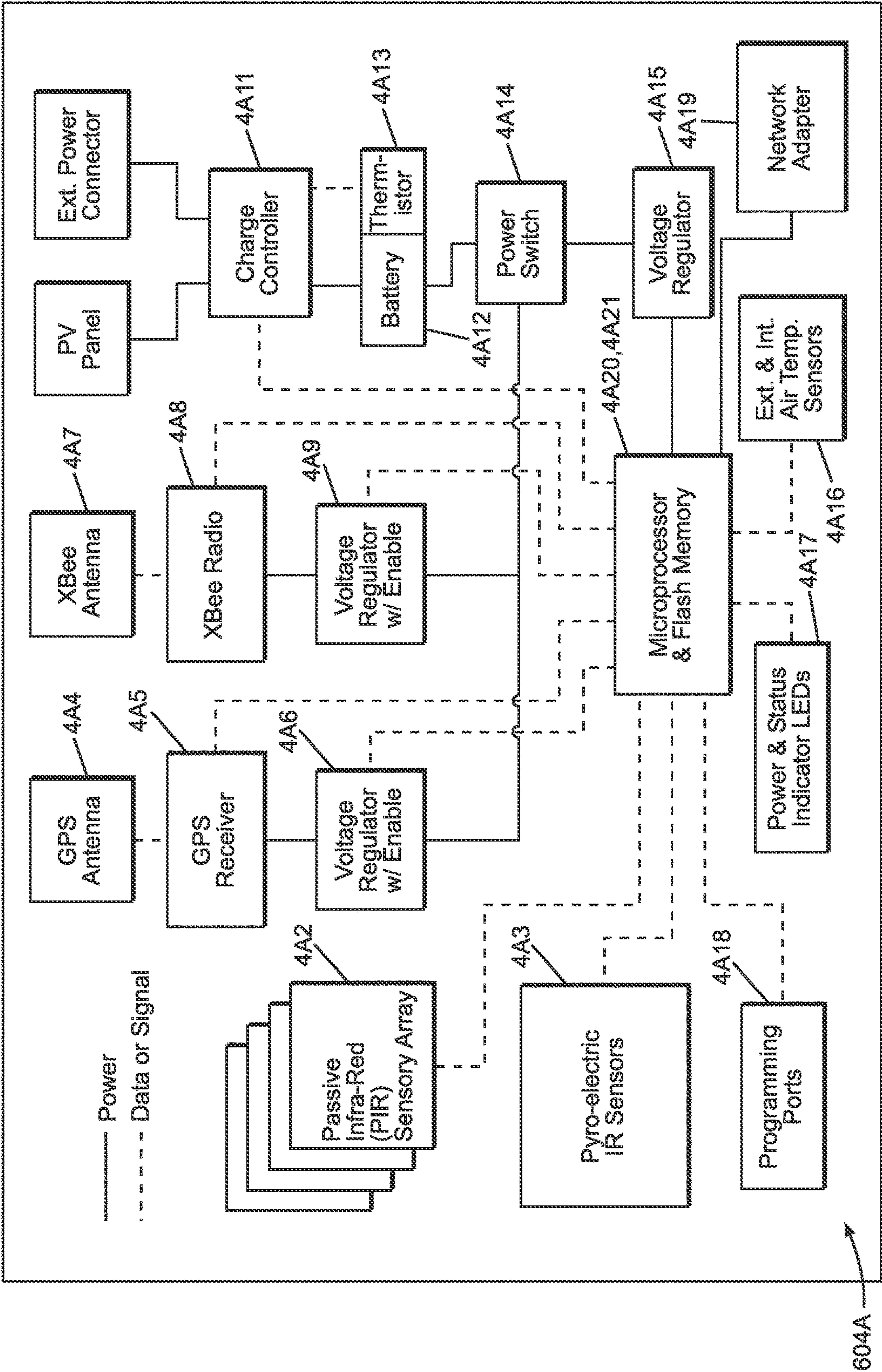
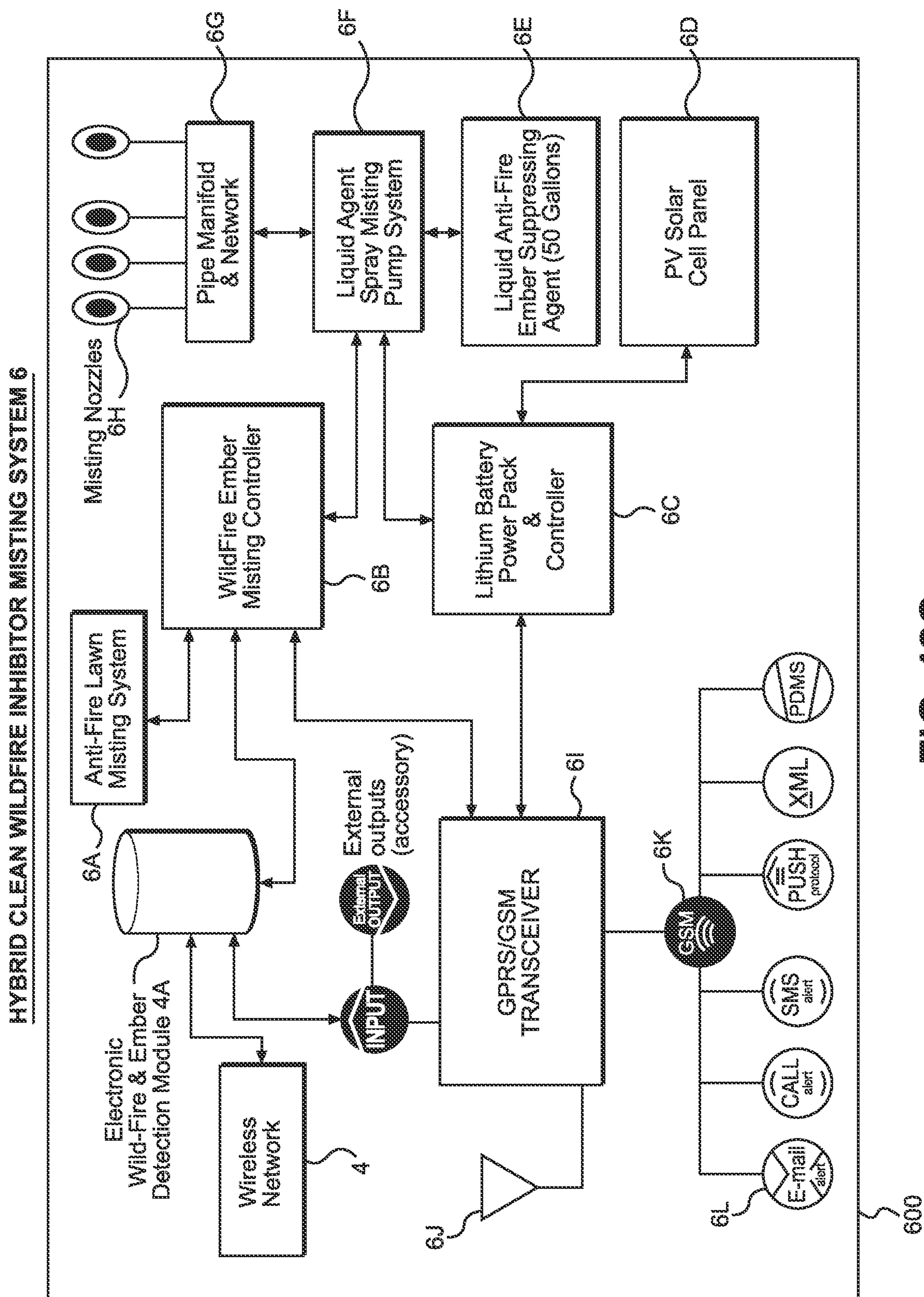
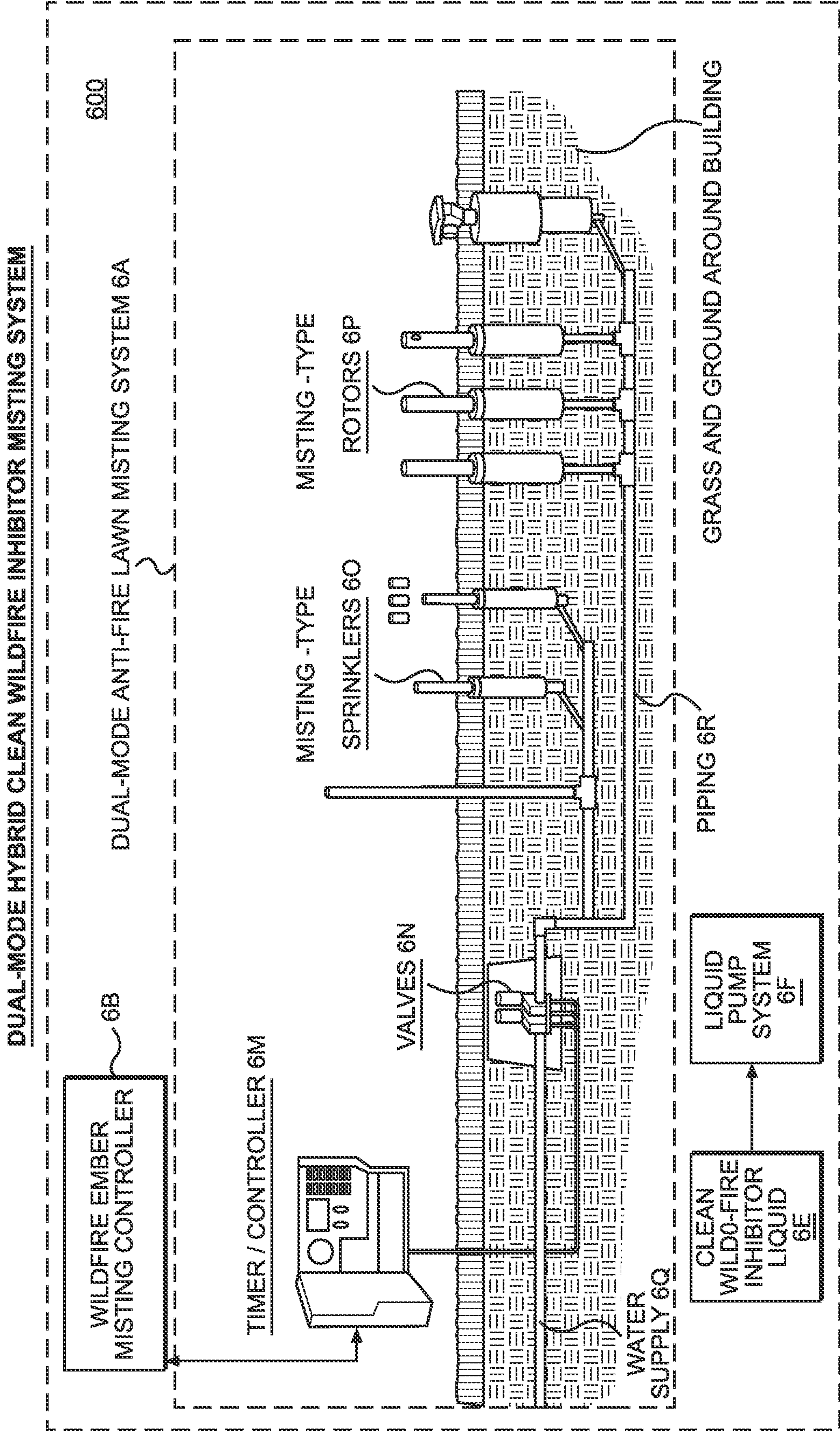


FIG. 42B









**FIG. 42D**



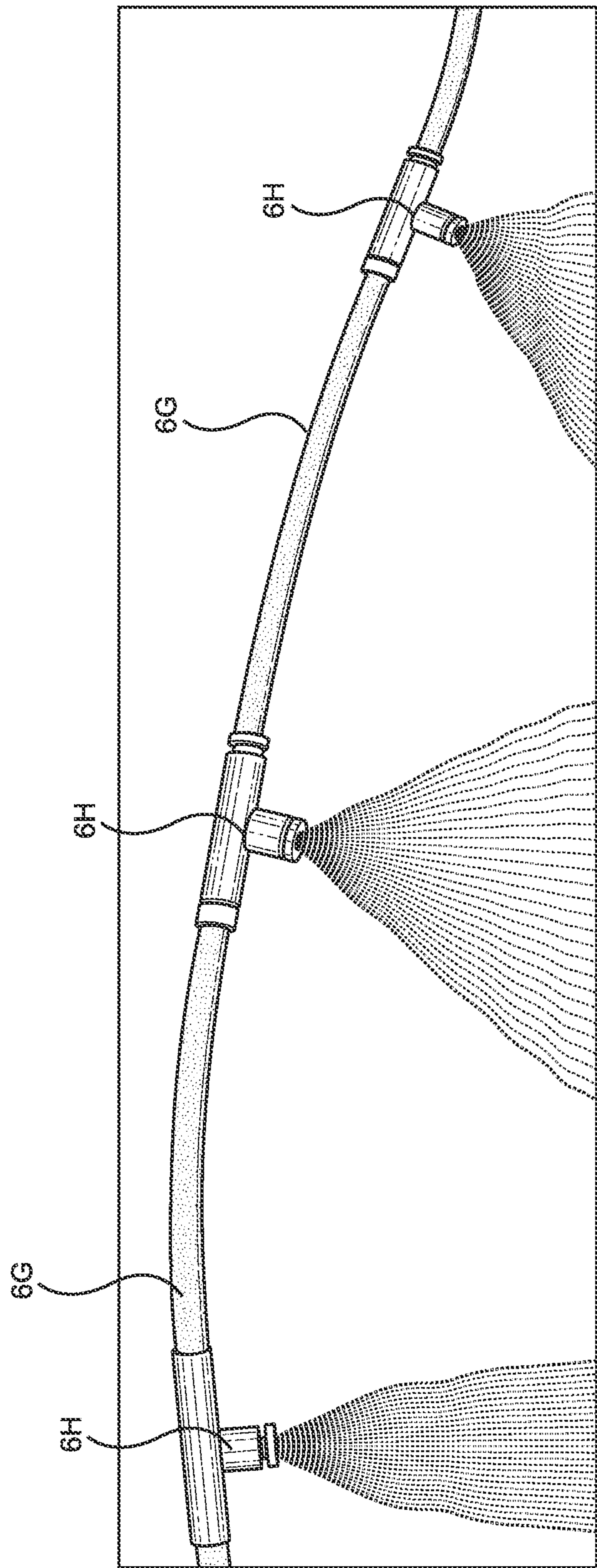


FIG. 42E



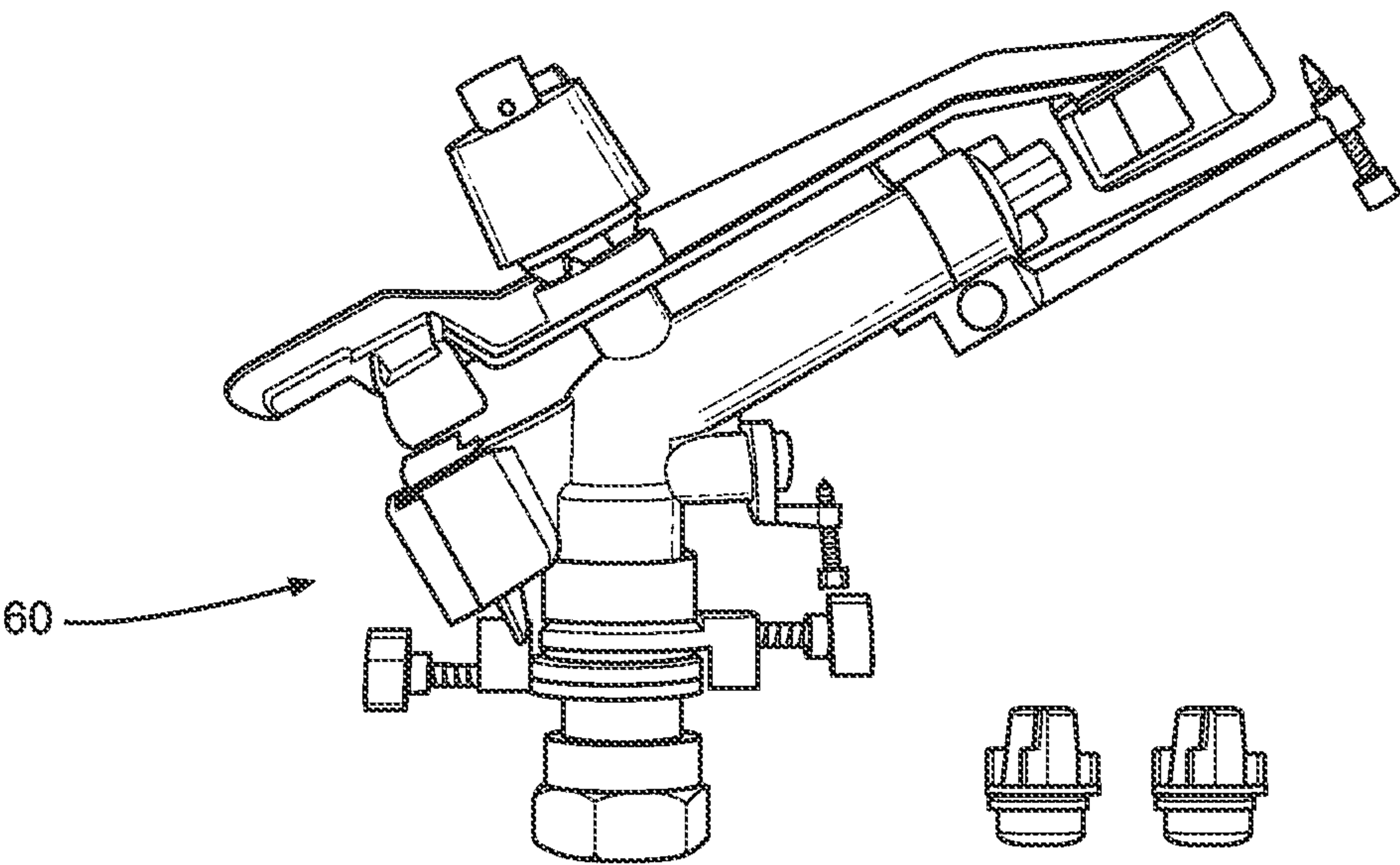


FIG. 42F1

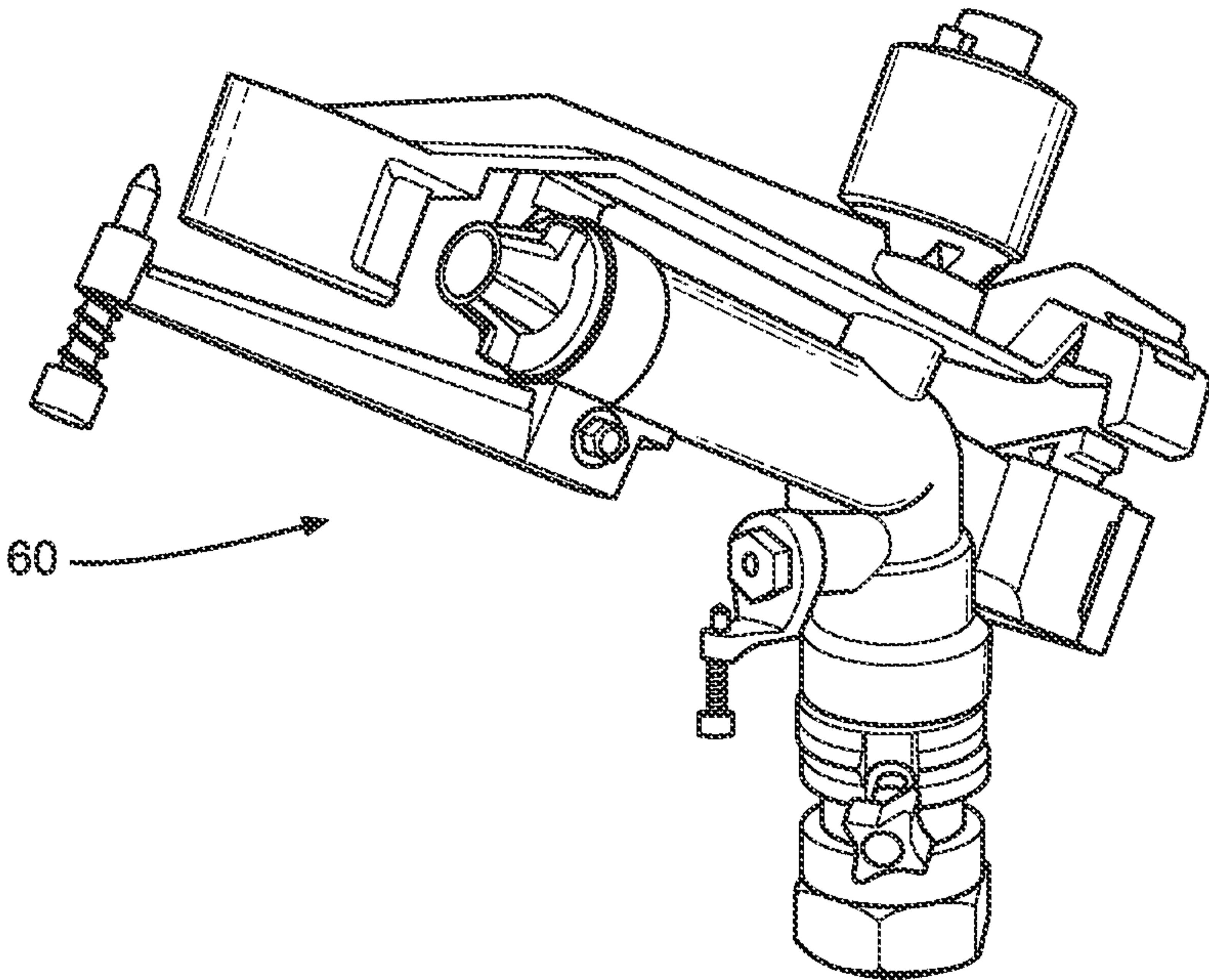


FIG. 42F2



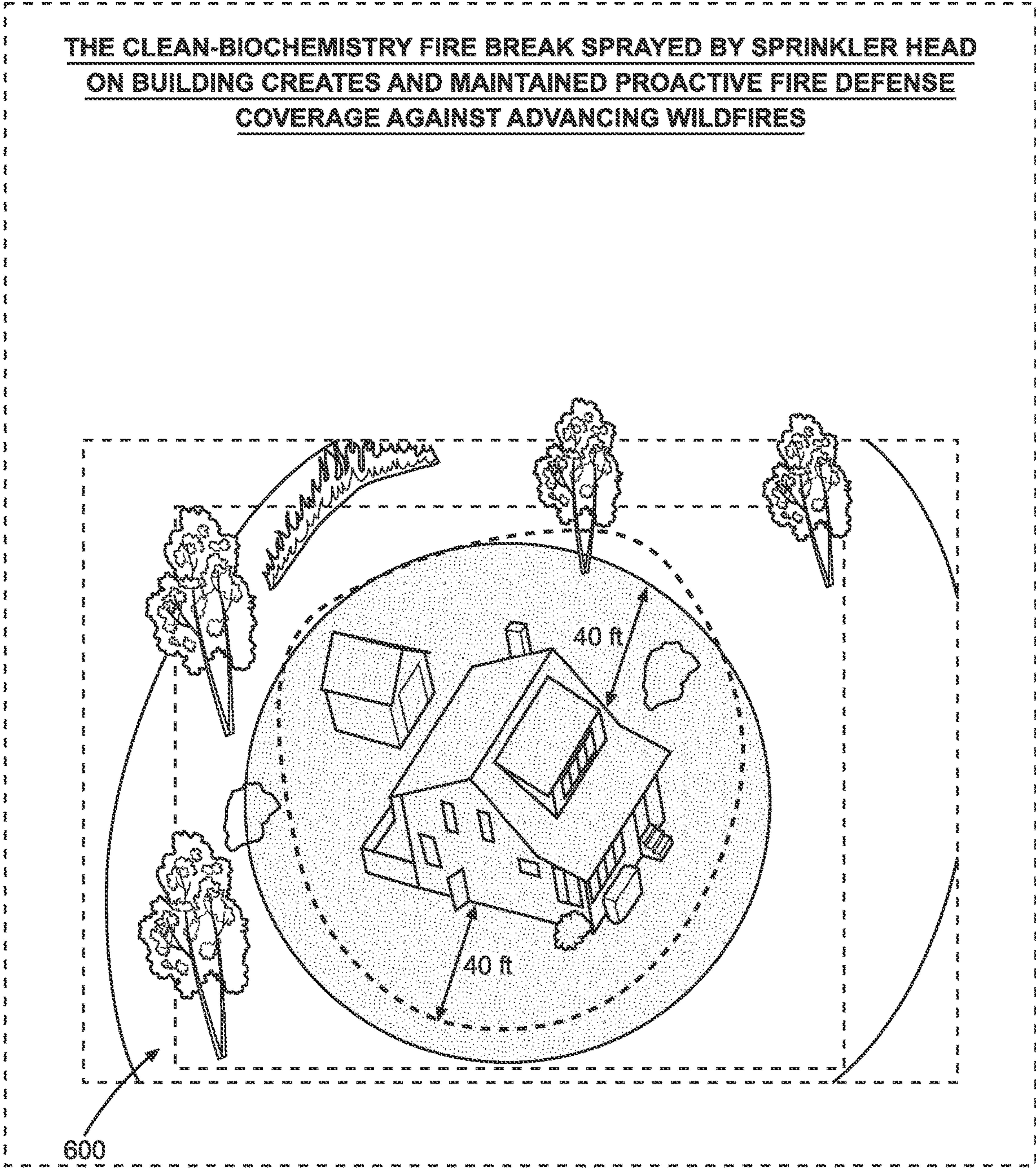


FIG. 42G



MOBILE COMPUTING DEVICES DEPLOYED ON THE SYSTEM NETWORK OF THE PRESENT INVENTION

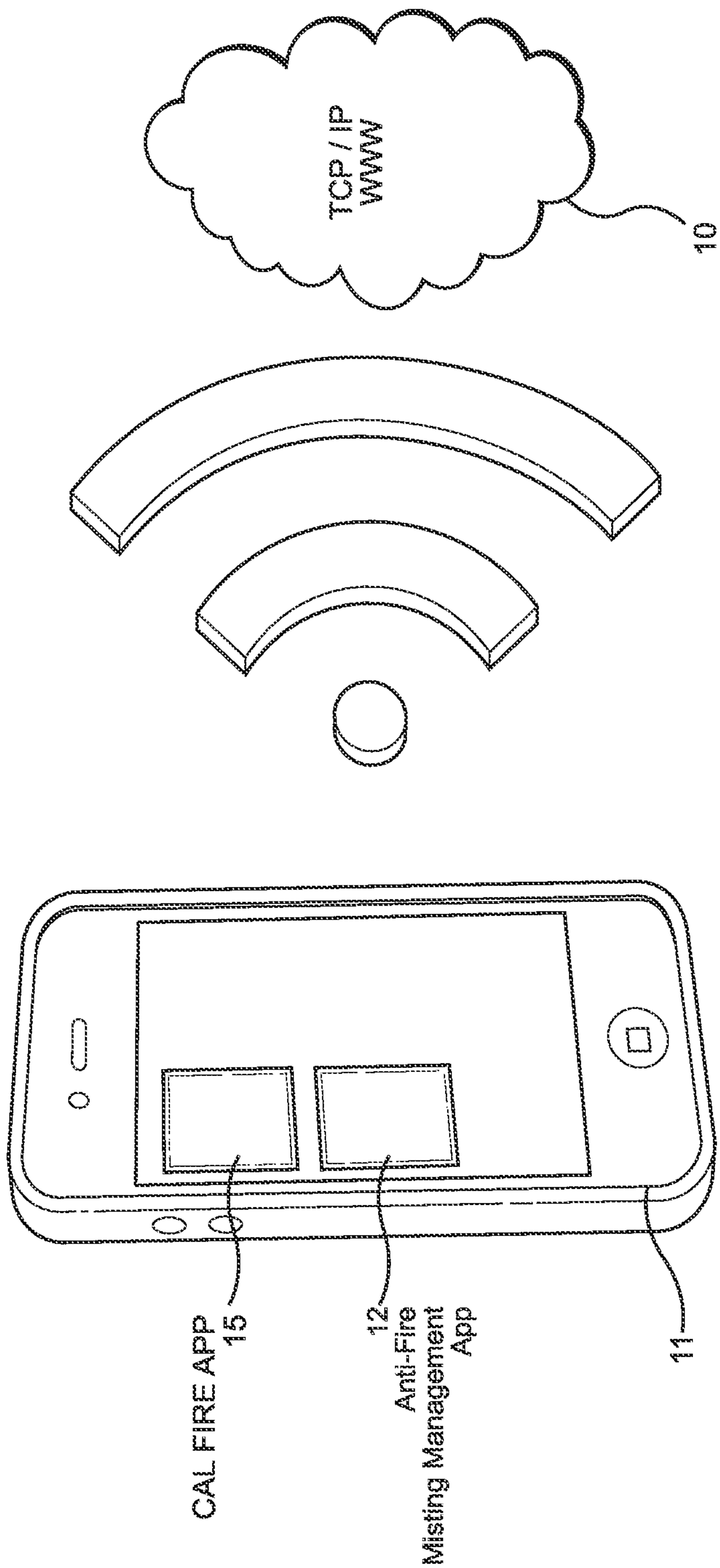
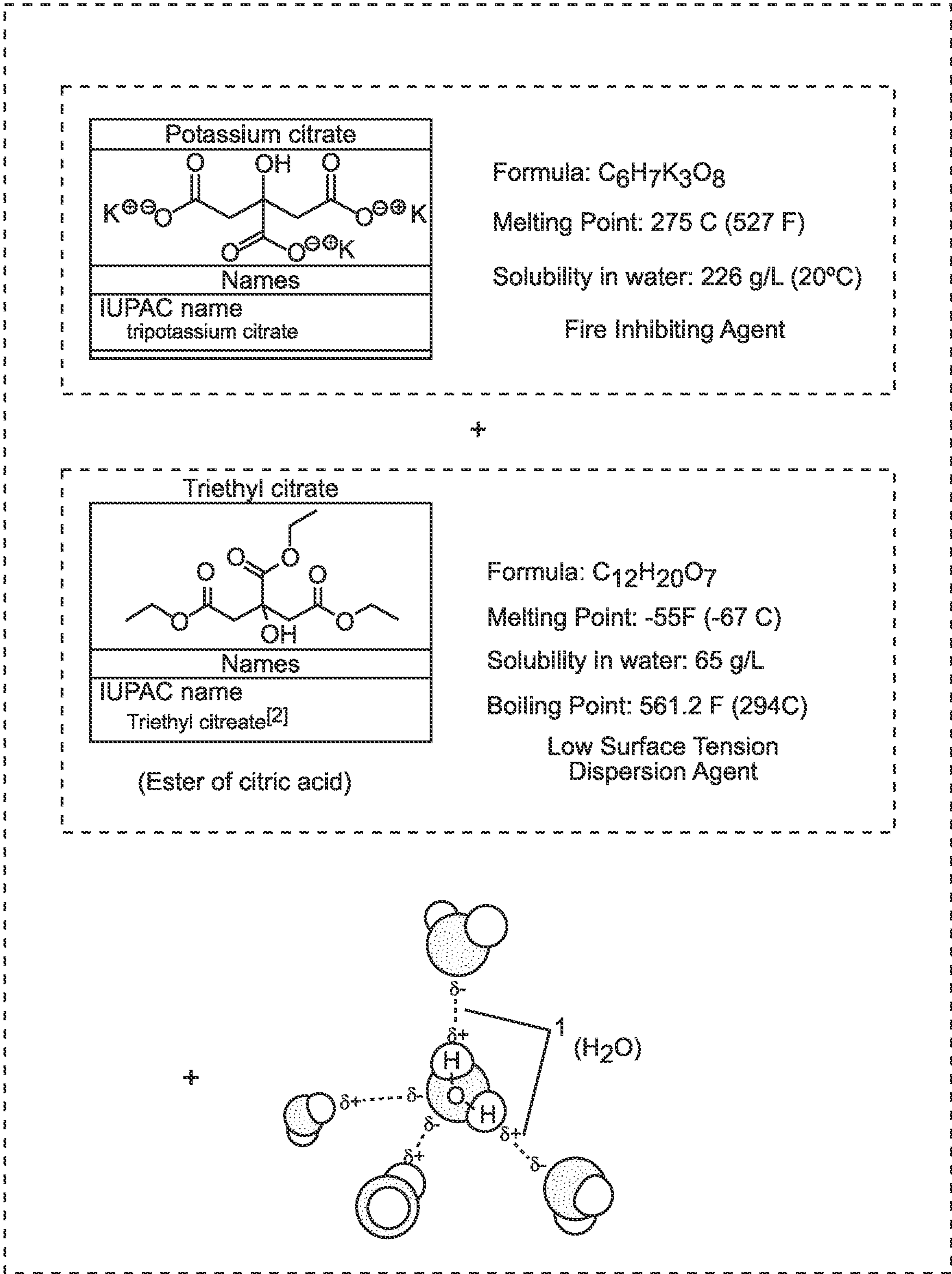


FIG. 43



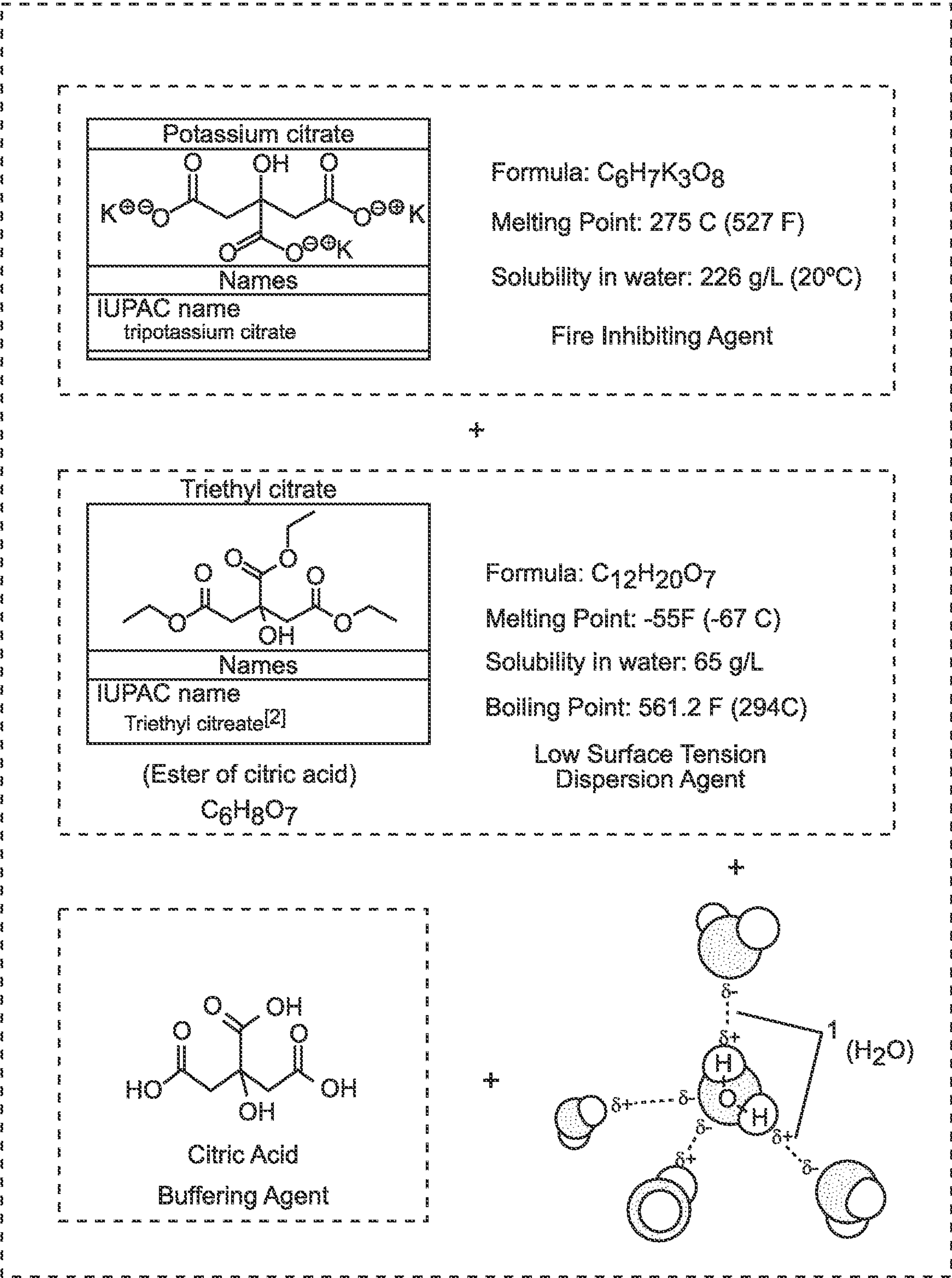
**FIRE EXTINGUISHING LIQUID CONCENTRATE COMPOSITIONS FOR  
ACTIVELY EXTINGUISHING FIRE ON CLASS-A AND CLASS B FIRES**



**FIG. 44**

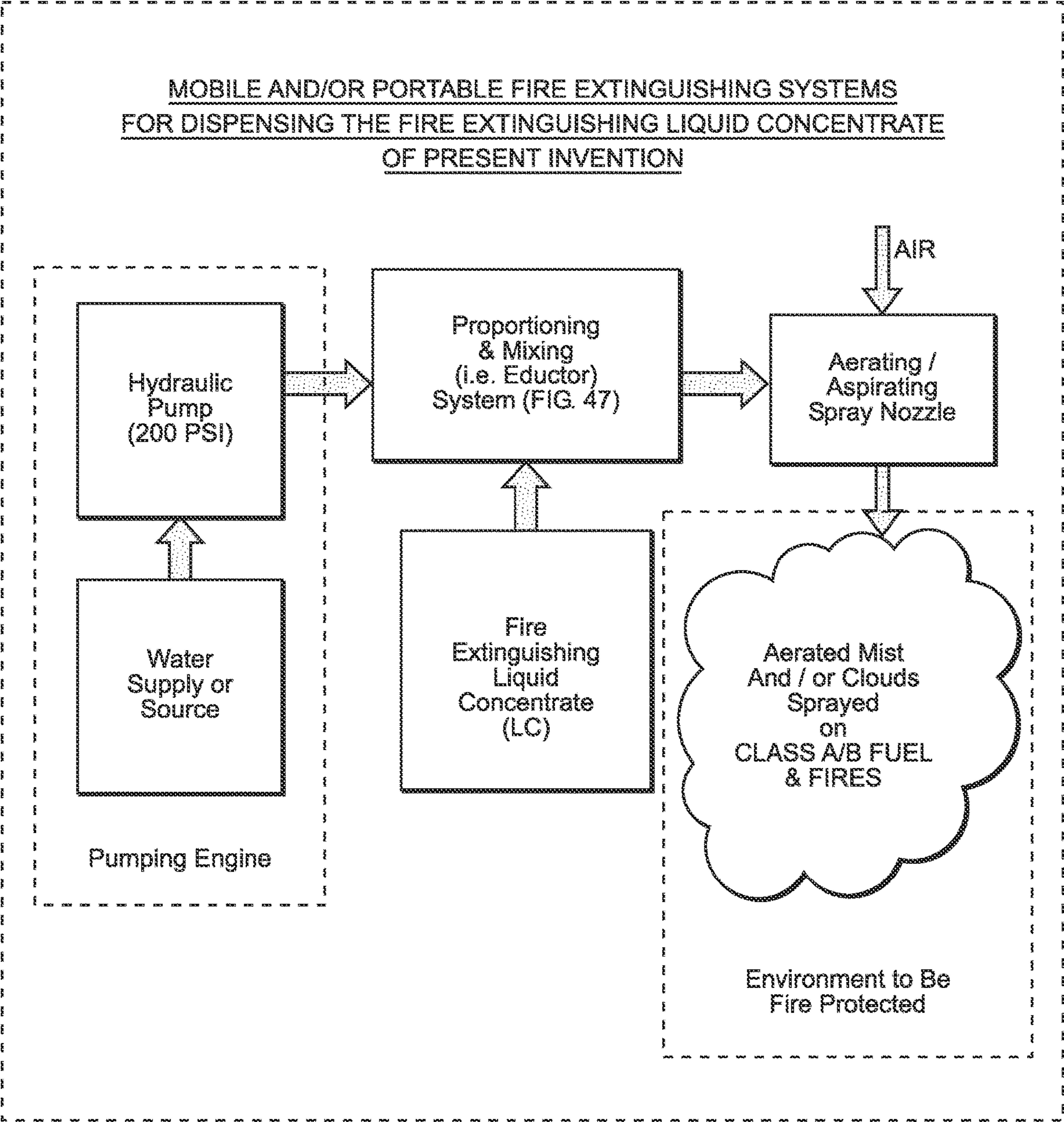


**FIRE EXTINGUISHING LIQUID CONCENTRATE COMPOSITIONS FOR  
ACTIVELY EXTINGUISHING FIRE INVOLVING CLASS-A AND CLASS B FUELS**



**FIG. 45**





**FIG. 46**



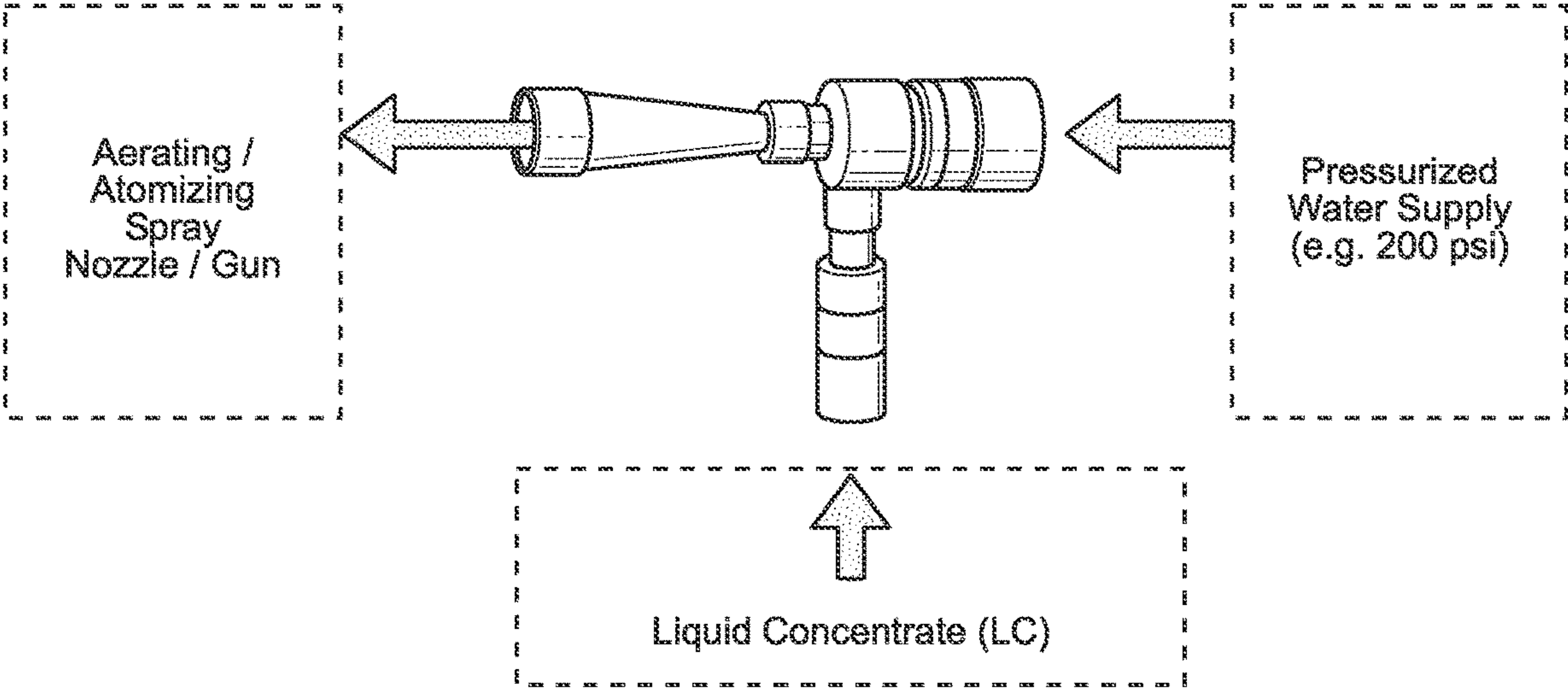


FIG. 47

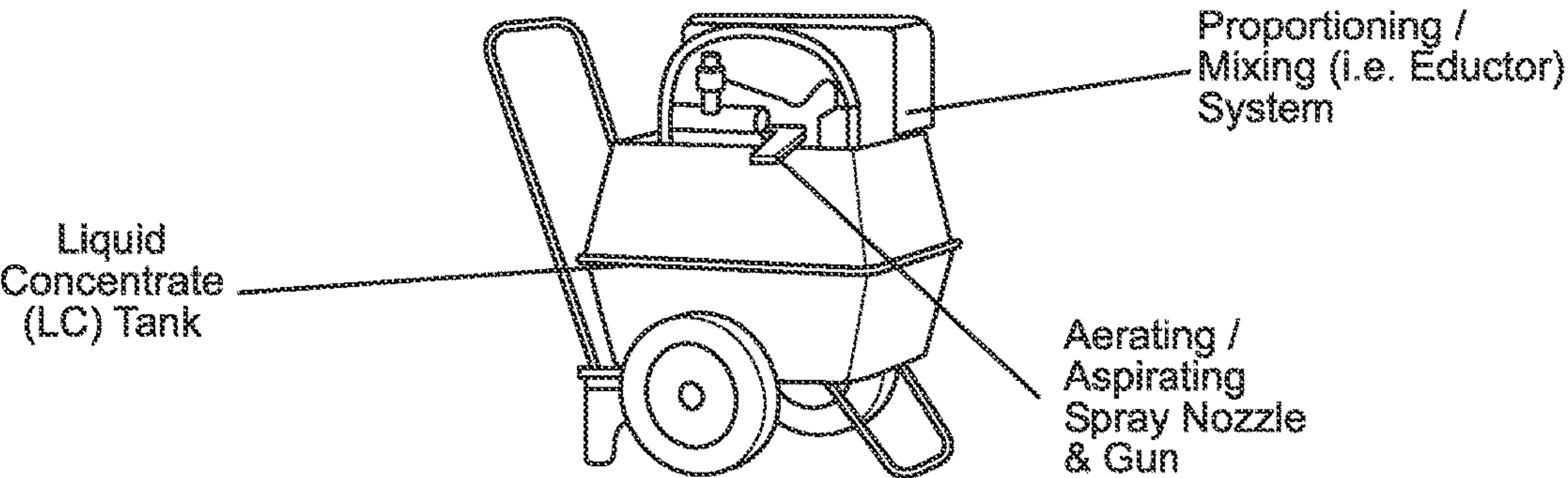


FIG. 48

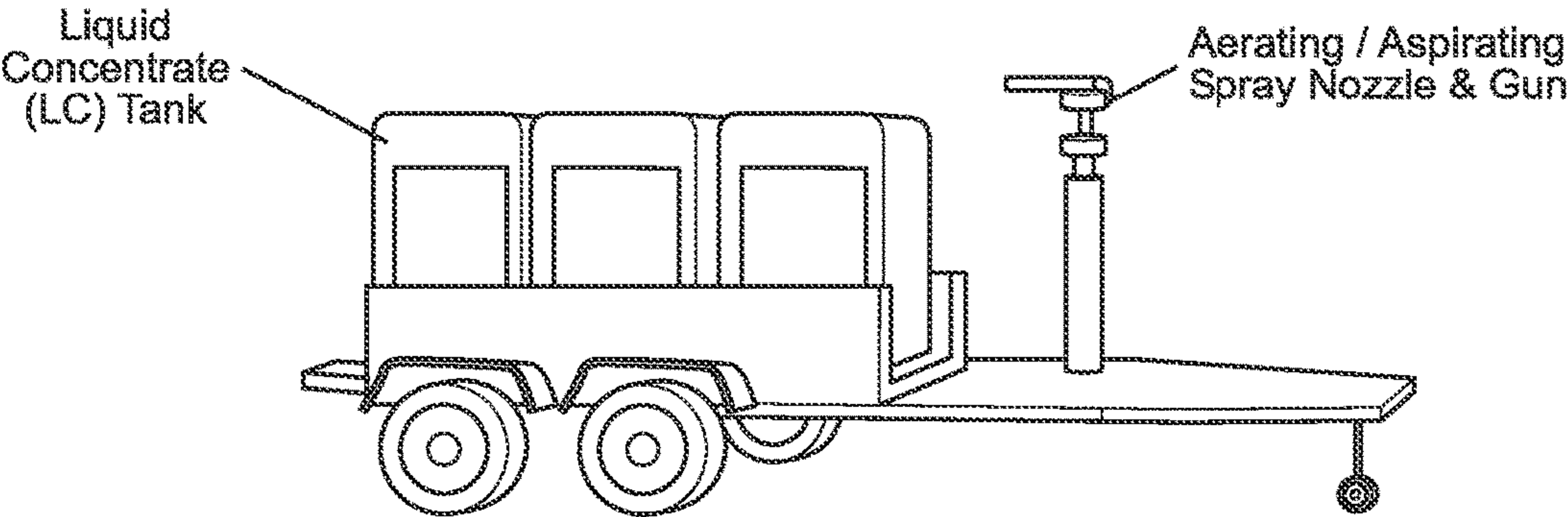
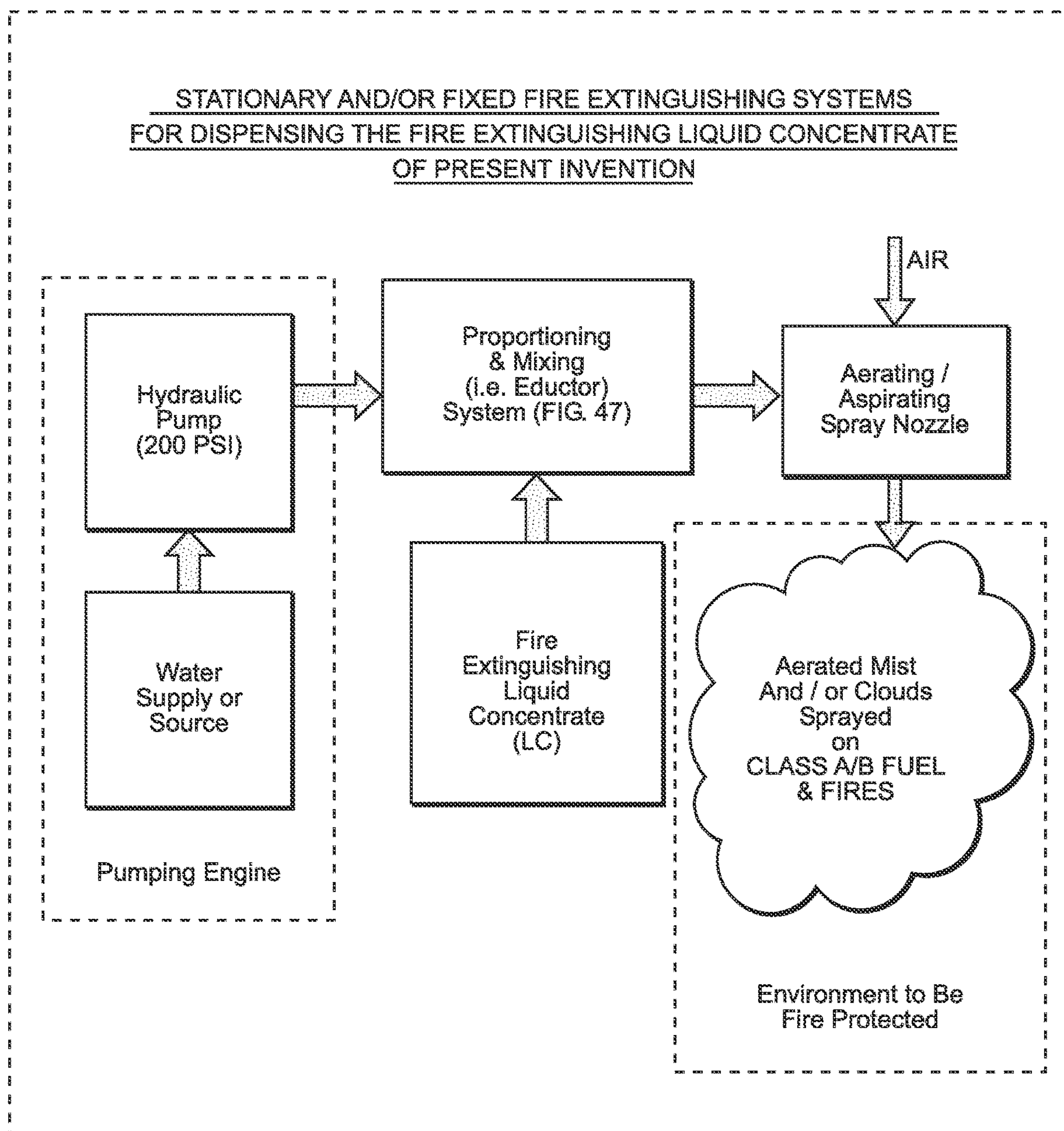


FIG. 49

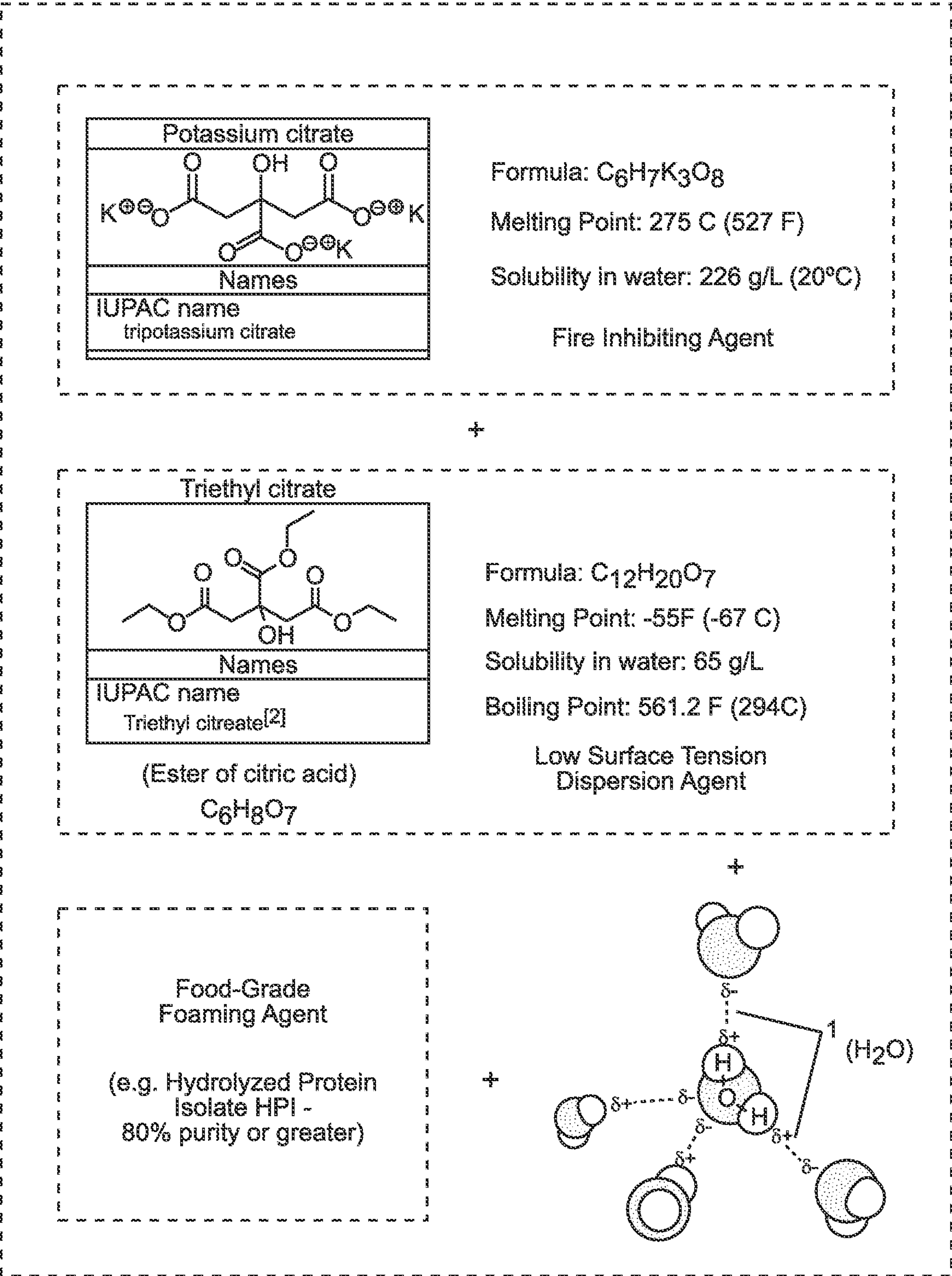




**FIG. 50**



**FIRE EXTINGUISHING FOAM CONCENTRATE FOR FIGHTING CLASS A/B FIRES**



**FIG. 51**



Biodegradable Fire Extinguishing Foam Concentrate  
For Producing Environmentally-Safe, Biodegradable  
Firefighting Foam For Use in Extinguishing Fires Involving  
Class A & B Fuels:

Performance Characteristics:

High Foam Stability

Fire Extinguishing Capability

Formulated Using Generally Safe Food-Grade Chemicals:

Food-Grade Foaming Agents

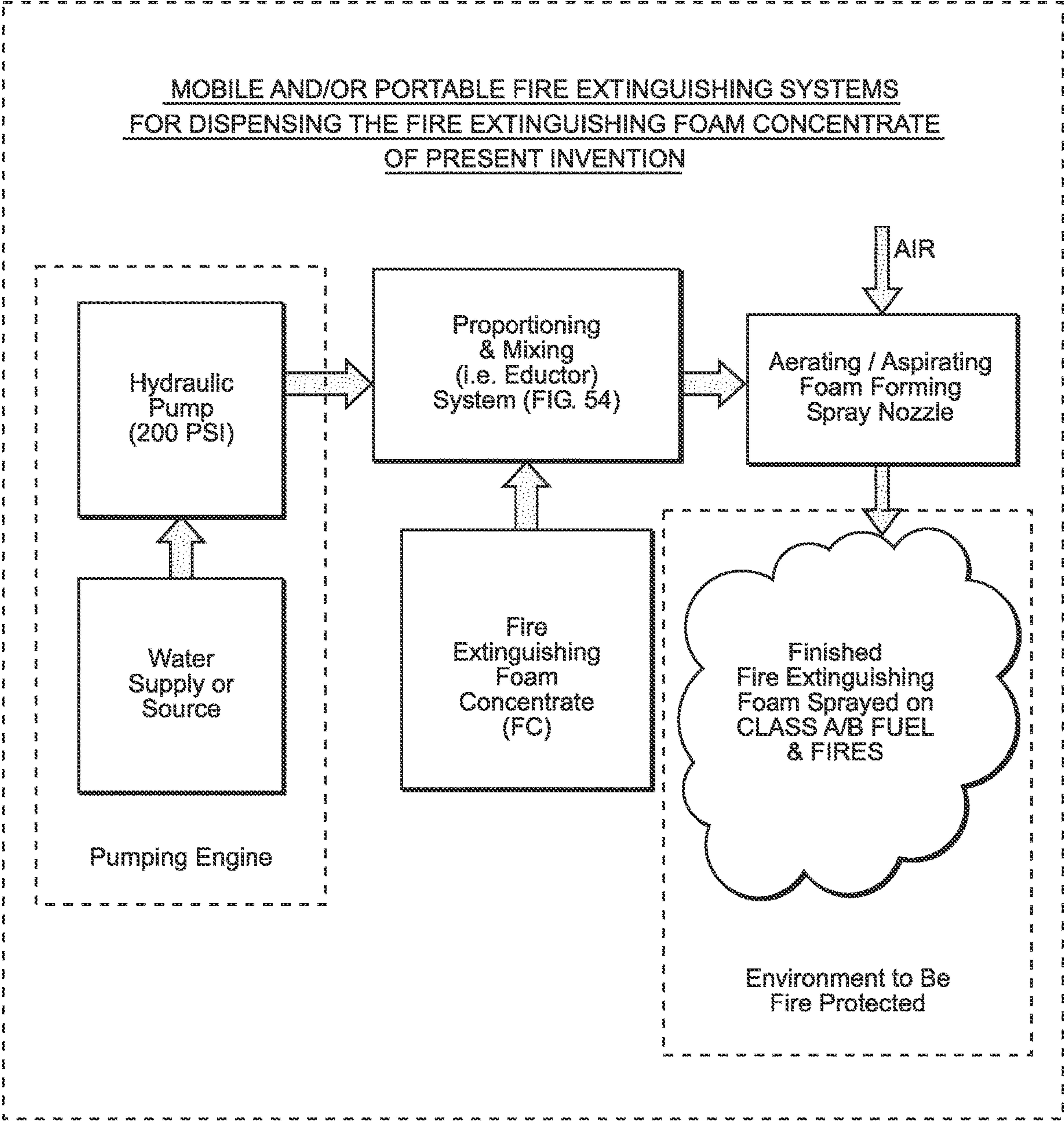
Fluorine-Free Low-Surface Tension Surfactants

Environmentally-Safe Free-Radical Fire Inhibitors

Free of Phosphates and Ammonia Compounds

**FIG. 52**





**FIG. 53**



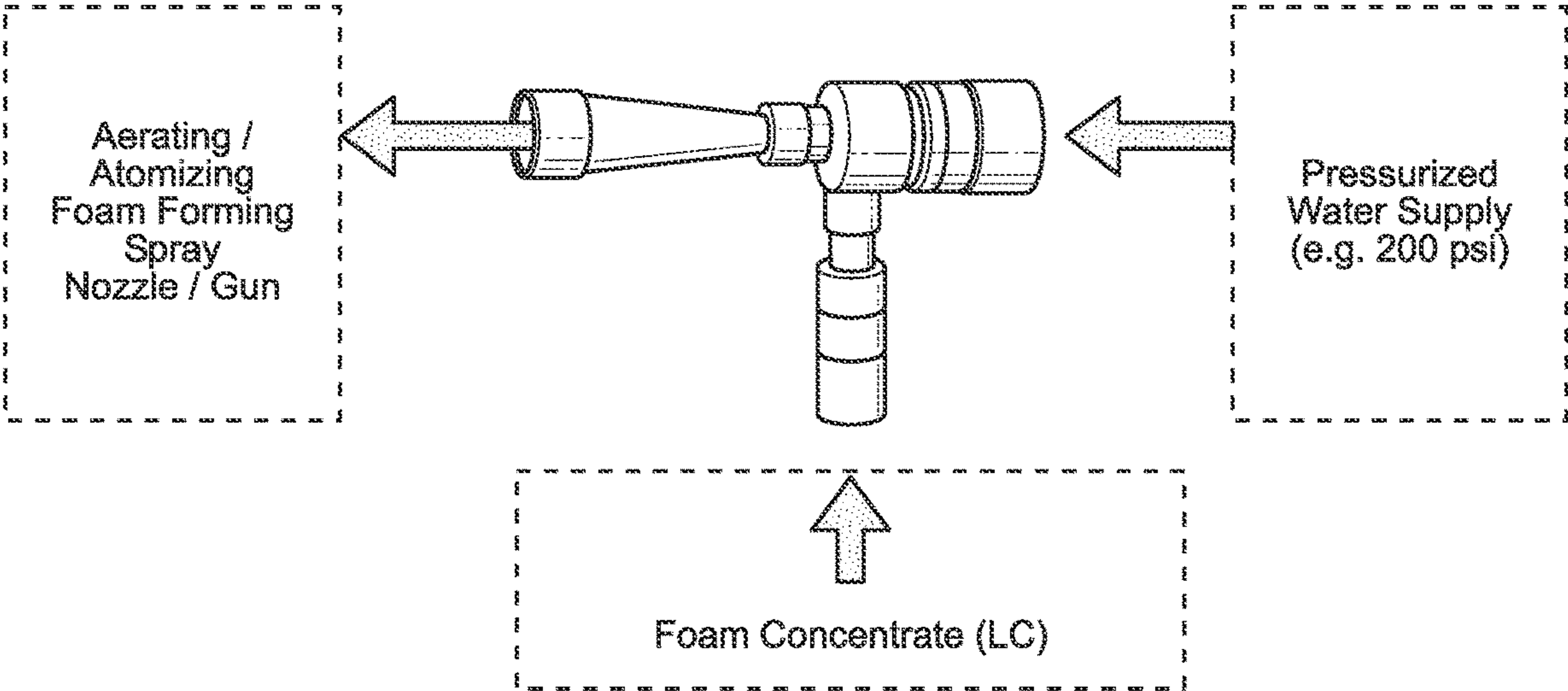


FIG. 54

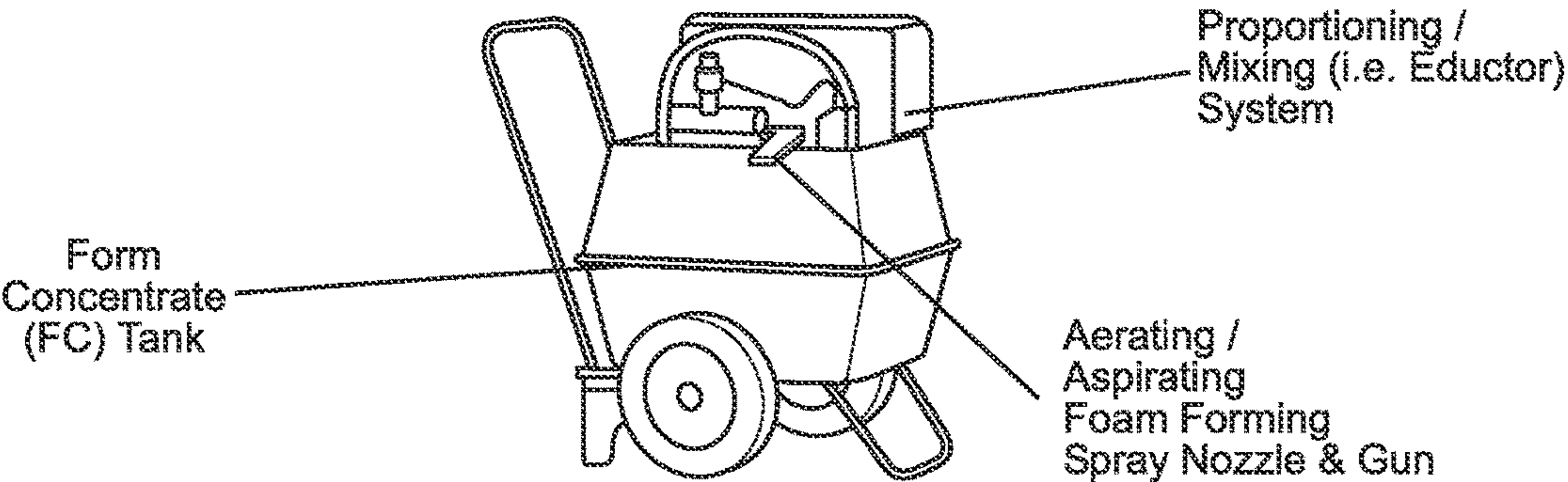


FIG. 55

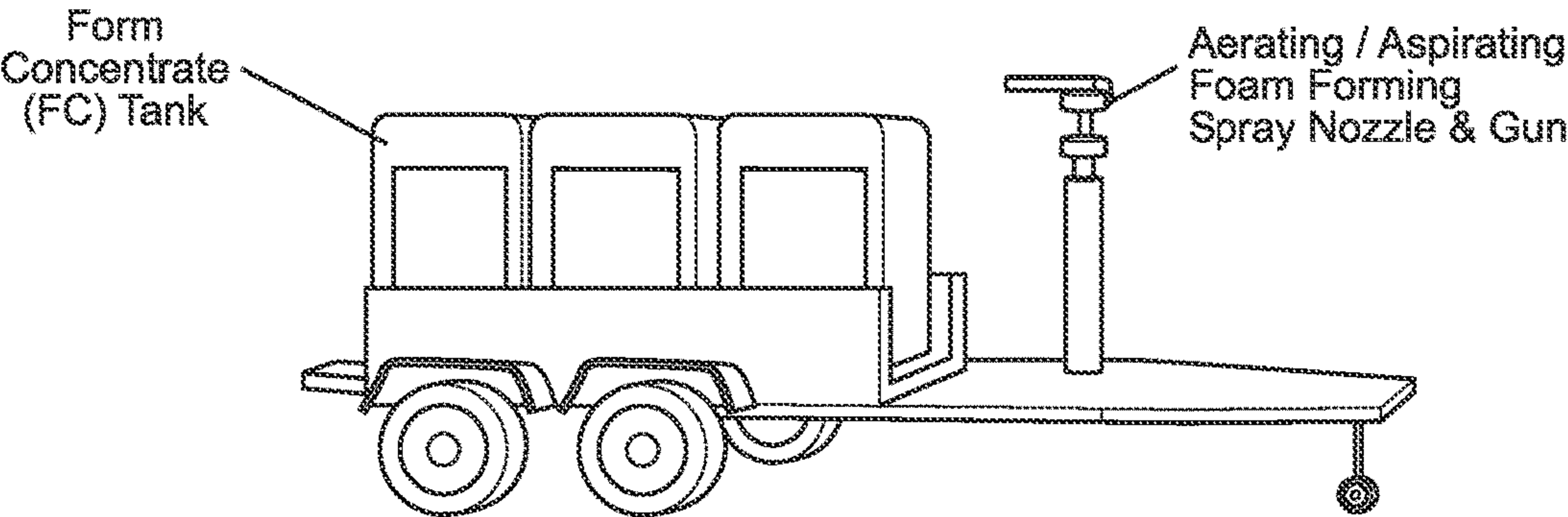
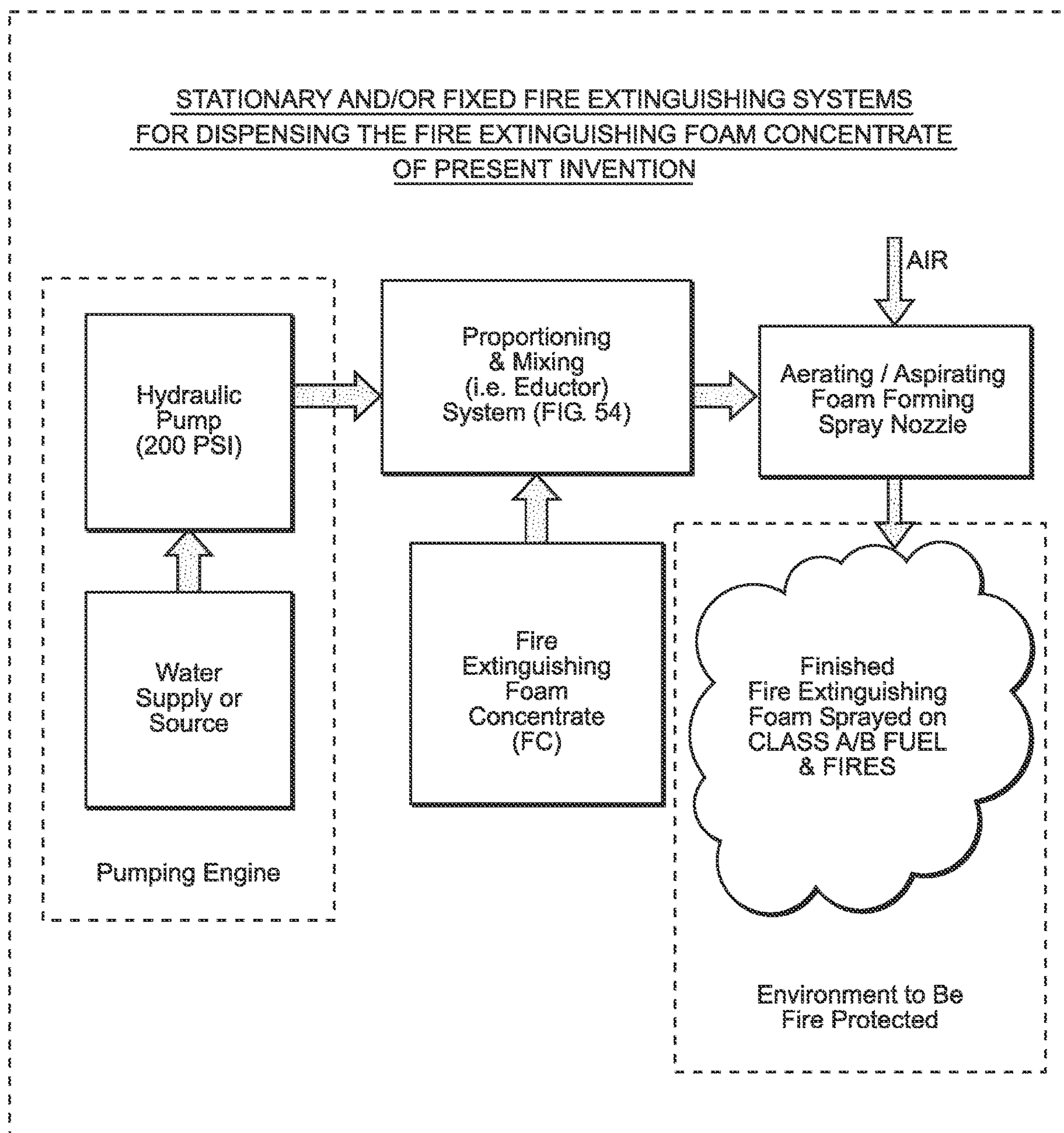


FIG. 56





**FIG. 57**



**ENVIRONMENTALLY-CLEAN  
BIODEGRADABLE WATER-BASED  
CONCENTRATES FOR PRODUCING FIRE  
INHIBITING AND FIRE EXTINGUISHING  
LIQUIDS AND FOAMS FOR FIGHTING  
CLASS A AND CLASS B FIRES**

RELATED CASES

**[0001]** The present patent application is a Continuation-in-Part of co-pending: U.S. patent application Ser. No. 17/167,084 filed Feb. 4, 2021; U.S. patent application Ser. No. 16/805,811 filed Mar. 1, 2020; U.S. patent application Ser. No. 16/449,389 filed Jun. 22, 2019; U.S. patent application Ser. No. 15/829,944 filed Dec. 3, 2017; U.S. patent application Ser. No. 16/914,067 filed Jun. 26, 2020; and U.S. patent application Ser. No. 16/029,861 filed Jun. 9, 2019; wherein each said US patent application is commonly owned by M-Fire Holdings, LLC and incorporated herein by reference as if fully set forth herein.

BACKGROUND OF INVENTION

Field of Invention

**[0002]** The present invention is directed towards improvements in science and technology applied in the defense of human and animal life and property, against the ravaging and destructive forces of fire caused by lightning, accident, arson and terrorism.

Brief Description of the State of Knowledge in the Art

**[0003]** Throughout the ages, mankind has had a complex relationship with fire. On one hand, mankind has feared fire for its power to damage and destroy property and life during warfare and acts of terrorism. On the other hand, mankind has worshipped before fire giving thanks to the power of fire to generate heat energy to keep us warm, cook foods to provide nourishment, make medicines to heal, make tools to abridge labor, and power machines to do physical work. Thus, there has been a great need to discover new and improved ways of controlling the ignition and spread of fire, and prevent the accidental and intentional damage and destruction of property and life by fire.

**[0004]** While most fear the thought of wildfire raging in a forest, in modern times, there is general agreement throughout the forest management industry that wildfires have positive ecological and environmental functions when they occur deep in the forests, far away from human inhabitants and human society at large. However, when wildfires rage close to where people are living and working in towns and communities, there is strong agreement that such wildfires need to be brought under quick control and containment to minimize the risk of damage to property and lives, and mitigate the production of air, water and other forms of environmental pollution caused by wildfires.

**[0005]** Unfortunately, over the past century, tens of millions of people have developed and settled towns, counties and neighborhoods in regions that today are called the Wildfire Urban Interface (WUI), which are at high risk to wildfires, and this is impacting home owners and property insurance industry. In order for man to live and survive a

sustainable future in the urban-wildfire interface, human society must quickly adapt in order to survive the destructive effects of wildfires.

**[0006]** Currently, conventional methods of wildfire fighting defense are proving inadequate because demographics have changed where people live and work relative to presence of wildfire:

**[0007]** Making firebreaks with bulldozers and shovels have not viable in most urbanized communities;

**[0008]** Making firebreaks with backfires provide ineffective and often dangerous as wildfires themselves;

**[0009]** Dropping PhosChek® AMP from 5000 Feet in urban areas is dangerous and not viable or effective in wildfire defense;

**[0010]** Thinning forests of dead trees and debris is effective in urban regions, especially near power poles, buildings and structures.

**[0011]** Current methods of wildfire defense and fighting are becoming unsustainable because the financial losses due to wildfire are exceeding what the insurance industry is willing to insure, as the damage caused by wildfire to the environment is typically catastrophic and total destruction.

**[0012]** FIG. 1 provides a table summarizing the primary conventional methods currently being used when fighting and defending against wild fires and forest fires, alike: aerial water dropping illustrated in FIG. 2A; aerial fire retardant chemical (e.g. PhosChek® Fire Retardant) dropping illustrated in FIGS. 2B1, 2B2 and 2B3; physical fire break by bulldozing, to stall the advance of wild fire; physical fire break by pre-burning, to stall the advance of wild fire; and chemical fire breaks by dropping fire retardant chemical such as PhosChek® chemical over land, to stall the advance of wild fire. While these methods are used, the results have not been adequate in most instances where wild fires rage across land under strong winds.

**[0013]** Except for spraying fire retardant foams and gels, all of the methods described above are generally “reactive” in nature, because they are either applied or practiced in response to the presence or incidence of wildfire, in effort to suppress and extinguish the wildfire, rather than proactively inhibiting wildfire from igniting combustible material along a wildfire’s tracks moving in the direction of prevailing winds. Consequently, PhosChek® water airdrops are generally reactive methods, because these methods are applied too often when it’s too late to suppress and extinguish a wildfire, and at best, airdropping this water-based fire retardant generates enormous quantities of smoke and noxious ammonia gases as well. Also, PhosChek® airdrops are very risky when applied to wildfires raging in wildfire urban interface (WUI) regions where people are living and working, because airdrops involve many tons of water falling to earth at high speed and with great force. This is little surprise when one understands the composition of this phosphorous-based chemical blended with tons of water.

Composition of PhosChek™ Fire Retardant: MAP, DAP, Gum Thickener & Coloring Agent

**[0014]** FIG. 2B4 describes the primary components of the PhosChek® fire retardant chemical, namely monoammonium phosphate (MAP), diammonium hydrogen phosphate (DAP) and water. PhosChek® MVP-F is a dry concentrate formulation that uses a combination of monoammonium phosphate [MAP; NH<sub>4</sub>H<sub>2</sub>PO<sub>4</sub>] and diammonium phosphate [DAP; (NH<sub>4</sub>)<sub>2</sub>HPO<sub>4</sub>] as the fire retardant salts. PhosChek®



MVP-F fire retardant also contains a gum thickener to provide a medium viscosity product for improved drop characteristics. The formulation contains a coloring agent having an alarming red color. The color fades over time with exposure to sunlight. A quick look at the chemical composition of the MAP and DAP components of PhosChek® fire retardant will be illuminating.

**[0015]** Monoammonium phosphate (MAP) is soluble in water and crystallizes as the anhydrous salt in the tetragonal system, as elongated prisms or needles. It is practically insoluble in ethanol. Solid monoammonium phosphate (MAP) can be considered stable in practice for temperatures up to 200° C., when it decomposes into gaseous ammonia NH<sub>3</sub> and molten phosphoric acid H<sub>3</sub>PO<sub>4</sub>. At 125° C. the partial pressure of ammonia is 0.05 mm Hg. A solution of stoichiometric monoammonium phosphate is acidic (pH 4.7 at 0.1% concentration, 4.2 at 5%).

**[0016]** According to the diammonium phosphate MSDS from CF Industries, Inc., decomposition starts as low as 70° C. “Hazardous Decomposition Products: Gradually loses ammonia when exposed to air at room temperature. Decomposes to ammonia and monoammonium phosphate at around 70° C. (158° F.). At 155° C. (311° F.), DAP emits phosphorus oxides, nitrogen oxides and ammonia.”

**[0017]** When airdropped from planes, the gum thickener contained in PhosChek® fire retardant binds MAP and DAP to water to provide mass and help drop the water onto the raging wildfire in effort to extinguish it. When airdropping, most firefighters understand that they have lost control of the wildfire, and that the target wildfire is destined to rage across property populated with buildings structures including homes, then PhosChek® airdrops are made on targeted property of home owners and towns—which can be observed by the red-colored PhosChek® fire retardant coating all over ground surfaces, in effort to protect the targeted property against wildfire. Recently, US Patent Application Publication No. US2020/0254290 describes improvements in liquid concentrate fire retardant compositions containing mixtures of ammonium phosphates (i.e. MAP, DAP and ammonium polyphosphate APP), in effort to reduce the toxicity and corrosive properties of such chemical components.

**[0018]** Many photographs are posted on the WWW showing the airdropping of PhosChek® fire retardant from air-planes. However, these firefighting operations should be viewed as a last ditch effort to save property and lives from a raging wildfire.

**[0019]** Airdropping PhosChek® infused water over wildfires is not a proactive measure of any sort, and it’s often too late, too expensive, and too in-effective to be continued as a best practice to contain and subdue wildfires raging across the WUI regions of America.

**[0020]** Also, the use of water-based phosphorous-rich fire retardants, and pick & shovel and bulldozer methods for defending against wildfires, does not represent technological advancement, progress and firefighter and environmental safety, within the rapidly expanding wildfire urban interface (WUI) regions of America and around the world. The world must do significantly better in response to the growing threat of climate-change driven wildfires, mixed with the challenges of a viral pandemic.

Smoke-Induced Asthma is Now Presenting a Great Health Risk to Wildfire Fighters and Citizens Alike

**[0021]** This past year, the Centers for Disease Control and Prevention (CDC) stated “when wildfires burn either in your area or many miles away, they produce smoke that may reach your community. Wildfire smoke is a mixture of gases and fine particles from burning trees and other plant materials. This smoke can hurt your eyes, irritate your respiratory system, and worsen chronic heart and lung diseases.”

**[0022]** Also, Asthma and Allergy Foundation of America (AAFA) stated that “each year, wildfires rage across the U.S. producing smoke in the air containing tiny particles that affect air quality. These particles can irritate your eyes, nose, throat and lungs. Poor air quality can worsen asthma symptoms. Children and those with respiratory disease like asthma are at high risk for asthma episodes when the air quality is poor. Wildfires do not only affect those in the immediate fire area. Smoke can blow many miles away and impact people hundreds of miles away.”

**[0023]** The American Lung Organization stated that “wildfires, including grassland fires and forest fires, are an ongoing concern where there is dry, hot weather. During a wildfire, people throughout the surrounding area may suffer the effects of the smoke. Talk with your doctor about how to prepare for this smoke, especially if you or someone in the family fits into one of these categories: works outdoors; is under age 18 or over age 65; or has asthma, COPD or other lung diseases, chronic heart disease, or diabetes. Monitor your breathing and exposure to the smoke.”

**[0024]** Clearly, the message from these health and health policy organizations is to “protect yourself from wildfire smoke”, and that includes those wildfire fighters trying to contain and suppress raging wildfires all across the WUI regions across our Nation. Also, it is well known that, in high doses, irritants, such hydrochloric acid, sulfur dioxide and ammonia, will induce occupational asthma, and this is something that wildfire fighters should be thinking about as well. On this point, it should be noted that PhosChek® fire retardant, when used to fight against raging forest fires rapidly decomposes at 200 C into gaseous ammonia NH<sub>3</sub> and molten phosphoric acid H<sub>3</sub>PO<sub>4</sub>. Thus, when such phosphorous agents are dropped onto wildfires, in effort to suppress or quell wildfire, decomposition into gaseous ammonia will only increase the toxic effects of smoke production from wildfires.

Increased Risks of Covid-19 with Asthma

**[0025]** It is no secret that individuals with asthma are at substantially higher risks when exposed or infected by the Covid-19 virus. For those with asthma, there is great fear that they will have a worse outcome or be more likely to get SARS-CoV-2 (the virus that causes COVID-19). While there is currently no evidence of increased infection rates in those with asthma, the Centers for Disease Control and Prevention has stated that patients with moderate-severe asthma could be at greater risk for more severe disease.

**[0026]** In the May 6, 2020 N Y Times article “Will Smoke From Controlled Burns Hurt Covid-19 Patients?”, Cal Fire spokesman, Scott McLean, said “What is Covid-19? A respiratory issue”. And then continued by stating “We’re not naïve to that, but we have to provide for the well-being of the public.” The NY Times article also reported that “Forest Service officials said they were concerned that assembling a work force to conduct the burns would expose traveling employees to the virus and potentially contribute to its



spread. They also raised doubts about how their fire crews could tend to burns while also abiding by social distancing directives. They said they would continue to use other methods—such as removing brush by hand and with heavy machinery—that reduce combustible forest fuel without generating smoke.”

#### Searching for Better Solutions to Fight Wildfires and Forest Fires

**[0027]** U.S. Pat. No. 8,273,813 assigned to BASF Aktiengesellschaft provides a comprehensive overview of the state of the art in 2012, of worldwide efforts to develop and deliver chemical solutions for preventing and fighting wildfires and forest fires around the world.

**[0028]** As disclosed, firefighters have long utilized solutions of inorganic salts, for example, alkali metal or alkaline earth metal salts of carbonic acid, phosphoric acid or boric acid. The salts augment the extinguishing action of water and are used as concentrated solutions. These salts are effective because they release inert gases, for example carbon dioxide from carbonates, or melt and so form an air-impervious layer on combustible materials. In either case, access of air to combustible material is controlled. The disadvantage with this approach is the formation of a coating which is later difficult to remove. They have no cooling effect and are barely able to extinguish burning matter, since the latter, like water as well, runs off very rapidly. Any protective effect is solely due to preceding and repeated spraying of objects. A salt solution does not adhere to smooth or waxy objects, such as leaves, planks or glass panes, to any significant extent, if at all.

**[0029]** The use of salts of organic carboxylic acids, for example oxalic acid, tartaric acid or citric acid, in firefighting has been known since the 1970s. In contradistinction to inorganic salts mentioned above, the coatings formed from the salts of organic carboxylic acids are easy to remove after the fire has been extinguished. Examples of the use of salts of organic carboxylic acids in firefighting are identified in the following patent documents: DE-C 13 02 520, DE-A 35 25 684, EP-A 059 178, EP-A 426 178, U.S. Pat. Nos. 1,278,718, 4,888,136, 5,945,025 and WO 88/00482. A brief overview of these prior art references will be useful at this juncture.

**[0030]** DE-C 13 02 520 discloses the use of alkali metal salts of oxy carboxylic acids in dry extinguishing powders.

**[0031]** DE-A 35 25 684 describes solutions consisting of citric acid/citrate, potassium hydroxide and water that are useful for firefighting and for impregnating combustible materials. More particularly, the solution is said to be capable of binding acidic gases generated in a fire.

**[0032]** EP-A 059 178 describes the use of concentrated solutions of alkali metal salts of citric acid as extinguishing compositions.

**[0033]** EP-A 426 178 discloses fire-retardant asphalt compositions, the fire-retarding component comprising potassium citrate and a silicone polymer.

**[0034]** U.S. Pat. No. 1,278,718 discloses compositions consisting of concentrated solutions of alkali metal salt of citric acid and alkali metal bicarbonate, as filling for fire extinguishers.

**[0035]** U.S. Pat. No. 4,888,136 describes the use of aluminum salts of citric acid and of lactic acid for fire-retarding impregnations of cellulosic fibers.

**[0036]** U.S. Pat. No. 5,945,025 describes compositions of potassium citrate and sodium bicarbonate for firefighting.

**[0037]** WO 88/00482 discloses compositions of matter for firefighting and for producing fire-retarding coatings based on alkali metal salts of citric acid.

**[0038]** The compositions mentioned above can be applied as aqueous solutions and retain their fire-retarding effect even after drying, and therefore, have a pronounced long-term effect.

**[0039]** The use of hydrogels was proposed more than 35 years, for example in U.S. Pat. Nos. 3,229,769 and 5,849,210, for the purpose of cooling the source of the fire by retaining water close to the flame. These hydrogels are produced from a water-absorbing polymer and water. The hydrogel binds the water and so stops the water from flowing away from the source of the fire. Because hydrogels are capable of maintaining a large amount of water near the fire, hydrogels have a good immediate extinguishing effect. In contrast, the long-term effect of hydrogels is poor. Hydrogels can dry and thereby rapidly lose their effect. The remaining salt-like dried hydrogels have a very low fire-retarding effect. More recent improvements in fire suppressing gel compositions employing super absorbing polymers (SAPs) are discussed in US Patent Application Publication No. US2021/0052928.

**[0040]** U.S. Pat. No. 8,273,813 (assigned to BASF) proposed combining water-absorbing polymers with fire-retarding salts to form fire-retarding compositions having a good immediate extinguishing effect and a good long-term effect. This fire retarding chemical solution is schematically depicted in FIG. 3A.

**[0041]** As illustrated in FIG. 3B, Hartindo’s aqueous-based anti-fire (AF) chemical solution AF31 employs as its active ingredient, Potassium Citrate, or TPC, dissolved in water, with minor amounts of a natural gum added to provide some cling. Tripotassium citrate (TPC) is considered Generally Recognized As Safe or “GRAS” by the United States Food and Drug Administration without restriction as to the quantity of use within good manufacturing practice.

**[0042]** Hartidino’s AF31 chemical solution has been used by others in many diverse applications, namely: (i) preventing and suppressing peat fires in Malaysia, as disclosed on Hartindo’s WWW site, (ii) treating wood to provide Class-A fire-protection as taught in U.S. patent Ser. No. 10/260,232 (Conboy); and (iii) proactively treating native fuel, ground cover and fixtures and buildings on real property, for the purpose of defending life and property against the threat of wildfires, as taught in U.S. patent Ser. No. 10/653,904 assigned to Applicant/Assignee (M-Fire Holdings, LLC) employing new and innovative technologies for proactively-protecting property and life against wildfires in the WUI region. These technologies include the use of a cloud-based GPS-tracking/mapping wildfire defense system network designed to support many different methods of proactively spraying equipment for efficient GPS-tracking and mapping of environmentally-clean wildfire inhibitor spraying operations, within a secure global database, to manage the strategic creation and maintenance of clean-chemistry wildfire breaks, created out in front of and around property and life to be proactively protected from wildfires.

**[0043]** Applicant/Assignee’s methods operates in stark contrast to conventional methods of reactively-fighting wildfires by air-dropping tons of PhosChek® containing



agricultural-grade fertilizer onto raging wild fires while brave fire fighters manually create wildfire breaks using picks, shovels and bulldozers, and are exposed to life threatening risks of fire, smoke and COVID-19 viral infection. Notably, Applicant's wildfire defense methods include the use of: GPS-guided, tracking and mapping spray drones; GPS-tracking mobile/backpack sprayers; GPS-tracking vehicle-supported high-pressure sprayers; mobile computing devices; data centers; wireless networking infrastructure; and the like. Each of these GPS-tracking mobile spraying systems is deployed on and supported by the GPS-tracking/mapping wild fire defense network illustrated in U.S. Pat. No. 10,260,232.

**[0044]** In addition to proactive fire inhibiting agents illustrated in FIG. 3B, various kinds of fire-fighting foams and gels have been developed over the years in effort to gain advantage against structural fires and wildfires alike, while using significantly less water as the fire extinguishing agent, to reduce water damage and/or environmental pollution. This class of prior art firefighting foams is illustrated in FIG. 3C, wherein the foaming agent may include hydrolyzed protein, and surfactants have included non-biodegradable fluoro-carbon compounds. Exemplary prior art foam concentrates for producing firefighting foams for Class A and B fires include: Phoschek® 1% Fluorine-Free Class AB Foam Concentrate; BioEx® Fluorine-Free Foaming Additive For Class A (Solid) and Hydrocarbon Fires; ChemGuard® DIRECTATTACK Foam Concentrate for Class A Fuel Fires; and ChemGuard® Fluoroprotein Foam Concentrate for fire and vapor suppression of Class B hydrocarbon fuel fires.

**[0045]** Recently, US Patent Application Publication No. 2020/0181328 describes twin-tail hydrocarbon surfactants for firefighting foam compositions and as additives for aqueous film forming foam (AFFF) agents, in hope of providing partial or complete replacements for fluorosurfactants and/or fluorinated foam stabilizers used in fire fighting foams, which have been under strict scrutiny by the EPA for known toxicity issues and human health and safety concerns.

**[0046]** Clearly, in these times of climate change and narrowing gaps between wildfire regions and urbanized areas, we must adapt to and defend against wildfires in smarter and better, and more proactive and less reactive ways—because “an ounce of prevention is worth a pound of cure,” as Benjamin Franklin taught the world over back in the mid-1750's.

**[0047]** Thus, there needs to be better, safer and more effective fire inhibiting and extinguishing chemical compositions, and methods and technology for applying the same to proactively defend property and life from fires of kinds, across all industrial applications, including wildfires devastating the rapidly expanding WUI region, and for humanity to do so, without creating risks of smoke and injury to firefighters, property owners, animals, and the human population at large, while overcoming the shortcomings and drawbacks of prior art compositions, apparatus and methodologies.

#### OBJECTS AND SUMMARY OF THE PRESENT INVENTION

**[0048]** Accordingly, a primary object of the present is to provide new and improved method of and system and network for managing the supply, delivery and spray-application of safer and more effective environmentally-clean

biodegradable water-based biochemical compositions and materials to reduce the risks of damage and/or destruction to property and life caused by fires of all kinds, across all industries, while overcoming the shortcomings and drawbacks of prior art methods and apparatus.

**[0049]** Another object of the present invention is to provide new and improved environmentally-clean aqueous-based fire inhibiting biochemical compositions in liquid phase over a broad ambient working temperature range, that can be atomized and sprayed as a fine mist over ground surfaces, native ground fuel, living plants, trees and shrubs and being an effective wildfire inhibitor, when dried forming a durable gas pervious coating having improved surface coverage.

**[0050]** Another object of the present invention is to provide new and improved environmentally-clean aqueous-based fire inhibiting biochemical compositions in liquid phase over its wide ambient working temperature ranges and pressure conditions.

**[0051]** Another object of the present invention is to provide new and improved fire inhibiting liquid biochemical compositions that allows its active fire inhibiting chemistry (e.g. potassium mineral salts) to efficiently penetrate into the combustible surfaces of natural fuels during atomization spraying and quick drying operations, in effort to improve the duration of fire protection offered by potassium mineral salts contained in the new and improved fire inhibiting compositions formed on the surfaces when dried, and when exposed to moisture and/or high levels of relative humidity.

**[0052]** Another object of the present invention is to provide a new and improved environmentally-clean fire inhibiting liquid biochemicals formulated by (i) dissolving in a quantity of water, a first quantity of tripotassium citrate (TPC) functioning as a fire inhibitor, with a second quantity of triethyl citrate (TEC) functioning as a coalescent agent, to form a clear wildfire inhibitor solution, and after spraying or otherwise applying the wildfire inhibitor solution to a surface to be protected against wildfire, (ii) allowing potassium cations dissolved in the solution to disperse and participate in the formation of thin relatively uniform potassium citrate salt crystal film structures on the treated surface and functioning as an optically transparent wildfire inhibitor coating, which once dried, will absorb water at its surface without rapid dissolution, to improve the duration of fire protection offered by the fire inhibiting composition in the presence of rain and ambient moisture levels.

**[0053]** Another object of the present invention is to provide apparatus for spraying the new and improved fire inhibiting liquid having a coalescent agent that promotes the formation of ultra-thin potassium mineral salt crystal film structures deposited onto the organic fuel surfaces to be protected against the threat of ignition by fire, providing optimized methods of wildfire inhibitor deposition in outdoor environments.

**[0054]** Another object of the present invention is to provide a fire extinguishing and/or fire inhibiting biochemical composition of matter, comprising: (a) a dispersing agent in the form of a quantity of water, for dispersing metal ions dissolved in water; (b) a fire inhibiting agent in the form of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid, for providing metal ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; and (c) a coalescing agent in the form of an organic compound containing three carboxylic acid groups,



or salt/ester derivatives thereof, for dispersing and coalescing the metal ions when the fire inhibiting liquid composition is applied to a surface to be protected against fire, while water molecules in the water evaporate during drying, and the metal ions cooperate to form metal salt crystal structure on the surface.

**[0055]** Another object of the present invention is to provide such fire extinguishing and/or fire inhibiting biochemical compositions, wherein the alkali metal salt is a sodium or potassium salt, and wherein the alkali metal salt is tripotassium citrate.

**[0056]** Another object of the present invention is to provide such fire extinguishing and/or fire inhibiting biochemical compositions, wherein said coalescing agent is triethyl citrate, an ester of citric acid. such fire extinguishing and/or fire inhibiting biochemical compositions.

**[0057]** Another object of the present invention is to provide a new and improved fire extinguishing and/or fire inhibiting biochemical composition, wherein a building material is coated with the fire retarding biochemical composition.

**[0058]** Another object of the present invention is to provide a new and improved fire extinguishing and/or fire inhibiting biochemical composition, wherein the biochemical composition comprises a major amount of tripotassium citrate dissolved in a major quantity of water, along with a minor amount of a coalescing agent such as triethyl citrate, an ester of citric acid.

**[0059]** Another object of the present invention is to provide a new and improved fire extinguishing and/or fire inhibiting biochemical composition, wherein an article of manufacture contains the biochemical composition, and the article of manufacture is selected from the group consisting of an extinguisher, an extinguishing fitting, and an extinguishing system.

**[0060]** Another object of the present invention is to provide a new and improved method of proactively fighting a fire comprising the steps of applying improved liquid fire inhibiting biochemical composition to the surfaces to be proactively protected from a fire.

**[0061]** Another object of the present invention is to provide a new and improved method of proactively fighting a fire such as a forest fire, a wildfire, a tire warehouse fire, a landfill fire, a coal stack fire, an oil field fire, a mine fire, a battlefield fire, a battleship fire, a fuel truck accident fire, or oil spill fire.

**[0062]** Another object of the present invention is to provide a new and improved method of proactively imparting fire resistance to an article comprising: (a) applying a liquid biochemical composition to the article; and (b) allowing the applied biochemical composition to dry on the article and form a fire inhibiting metal salt crystal coating on the article, wherein the article is a textile material, a building material, a structural component, or property to be proactively defended from a fire.

**[0063]** Another object of the present invention is to provide a new and improved environmentally-clean aqueous-based fire inhibiting liquid biochemical composition comprising: a major amount of tripotassium citrate (TPC) and a minor amount of triethyl citrate (TEC) added to and mixed with a major amount of water functioning as a solvent, carrier and dispersant of potassium salt ions dissolved in the water with the tripotassium citrate.

**[0064]** Another object of the present invention is to provide a new and improved inhibiting biochemical composition kit comprising: a major amount of dry tripotassium citrate monohydrate (TPC) and a minor amount of triethyl citrate (TEC), as components for mixing with a predetermined major amount of water functioning as a solvent, carrier and dispersant, to make up a predetermined quantity of environmentally-clean liquid fire inhibiting biochemical composition for proactively protecting combustible property and wood products.

**[0065]** Another object of the present invention is to provide a new and improved environmentally-clean aqueous-based fire inhibiting liquid biochemical composition comprising: a major amount of tripotassium citrate (TPC), a minor amount of triethyl citrate monohydrate (TEC), and a minor amounts of biocidal agent), added to and mixed with a major amount of water functioning as a solvent, carrier and dispersant.

**[0066]** Another object of the present invention is to provide a new and improved fire inhibiting biochemical composition kit comprising: a major amount of dry tripotassium citrate (TPC), a minor amount of triethyl citrate (TEC), and a minor amount of biocidal agent, as components for mixing with a predetermined major amount of water functioning as a solvent, carrier and dispersant, to make up a predetermined quantity of environmentally-clean liquid fire inhibiting biochemical composition for proactively protecting combustible property and wood products.

**[0067]** Another object of the present invention is to provide a new and improved environmentally-clean aqueous-based fire inhibiting liquid biochemical composition comprising: a major amount of tripotassium citrate monohydrate (TPC), a minor amount of triethyl citrate (TEC), and a minor amount of citric acid (CA) for adding to and mixing with a major quantity of water functioning as a solvent, carrier and dispersant.

**[0068]** Another object of the present invention is to provide a new and improved fire inhibiting biochemical composition kit comprising: a major amount of dry tripotassium citrate (TPC), a minor amount of triethyl citrate (TEC), and a minor amount of citric acid (CA), as components for mixing with a predetermined major amount of water functioning as a solvent, carrier and dispersant, to make up a predetermined quantity of environmentally-clean liquid fire inhibiting biochemical composition for proactively protecting combustible property and wood products.

**[0069]** Another object of the present invention is to provide a new and improved environmentally-clean aqueous-based fire inhibiting biochemical composition for producing good immediate extinguishing effects when applied to extinguish a burning or smoldering fire, and very good long-term fire inhibiting effects when being proactively applied to protect combustible surfaces against the threat of fire, comprising: (a) a dispersing agent realized in the form of a quantity of water, for dispersing metal ions dissolved in water; (b) a fire inhibiting agent in the form of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid, for providing metal ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (c) a coalescing agent in the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), such as triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal ions when the fire inhibiting liquid composition is applied to a surface to be



protected against fire, while water molecules in the water evaporate during drying, and the metal ions cooperate to form metal salt crystal structure on the surface; (d) if appropriate, at least one biocide (e.g. Polyphase® PW40 Biocide from Troy Corporation or citric acid) dissolved in water; and (e) if appropriate at least one colorant.

**[0070]** Another object of the present invention is to provide a new and improved environmentally-clean aqueous-based fire inhibiting biochemical composition, wherein the alkali metal salts of nonpolymeric saturated carboxylic acids for inclusion in the biochemical composition comprises: alkali metal salts of oxalic acid; alkali metal salts of gluconic acid; alkali metal salts of citric acid; and also alkali metal salts of tartaric acid.

**[0071]** Another object of the present invention is to provide a new and improved environmentally-clean aqueous-based fire inhibiting biochemical composition, wherein said alkali metal salts of nonpolymeric saturated carboxylic acids comprise potassium carboxylates.

**[0072]** Another object of the present invention is to provide a new and improved environmentally-clean aqueous-based fire inhibiting biochemical composition, wherein said alkali metal salts of nonpolymeric saturated carboxylic acids comprise tripotassium citrate monohydrate (TPC).

**[0073]** Another object of the present invention is to provide a new and improved fire inhibiting liquid biochemical composition comprising: (a) a dispersing agent realized in the form of a quantity of water, for dispersing metal ions dissolved in the water; (b) the fire inhibiting agent is realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate, for providing metal potassium ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; and (c) a coalescing agent realized the form of an organic compound containing three carboxylic acid groups or salt/ester derivatives thereof, specifically triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal potassium ions when the fire inhibiting liquid composition is applied to a surface to be protected against fire, and while water molecules in the water evaporate during drying, the metal potassium ions cooperate to form a thin potassium citrate salt crystal film structure on the treated surface to be protected against ignition by fire.

**[0074]** Another object of the present invention is to provide a new and improved fire inhibiting liquid biochemical composition comprising: (a) a dispersing agent realized in the form of a quantity of water, for dispersing metal ions dissolved in the water; (b) the fire inhibiting agent is realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate, for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (c) a coalescing agent realized the form of an organic compound containing three carboxylic acid groups or salt/ester derivatives thereof, specifically triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal potassium ions when the fire inhibiting liquid composition is applied to a surface(s) to be proactively protected against fire, and while water molecules in the water evaporate during drying, the metal potassium ions cooperate to form potassium citrate salt crystal film structures on the treated surfaces; and (d) at least one biocide agent dissolved in the quantity of water.

**[0075]** Another object of the present invention is to provide a new and improved fire inhibiting liquid biochemical composition comprising: (a) a dispersing agent realized in the form of a quantity of water, for dispersing metal ions dissolved in the water; (b) the fire inhibiting agent is realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate, for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (c) a coalescing agent realized the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), specifically triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal potassium ions when the fire inhibiting liquid composition is applied to a surface to be protected against fire, and while water molecules in the water evaporate during drying, the metal potassium ions cooperate to form potassium citrate salt crystal film structure on the treated surface; and (d) at least one biocide agent in the form of citric acid dissolved in the quantity of water.

**[0076]** Another object of the present invention is to provide a new and improved environmentally-clean aqueous-based fire inhibiting liquid biochemical composition of the present invention consisting of tripotassium citrate (TPC) and triethyl citrate (TEC) formulated with water functioning as a solvent, carrier and dispersant in the biochemical composition.

**[0077]** Another object of the present invention is to provide a new and improved fire-extinguishing and/or fire-retarding biochemical composition produced by stirring components into water, in amounts substantially proportional to, the formulation comprising: 0.05 pounds by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume); and 4.4 pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 9.61 pounds having 128 ounces or 1 gallon of volume.

**[0078]** Another object of the present invention is to provide a new and improved fire-extinguishing and/or fire-retarding biochemical composition kit comprising components, in amounts substantially proportional to: 0.05 pounds by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume). for blending and mixing together with 4.4 pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 9.61 pounds having 128 ounces or 1 gallon of volume.

**[0079]** Another object of the present invention is to provide a new and improved fire-extinguishing and/or fire-retarding biochemical composition produced by stirring components into water, in amounts substantially proportional to, the formulation comprising: 0.05 pounds by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume); 4.0 ounces by weight of a biocide (e.g. Polyphase® PW40 by Troy Chemical); and 4.4 pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 10.00 pounds having 128 ounces or 1 gallon of volume.

**[0080]** Another object of the present invention is to provide a new and improved fire-extinguishing and/or fire-retarding biochemical composition kit comprising components, in amounts substantially proportional to: 0.05 pounds



by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume); 4.0 ounces by weight of a biocide agent (e.g. Polyphase® PW40 by Troy Chemical), for blending and mixing together with 4.4 pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 10.0 pounds having 128 ounces or 1 gallon of volume.

**[0081]** Another object of the present invention is to provide a new and improved fire-extinguishing and/or fire-retarding biochemical composition produced by stirring components into water, in amounts substantially proportional to, the formulation comprising: 0.05 pounds by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume); 4.0 ounces by weight of a biocide agent in the form of citric acid; and 4.4 pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 10.00 pounds having 128 ounces or 1 gallon of volume.

**[0082]** Another object of the present invention is to provide a new and improved fire-extinguishing and/or fire-retarding biochemical composition kit comprising components, in amounts substantially proportional to: 0.05 pounds by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume); and 4.0 ounces by weight of a biocide agent in the form of citric acid, for blending and mixing together with 4.4 pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 10.0 pounds having 128 ounces or 1 gallon of volume.

**[0083]** Another object of the present invention is to provide a new and improved fire inhibiting biochemical composition comprising: a major amount from 1% to 65% by weight, preferably from 20% to 50% by weight and more preferably from 30% to 55% by weight, of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid such as tripotassium citrate monohydrate; and a minor amount from 0.08% to 5% by weight, preferably from 0.5% to 2% by weight and more preferably from 0.1% to 1.0% by weight, of triethyl citrate, an ester of citrate acid; wherein the sum by % weight of the components above should not exceed 100% by weight.

**[0084]** Another object of the present invention is to provide a new and improved fire inhibiting biochemical composition, wherein the water content is present in a major amount and is typically not less than 30% by weight, preferably not less than 40% by weight, more preferably not less than 50% by weight and most preferably not less than 60% by weight and preferably not more than 60% by weight and more preferably not more than 70% by weight, all based on the fire inhibiting biochemical composition.

**[0085]** Another object of the present invention is to provide a GPS-tracking, mapping and recording techniques that enable a population to know where environmentally-clean-wildfire chemistry-based wildfire breaks and zones have been formed by whom, and when using the principles of the present invention.

**[0086]** Another object of the present invention is to provide a wildfire defense network supporting integrated GPS-tracking, mapping and recording techniques, that enable fire jurisdictions to plan and implement clean-chemistry wildfire breaks and zones (e.g. around telephone poles, homes and

other building structures) to proactively protect property and life from raging wildfires—by effectively inhibiting specific regions of combustible fuel from ignition and flame spread, along the path towards targeted property and life to be protected from the incidence of wildfire.

**[0087]** Another object of the present invention is to provide a method of spraying an ultra-thin layer of wildfire inhibiting liquid biochemical compositions onto combustible ground cover and surfaces to be proactively protected against the presence of wildfire, so that when the water molecules in the wildfire inhibiting liquid chemicals evaporate during drying operations, ultra-thin potassium salt crystal structures form on the surfaces, to provide potassium cations available to inhibit the wildfire along one or more pathways including, for example, interruption of free radical chain reactions driving the combustible phase of wildfire, and thereby taking the energy out of the wildfire, reducing the production of smoke, and protecting property treated in advance of a wildfire incidence.

**[0088]** Another object of the present invention is to provide a new and improved environmentally-clean wildfire inhibiting liquid biochemical compositions formulated so that, when applied in hot dry climates, conditioned by hot dry prevailing winds, the relative humidity will be expectedly low, and in the absence of rain, the all-natural wild fire inhibiting liquid of the present invention sprayed over wild fire break and zone regions, will last for durations into weeks and months in many situations.

**[0089]** Another object of the present invention is to provide wireless network for GPS-tracking when and where the new and improved environmentally-clean wildfire inhibiting liquid biochemical composition is spray applied, and documenting the same in a wireless network database, so that, whenever rain occurs, the wireless network can inform and advise fire departments and homeowners using mobile phones or computing systems that certain GPS-specified environmentally-clean wildfire breaks and zones require maintenance by an additional spraying of the wildfire inhibitor liquid, while GPS-tracking, mapping and recording the spraying operations on the wireless network, for management purposes.

**[0090]** Another object of the present invention is to provide a new and improved methods for spraying environmentally-clean wildfire inhibiting liquid biochemical compositions to form GPS-tracked clean chemistry wildfire breaks—well in advance of the incidence of wild fires moving in the direction of prevailing winds.

**[0091]** Another object of the present invention is to provide a novel method of proactive wildfire defense in the WUI region using natural safe potassium mineral salts that pose zero to little threat to our natural environments or human beings and animals living in these WUI regions, where homes and businesses exist.

**[0092]** Another object of the present invention is to provide a new and improved methods of spraying utility poles and infrastructure with new and improved environmentally-clean wildfire inhibiting liquid biochemical compositions and tracking and documenting the same on a GPS-based wireless system network so that fire jurisdictions can plan and implement clean-chemistry wildfire breaks and zones (e.g. around telephone poles) to proactively protect property and life from a raging wildfire seeking combustible fuel by interrupting the combustible phase of the wildfire, reducing



the production of smoke, and protecting property that has been treated in advance of a wildfire incidence.

**[0093]** Another object of the present invention is to provide a new and improved method of and apparatus for GPS-tracking and mapping operations involving the spraying of an environmentally-clean aqueous-based wildfire inhibiting biochemicals on property surfaces having native fuel, and other combustible structures, including wood buildings, decks, fences, etc. prior to the arrival or outbreak of a wildfire.

**[0094]** Another object of the present is to provide method of reducing the risks of damage to private property due to wild fires by centrally managed application of wildfire inhibiting biochemical liquid spray to ground cover and building surfaces prior to arrival of the wild fires.

**[0095]** Another object of the present is to provide method of reducing the risks of damage to private property due to wild fires using a global positioning satellite (GPS) system and mobile communication messaging techniques, to help direct the application of AF chemical liquid prior to the arrival of wild fires.

**[0096]** Another object of the present invention is to provide a new and improved system for wild fire suppression and neighborhood and home defense comprising a platoon of small planes, all-terrain vehicles (ATVs) and other mobile systems adapted for spraying an environmentally-clean fire inhibiting and extinguishing chemical liquid that clings to the ground cover, and buildings, where applied in regions of high wild fire risk, that operates in both wet and dry states of application.

**[0097]** Another object of the present invention is to provide a new and improved system for wild fire suppression and home defense system comprising (i) a plurality of home wild-fire defense systems assigned to each home or building in the strategic area, for spraying the outside of their homes and surrounding ground cover with the environmentally-clean fire inhibiting and extinguishing biochemical spray liquid, (ii) a command center for managing wild fire pre-defense operations in the region, involving the spray application of the environmentally-clean fire inhibiting and extinguishing biochemical spray liquid to create and maintain strategic fire breaks in the region in advance of the outbreak of wild fires, and protection of homes and property in the region against wild fires breaking out in the region, and sending messages and instructions to home owners in the region as well as operators of the small planes and ATVs deployed in the system, and (iii) a mobile application installed on the mobile phone of each home owner in the strategic region, and configured for receiving email and/or SMS messages from a command center managing the system, and instructing home owners to pre-defend their homes using the environmentally-clean fire inhibiting and extinguishing biochemical spray liquid.

**[0098]** Another object of the present invention is to provide a new and improved system for wild fire suppression and home defense system, wherein each home defense spray system includes a GPS-tracking and radio-controlled circuit board to remotely monitor the location of each location-deployed home defense spray system and automatically monitor the fire inhibiting and extinguishing chemical liquid level in its storage tank, and automatically generate electronic refill orders sent to the command center, so that a third-party service can automatically replenish the tanks of such home-based systems with a fire inhibiting and extin-

guishing biochemical liquid when the fluid level falls below a certain level in the GPS-tracked tank.

**[0099]** Another object of the present invention is to provide a new and improved system for wild fire suppression and home defense system, wherein the mobile application supporting the following functions: (i) sends automatic notifications from the command center to home owners with the mobile application, instructing them to spray their property and home at certain times with fire inhibiting and extinguishing chemical liquid in their tanks; (ii) the system will automatically monitor consumption of sprayed fire inhibiting and extinguishing biochemical liquid and generate auto-replenish order via its onboard GSM-circuits so as to achieve compliance with the home spray-based wild-fire-defense program, and report fire inhibiting biochemical liquid levels in each home-owner tank; and (iii) show status of wild fire risk in the region, and actions to be taken before wild fire outbreak.

**[0100]** Another object of the present invention is to provide a GPS-guided method of suppressing a wild fire raging towards a target region of land in a direction determined by currently blowing winds and other environmental and weather factors.

**[0101]** Another object of the present invention is to provide a method of reducing the risks of damage to public property due to wild fires by managed application of fire inhibiting and extinguishing biochemical liquid spray to ground cover and building surfaces prior to arrival of the wild fires.

**[0102]** Another object of the present invention is to provide a wireless system for managing the supply, delivery and spray-application of environmentally-clean fire inhibiting and extinguishing biochemical liquid on private and public property to reduce the risks of damage and/or destruction caused by wild fires.

**[0103]** Another object of the present invention is to provide a new and improved system for spraying a defensive path around vulnerable neighborhoods out in front of wild fires to make sure that an environmentally-safe fire break, created by the spray application of fire inhibiting and extinguishing biochemical liquid, defends homes from the destructive forces of raging wild fires.

**[0104]** Another object of the present invention is to provide a new and improved system and method of mitigating the damaging effects of wild fires by spraying environmentally-clean fire inhibiting and extinguishing biochemical liquid in advance of wild fires, that do not depend on water to extinguish fire, such that, even after a month or two after spray application on dry brush around the neighborhood, the fire inhibiting and extinguishing chemical continues to work by stalling the ability of a fire to advance and consume homes.

**[0105]** Another object of the present invention is to provide new and improved methods of and apparatus for protecting wood-framed buildings from wild fires by automatically spraying water-based environmentally-clean fire inhibiting and extinguishing chemical liquid over the exterior surfaces of the building, surrounding ground surfaces, shrubs, decking and the like, prior to wild fires reaching such buildings.

**[0106]** Another object of the present invention is to provide new and improved method of suppressing a wild fire raging across a region of land in the direction of the prevailing winds.



[0107] Another object of the present invention is to provide a method of and apparatus for applying fire and smoke inhibiting compositions on ground surfaces before the incidence of wild-fires, and also thereafter, upon smoldering ambers and ashes to reduce smoke and suppress fire re-ignition.

[0108] Another object of the present invention is to provide a method of and apparatus applying by an aqueous-based fire and smoke inhibiting slurry formulation that can hydraulically sprayed around whole neighborhoods to create strategic chemical-type fire breaks that remove wild fire energy before such wildfires arrive at the doors of homes and businesses.

[0109] Another object of the present invention is to provide a method of spraying a clean fire and smoke inhibiting slurry composition containing clean fire inhibiting chemicals, and cellulose or wood fiber, mixed with water and other additives, for application to ground surfaces in advance of wild fire, to blanket grounds from wildfire ignition, and also application over smoldering ambers and ashes to prevent resignation while saving millions of gallons of water, and producing considerable waste water and reducing toxic run off, while reducing toxic smoke.

[0110] Another object of the present invention is to provide equipment for applying such fire and smoke inhibiting slurry mixtures to ground surfaces, after the presence of wildfire, to prevent smoke smoldering and resignation of fires, without creating toxic water runoff which occurs using conventional methods based on the application of water by fire hoses.

[0111] Another object of the present is to provide a new and improved method of generating biodegradable fire extinguishing liquid sprays on fires involving Class A and/or B fuels, using the biodegradable clean liquid concentrate of the present invention, and Venturi-type proportioning and mixing technology.

[0112] Another object of the present is to provide a new and improved method of generating biodegradable fire extinguishing foam on fires involving Class A and/or B fuels using the biodegradable clean foam concentrate of the present invention, and proportioning and mixing technology.

[0113] Another object of the present is to provide a new and improved method of producing highly-effective fire extinguishing water streams, as well as fine mists and vapor clouds, for inhibiting and extinguishing fires with the least water possible to minimize water damage.

[0114] Another object of the present is to provide a new and improved method of generating clean and safe biodegradable water-based fire extinguishing foams that are highly effective in extinguishing fires involving Class A and B fuels.

[0115] Another object of the present is to provide a new and improved biodegradable liquid concentrates for proportioning and mixing with pressurized streams of water to produce proactive fire inhibiting liquids for use in practicing wildfire defense, and proactive fire protection of lumber carbon, wood products, property, etc.

[0116] Another object of the present is to provide a new and improved biodegradable liquid concentrates for proportioning and mixing with pressurized streams of water to produce fire extinguishing liquids, mists, and vapors for use in fighting Class A/B fuel fires.

[0117] Another object of the present is to provide a new and improved biodegradable foam concentrates for propor-

tioning and mixing with pressurized streams of water to produce liquid foam solutions that are aerated to generate finished firefighting foam material for use in fighting Class A/B fuel fires.

[0118] Another object of the present is to provide a new and improved biodegradable foam concentrate comprising a major amount of hydrolyzed protein isolate (HPI) as the foaming agent, a minor amount of triethyl citrate (TEC) as a dispersant and surfactant, a major amount of tripotassium citrate (TPC) as the fire extinguishing agent, and a major amount of water as a solvent, wherein the liquid foam concentrate is mixed and proportioned with water, and thereafter is aerated to generated a firefighting foam material for extinguishing fires involving Class A and/or B fuels.

[0119] Another object of the present is to provide a new and improved biodegradable foam concentrate which is made from food-grade chemistry, and 100% free of fluorocemical, phosphates, and ammonia compounds.

[0120] Another object of the present is to provide a new and improved biodegradable foam concentrate for proportioning and mixing with pressurized streams of water to produce liquid foam solution that is aerated within an aerating/aspirating foam producing nozzles, for use in inhibiting and extinguishing fires involving Class A and B fuels. Another object of the present invention is to provide a new and improved fire extinguishing biochemical liquid foam concentrate to be mixed with a proportioned quantity of water, and then mixed with air within an aerating/aspirating foam forming nozzle to generate finished fire extinguishing foam material.

[0121] Another object of the present invention is to provide a new and improved fire extinguishing biochemical liquid foam concentrate, comprising: a dispersing agent in the form of a quantity of water, for dispersing metal ions dissolved in said quantity of water; a fire inhibiting agent in the form of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid, for providing metal ions dispersed in the water when the at least one alkali metal salt is dissolved in said quantity of water; a foaming agent including hydrolyzed protein isolate (HPI) material dissolved in the quantity of water; and a dispersing agent in the form of an organic compound containing three carboxylic acid groups, or salt/ester derivatives thereof, for dispersing the metal ions in the quantity of water, and lowering the surface tension of the liquid solution formed by the fire inhibiting agent, the foaming agent and the dispersing agent dissolved in the quantity of water, to enable the forming of a fire extinguishing foam material when the liquid solution is mixed with air within an aerating/aspirating foam forming nozzle.

[0122] Another object of the present invention is to provide a new and improved aqueous-based fire extinguishing biochemical liquid concentrate for mixing with a prespecified quantity of water to produce a fire extinguishing liquid solution that produces good immediate extinguishing effects when applied to extinguish a burning or smoldering fire, and very good long-term fire inhibiting effects when being proactively applied to protect combustible surfaces against the threat of fire.

[0123] Another object of the present invention is to provide a new and improved aqueous-based fire extinguishing biochemical liquid concentrate comprising: a dispersing agent realized in the form of a quantity of water, for dispersing metal ions dissolved in water; a fire inhibiting



agent in the form of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid, for providing metal ions dispersed in the water when the at least one alkali metal salt is dissolved in the quantity of water; and a dispersing agent in the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), such as triethyl citrate, an ester of citric acid, for dispersing the metal ions in the quantity of water, and lowering the surface tension of the liquid solution formed by the fire inhibiting agent, and the dispersing agent dissolved in the quantity of water, and forming of a fire extinguishing liquid solution that produces good immediate extinguishing effects when applied to extinguish a burning or smoldering fire, and very good long-term fire inhibiting effects when being proactively applied to protect combustible surfaces against the threat of fire.

**[0124]** Another object of the present invention is to provide such a new and improved aqueous-based fire extinguishing biochemical liquid concentrate, wherein the alkali metal salts of nonpolymeric saturated carboxylic acids for inclusion in the biochemical composition comprises: alkali metal salts of oxalic acid; alkali metal salts of gluconic acid; alkali metal salts of citric acid; and also alkali metal salts of tartaric acid.

**[0125]** These and other benefits and advantages to be gained by using the features of the present invention will become more apparent hereinafter and in the appended Claims to Invention.

#### BRIEF DESCRIPTION OF THE DRAWINGS

**[0126]** The following Objects of the Present Invention will become more fully understood when read in conjunction of the Detailed Description of the Illustrative Embodiments, and the appended Drawings, wherein:

**[0127]** FIG. 1 is a table listing conventional prior art methods for fighting and defending against wild fires including (i) aerial water drop methods using airplanes and helicopters, (ii) aerial fire retardant chemical (e.g. PhosChek® Fire Retardant) drop using airplanes and helicopters, (iii) physical fire breaks formed by bulldozing land and other landscaping methods to remove combustible vegetation from the land, (iv) physical fire breaks by pre-burning combustible material on the land, and (v) chemical fire break by fire retardant chemical drop;

**[0128]** FIG. 2A is a first image illustrating a prior art method of wild fire suppression involving an airplane dropping water on a wild fire from the sky;

**[0129]** FIG. 2B1 is a second image illustrating a prior art method of wild fire suppression involving an airplane dropping chemical fire retardant (e.g. PhosChek®) on a wild fire, from the sky;

**[0130]** FIG. 2B2 is third image showing a prior art ground-based tank containing the chemical fire retardant (e.g. PhosChek® fire retardant chemical) that is shown being contained in a storage tank in FIG. 2B2, and dropped from an airplane in FIG. 2B1;

**[0131]** FIG. 2B3 is a fourth image showing a prior art ground-based tank containing a supply of PhosChek® fire retardant chemical mixed in the tank shown in FIG. 2B3, and dropped from an airplane in FIG. 2B1;

**[0132]** FIG. 2B4 is a schematic representation illustrating the primary components of the PhosChek® fire retardant chemical, namely monoammonium phosphate (MAP), diammonium hydrogen phosphate (DAP) and water;

**[0133]** FIG. 3A is a schematic representation illustrating the primary active components of the fire retardant chemical disclosed and claimed in BASF's U.S. Pat. No. 8,273,813 to Beck et al., namely tripotassium citrate (TPC), and a water-absorbing polymer dissolved water;

**[0134]** FIG. 3B is a schematic representation illustrating the primary components of Hartidino's AF-31 fire retardant chemical, namely, potassium citrate and, a natural gum dissolved water as described in the Material Safety Data Sheet for Hartindo AF31 (Eco Fire Break) dated Feb. 4, 2013 (File No. DWMS2013);

**[0135]** FIG. 3C is a schematic representation describing a class of prior art firefighting foams, wherein the foaming agent may include hydrolyzed protein, and surfactants have included non-biodegradable fluorocarbon compounds;

**[0136]** FIG. 4A is schematic representation of the wireless system network of the present invention designed for managing the supply, delivery and spray-application of the environmentally-clean fire inhibiting (e.g. anti-fire or AF) liquid composition of the present invention, on private and public property to reduce the risks of property damage and/or destruction and harm to life caused by wild fires, and shown comprising GPS-tracked fire inhibiting and extinguishing liquid spray ground vehicles, GPS-tracked fire inhibiting liquid spray air vehicles, GPS-tracked fire inhibiting and extinguishing liquid spray backpack systems for spraying houses and surrounding properties, GPS-tracked fire inhibiting liquid spraying systems for spraying private real property and buildings, GPS-tracked fire inhibiting liquid spraying systems for spraying public real property and buildings, mobile computing systems running the mobile application of the present invention and used by property owners, residents, fire departments, insurance underwriters, government officials, medical personal and others, remote data sensing and capturing systems for remotely monitoring land and wild fires wherever they may break out, a GPS system for providing GPS-location services to each and every system components in the system network, and one or more data center containing clusters of web, application and database servers for supporting wire wild alert and notification systems, and microservices configured for monitoring and managing the system and network of GPS-tracking anti-fire fire inhibiting liquid spraying systems and mobile computing and communication devices configured in accordance with the principles of the present invention;

**[0137]** FIG. 4B is a schematic representation illustrating exemplary multi-spectral imaging (MSI) and hyper-spectral imaging (HSI) based remote sensing technology platforms supported by the US Geological Survey (USGS) Agency including, for example, the MODIS (Moderate Resolution Imaging Spectroradiometer) satellite system, the World View 2 Satellite System, the Octocopter unmanned airborne system (UAS) (e.g. OnyxStar Hydra-12 heavy-lifting drone), and the SenseFly eBee SQ UAS, for use in supporting and practicing the system network of the present invention;

**[0138]** FIG. 4C is a perspective view of the OnyxStar Hydra-12 heavy lifter drone supporting MSI and HSI camera systems, and providing remote data sensing services that can be used to help carry out the GPS-directed methods of wild fire suppression disclosed herein in accordance with the principles of the present invention;

**[0139]** FIG. 5A is a perspective view of an exemplary mobile computing device deployed on the system network



of the present invention, supporting (i) the mobile fire inhibiting spray management application of the present invention deployed as a component of the system network of the present invention as shown in FIGS. 4A and 4B, as well as (ii) conventional wildfire alert and notification systems as shown in FIGS. 3A through 3E;

[0140] FIG. 5B shows a system diagram for an exemplary mobile client computer system deployed on the system network of the present invention;

[0141] FIG. 6A1 is a schematic representation illustrating the primary components of a first environmentally-clean aqueous-based fire inhibiting liquid biochemical composition of the present invention consisting of major amounts of tripotassium citrate (TPC) and minor amounts of triethyl citrate (TEC) formulated with water functioning as a solvent, carrier and dispersant;

[0142] FIG. 6A2 is a schematic representation illustrating the primary components of a first fire inhibiting biochemical composition kit of the present invention, consisting of major amounts of dry tripotassium citrate monohydrate (TPC) and minor amounts of triethyl citrate (TEC), as components in a package prepared and ready for mixing with a predetermined quantity of water functioning as a solvent, carrier and dispersant, to make up a predetermined quantity of environmentally-clean liquid fire inhibiting biochemical composition for proactively treating and protecting wood products;

[0143] FIG. 6B1 is a schematic representation illustrating the primary components of a second environmentally-clean aqueous-based fire inhibiting liquid biochemical composition of the present invention consisting of major amounts of tripotassium citrate (TPC), minor amounts of triethyl citrate monohydrate (TEC), and minor amounts of biocidal agent (e.g. Polyphase® PW40 biocide), formulated with water functioning as a solvent, carrier and dispersant;

[0144] FIG. 6B2 is a schematic representation illustrating the primary components of the second fire inhibiting biochemical composition kit of the present invention, consisting of a major amount of dry tripotassium citrate (TPC), a minor amount of triethyl citrate (TEC), and a minor amount of biocidal agent (e.g. Polyphase® PW40 biocide), as components in a package prepared and ready for mixing with a predetermined quantity of water functioning as a solvent, carrier and dispersant, to make up a predetermined quantity of environmentally-clean liquid fire inhibiting biochemical composition for proactively treating and protecting wood products;

[0145] FIG. 6C1 is a schematic representation illustrating the primary components of a second environmentally-clean aqueous-based fire inhibiting liquid biochemical composition of the present invention consisting of a major amount of tripotassium citrate monohydrate (TPC), a minor amount of triethyl citrate (TEC), and a minor amount of citric acid (CA) formulated with water functioning as a solvent, carrier and dispersant;

[0146] FIG. 6C2 is a schematic representation illustrating the primary components of the second fire inhibiting biochemical composition kit of the present invention, consisting of a major amount of dry tripotassium citrate (TPC), a minor amount of triethyl citrate (TEC), and a minor amount of citric acid (CA), as components in a package prepared and ready for mixing with a predetermined quantity of water functioning as a solvent, carrier and dispersant, to make up a predetermined quantity of environmentally-clean liquid

fire inhibiting biochemical composition for proactively treating and protecting wood products;

[0147] FIG. 7A is a schematic representation illustrating a process of forming a tripotassium citrate (TPC) crystalline structures on combustible surfaces, such as ground cover, native fuel, lumber, living plant tissue, tree bark, and other combustible tissue and like materials that are sprayed with atomized sprays, or otherwise coated, with the chemical material comprising the aqueous-based fire inhibiting solutions of the present invention;

[0148] FIG. 7B is a schematic representation illustrating the atoms and atom numbering in the crystal structure of the compound, tripotassium citrate ( $K_3C_6H_5O_7$ ) formed on treated surfaces in accordance with the principles of the present invention;

[0149] FIG. 7C is a schematic representation of the atomic crystal structure of a small piece of the crystalline structure of tripotassium citrate ( $K_3C_6H_5O_7$ ) salt structure formed on a substrate to be protected against fire by way of application of the fire inhibiting chemical solution of the present invention, graphically illustrated the stage C illustration of FIG. 7A when water molecules mixed therein have evaporated to the ambient environment during air-drying;

[0150] FIG. 8A is a perspective view of a mobile GPS-tracked fire inhibiting liquid spraying system supported on a set of wheels (or supported on a back-rack), with integrated supply tank and rechargeable-battery operated electric spray pump, for deployment at private and public properties having building structures, for spraying the same with environmentally-clean fire inhibiting liquid in accordance with the principles of the present invention;

[0151] FIG. 8B is a schematic representation of the GPS-tracked mobile anti-fire (AF) fire inhibiting chemical liquid spraying system shown in FIG. 8A, comprising a GPS-tracked and remotely-monitored fire inhibiting chemical liquid spray control subsystem interfaced with a micro-computing platform for monitoring the spraying of fire inhibiting chemical liquid from the system when located at specific GPS-indexed location coordinates, and automatically logging and recording such fire inhibiting spray application operations within the network database system;

[0152] FIG. 9A is a perspective view of a GPS-tracked manned or autonomous vehicle system for spraying fire inhibiting chemical liquid on building and ground surfaces with environmentally-clean fire inhibiting chemical liquid in accordance with the principles of the present invention;

[0153] FIG. 9B is a schematic representation of the manned or autonomously-driven vehicle system shown in FIG. 9A, comprising a GPS-tracked and remotely-monitored fire inhibiting chemical liquid spray control subsystem interfaced with a micro-computing platform for monitoring the spraying of fire inhibiting chemical liquid from the vehicle when located at any specific GPS-indexed location coordinates, and automatically logging and recording such fire inhibiting spray application operations within the network database system;

[0154] FIG. 10A is a perspective view of an autonomously-driven or remotely-controlled unmanned airborne system (i.e. UAS or “drone”) adapted for spraying fire inhibiting chemical liquid on building and ground surfaces for spraying the same with environmentally-clean fire inhibiting liquid in accordance with the principles of the present invention;



**[0155]** FIG. 10B is a schematic representation of the autonomously-driven or remotely-controlled aircraft system (i.e. drone) shown in FIG. 10A, comprising a GPS-tracked and remotely monitored fire inhibiting chemical liquid spray control subsystem interfaced with a micro-computing platform for monitoring the spraying of fire inhibiting chemical liquid from the aircraft when located at specific GPS-indexed location coordinates, and automatically logging and recording such fire inhibiting spray application operations within the network database system;

**[0156]** FIG. 11A is a perspective view of a GPS-tracked aircraft system (i.e. helicopter) adapted for spraying an environmentally-clean fire inhibiting biochemical liquid of the present invention, from the air onto ground and property surfaces in accordance with the principles of the present invention;

**[0157]** FIG. 11B is a schematic representation of the GPS-tracked aircraft system (i.e. helicopter) shown in FIG. 11A, comprising a GPS-tracked and remotely monitored fire inhibiting chemical liquid spray control subsystem interfaced with a micro-computing platform for monitoring the spraying of fire inhibiting chemical liquid from the aircraft when located at specific GPS-indexed location coordinates, and automatically logging and recording such fire inhibiting spray application operations within the network database system;

**[0158]** FIG. 12A is a GPS-tracked all-terrain vehicle (ATV) system adapted for spraying ground surfaces with environmentally-clean fire inhibiting liquid in accordance with the principles of the present invention;

**[0159]** FIG. 12B is the GPS-tracked all-terrain vehicle (ATV) system shown in FIG. 12A, comprising a GPS-tracked and remotely-monitored fire inhibiting chemical liquid spray control subsystem interfaced with a micro-computing platform for monitoring the spraying of fire inhibiting chemical liquid from the system when located at specific GPS-indexed location coordinates, and automatically logging and recording such fire inhibiting spray application operations within the network database system;

**[0160]** FIG. 13A is a GPS-tracked portable backpack-mounted atomizing spray “cannon” system adapted for spraying ground and building surfaces with an environmentally-clean fire inhibiting liquid formulated in accordance with the principles of the present invention;

**[0161]** FIG. 13B shows the GPS-tracked portable backpack-mounted atomizing spray “cannon” system of FIG. 13A being worn by a person who is using it with the system network GPS-track and record the spraying of ground and building surfaces with the environmentally-clean fire inhibiting liquid biochemical composition formulated in accordance with the principles of the present invention;

**[0162]** FIG. 13C is the GPS-tracked backpack mounted atomizing spray cannon system shown in FIG. 13A, comprising a GPS-tracked and remotely-monitored fire inhibiting chemical liquid spray control subsystem interfaced with a micro-computing platform for monitoring the spraying of environmentally-clean fire inhibiting chemical liquid from the system when located at specific GPS-indexed location coordinates, and automatically logging and recording such fire inhibiting spray application operations within the network database system;

**[0163]** FIG. 14A is a perspective view of a GPS-tracked mobile atomizing spray cannon vehicle (SCV) system adapted for spraying ground surfaces with environmentally-

clean fire inhibiting biochemical liquid in accordance with the principles of the present invention;

**[0164]** FIG. 14B is perspective view of the GPS-tracked spray cannon vehicle (SPV) system shown in FIG. 14A, adapted for spraying ground surfaces with anti-fire (AF) fire inhibiting biochemical liquid in accordance with the principles of the present invention;

**[0165]** FIG. 14C is a perspective view of the atomizing spray cannon component of the GPS-tracked spray cannon vehicle (SPV) system shown in FIGS. 14A and 14B, showing a ring of atomizing spray nozzles mounted in a ring disposed about the inner aperture of the spray cannon, through is driven high velocity air streaming past the nozzles during spray atomizing operations using the environmentally-clean fire inhibiting biochemical liquid of the present invention;

**[0166]** FIG. 14D is a schematic block diagram of the GPS-tracked spray cannon vehicle (ASPV) system shown in FIGS. 14A, 14B and 14C, comprising a GPS-tracked and remotely-monitored fire inhibiting chemical liquid spray control subsystem interfaced with a micro-computing platform for monitoring the spraying of fire inhibiting biochemical liquid from the system when located at specific GPS-indexed location coordinates, and automatically logging and recording such fire inhibiting spray application operations within the network database system;

**[0167]** FIG. 15A is a GPS-tracked portable wheel-mounted atomizing spray “cannon” system, configured as a trailer and adapted for towing behind a powered vehicle (e.g. truck), and supporting atomization spraying of ground and property surfaces with an environmentally-clean fire inhibiting biochemical liquid formulated in accordance with the principles of the present invention;

**[0168]** FIG. 15B is the GPS-tracked portable wheel-mounted atomizing spray cannon system shown in FIG. 15A, comprising a GPS-tracked and remotely-monitored fire inhibiting chemical liquid spray control subsystem interfaced with a micro-computing platform for monitoring the spraying of environmentally-clean fire inhibiting biochemical liquid from the system when located at specific GPS-indexed location coordinates, and automatically logging and recording such fire inhibiting spray application operations within the network database system;

**[0169]** FIG. 16A is a GPS-tracked portable backpack-mounted atomizing spraying system adapted for spraying ground surfaces with environmentally-clean fire inhibiting biochemical liquid in accordance with the principles of the present invention;

**[0170]** FIG. 16B is the GPS-tracked backpack-mounted atomizing spraying system shown in FIG. 13A, comprising a GPS-tracked and remotely-monitored fire inhibiting chemical liquid spray control subsystem interfaced with a micro-computing platform for monitoring the spraying of environmentally-clean biochemical liquid from the system when located at specific GPS-indexed location coordinates, and automatically logging and recording such fire inhibiting spray application operations within the network database system;

**[0171]** FIG. 17A is a GPS-traced mobile remotely-controllable atomizing spray “cannon” system adapted for spraying ground surfaces with environmentally-clean fire inhibiting liquid in accordance with the principles of the present invention;



[0172] FIG. 17B is the GPS-tracked mobile remotely-controllable atomizing spray cannon system shown in FIG. 13A, comprising a GPS-tracked and remotely-monitored fire inhibiting chemical liquid spray control subsystem interfaced with a micro-computing platform for monitoring the spraying of environmentally-clean AF chemical liquid from the system when located at specific GPS-indexed location coordinates, and automatically logging and recording such fire inhibiting spray application operations within the network database system;

[0173] FIG. 18 is a schematic representation of a schema for the network database (RDBMS) supported by the system network of the present invention, showing the primary enterprise level objects supported in the database tables created in the network database using the schema, and the relationships that are specified or indicated, to support all of the enterprise-level objects defined and managed on the system network;

[0174] FIG. 19 is an exemplary wire-frame model of a graphical user interface supported by mobile application configured for use by a first specific class of registered users (e.g. property parcel owners, contractors and/or agents, residents, government officials, and others) to request and receive services, including notices and orders, supported by the system network of the present invention;

[0175] FIG. 19A is an exemplary wire-frame model of a graphical user interface supported by the mobile application showing a user updating the registration profile as a task on the system network;

[0176] FIG. 19B is an exemplary wire-frame model of a graphical user interface supported by the mobile application showing a user receiving a message request (via email, SMS messaging and/or push-notifications) issued from the command center to spray GPS-specified private property parcel (s) with clean fire inhibiting biochemical liquid and registered equipment;

[0177] FIG. 19C is an exemplary wire-frame model of a graphical user interface supported by the mobile application showing a user receiving a request/notice of order (via email, SMS messaging and/or push-notifications) to wild-fire spray-protect GP S-specified public property parcel(s) with clean fire inhibiting biochemical liquid to create and maintain a GPS-specified public firebreak, maintained on public property;

[0178] FIG. 19D is an exemplary wire-frame model of a graphical user interface supported by the mobile application showing a user requesting a refill supply of clean fire inhibiting biochemical liquid for supply to GPS-specified spray equipment registered on the system network;

[0179] FIG. 20 is an exemplary wire-frame model of a graphical user interface supported by the mobile application configured for second specific class of registered users, namely, command center administrators, enabling such users to issue wild-fire protection orders, plan wild-fire protection tasks, generate wild-fire and protection reports, and send and receive messages to users on the system network;

[0180] FIG. 20A is an exemplary wire-frame model of a graphical user interface supported by the mobile application for use by command center administrators to issue wild-fire protection orders using the system network of the present invention;

[0181] FIG. 20B exemplary wire-frame model of a graphical user interface supported by the mobile application for

use by command center administrators to issue wild-fire protection orders involving the creation and maintenance of a clean biochemical firebreak using the methods of the present invention, as illustrated in FIGS. 24 through 32B;

[0182] FIG. 20C is an exemplary wire-frame models of a graphical user interface supported by the mobile application for use by command center administrators to order the creation and/or maintenance of a GPS-specified environmentally-clean biochemical firebreak on one or more public/private property parcels, using the methods of the present invention;

[0183] FIG. 20D is an exemplary wire-frame models of a graphical user interface for the mobile application used by command center administrators to receive messages from users including property owners and contractors requesting refills for clean fire inhibiting biochemical liquid for GPS-specified spray system equipment;

[0184] FIG. 21 is a graphical representation of an exemplary fire hazard severity zone (FHSZ) map generated by the CAL FIRE™ System in state responsibility areas of the State of California, and accessible through the mobile application, for use while informing the strategic application of environmentally-clean fire inhibiting biochemical liquid spray onto specified regions of property prior to the arrival of wild fires, using the system network of the present invention;

[0185] FIG. 22 is an exemplary anti-fire (AF) spray protection map generated by the system network of the present invention, showing houses and buildings that have been sprayed, and not-sprayed, with state/county-issued clean anti-fire biochemical liquid as of the report date 15 Dec. 2017;

[0186] FIG. 23 is an exemplary anti-fire spray protection task report generated by the system of the present invention for state/county xxx on 15 Dec. 2017, indicating which properties on what streets, in what town, county, state, requires the reapplication of fire inhibiting chemical liquid spray treatment in view of factors such as weather (e.g. rainfall, sunlight) and passage of time since last fire inhibiting biochemical spray application;

[0187] FIG. 24 is a schematic representation showing a plan view of a wild fire emerging from a forest region and approaching a neighboring town moving in the direction of prevailing winds;

[0188] FIG. 25 is a graphical representation illustrating a method of suppressing a wild fire raging across a region of land in the direction of the prevailing winds, by forming a multi-stage anti-fire biochemical fire-break system, by GPS-controlled application of fire inhibiting liquid mist and spray streams;

[0189] FIGS. 26A and 26B set forth a flow chart describing the high level steps of the method of suppressing a wild fire raging towards a target region of land in a direction determined by prevailing winds and other environmental and weather factors, as schematically illustrated in FIG. 25;

[0190] FIG. 27 is a graphical representation illustrating a method of reducing the risks of damage to private property due to wild fires by GPS-controlled application of fire inhibiting biochemical liquid spray, using the system network of the present invention;

[0191] FIGS. 28A, 28B and 28C, taken together, set forth a flow chart describing the high level steps carried out by the method of reducing the risks of damage to private property due to wild fires by managed application of fire inhibiting



biochemical liquid spray, using the system network and methods of the present invention, as illustrated in FIG. 27; [0192] FIG. 29 is a graphical illustration showing a method of reducing the risks of damage to public property due to wild fires, by GPS-controlled application of fire inhibiting biochemical liquid spray over ground cover and building surfaces prior to the arrival of wild fires, using the system network and methods of the present invention;

[0193] FIGS. 30A, 30B and 30C, taken together, set forth a flow chart describing the high level steps carried out by the method of reducing the risks of damage to public property due to wild fires by GPS-controlled application of fire inhibiting biochemical liquid spray, using the system network and methods of the present invention, as illustrated in FIG. 29;

[0194] FIG. 31 is a graphical illustration showing a method of remotely managing the GPS-controlled application of environmentally-clean fire inhibiting biochemical liquid spray of the present invention to ground cover and buildings so as to reduce the risks of damage due to wild fires, using the system network and methods of the present invention;

[0195] FIGS. 32A and 32B, taken together, set forth a flow chart describing the high level steps carried out by the method of GPS-controlled application of fire inhibiting biochemical liquid spray to ground cover and buildings so as to reduce the risks of damage due to wild fires, using the system network and methods of the present invention;

[0196] FIG. 33A is a perspective view of the clean fire and smoke inhibiting slurry spray application vehicle of the present invention carrying a high-capacity (e.g. 3000 gallon) stainless steel mixing tank with an integrated agitator mechanism (e.g. motor driven mixing paddles) for mixing the fire and smoke inhibiting slurry spray mixture of the present invention, and a hydraulic pumping apparatus and spray nozzle for spraying the clean aqueous-based clean fire and smoke inhibiting slurry of the present invention, on ground surfaces to create clean biochemical fire breaks around regions to be protected from wildfires, and also to cover smoldering ambers and ash after the present of wildfires to reduce toxic waste water runoff and smoke production;

[0197] FIG. 33B is a rear view of the vehicle shown in FIG. 33A;

[0198] FIG. 33C is a side view of the vehicle shown in FIG. 33A;

[0199] FIG. 34 is a schematic system block diagram of the fire and smoke inhibiting slurry spray vehicle system shown in FIGS. 33A, 33B and 33C;

[0200] FIG. 35 is a flow chart describing the method of applying fire and smoke inhibiting slurry compositions of the present invention on ground surfaces before the incidence of wild-fires, and also thereafter, upon smoldering ambers and ashes to reduce smoke and suppress fire reignition;

[0201] FIG. 36 is a base hydraulic mulch loading chart for making the fire and smoke inhibiting slurry mixture of the present invention, using Profile® brand mulch fiber, for several different application rates measured in lbs./acre (e.g. 1500 lbs./acre, 2000 lb./acre, and 2500 lb./acre);

[0202] FIG. 37 is a schematic representation of a neighborhood of houses surrounded by a high-risk wildfire region, wherein a clean biochemical wild-fire break region is hydraulically sprayed on the ground surface region all

around the houses using the clean fire and smoke inhibiting slurry composition of the present invention;

[0203] FIG. 38 is a schematic representation of a highway surrounded by a high-risk wildfire region on both sides, wherein a clean biochemical wild-fire break region is hydraulically sprayed on both sides of the highway using the clean fire and smoke inhibiting slurry composition of the present invention;

[0204] FIG. 39 is a schematic representation of a highway off ramp that has been sprayed with the clean fire and smoke inhibiting slurry composition of the present invention, to provide a safe way to exit a wildfire burning region, while suppressing and preventing reignition of the fire, and reducing the production of smoke and creation of toxic water runoff during post fire management operations;

[0205] FIG. 40 is a schematic representation of a wood-framed or mass timber building that just burned to the ground after a wildfire passed through an unprotected neighborhood, wherein the clean fire and smoke inhibiting slurry composition is hydraulically sprayed over the glowing ambers and fire ash to suppress and prevent reignition of the fire, and reduce the production of smoke and creation of toxic water runoff during post fire management operations;

[0206] FIG. 41 is a schematic representation of a wood-framed or mass timber building that is burning due to a fire within the building, wherein the wet fire and smoke inhibiting slurry composition of the present invention is hydraulically sprayed on and over the fire to suppress it, while reducing the production of smoke during the fire suppression process;

[0207] FIG. 42A is a schematic representation of an automated wireless wildfire ember detection and suppression system of present invention, showing a wildfire ember detection module mounted on the top of each building in the wireless network receiving wildfire alerts and messages from neighboring modules which can scout for wildfires and alert other modules in the network in terms of GPS coordinates so that the individual properties can timely prepare for any such wildfire outbreaks in the vicinity, using the hybrid wildfire misting system of the present invention shown in FIGS. 42C and 42D;

[0208] FIG. 42B is a schematic block diagram showing the components used to construct the wireless GPS-tracked wildfire ember detection module of the present invention, shown in FIG. 42A;

[0209] FIGS. 42C and 42D, taken together, set forth a schematic diagram showing the automated hybrid clean wildfire inhibiting misting system of the present invention, providing both a fire inhibiting chemical misting system for suppressing wildfire embers impacting a building as shown in FIG. 42C and a lawn and ground fire-inhibiting biochemical liquid misting system impacting the lawn and ground around the building as shown in FIG. 42D, both automatically controlled by an automated wildfire ember detection and notification network shown in FIGS. 42A and 42B, all being integrated into the system network shown in FIGS. 4A and 4B;

[0210] FIG. 42E shows several spray misting nozzles used in the system of the FIGS. 42A through 43D;

[0211] FIGS. 42F1 and 42F2 show side and front perspective views of 360 degree rotating sprinkler heads for mounting on building rooftops and integrated into the system of FIG. 42A, to spray fire inhibiting biochemical liquid accord-



ing to the present invention, and treated combustible surfaces before wildfire embers arrival;

**[0212]** FIG. 42G is a schematic representation illustrating that the clean-biochemistry fire break sprayed by sprinkler-type head(s) shown in FIGS. 42F1 and 42F2, mounted on home building rooftops and driven by automated pumps, automatically creates and maintains a proactive fire defense coverage against an advancing wildfire, so as to help reduce risk of destruction of property and life by wildfire;

**[0213]** FIG. 43 is a perspective view of an exemplary mobile computing device deployed on the system network of the present invention, supporting (i) the mobile fire inhibiting biochemical spray management application of the present invention deployed as a component of the system network of the present invention, as well as (ii) conventional wildfire alert and notification systems, as shown in FIGS. 42A through 42D;

**[0214]** FIG. 44 is a schematic representation illustrating the primary components of a first environmentally-clean aqueous-based fire extinguishing biochemical liquid concentrate (i.e. fire extinguishing additive) of the present invention consisting of (i) major amounts of a fire inhibiting agent realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate (TPC), for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (ii) minor amounts of triethyl citrate (TEC) as a low-surface tension surfactant and dispersing agent, and (iii) major amounts of water functioning as a solvent, carrier and dispersant, to form the fire extinguishing biochemical liquid concentrate (LC) designed to be added to and mixed in-line with a pressurized supply of water in pre-specified proportions so as to produce an environmentally-clean fire extinguishing aqueous liquid for spraying onto an actively combusting fire involving Class A fuel and/or Class B fuel;

**[0215]** FIG. 45 is a schematic representation illustrating the primary components of a second environmentally-clean aqueous-based fire extinguishing biochemical liquid concentrate (i.e. fire extinguishing additive) of the present invention consisting of (i) major amounts of a fire inhibiting agent realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate (TPC), for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (ii) minor amounts of triethyl citrate (TEC) as a low-surface tension surfactant and dispersing agent, (iii) minor amounts of citric acid as a buffering agent, and (iv) major amounts of water functioning as a solvent, carrier and dispersant, to form the fire extinguishing biochemical liquid concentrate (LC) designed to be added to and mixed in-line with a pressurized supply of water in pre-specified proportions so as to produce an environmentally-clean fire extinguishing aqueous liquid for spraying onto an actively combusting fire involving Class A fuel and/or Class B fuel;

**[0216]** FIG. 46 is a block schematic representation of a mobile and/or portable fire extinguishing system for mixing and proportioning the fire extinguishing liquid concentrate of present invention with pressurized water to produce a fire extinguishing water solution for use in fighting active fires involving Class A and/or Class B fuels, comprising a venturi-based fluid mixing/proportioning device operably connected to (i) a pressurized supply of water output at 200+

PSI pressure from a hydraulic pumping engine connected to a supply of water and pressurized by a hydraulic pump system driven by gasoline, diesel or electric engine, (ii) a supply of fire extinguishing liquid concentrate (LC) of the present invention contained within a 20+ gallon container, and (iii) one or more aerating/atomizing-type fire hose spray nozzles manually-actuatable for producing a manually-adjustable water stream containing a proportioned quantity of fire extinguishing additive for every proportioned quantity of water, and comprising fine water droplets in the range of about 1500 microns to about 50 microns, as required for extinguishing an particular fire involving Class A and/or Class B fuels;

**[0217]** FIG. 47 is a perspective view of a conventional in-line type venturi-based eductor device for proportioning and mixing the fire extinguishing concentrate (e.g. additive) of the present invention, into a pressurized water stream flowing into the eductor device while spraying a pressurized stream of water from an atomizing-type or spray-type nozzle assembly connected to a length of fire hose, as schematically illustrated in FIG. 46;

**[0218]** FIG. 48 is a perspective view of a portable spray cart containing a supply of fire extinguishing liquid concentrate additive in a tank, supported on a set of wheels, and equipped with an in-line eductor device for drawing liquid concentrate into a pressurized water stream, as shown in FIG. 46, and being (i) operably connected to a length of fire hose terminated with an adjustable aerating/aspirating-type spray nozzle, and (ii) operably connected to a pressurized water pumping engine as illustrated in FIG. 46, to mix a proportioned quantity (i.e. %) of fire extinguishing liquid concentrate with a pressurized supply of water flowing through the eductor device, along the length of fire hose to the adjustable spray nozzle, spraying an active fire involving a Class A and/or B fuel;

**[0219]** FIG. 49 is a perspective view of a portable triple tote spray trailer designed to be pulled and driven by a pressurized water pumping firetruck, and having a trailer platform supporting three liquid concentrate totes, each containing 200 gallons of fire extinguishing liquid concentrate additive of the present invention, and being operably connected to an in-line eductor device as shown in FIG. 46, and also to an adjustable spray nozzle gun assembly mounted for spraying operations, and also being operably connectable to the pressurized water pumping engine aboard the water pumping firetruck, as illustrated in FIG. 46, so as to mix a proportioned quantity (i.e. 1%, 3% or 6%) of fire extinguishing liquid concentrate with a pressurized supply of water flowing through the eductor device, to the adjustable spray nozzle, while the spraying pressurized water with fire extinguishing chemical additive, from the spray gun nozzle during an actively combustible fire;

**[0220]** FIG. 50 is a block schematic representation of a stationary and/or fixed fire extinguishing system for mixing and proportioning the fire extinguishing liquid concentrate of present invention with pressurized water to produce a fire extinguishing water solution for use in fighting active fires involving Class A and/or Class B fuels, comprising a venturi-based fluid eductor-type mixing/proportioning device operably connected to (i) a pressurized supply of water output at 200+ PSI pressure from a hydraulic pumping engine connected to a supply of water and pressurized by a hydraulic pump system driven by an electric, propane or other engine, (ii) a supply of fire extinguishing liquid



concentrate (LC) of the present invention contained within a 20+ gallon container, and (iii) one or more aerating-type spray nozzles, typically triggered by electronic-controlled sensors, IR cameras and/or controllers, for automatically producing a cloud of water mist or vapor comprising fine water microdroplets in the range of about 500 microns to about 50 microns, with proportioned quantities of fire extinguishing biochemical additives, as required for extinguishing a particular fire involving Class A and/or Class B fuels, with improved fire extinguishing efficacy and efficiency using reduced quantities of water to minimize water damage to property during a fire outbreak;

[0221] FIG. 51 is a schematic representation illustrating the primary components of a first environmentally-clean aqueous-based fire extinguishing biochemical foam concentrate (i.e. fire extinguishing additive) of the present invention consisting of (i) major amounts of a fire inhibiting agent realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate (TPC), for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (ii) minor amounts of triethyl citrate (TEC) as a low-surface tension surfactant agent, (iii) major amounts of soy protein isolate as a foaming agent, and (iv) major amounts of water functioning as a solvent, carrier and dispersant, to form the fire extinguishing biochemical foam concentrate (LC) designed to be added to and mixed in-line with a pressurized supply of water in pre-specified proportions so as to produce an environmentally-clean aqueous fire extinguishing foam for spraying onto an actively combusting fire involving Class A fuel and/or Class B fuel;

[0222] FIG. 52 is a table schematically illustrating the performance characteristics and chemical components associated with the bio-degradable Class AB firefighting foam concentrate specified in FIG. 51;

[0223] FIG. 53 is a block schematic representation of a mobile and/or portable fire extinguishing system for mixing and proportioning the fire extinguishing foam concentrate of present invention with pressurized water, and then injecting air into the foam liquid to produce a finished fire extinguishing foam for use in fighting active fires involving Class A and/or Class B fuels, comprising a venturi-based fluid eductor-type mixing/proportioning device/system operably connected to (i) a pressurized supply of water output at 200+ PSI pressure from a hydraulic pumping engine connected to a supply of water and pressurized by a hydraulic pump system driven by gasoline, diesel or electric engine, (ii) a supply of fire extinguishing foam concentrate (FC) of the present invention contained within a 20+ gallon container, and (iii) one or more aerating/aspirating foam spray nozzles for generating finished fire extinguishing foam material for use in extinguishing any particular fire involving Class A and/or Class B fuels;

[0224] FIG. 54 is a perspective view of a conventional in-line type eductor device for proportioning and mixing the liquid foam concentrate (e.g. biochemical additive) to a pressurized water stream, while providing the liquid foam concentrate solution to an aerating/aspirating foam forming nozzle assembly connected thereto, as schematically illustrated in FIG. 53;

[0225] FIG. 55 is a perspective view of a portable spray cart, containing a supply of fire extinguishing liquid concentrate additive, with an in-line proportioning/mixing system (i.e. eductor device) of FIG. 54, connected to a length

of fire hose and an adjustable spray nozzle, and operably connectable to a pressurized water pumping engine as illustrated in FIG. 53, to mix a proportioned quantity (i.e. %) of fire extinguishing foam concentrate with a pressurized supply of water flowing through the eductor device, along the length of fire hose to the aerating/aspirating foam producing nozzle, for generating finished firefighting foam material for application to an active fire;

[0226] FIG. 56 is a perspective view of a portable triple tote spray trailer designed to be pulled and driven by a pressurized water pumping firetruck, and having a trailer platform supporting three foam concentrate totes, each containing 200 gallons of fire extinguishing foam concentrate additive of the present invention, and being operably connected to an in-line eductor device as shown in FIG. 54, and also to an aerating/aspirating foam forming nozzle gun assembly mounted for spraying operations, and also being operably connectable to the pressurized water pumping engine aboard the water pumping firetruck, as illustrated in FIG. 53, so as to mix a proportioned quantity (i.e. 1%, 3% or 6%) of fire extinguishing foam concentrate (FC) with a pressurized supply of water flowing through the venturi-based eductor device that is continuously proportions and mixes foam concentrate and water to produce a liquid foam solution that is supplied to the input port of an aerating-type foam spray nozzle so as to generate a finished fire extinguishing foam from its nozzle for application to Class A and/or B fuel sources during an active fire; and

[0227] FIG. 57 is a block schematic representation of a stationary and/or fixed fire extinguishing system for mixing and proportioning the fire extinguishing foam concentrate of present invention with pressurized water, and then injecting air into the foam liquid to produce a finished fire extinguishing foam for use in fighting active fires involving Class A and/or Class B fuels, comprising a venturi-based fluid eductor-type mixing/proportioning device operably connected to (i) a pressurized supply of water output (e.g. at 200+ psi pressure) produced from a hydraulic pumping engine connected to a supply of water and pressurized by a hydraulic pump system driven by an electric, propane or other engine, (ii) a supply of fire extinguishing foam concentrate (FC) of the present invention contained within a 20+ gallon container, and (iii) one or more aerating/aspirating foam forming nozzles, for automatically producing fire extinguishing foam for extinguishing a particular fire involving Class A and/or Class B fuels, with improved fire extinguishing efficacy and efficiency using reduced quantities of water to minimize water damage to property during a fire outbreak.

#### DETAILED DESCRIPTION OF THE ILLUSTRATIVE EMBODIMENTS OF THE PRESENT INVENTION

[0228] Referring to the accompanying Drawings, like structures and elements shown throughout the figures thereof shall be indicated with like reference numerals.

Bio-Degradable Fire Inhibiting Liquid Concentrate (LC) Compositions of the Present Invention in Connection with Proactively Defending Against Virtually any Kind of Fires Involving Class A/B Fuels, Including Wildfires Raging Across the Rapidly Expanding Wildfire Urban Interface (WUI) Region of the Planet Earth

[0229] Referring to FIGS. 4 through 43, technical details will now be described teaching how to practice the bio-degradable fire inhibiting liquid concentrate (LC) composi-



tions of the present invention in connection with proactively defending against virtually any kind of fires involving Class A/B fuels, including wildfires raging across the rapidly expanding wildfire urban interface (WUI) region of the planet Earth.

Wireless System Network for Managing the Supply, Delivery and Spray-Application of Environmentally-Clean Anti-Fire/Fire-Inhibiting Biochemical Liquid on Private and Public Property to Reduce the Risks of Damage and/or Destruction Caused by Wild Fires

**[0230]** FIG. 4A shows the wireless system network of the present invention 1 designed for managing the supply, delivery and spray-application of environmentally-clean anti-fire (AF) (i.e. fire inhibiting) biochemical liquid composition of the present invention, on private and public property to reduce the risks of damage and/or destruction caused by wild fires.

**[0231]** As shown, the wireless system network 1 comprises a distribution of system components, namely: GPS-tracked anti-fire (AF) liquid spray ground vehicles 2 (e.g. all-terrain vehicles or ATVs), as shown in FIGS. 9A, 9B, 12A, 12B, 14A, 14B, 14C, 14D, 15A, 15B, 17A and 17B, for applying AF chemical liquid spray fire inhibitor chemical, formulated according to the present invention, to ground surfaces, brush surfaces, and the surfaces of other forms of organic combustible material on property; GPS-tracked anti-fire liquid spray air-based vehicles 3, as shown in FIGS. 10A, 10B, 11A, and 11B, for applying AF chemical liquid spray of the present invention (formulated as illustrated in FIGS. 6 and 7 and specified herein) from the air to ground surfaces, brush, bushes and other forms of organic material; GPS-tracked mobile anti-fire liquid back-pack spraying systems 4 (e.g. including wheel supported, and backpack-carried systems), as shown in FIGS. 8A, 8B, 13A, 13B, 16A and 16B, for applying AF chemical liquid spray to combustible ground surfaces, brush, bushes, decks, houses, buildings, and other forms of organic material and property surrounding houses; GPS-tracked/GSM-linked anti-fire liquid spraying systems 5, as shown in FIGS. 8A through 17B, for applying AF chemical liquid spray to combustible surfaces on private real property, buildings and surrounding areas; GPS-tracked/GSM-linked liquid spraying systems 6, as shown in FIGS. 8A through 17B, for applying AF chemical liquid spray to combustible surfaces on public real property and buildings and surrounding properties; a GPS-indexed real-property (land) database system 7 for storing the GPS coordinates of the vertices and maps of all land parcels, including private property and building 17 and public property and building 18, situated in every town, county and state in the region over which the system network 1 is used to manage wild fires as they may occur; a cellular phone, GSM, and SMS messaging systems and email servers, collectively 16; and one or more data centers 8 for monitoring and managing GPS-tracking/GSM-linked anti-fire (AF) liquid supply and spray systems, including web servers 9A, application servers 9B and database servers 9C (e.g. RDBMS) operably connected to the TCP/IP infrastructure of the Internet 10, and including a network database 9C1, for monitoring and managing the system and network of GPS-tracking anti-fire liquid spraying systems and various functions supported by the command center 19, including the management of wild fire suppression and the GPS-guided application of anti-fire (AF) chemical liquid over public and private property, as will be described in

greater technical detail hereinafter. As shown, each data center 8 also includes an SMS server 9D and an email message server 9E for communicating with registered users on the system network 1 who use a mobile computing device (e.g. an Apple® iPhone or iPad tablet) 11 with the mobile application 12 installed thereon and configured for the purposes described herein. Such communication services will include SMS/text, email and push-notification services known in the mobile communications arts.

**[0232]** As shown in FIG. 4A, the GPS-indexed real-property (land) database system 7 will store the GPS coordinates of the vertices and maps of all land parcels contained in every town, county and state of the region over which the system network is deployed and used to manage wild fires as they may occur. Typically, databases and data processing methods, equipment and services known in the GPS mapping art, will be used to construct and maintain such GPS-indexed databases 7 for use by the system network of the present invention, when managing GPS-controlled application of clean anti-fire (AF) chemical liquid spray and mist over GPS-specified parcels of land, at any given time and date, under the management of the system network of the present invention. Examples of such GPS-indexed maps of land parcels are reflected by the task report shown in FIG. 23, and examples of GPS-indexed maps are shown in the schematic illustrations depicted in FIGS. 18, 20, 22 and 24.

**[0233]** As shown in FIG. 4A, the system network 1 also includes a GPS system 100 for transmitting GPS reference signals transmitted from a constellation of GPS satellites deployed in orbit around the Earth, to GPS transceivers installed aboard each GPS-tracking ground-based or air-based anti-fire (AF) liquid misting/spraying system of the present invention, shown in FIGS. 6A through 10B, as part of the illustrative embodiments. From the GPS signals it receives, each GPS transceiver aboard such AF liquid spraying/misting systems is capable of computing in real-time the GPS location of its host system, in terms of longitude and latitude. In the case of the Empire State Building in NYC, N.Y., its GPS location is specified as: N40° 44.9064', W073° 59.0735'; and in number only format, as: 40.748440, -73.984559, with the first number indicating latitude, and the second number representing longitude (the minus sign indicates "west").

**[0234]** As shown in FIG. 4B, the system network 1 further includes multi-spectral imaging (MSI) systems and/or hyper-spectral-imaging (HSI) systems 14 for remotely data sensing and gathering data about wild fires and their progress. Such MSI and HSI systems may be space/satellite-based and/or drone-based (supported on an unmanned airborne vehicle or UAV). Drone-based systems can be remotely-controlled by a human operator, or guided under an artificial intelligence (AI) navigation system. Such AI-based navigation systems may be deployed anywhere, provided access is given to such remote navigation system the system network and its various systems. Typically, the flight time will be limited to under 1 hour using currently available battery technology, so there will be a need to provide provisions for recharging the batteries of such drones/UASs in the field, necessitating the presence of human field personnel to support the flight and remote data sensing and mapping missions of each such deployed drone, flying about raging wild fires, in connection with the system network of the present invention.



[0235] During each wild fire data sensing and mapping mission, carried out by such UAS, a series of MSI images and HSI images can be captured during a wild fire, and mapped to GPS-specific coordinates, and this mapped data can be transmitted back to the system network for storage, analysis and generation of GPS-specified flight plans for anti-fire (AF) chemical liquid spray and misting operations carried out using the methods illustrated in FIGS. 24, 25, 26A and 26B seeking to stall and suppress such wild fires, and mitigate risk of damage to property and harm to human and animal life.

[0236] FIG. 4B shows a suite of MSI and HSI remote sensing and mapping instruments and technology 14 that is currently being used by the US Geological Survey (USGS) Agency to collect, monitor, analyze, and provide science about natural resource conditions, issues, and problems on Earth. It is an object of the present invention to exploit such instruments and technology when carrying out and practicing the various methods of the present invention disclosed herein. As shown in FIG. 4B, these MSI/HSI remote sensing technologies 14 include: MODIS (Moderate Resolution Imaging Spectro-radiometer) satellite system 14A for generating MODIS imagery subsets from MODIS direct readout data acquired by the USDA Forest Service Remote Sensing Applications Center, to produce satellite fire detection data maps and the like <https://fsapps.nwcg.gov/afm/activefire-maps.php>; the World View 2 Satellite System 14B manufacture from the Ball Aerospace & Technologies and operated by DigitalGlobe, for providing commercially available panchromatic (B/W) imagery of 0.46 meter resolution, and eight-band multi-spectral imagery with 1.84 meter resolution; Octocopter UAS (e.g. OnyxStar Hyra-12 heavy lifting drone) 14C as shown in FIG. 4B supporting MSI and HSI camera systems for spectral imaging applications, <http://www.onyxstar.net> and <http://www.genidrone.com>; and SenseFly eBee SQ UAS 14D for capturing and mapping high-resolution aerial multi-spectral images <https://www.sensefly.com/drones/ebec-sq.html>.

[0237] Any one or more of these types of remote data sensing and capture instruments, tools and technologies can be integrated into and used by the system network 1 for the purpose of (i) determining GPS-specified flight/navigation plans for GPS-tracked anti-fire (AF) chemical liquid spraying and misting aircraft and ground-based vehicle systems, described above, and (ii) practicing the various GPS-guided methods of wild fire suppression illustrated in FIGS. 24 through 32B, and described in detail herein.

#### Specification of the Network Architecture of the System Network of the Present Invention

[0238] FIG. 4A illustrates the network architecture of the system network 1 implemented as a stand-alone platform deployed on the Internet. As shown, the Internet-based system network comprises: cellular phone and SMS messaging systems and email servers 16 operably connected to the TCP/IP infrastructure of the Internet 10; a network of mobile computing systems 11 running enterprise-level mobile application software 12, operably connected to the TCP/IP infrastructure of the Internet 10; an array of mobile GPS-tracked anti-fire (AF) liquid spraying systems (20, 30, 40, 50), each provided with GPS-tracking and having wireless internet connectivity with the TCP/IP infrastructure of the Internet 10, using various communication technologies (e.g. GSM, Bluetooth, WIFI, and other wireless networking

protocols well known in the wireless communications arts); and one or more industrial-strength data center(s) 8, preferably mirrored with each other and running Border Gateway Protocol (BGP) between its router gateways, and operably connected to the TCP/IP infrastructure of the Internet 10.

[0239] As shown in FIG. 4A, each data center 8 comprises: the cluster of communication servers 9A for supporting http and other TCP/IP based communication protocols on the Internet (and hosting Web sites); a cluster of application servers 9B; the cluster of RDBMS servers 9C configured within a distributed file storage and retrieval ecosystem/system, and interfaced around the TCP/IP infrastructure of the Internet well known in the art; the SMS gateway server 9D supporting integrated email and SMS messaging, handling and processing services that enable flexible messaging across the system network, supporting push notifications; and the cluster of email processing servers 9E.

[0240] Referring to FIG. 4A, the cluster of communication servers 9A is accessed by web-enabled mobile computing clients 11 (e.g. smart phones, wireless tablet computers, desktop computers, computer workstations, etc.) used by many stakeholders accessing services supported by the system network 1. The cluster of application servers 9A implement many core and compositional object-oriented software modules supporting the system network 1. Typically, the cluster of RDBMS servers 9C use SQL to query and manage datasets residing in its distributed data storage environment, although non-relational data storage methods and technologies such as Apache's Hadoop non-relational distributed data storage system may be used as well.

[0241] As shown in FIG. 4A, the system network architecture shows many different kinds of users supported by mobile computing devices 11 running the mobile application 12 of the present invention, namely: the plurality of mobile computing devices 11 running the mobile application 12, used by fire departments and firemen to access services supported by the system network 1; the plurality of mobile computing systems 11 running mobile application 12, used by insurance underwriters and agents to access services on the system network 1; the plurality of mobile computing systems 11 running mobile application 12, used by building architects and their firms to access the services supported by the system network 1; the plurality of mobile client systems 11 (e.g. mobile computers such as iPad, and other Internet-enabled computing devices with graphics display capabilities, etc.) used by spray-project technicians and administrators, and running a native mobile application 12 supported by server-side modules, and the various illustrative GUIs shown in FIGS. 19 through 19D, supporting client-side and server-side processes on the system network of the present invention; and a GPS-tracked anti-fire (AF) liquid spraying systems 20, 30, 40 and 50 for spraying buildings and ground cover to provide protection and defense against wild-fires.

[0242] In general, the system network 1 will be realized as an industrial-strength, carrier-class Internet-based network of object-oriented system design, deployed over a global data packet-switched communication network comprising numerous computing systems and networking components, as shown. As such, the information network of the present invention is often referred to herein as the "system" or "system network". The Internet-based system network can be implemented using any object-oriented integrated devel-



opment environment (IDE) such as for example: the Java Platform, Enterprise Edition, or Java EE (formerly J2EE); Websphere IDE by IBM; Weblogic IDE by BEA; a non-Java IDE such as Microsoft's .NET IDE; or other suitably configured development and deployment environment well known in the art. Preferably, although not necessary, the entire system of the present invention would be designed according to object-oriented systems engineering (DOSE) methods using UML-based modeling tools such as ROSE by Rational Software, Inc. using an industry-standard Rational Unified Process (RUP) or Enterprise Unified Process (EUP), both well known in the art. Implementation programming languages can include C, Objective C, C++, Java, PHP, Python, Google's GO, and other computer programming languages known in the art. Preferably, the system network is deployed as a three-tier server architecture with a double-firewall, and appropriate network switching and routing technologies well known in the art. In some deployments, private/public/hybrid cloud service providers, such as Amazon Web Services (AWS), may be used to deploy Kubernetes, an open-source software container/cluster management/orchestration system, for automating deployment, scaling, and management of containerized software applications, such as the mobile enterprise-level application 12 of the present invention, described above.

#### Specification of System Architecture of an Exemplary Mobile Smartphone System Deployed on the System Network of the Present Invention

[0243] FIG. 5A shows an exemplary mobile computing device 11 deployed on the system network of the present invention, supporting conventional wildfire alert and notification systems (e.g. CAL FIRE® wild fire notification system 14), as well as the mobile anti-fire spray management application 12 of the present invention, that is deployed as a component of the system network 1.

[0244] FIG. 5B shows the system architecture of an exemplary mobile client computing system 11 that is deployed on the system network 1 and supporting the many services offered by system network servers 9A, 9B, 9C, 9D, 9E. As shown, the mobile smartphone device 11 can include a memory interface 202, one or more data processors, image processors and/or central processing units 204, and a peripherals interface 206. The memory interface 202, the one or more processors 204 and/or the peripherals interface 206 can be separate components or can be integrated in one or more integrated circuits. The various components in the mobile device can be coupled by one or more communication buses or signal lines. Sensors, devices, and subsystems can be coupled to the peripherals interface 206 to facilitate multiple functionalities. For example, a motion sensor 210, a light sensor 212, and a proximity sensor 214 can be coupled to the peripherals interface 206 to facilitate the orientation, lighting, and proximity functions. Other sensors 216 can also be connected to the peripherals interface 206, such as a positioning system (e.g. GPS receiver), a temperature sensor, a biometric sensor, a gyroscope, or other sensing device, to facilitate related functionalities. A camera subsystem 220 and an optical sensor 222, e.g. a charged coupled device (CCD) or a complementary metal-oxide semiconductor (CMOS) optical sensor, can be utilized to facilitate camera functions, such as recording photographs and video clips. Communication functions can be facilitated through one or more wireless communication subsystems 224, which can

include radio frequency receivers and transmitters and/or optical (e.g. infrared) receivers and transmitters. The specific design and implementation of the communication subsystem 224 can depend on the communication network(s) over which the mobile device is intended to operate. For example, the mobile device 11 may include communication subsystems 224 designed to operate over a GSM network, a GPRS network, an EDGE network, a Wi-Fi or WiMax network, and a Bluetooth™ network. In particular, the wireless communication subsystems 224 may include hosting protocols such that the device 11 may be configured as a base station for other wireless devices. An audio subsystem 226 can be coupled to a speaker 228 and a microphone 230 to facilitate voice-enabled functions, such as voice recognition, voice replication, digital recording, and telephony functions. The I/O subsystem 240 can include a touch screen controller 242 and/or other input controller(s) 244. The touch-screen controller 242 can be coupled to a touch screen 246. The touch screen 246 and touch screen controller 242 can, for example, detect contact and movement or break thereof using any of a plurality of touch sensitivity technologies, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with the touch screen 246. The other input controller(s) 244 can be coupled to other input/control devices 248, such as one or more buttons, rocker switches, thumb-wheel, infrared port, USB port, and/or a pointer device such as a stylus. The one or more buttons (not shown) can include an up/down button for volume control of the speaker 228 and/or the microphone 230. Such buttons and controls can be implemented as a hardware objects, or touch-screen graphical interface objects, touched and controlled by the system user. Additional features of mobile smartphone device 11 can be found in U.S. Pat. No. 8,631,358 incorporated herein by reference in its entirety.

#### Different Ways of Implementing the Mobile Client Machines and Devices on the System Network of the Present Invention

[0245] In one illustrative embodiment, the enterprise-level system network is realized as a robust suite of hosted services delivered to Web-based client subsystems 1 using an application service provider (ASP) model. In this embodiment, the Web-enabled mobile application 12 can be realized using a web-browser application running on the operating system (OS) (e.g. Linux, Application IOS, etc.) of a mobile computing device 11 to support online modes of system operation, only. However, it is understood that some or all of the services provided by the system network 1 can be accessed using Java clients, or a native client application, running on the operating system of a client computing device, to support both online and limited off-line modes of system operation. In such embodiments, the native mobile application 12 would have access to local memory (e.g. a local RDBMS) on the client device 11, accessible during off-line modes of operation to enable consumers to use certain or many of the system functions supported by the system network during off-line/off-network modes of operation. It is also possible to store in the local RDBMS of the mobile computing device 11 most if not all relevant data collected by the mobile application for any particular fire-protection spray project, and to automatically synchronize



the dataset for user's projects against the master datasets maintained in the system network database 9C1, within the data center 8 shown in FIG. 4A. This way, when using a native application, during off-line modes of operation, the user will be able to access and review relevant information regarding any building spray project, and make necessary decisions, even while off-line (i.e. not having access to the system network).

[0246] As shown and described herein, the system network 1 has been designed for several different kinds of user roles including, for example, but not limited to: (i) public and private property owners, residents, fire departments, local, county, state and federal officials; and (ii) wild fire suppression administrators, contractors, technicians et al registered on the system network. Depending on which role, for which the user requests registration, the system network will request different sets of registration information, including name of user, address, contact information, etc. In the case of a web-based responsive application on the mobile computing device 11, once a user has successfully registered with the system network, the system network will automatically serve a native client GUI, or an HTML5 GUI, adapted for the registered user. Thereafter, when the user logs into the system network, using his/her account name and password, the system network will automatically generate and serve GUI screens described below for the role that the user has been registered with the system network.

[0247] In the illustrative embodiment, the client-side of the system network 1 can be realized as mobile web-browser application, or as a native application, each having a "responsive-design" and adapted to run on any client computing device (e.g. iPhone, iPad, Android or other Web-enabled computing device) 11 and designed for use by anyone interested in managing, monitoring and working to defend against the threat of wild fires.

Specification of Environmentally-Clean Aqueous-Based Liquid Fire Inhibiting Bio-Chemical Compositions and Formulations, and Methods of Making the Same in Accordance with the Principles of the Present Invention

[0248] Another object of the present invention is to provide new and improved environmentally-clean aqueous-based fire inhibiting biochemical solutions (i.e. wet liquid compositions, liquid concentrate for proportioning and mixing with supplies of water, and dry powder composition formulation kits) for producing biochemical products that demonstrate good immediate extinguishing effects when applied to extinguish a burning or smoldering fire, and very good long-term fire inhibiting effects when being proactively applied to protect combustible surfaces against the threat of fire.

[0249] In general, the novel fire inhibiting liquid biochemical compositions of the present invention comprise: (a) a dispersing agent in the form of a quantity of water, for dispersing metal ions dissolved in water; (b) a fire inhibiting agent in the form of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid, for providing metal ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (c) a coalescing agent in the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), such as triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal ions when the fire inhibiting liquid composition is applied to a surface to be protected against fire, while water molecules in the water evaporate during

drying, and the metal ions cooperate to form metal salt crystal structure on the surface; (d) if appropriate, at least one biocide (e.g. Polyphase® PW40 Biocide from Troy Corporation or citric acid) dissolved in water; (e) if appropriate, at least one colorant; and (f) if appropriate, an adhesive agent (e.g. natural gum) for adding cling factor or adhesion properties to the biochemical liquid composition when applied to a surface to be protected against fire.

[0250] As will be described herein, the fire inhibiting biochemical liquid compositions of the present invention can be premixed and bottled/containerized at full strength for final usage and application, as illustrated for use in applications shown in FIGS. 7A, 8A, 10A, 11A, 12A, 13A, and 16A.

[0251] The dry chemical (DC) components of the fire inhibiting compositions can be premixed and packaged in a container, for subsequent mixing with water to produce final liquid composition. The biochemical liquid compositions can be made into a liquid concentrate, and then bottled/containerized, and transported to an intermediate, or end user location, for mixing with a supply of clean water in correct proportions, to produce fire inhibiting liquid compositions in a batch mode, with proper chemical constituent proportions maintained, as described herein, as illustrated for use in applications shown in FIGS. 7A, 8A, 10A, 11A, 12A, 13A, and 16A.

[0252] Alternatively, these biochemical liquid compositions can be made into a liquid concentrate (LC), and then bottled/containerized, and transported to end user location, for mixing with a supply of clean water in correct proportions using a hydraulic inductor device, to produce fire inhibiting liquid compositions in an in-line proportioning/mixing mode, with proper chemical constituent proportions maintained, as described herein, as illustrated for use in applications shown in FIGS. 14A, 14B, 15A, 17A, 42A, and 42C.

[0253] In general, useful alkali metal salts of nonpolymeric saturated carboxylic acids for inclusion in the compositions of the present invention preferably comprise: alkali metal salts of oxalic acid; alkali metal salts of gluconic acid; alkali metal salts of citric acid; and also alkali metal salts of tartaric acid. Alkali metal salts of citric acid are particularly preferred, as will be further explained hereinafter.

[0254] Notably, while the efficacy of the alkali metal salts increases in the order of lithium, sodium, potassium, cesium and rubidium, the salts of sodium and salts of potassium are preferred for cost of manufacturing reasons. Potassium carboxylates are very particularly preferred, but tripotassium citrate monohydrate (TPC) is the preferred alkali metal salt for use in formulating the environmentally-clean fire inhibiting biochemical compositions of the present invention.

[0255] While it is understood that other alkali metal salts are available to practice the biochemical compositions of the present invention, it should be noted that the selection of tripotassium citrate as the preferred alkali metal salt, includes the follow considerations: (i) the atomic ratio of carbon to potassium (the metal) in the utilized alkali metal salt (i.e. tripotassium citrate); (ii) that tripotassium citrate is relatively stable at transport and operating temperatures; (iii) tripotassium citrate is expected to be fully dissociated to citrate and potassium when dissolved in water, and that the dissociation constant is not relevant for the potassium ions, while citric acid/citrate has three ionizable carboxylic acid groups, for which pKa values of 3.13, 4.76 and 6.4 at 25° C.



are reliably reported the European Chemicals Agency (ECHA) handbook; and (iv) that tripotassium citrate produces low carbon dioxide levels when dissolved in water.

**[0256]** Tripotassium citrate is an alkali metal salt of citric acid (a weak organic acid) that has the molecular formula  $C_6H_8O_7$ . While citric acid occurs naturally in citrus fruit, in the world of biochemistry, citric acid is an intermediate in the celebrated “Citric Acid cycle, also known as the Krebs Cycle (and the Tricarboxylic Acid Cycle), which occurs in the metabolism of all aerobic organisms. The role that citric acid plays in the practice of the biochemical compositions of the present invention will be described in greater detail hereinafter.

**[0257]** Preferably, the water soluble coalescing agent should have a melting point at least 32 F (0 C) or lower in temperature, and be soluble in water. Triethyl citrate (TEC) is a preferred coalescing agent when used in combination with tripotassium citrate (TPC) having excellent compatibility given that both chemical compounds are derived from citric acid.

**[0258]** Ideally, the biocidal agent should help increase stability in storage, especially of the aqueous preparations, and also prevent or inhibit growth of mildew, mold and fungus when the biochemical liquid compositions are sprayed or otherwise applied to the surfaces of wood products that to be treated therewith, to produce Class-A fire-protected wood products with resistance to mold, mildew and fungus growth. This is important when wood products are shipped and stored in lumber yard and allowed to be exposed to the natural elements for months on a construction site, where moisture is present and conditions are excellent for such microbial growth. Mold, mildew and fungus growth not only detracts from the appearance of the wood product, but also can adversely decrease wood fiber strength and other mechanical properties for which wood products are used in specific construction applications.

**[0259]** In some applications, the use of colorants may be advantageous with or without opacifying assistants, to the fire inhibiting biochemical liquid compositions of the present invention. Opacifying assistants make the fire-retarding biochemical composition cloudy and prevent any interaction between the color of the added colorant used and the background color.

**[0260]** The preferred colorant is mica, especially natural mica. Mica also acts as an opacifying assistant, so that a separate opacifying assistant can be omitted. Areas which have already been treated are easier to identify, for example from the air. In addition, mica is capable of reflecting direct thermal radiation.

**[0261]** The concentration of the dye in the fire-retarding biochemical composition is preferably in the range from 0.005% to 10% by weight, more preferably in the range from 0.01% to 5% by weight and most preferably in the range from 0.015% to 2% by weight.

**[0262]** Of particular advantage are dyes, food dyes for example, which fade as the fire-retarding composition dries and gradually decompose or are otherwise easily removable, for example by flushing with water.

**[0263]** Also, if appropriate for any particular fire inhibiting application at hand, an adhesion agent can be added to the biochemical composition, and realized in the form of a natural gum or starch in minor amounts to promote cling factor or adhesion properties between the metal salt crystal structures formed within liquid biochemical and the surface

to which it has been applied, preferably by spraying, for proactive fire protection. Preferably, the concentration of the adhesion agent in the fire-retarding biochemical composition is preferably in the range from 0.005% to 10% by weight, more preferably in the range from 0.01% to 5% by weight and most preferably in the range from 0.015% to 2% by weight.

**[0264]** The fire inhibiting liquid biochemical compositions of the present invention are producible and prepared by mixing the components in specified amounts with water to produce the fire inhibiting composition. The order of mixing is discretionary. It is advantageous to produce aqueous preparations by mixing the components other than water, into water.

**[0265]** The fire-retarding biochemical compositions of the present invention have a good fire inhibiting effect and, a good immediate fire extinguishing effect. This mixing of the constituent biochemical compounds can take place before or during their use. For example, an aqueous preparation may be set and kept in readiness for fire inhibiting use. However, it is also possible for the aqueous preparation not to be produced until it is produced, by diluting with water, during a fire defense deployment application.

**[0266]** The compositions of the present invention are also useful as a fire extinguishing agent for fighting fires of Class A, B, C and D. For example, an aqueous biochemical solution of the present invention may be prepared and deployed for firefighting uses in diverse applications. However, it is also possible for the aqueous biochemical composition to not to be produced until it is needed, and when so, by diluting and dissolving its components with a pre-specified quantity of water, during firefighting deployments.

#### Specification of Preferred Embodiments of Aqueous-Based Fire Inhibiting Biochemical Compositions of Matter

**[0267]** In the first preferred embodiment of the fire inhibiting liquid biochemical composition of the present invention, the components are realized as follows: (a) the dispersing agent is realized in the form of a quantity of water, for dispersing metal ions dissolved in the water; (b) the fire inhibiting agent is realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate, for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; and (c) a coalescing agent realized in the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), specifically triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal potassium ions when the fire inhibiting liquid composition is applied to a surface to be protected against fire, and while water molecules in the water evaporate during drying, the metal potassium ions cooperate to form potassium citrate salt crystal structure on the treated surface.

**[0268]** In the second preferred embodiment of the fire inhibiting liquid biochemical composition of the present invention, the components are realized as follows: (a) the dispersing agent is realized in the form of a quantity of water, for dispersing metal ions dissolved in the water; (b) the fire inhibiting agent is realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate, for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (c) a coalescing agent



realized the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), specifically triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal potassium ions when the fire inhibiting liquid composition is applied to a surface to be protected against fire, and while water molecules in the water evaporate during drying, the metal potassium ions cooperate to form potassium citrate salt crystal structure on the treated surface; and (d) at least one biocide agent dissolved in the quantity of water.

**[0269]** In the third preferred embodiment of the fire inhibiting liquid biochemical composition of the present invention, the components are realized as follows: (a) the dispersing agent is realized in the form of a quantity of water, for dispersing metal ions dissolved in the water; (b) the fire inhibiting agent is realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate, for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (c) a coalescing agent realized the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), specifically triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal potassium ions when the fire inhibiting liquid composition is applied to a surface to be protected against fire, and while water molecules in the water evaporate during drying, the metal potassium ions cooperate to form potassium citrate salt crystal structure on the treated surface; and (d) at least one biocide agent in the form of citric acid dissolved in the quantity of water.

**[0270]** In the fourth preferred embodiment of the fire inhibiting liquid biochemical composition of the present invention, the components are realized as follows: (a) the dispersing agent is realized in the form of a quantity of water, for dispersing metal ions dissolved in the water; (b) the fire inhibiting agent is realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate, for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (c) a coalescing agent realized the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), specifically triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal potassium ions when the fire inhibiting liquid composition is applied to a surface to be protected against fire, and while water molecules in the water evaporate during drying, the metal potassium ions cooperate to form potassium citrate salt crystal structure on the treated surface; (d) at least one biocide agent dissolved in the quantity of water; and (e) at least one colorant.

**[0271]** Once prepared using any of formulations specified above, the liquid biochemical composition is then stored in a container, bottle or tote (i.e. its package) suitable for the end user application in mind. Then, the filled package should be sealed with appropriate sealing technology and immediately labeled with a specification of (i) its biochemical components, with weight percent measures where appropriate, and the date and time of manufacture, printed and recorded in accordance with good quality control (QC) practices well known in the art. Where necessary or desired, barcode symbols and/or barcode/RFID identification tags and labels can be produced and applied to the sealed package to efficiently track each barcoded package containing a specified quantity of clean fire inhibiting biochemical com-

position. All product and QC information should be recorded in globally accessible network database, for use in tracking the movement of the package as it moves along the supply chain from its source of manufacture, toward its end use at a GPS specified location.

Specification of Preferred Embodiments of the Dry Fire Inhibiting Biochemical Compositions of Matter Assembled as a Fire Inhibiting Biochemical Composition Kit for Use with Specified Quantities of Water

**[0272]** In the fifth preferred embodiment of the fire inhibiting liquid biochemical composition of the present invention, the components are realized as follows: (a) the fire inhibiting agent is realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate, for providing metal potassium ions to be dissolved and dispersed in a quantity of water; (b) a coalescing agent realized the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), specifically triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal potassium ions when the fire inhibiting liquid composition is applied to a surface to be protected against fire, and while water molecules in the water evaporate during drying, the metal potassium ions cooperate to form potassium citrate salt crystal structure on the treated surface; (c) if appropriate, at least one biocide in the form of citric acid, dissolved in the quantity of water; and (d) if appropriate, at least one colorant.

**[0273]** In the sixth preferred embodiment of the fire inhibiting liquid biochemical composition of the present invention, the components are realized as follows: (a) the fire inhibiting agent is realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate, for providing metal potassium ions to be dissolved and dispersed in a quantity of water; (b) a coalescing agent realized the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), specifically triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal potassium ions when the fire inhibiting liquid composition is applied to a surface to be protected against fire, and while water molecules in the water evaporate during drying, the metal potassium ions cooperate to form potassium citrate salt crystal structure on the treated surface; (c) at least one biocide in the form of citric acid, dissolved in the quantity of water; and (d) if appropriate, at least one colorant.

Selecting Tripotassium Citrate (TCP) as a Preferred Fire Inhibiting Agent for Use in the Fire Inhibiting Biochemical Compositions of the Present Invention

**[0274]** In the preferred embodiments of the present invention, tripotassium citrate (TPC) is selected as active fire inhibiting chemical component in fire inhibiting biochemical composition. In dry form, TPC is known as tripotassium citrate monohydrate ( $C_6H_5K_3O_7 \cdot H_2O$ ) which is the common tribasic potassium salt of citric acid, also known as potassium citrate. It is produced by complete neutralization of citric acid with a high purity potassium source, and subsequent crystallization. Tripotassium citrate occurs as transparent crystals or a white, granular powder. It is an odorless substance with a cooling, salty taste. It is slightly deliquescent when exposed to moist air, freely soluble in water and almost insoluble in ethanol (96%).



**[0275]** Tripotassium citrate is a non-toxic, slightly alkaline salt with low reactivity. It is chemically stable if stored at ambient temperatures. In its monohydrate form, TPC is very hygroscopic and must be protected from exposure to humidity. Care should be taken not to expose tripotassium citrate monohydrate to high pressure during transport and storage as this may result in caking. Tripotassium citrate monohydrate is considered “GRAS” (Generally Recognized As Safe) by the United States Food and Drug Administration without restriction as to the quantity of use within good manufacturing practice. CAS Registry Number: [6100-05-6]. E-Number: E332.

**[0276]** Tripotassium citrate monohydrate (TPC) is a non-toxic, slightly alkaline salt with low reactivity. It is a hygroscopic and deliquescent material. It is chemically stable if stored at ambient temperatures. In its monohydrate form, it is very hygroscopic and must be protected from exposure to humidity. Its properties are:

**[0277]** Monohydrate

**[0278]** White granular powder

**[0279]** Cooling, salty taste profile, less bitter compared to other potassium salts

**[0280]** Odorless

**[0281]** Very soluble in water

**[0282]** Potassium content of 36%

**[0283]** Slightly alkaline salt with low reactivity

**[0284]** Hygroscopic

**[0285]** Chemically and microbiologically stable

**[0286]** Fully biodegradable

**[0287]** Allergen and GMO free

**[0288]** Jungbunzlauer (JBL), a leading Swiss manufacturer of biochemicals, manufactures and distributes TPC for food-grade, healthcare, pharmaceutical and over the counter (OTC) applications around the world. As disclosed in JBL’s product documents, TPC is an organic mineral salt which is so safe to use around children and adults alike. Food scientists worldwide have added TPC to (i) baby/infant formula powder to improve the taste profile, (ii) pharmaceuticals/OTC products as a potassium source, and (iii) soft drinks as a soluble buffering salt for sodium-free pH control in beverages, improving stability of beverages during processing, heat treatment and storage.

Selecting Triethyl Citrate (TEC) as a Preferred Coalescing Agent with Surface Tension Reducing and Surfactant Properties for Use in the Fire Inhibiting Biochemical Compositions of the Present Invention

**[0289]** In the preferred illustrative embodiments of the present invention, the coalescing agent used in the fire inhibitor biochemical compositions of the present invention is realized as a food-grade additive component, namely, triethyl citrate (TEC) which functions as a coalescing agent with surface tension reducing properties and surfactant properties as well. Triethyl citrate belongs to the family of tricarboxylic acids (TCAs) and derivatives, organic compounds containing three carboxylic acid groups (or salt/ester derivatives thereof).

**[0290]** In the aqueous-based fire inhibiting liquid composition, the coalescing agent functions as temporary dispersing agent for dispersing the metal ions dissolved and dissociated in aqueous solution. As water molecules evaporate from a coating of the biochemical composition, typically spray/atomized applied to a surface to be protected from fire, the coalescing agent allows the formation of thin metal (e.g. potassium citrate) salt crystal structure/films at ambient

response temperature conditions of coating application. The coalescent agent promotes rapid metal salt crystal structure formation on surfaces to be protected against wildfire, and have a hardness evolution that promotes durability against rain and ambient moisture, while apparently allowing vital oxygen and CO<sub>2</sub> gas transport to occur, without causing detrimental effects to the vitality of living plant tissue surfaces sought to be protected against wildfire.

**[0291]** A relatively minor quantity of triethyl citrate (TEC) liquid is blended with a major quantity of TCP powder in specific quantities by weight and dissolved in a major quantity of water to produce a clear, completely-dissolved liquid biochemical formulation consisting of food-grade biochemicals mixed with water and having highly effective fire inhibiting properties, as proven by testing. The resulting aqueous biochemical solution remains stable without the formation of solids at expected operating temperatures (e.g. 34 F to 120 F).

**[0292]** Jungbunzlauer (JBL) also manufactures and distributes its CITROFOL® A1 branded bio-based citrate esters for food-grade, healthcare, pharmaceutical and over the counter (OTC) applications around the world. CITROFOL® A1 triethyl citrate (TEC) esters have an excellent toxicological and eco-toxicological profile, and provide good versatility and compatibility with the tripotassium citrate (TPC) component of the biochemical compositions of the present invention. CITROFOL® A1 branded citrate esters are particularly characterized by highly efficient solvation, low migration and non-VOC (volatile organic compound) attributes. As an ester of citric acid, triethyl citrate is a colorless, odorless liquid which historically has found use as a food additive (E number E1505) to stabilize foams, especially as a whipping aid for egg whites.

**[0293]** Broadly described, the fire inhibiting biochemical liquid coatings of the present invention consist of an aqueous dispersion medium such as water which carries dissolved metal salt cations that eventually form a thin metal salt crystalline structure layer on the surface substrate to be protected from ignition of fire. The aqueous dispersion medium may be an organic solvent, although the preferred option is water when practicing the present invention. After the application of a coating onto the combustible surface to be protected against fire ignition and flame spread and smoke development, the aqueous dispersion medium evaporates, causing the metal salt (i.e. potassium salt) cations to draw together. When these metal salt particles come into contact, the coalescing agent, triethyl citrate, takes effect, uniformly dispersing the same while reducing liquid surface tension, and giving rise to the formation of a relatively homogeneous metal salt crystalline structure layer over the surface. In practice, this interaction is more complex and is influenced by various factors, in particular, the molecular interaction of the potassium salt cations and the coalescing agent, triethyl citrate, as the water molecules are evaporating during the drying process.

**[0294]** While offering some surface tension reducing effects, the main function of the coalescing agent in the biochemical composition of the present invention is to ensure a relatively uniform and optimal formation of the salt crystalline structure layers on the combustible surfaces to be protected, as well as desired mechanical performance (e.g. offering scrub resistance and crystal coating hardness) and aesthetic values (e.g. gloss and haze effects).



**[0295]** The fact that CITROFOL® A1 triethyl citrate (TEC) esters are bio-based, odorless, biodegradable and label-free, represents a great advantage over most other coalescing agents, and fully satisfies the toxicological and environmental safety requirements desired when practicing the biochemical compositions of the present invention.

**[0296]** In the preferred embodiments of the present invention, the use of CITROFOL® A1 triethyl citrate (TEC) esters with tripotassium citrate monohydrate (TPC) dissolved in water as a dispersion solvent, produce fire inhibiting biochemical formulations that demonstrate excellent adhesion, gloss and hardness properties. The chemical and colloidal nature of potassium salt ions (which are mineral salt dispersions) present in TPC dissolved in water, is highly compatible with the CITROFOL® A1 triethyl citrate (TEC) ester used as the coalescing agent in the preferred embodiments of the present invention. Also, CITROFOL® A1 triethyl citrate esters are REACH registered and are safe, if not ideal, for use in environmentally sensitive products such as fire and wildfire inhibitors which must not adversely impact human, animal and plant life, ecological systems, or the natural environment.

**[0297]** CITROFOL® triethyl citrate esters were selected because they are biodegradable, and exhibit an excellent toxicological and eco-toxicological profile for the applications of the present invention. These esters are also versatile and demonstrate very good compatibility with the TPC solution, and are characterized by a high solvating efficiency.

Selecting Citric Acid as a Natural and Safe Biocidal Agent for Use in the Fire Inhibiting Biochemical Compositions of the Present Invention

**[0298]** Polyphase® PW40 water-based biocidal agent from Troy Chemical can be added to the biochemical compositions of the present invention, as described and specified herein, to control and inhibit the growth of mold, mildew and fungus on wood products treated with the biochemical of the present invention. This biocidal agent (i.e. biocide) has shown to be effective in the applications described herein. However, the water-based Polyphase® PW40 biocide includes compounds (i.e.  $C_8H_{12}INO_2$  or IPBC) as active ingredients that have been shown to have a toxicity profile that is not as safe as common organic acids such as citric acid, which is ubiquitous in nature and all of nature's life processes. Thus, it would be highly desirable to use organic food grade compounds to provide effective biocidal properties to the biochemical compositions of the present invention, to control and inhibit the growth of mold, mildew and fungus on wood surfaces that are (i) proactively treated with the biochemical compositions of the present invention, and (ii) later exposed to rain, moisture and natural elements while in storage at lumber yards, and/or on wet damp building construction sites where projects may last for at least 3-6 or more months before the buildings under construction are closed in and protected from the natural elements.

**[0299]** As an alternative biocidal agent, an object of the present invention is to add a minor amount of citric acid to the biochemical compositions of the present invention to effectively realize a natural and safe biocidal agent in the fire inhibitor biochemical compositions of the present invention, based on a food-grade additive component, namely, citric acid, which functions to control and inhibit the growth of

mold, mildew and fungus on the surface coated with the fire inhibiting biochemical composition of the present invention.

**[0300]** It is well known that citric acid also belongs to the family of tricarboxylic acids (TCA) and derivatives, organic compounds containing three carboxylic acid groups (or salt/ester derivatives thereof). Citric acid is a weak organic acid found in citrus fruits. In biochemistry, citric acid is important as an intermediate in the citric acid cycle (i.e. tricarboxylic acid (TCA) cycle), and therefore occurs in the metabolism of almost all living things. The tricarboxylic acid (TCA) cycle is also called the Krebs cycle which functions in the second stage of cellular respiration, a three-stage process by which living cells break down organic fuel molecules in the presence of oxygen to harvest the energy they need to grow and divide and maintain cellular vitality. TCA cycle is the predominant source in all aerobic organisms to generate NADH and FADH<sub>2</sub> from acetyl CoA, a product obtained by the decarboxylation of pyruvate. In addition, TCA cycle is also a major pathway for interconversion of metabolites and provides substrates for amino acid synthesis by transamination as well as for fatty acid synthesis and gluconeogenesis. The cycle starts with the condensation of acetyl-CoA with oxaloacetate to form citrate, a reaction catalyzed by citrate synthase. The entire cycle can be divided into two stages: (a) a decarboxylating stage involving conversion of citrate to succinyl-CoA; and (b) a reductive stage involving successive oxidation of succinate to fumarate, fumarate to malate, and then malate to oxaloacetate.

**[0301]** Through control of PH and oxidation in the biochemical compositions of the present invention, the citric acid is used in minor amounts in these biochemical compositions of matter for the purpose of controlling, inhibiting and preventing the grow of mold, mildew and fungus without the use of toxic chemical compounds known to pose health effects to humans and animals alike.

Specification of Preferred Formulations for the Fire Inhibiting Biochemical Compositions of Matter According to the Present Invention

#### Example #1: Liquid-Based Fire Inhibiting Biochemical Composition

**[0302]** FIG. 6A1 illustrates the primary components of a first environmentally-clean aqueous-based fire inhibiting liquid biochemical composition of the present invention consisting of tripotassium citrate (TPC) and triethyl citrate (TEC) formulated with water functioning as a solvent, carrier and dispersant in the biochemical composition.

**[0303]** Example 1: Schematically illustrated in FIG. 6A1: A fire-extinguishing and/or fire-retarding biochemical composition was produced by stirring the components into water. The composition comprising: 0.05 pounds by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume); and 4.4 pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 9.61 pounds having 128 ounces or 1 gallon of volume.

#### Example #2: Dry-Powder Fire Inhibiting Biochemical Composition

**[0304]** FIG. 6A2 illustrates the primary components of a first fire inhibiting biochemical composition kit of the pres-



ent invention, consisting of dry tripotassium citrate (TPC) and triethyl citrate (TEC) components for mixing with a predetermined quantity of water functioning as a solvent, carrier and dispersant, to make up a predetermined quantity of environmentally-clean liquid fire inhibiting biochemical composition for proactively protecting wood products.

**[0305]** Example 2: Schematically Illustrated in FIG. 6A2: A fire-extinguishing and/or fire-retarding biochemical composition was produced by blending the following components, in amounts proportional to the formulation comprising: 0.05 pounds by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume); packaging the blended components together in a container or package for mixing with 4.4 pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 9.61 pounds having 128 ounces or 1 gallon of volume.

Example #3: Liquid-Based Fire Inhibiting  
Biochemical Composition with  
Mold/Mildew/Fungus-Resistance

**[0306]** FIG. 6B1 illustrates the primary components of a second environmentally-clean aqueous-based fire inhibiting liquid biochemical composition of the present invention consisting of tripotassium citrate (TPC), triethyl citrate (TEC) and citric acid (CA) formulated with water functioning as a solvent, carrier and dispersant in the biochemical composition.

**[0307]** Example 3: Schematically Illustrated in FIG. 6B1: A fire-extinguishing and/or fire-retarding biochemical composition was produced by stirring the components into water. The composition comprising: 0.05 pounds by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume); 4.0 ounces by weight of a biocide (e.g. Polyphase® PW40 by Troy Chemical); and 4.4 pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 10.00 pounds having 128 ounces or 1 gallon of volume.

Example #4: Dry-Powder Fire Inhibiting  
Biochemical Composition with  
Mold/Mildew/Fungus-Resistance

**[0308]** FIG. 6B2 illustrates the primary components of the second fire inhibiting biochemical composition kit of the present invention, consisting of dry tripotassium citrate (TPC), triethyl citrate (TEC) and citric acid (CA) components for mixing with a predetermined quantity of water functioning as a solvent, carrier and dispersant, to make up a predetermined quantity of environmentally-clean liquid fire inhibiting biochemical composition for proactively protecting wood products.

**[0309]** Example 4: Schematically Illustrated in FIG. 6B2: A fire-extinguishing and/or fire-retarding biochemical composition was produced by blending the following components in amounts proportional to the formulation comprising: 0.05 pounds by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume); 4.0 ounces by weight of a biocide agent (e.g. Polyphase® PW40 by Troy Chemical); packaging the blended components together in a container or package for mixing with 4.4

pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 10.0 pounds having 128 ounces or 1 gallon of volume.

Example #5: Liquid-Based Fire Inhibiting  
Biochemical Composition with  
Mold/Mildew/Fungus-Resistance

**[0310]** FIG. 6C1 illustrates the primary components of a second environmentally-clean aqueous-based fire inhibiting liquid biochemical composition of the present invention consisting of tripotassium citrate (TPC), triethyl citrate (TEC) and citric acid (CA) formulated with water functioning as a solvent, carrier and dispersant in the biochemical composition.

**[0311]** Example 5: Schematically Illustrated in FIG. 6C1: A fire-extinguishing and/or fire-retarding biochemical composition was produced by stirring the components into water. The composition comprising: 0.05 pounds by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume); 4.0 ounces by weight of a biocide agent (e.g. citric acid); and 4.4 pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 10.00 pounds having 128 ounces or 1 gallon of volume.

Example #6: Dry-Powder Fire Inhibiting  
Biochemical Composition with  
Mold/Mildew/Fungus-Resistance

**[0312]** FIG. 6C2 illustrates the primary components of the second fire inhibiting biochemical composition kit of the present invention, consisting of dry tripotassium citrate (TPC), triethyl citrate (TEC) and citric acid (CA) components for mixing with a predetermined quantity of water functioning as a solvent, carrier and dispersant, to make up a predetermined quantity of environmentally-clean liquid fire inhibiting biochemical composition for proactively protecting wood products.

**[0313]** Example 6: Schematically Illustrated in FIG. 6C2: A fire-extinguishing and/or fire-retarding biochemical composition was produced by blending the following components in amounts proportional to the formulation comprising: 0.05 pounds by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume); 4.0 ounces by weight of a biocide agent (e.g. citric acid); packaging the blended components together in a container or package for mixing with 4.4 pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 10.0 pounds having 128 ounces or 1 gallon of volume.

Preferred Weights Percentages of the Components of the Fire Inhibiting Biochemical Formulation of the Present Invention

**[0314]** In the biochemical compositions of the present invention The ratio of the ester of citrate (e.g. triethyl citrate) to the alkali metal salt of a nonpolymeric carboxylic acid (e.g. tripotassium citrate) may be major amount between 1:100: to 1:1000 and is typically in the range from 1:1 to 1:100, preferably in the range from 1:2 to 1:50, more preferably in the range from 1:4 to 1:25 and most preferably in the range from 1:8 to 1:15.



**[0315]** A preferred biochemical composition according to the present invention comprises: a major amount from 1% to 65% by weight, preferably from 20% to 50% by weight and more preferably from 30% to 55% by weight, of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid (e.g. tripotassium citrate monohydrate or TPC); and minor amount from 0.08% to 5% by weight, preferably from 0.5% to 2% by weight and more preferably from 0.1% to 1.0% by weight, of triethyl citrate (an ester of citrate acid); wherein the sum by % weight of the components (a) and (b) should not exceed 100% by weight.

**[0316]** In a preferred embodiment, the fire inhibiting composition further comprises water. The water content is present in a major amount and is typically not less than 30% by weight, preferably not less than 40% by weight, more preferably not less than 50% by weight and most preferably not less than 60% by weight and preferably not more than 60% by weight and more preferably not more than 70% by weight, all based on the fire inhibiting biochemical composition.

**[0317]** The viscosity of the aqueous preparation is preferably at least 5 [mPas] (millipascal-seconds, in SI units, defined as the internal friction of a liquid to the application of pressure or shearing stress determined using a rotary viscometer), and preferably not more than 50 [mPas], or 50 centipois) [cps], for most applications.

#### Physical Examination and Fire-Performance Testing of the Thin Metal Salt Crystalline Structures Formed Using the Biochemical Compositions and Methods and Apparatus of the Present Invention

**[0318]** One method of viewing the resulting metal salt crystal structures formed upon a surface substrate to be protected against fire, as illustrated in FIG. 7A, would be by using atomic force microscope to form atomic force microscopy (AFM) images of the biochemical coatings applied in accordance with the principles of the present invention. Another method of viewing the resulting metal salt crystal structures would be to use a scanning electron microscope to form scanning electron microscopy (SEM) images. Expectedly, using either instrument, such images of metal salt crystal structures formed using a greater wt % of coalescent agent (e.g. triethyl citrate dissolved in water with tripotassium citrate) will show that the coalescent agent resulted in metal salt crystal structures that are more coalesced and smoother, and demonstrating higher hardness evolution and better water repulsion, than when the metal salt crystal structures are formed using a lower wt. % coalescent agent in the aqueous-based fire inhibiting liquid composition.

**[0319]** FIG. 7A illustrates the primary steps involved during the formation of tripotassium citrate salt crystalline structure coatings on spray treated surfaces to be proactively protected against ignition and flame spread of incident fire.

**[0320]** At Step A, a spray nozzle is used to spray a liquid coating of a biochemical composition of the present invention, and once applied, the water molecules being to evaporate at a rate determined by ambient temperature and wind currents, if any. When the minimum film formation temperature (MFT) is reached for the biochemical composition, the potassium cations can inter diffuse within the triethyl citrate (TEC) coalescent agent and water molecule matrix that is supported on the surface that has been sprayed and to be proactively treated with fire inhibiting properties by

virtue of a thin film deposition of tripotassium salt crystalline structure, modeled and illustrated in FIGS. 7B and 7C.

**[0321]** At Step B, potassium cations diffuse and the TPC crystalline structure deforms. During the coalescence of potassium cations, interparticle potassium cation diffusion (PCD) occurs within the TEC coalescing agent to produce a semi-homogenous tripotassium citrate salt crystalline structure.

**[0322]** At Step C, coalescence occurs to form the TPC salt crystalline structure. The mechanical properties of tripotassium citrate crystalline structures are highly dependent on the extent of PCD within the TEC coalescent agent.

**[0323]** Upon complete evaporation of water molecules from the biochemical liquid coating, the resulting fire inhibiting coating that is believed to be formed on the sprayed and dried surface comprises a thin film of tripotassium citrate salt crystalline structures formed on the structure, with substantially no water molecules present. The nature and character of such tripotassium citrate salt crystalline structures are believed to be reflected in models provided in FIGS. 7B and 7C, which were first reported in 2016 in a published research paper by Alagappa Rammahon and James A. Kaduk, titled "Crystal Structure of Anhydrous Tripotassium Citrate From Laboratory X-Ray Diffraction Data and DFT Comparison" cited in ACTA CRYSTL (2016) Vol. E72, Pages 1159-1162, and published by Crystallographic Communications.

**[0324]** To determine and confirm that the fire inhibiting liquid compositions of the present invention produce potassium citrate salt crystalline structures on treated surfaces that have attained certain standards of fire inhibiting protection, it is necessary to test such treated surface specimens according to specific fire protection standards. In the USA, ASTM E84 Flame Spread and Smoke Development Testing can be used to test how well surfaces made of wood, cellulose and other combustible materials perform during E84 testing, and then compared against industry benchmarks. The environmentally-clean fire inhibiting chemical liquid solutions disclosed herein are currently being tested according to ASTM E84 testing standards and procedures, and it is expected that these ASTM test will show that fire-protected surfaces made of Douglas Fir (DF) will demonstrate Flame Spread Indices and Smoke Development Index to qualify for Class-A fire protected certification, when treated by the fire inhibiting biochemical compositions of the present invention disclosed and taught herein.

#### Methods of Blending, Making and Producing the Biochemical Liquid Formulations

**[0325]** The fire inhibiting liquid chemical compositions illustrated in FIGS. 6A1, 6A2, 6B1, 6B2, 6C1 and 6C2 are reproducible by mixing the components described above. The order of mixing is discretionary. However, it is advantageous to produce aqueous preparations by mixing the components other than water, into the quantity of water.

#### Specification of the Methods of Preparing and Applying the Fire Inhibiting Biochemical Compositions of the Present Invention

**[0326]** Once the fire inhibiting biochemical compositions are prepared in accordance with the formations described above, the mixture is then stirred for several minutes at room temperature, and subsequently, the mixture is then pack-



aged, barcoded with chain of custody information and then either stored, or shipped to its intended destination for use and application in accordance with present invention. As described herein, preferred method of surface coating application is using, for example, an atomizing sprayer having a backpack form factor suitable and adapted for rapidly spraying the fire inhibiting biochemical compositions on property surfaces as shown in FIGS. 13 and 13B, and form ultra-thin potassium salt crystal structure coatings to treated surfaces of combustible material on a specific parcels of property. Any of the other methods of and apparatus for spraying and GPS-tracking of fire inhibiting biochemicals of the present invention taught herein, as shown in FIGS. 8A through 17B, can be used with excellent results.

[0327] During examination and testing protocols, all fire inhibiting biochemical formulations of the present invention are proactively applied to combustible wood surfaces, allowed to dry, and are then analyzed tested for hardness, gloss and adhesion properties in a conventional manner, as well as subjected to strict ASTM E84 fire protection testing to ensure the fire inhibiting metal salt crystal coatings meet Class A Fire Protection Standards.

#### Useful Applications for the Fire Inhibiting Biochemical Liquid Compositions of the Present Invention

[0328] As disclosed, the fire inhibiting biochemical compositions of the present invention are very useful in two ways: (i) producing fire inhibiting (i.e. retarding) coatings formed by ultra-thin alkali metal (potassium citrate) salt crystal structures on surfaces to be protected against fire as illustrated in FIGS. 8A, 8B, 9A, 9B, 10A, 10B, 11A, 11B, 12A, 12B, 13A, 13B, 14A, 14B, 14C, 14D, 15A, 15B, 16A, 16B, 17A and 17B; and (ii) extinguishing active fires by application of the fire inhibiting biochemical composition of the present invention onto the fire to suppress and extinguish the fire, as illustrated in FIGS. 16A and 16B.

[0329] The biochemical compositions of the present invention can be used for example for firefighting in forests, tire warehouses, landfill sites, coal stocks, oil fields, timberyards and mines, for proactively fighting wildfires from the air, by airplanes and helicopters and drone, as illustrated in FIGS. 10A, 10B, 11A and 11B.

[0330] The biochemical compositions of the present invention can be used as an fire extinguishing agent dispensed from a hand-held device or automated dispensing system under real-time sensor control. For example, an aqueous solution may be prepared and filled in a hand-operated fire extinguisher, and configured for readiness during firefighting use. However, the aqueous composition of the present invention need not be prepared in aqueous solution until it is produced by diluting with water, during a firefighting deployment operation.

[0331] The fire inhibiting biochemical compositions of the present invention can be used to treat and protect combustible wood building materials and/or structural components, such as wood products and engineered wood products (EWPs) including panels and structural members, using the fire inhibiting biochemical compositions of the present invention as disclosed and taught herein, and as illustrated in FIGS. 50 through 62.

[0332] When coated with the biochemical liquid compositions of the present invention, and allowed to dry and form ultra-thin fire inhibiting potassium salt crystal coatings over treated wood surfaces, these wood products remarkably

demonstrate Class-A fire protection characteristics that can be reliably proven using the ASTM E84 Testing Standards, having ultra-low flame spread and smoke development indices, as illustrated in FIGS. 50 through 62.

[0333] The fire inhibiting biochemical compositions of the present invention are effective even in the dry state (long-term action) in giving a distinctly delayed ignition on the surface of a flammable material (ignition time), an appreciably reduced smoke evolution and development, and almost no afterglow (anti-smoldering effect).

[0334] The fire inhibiting biochemical compositions of the present invention are useful in extinguishing Class A, B, C and D fires. Also, an aqueous preparation of the biochemical composition may be prepared and stationed as ready for firefighting use when the occasion calls. However, it is also possible for the aqueous preparation not to be produced until it is needed, and then by diluting and dissolving the biochemical components in water, during a firefighting deployment.

[0335] The fire inhibiting biochemical compositions of the present invention are further useful as an extinguishing agent in fire extinguishers and/or fire extinguishing systems, and also via existing fire extinguishing pumps and fittings. Such fire extinguishers include, for example, portable and/or mobile fire extinguishers, as well as fixed installations, such as sprinkler systems disclosed in Applicant's US Patent Application Publication No. US2019/168047, incorporated herein by reference.

[0336] The fire inhibiting liquid biochemical compositions of the present invention can be used to produce an aqueous-based fire and smoke inhibiting slurry mixture that can be sprayed on ground cover surfaces and allowed to dry to form Class-A fire-protected wildfire protected mulch to form wildfire breaks, barriers and protective zones around property, buildings and like structures, as illustrated in FIGS. 33A through 41.

[0337] In the preferred embodiments of the biochemical compositions of the present invention, potassium citrate salts are utilized in the biochemical formulations and are very readily biodegradable without harm or impact to the natural environment. This is highly advantageous especially in relation to the proactive defense of towns, communities, home owner associations (HOAs), homes, business buildings and other forms of public and private property, from the destructive impact of raging wildfires, using the systematic and organized application, tracking and mapping of fire inhibiting biochemical compositions of the present invention, over large property.

[0338] In such planned deployments of the present invention involving the proactive defense of a state, towns, communities and homes and property against the destructive effects of wildfires, as disclosed in FIGS. 9 through 32B, and FIGS. 33A through 43, various methods and apparatus will be used to proactively spray and GPS-track and map, the formation of ultra-thin coatings of potassium citrate salt crystal structures on treated surfaces of property (e.g. in the form of clean chemistry wildfire breaks and barriers) to be proactively protected against wildfires whenever they break out and arrive at and threaten a state, town, county, community, and/or homes and businesses. At the same time, these potassium citrate salt crystal structures of the present invention favorably allow and support the transport of oxygen and CO<sub>2</sub> gases across the fire protected surfaces (e.g. which may include living plant tissue on leaves of trees



in orchard and on vines in vineyards), without adversely affecting the vitality of such living plant tissue present and covering the ground of property. Also, the clean wildfire chemistry of the present invention can be used around animal such as horses, dogs, cats and other pets without posing any health risk to such creatures, while mitigating the risks that raging wildfires will present to their lives.

[0339] Also, and most significantly, the fire inhibiting biochemical compositions of the present invention are substantially free of the many disadvantages and dangers associated with the use of ammonium-based compounds historically used in forest fire fighting, and which may at the same time have an adverse effect as fertilizers in watercourses.

[0340] Furthermore, the biochemical compositions of the present invention are very resistant to freezing when used or applied in sub-zero temperatures (e.g. less than 32 F). Thus, it is possible to obtain an aqueous biochemical composition according to the present invention which is still sprayable at temperatures below 0 C.

[0341] Notably, the biochemical compositions of the present invention are non-corrosive, especially not with regard to aluminum and other metals that may be used as containers for the biochemical solutions of the present invention, especially during mixing, storage and application operations. This features is of particular importance in relation to the proactive defense of wild fires from both the ground using GPS-tracked ground based spraying vehicles of the present invention, and from the air using GPS-tracked aircraft-based spraying vehicles of the present invention, as disclosed in FIGS. 8 through 32B.

[0342] The biochemical compositions of the present invention can be used for proactively firefighting wildfires and fires that may break out in many places including, but not limited to, forests, WUI regions, tire warehouses, landfill sites, coal stocks, timberyards and even mines, as illustrated in FIGS. 24 through 32B.

[0343] The biochemical compositions of the present invention can also be used to proactively fight wildfire fires from the air, for example by airplanes and helicopters and drones, applying and GPS-tracking the spray application of environmentally-clean fire inhibiting biochemical liquid over ground and property surfaces to create and maintain clean-chemistry fire breaks and barriers where wildfire are not to be permitted to targeted property to be protected, in accordance with the principles of the present invention.

[0344] The biochemical compositions of the present invention can also be used to proactively protect, in factory environments, carbon-storing building materials and/or structural components, such wood panel and engineering wood products (EWPs), from fire outbreaks caused by nature, accident, arson or terrorism, by applying Class-A fire-protected metal salt crystalline coatings using the biochemical compositions and methods of the present invention, as illustrated in FIGS. 50 through 62B. The building materials and/or structural components coated with the biochemical compositions of the present invention are distinctly less flammable than uncoated building materials and/or structural components.

[0345] The biochemical liquid compositions of the present invention will also be useful as a rapid fire extinguishing agent, an illustrated in FIGS. 16A and 16B, showing GPS-tracking apparatus for hand spraying atomized clouds of the solution to rapidly extinguish, preferably, fires of classes A, B, C and D, more preferably for fires of classes A, B and C

and most preferably for fires of classes A and B. For example, an aqueous preparation may be set and kept in readiness for firefighting use. However, it is also possible for the aqueous preparation not to be produced until it is produced, by diluting with water, during a firefighting deployment.

[0346] The fire-extinguishing and/or fire-retarding composition of the present invention are further useful as an extinguishant in extinguishers and/or extinguishing systems and also via existing fire extinguishing pumps and fittings. Extinguishers are for example portable and/or mobile fire extinguishers. Extinguishing systems are fixed installations, such as sprinkler systems, as illustrated in FIGS. 16A through 16B, and in other building misting systems illustrated in Applicant's US Patent Application Publication No. US2019/168047, incorporated herein by reference.

[0347] The biochemical compositions of the present invention are effective even in the dry state (long-term action) in giving a distinctly delayed ignition on the surface of a flammable material (ignition time), an appreciably reduced smoke evolution (light absorption) and almost no afterglow (anti-smoldering effect), as illustrated in FIGS. 33A through 41.

#### Specification of the Mobile GPS-Tracked Anti-Fire (AF) Liquid Spraying System of the Present Invention

[0348] FIG. 8A shows a mobile GPS-tracked anti-fire (AF) liquid spraying system 20 supported on a set of wheels 20A, having an integrated supply tank 20B and rechargeable-battery operated electric spray pump 20C with portable battery module (20C), for deployment at private and public properties having building structures, for spraying the same with environmentally-clean anti-fire (AF) liquid using a spray nozzle assembly 20D connected to the spray pump 20C by way of a flexible 20E.

[0349] FIG. 8B shows the GPS-tracked mobile anti-fire liquid spraying system 20 of FIG. 6A as comprising a number of subcomponents, namely: a GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 20F; a micro-computing platform or subsystem 20G interfaced with the GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 20F by way of a system bus 201; and a wireless communication subsystem 20H interfaced to the micro-computing platform 20G via the system bus 201. As configured, the GPS-tracked mobile anti-fire liquid spraying system 20 enables and supports (i) the remote monitoring of the spraying of anti-fire (AF) chemical liquid from the system 20 when located at specific GPS-indexed location coordinates, and (ii) the logging of all such GPS-indexed spray application operations, and recording the data transactions thereof within a local database maintained within the micro-computing platform 20G, as well as in the remote network database 9C1 maintained at the data center 8 of the system network 1.

[0350] As shown in FIG. 8B, the micro-computing platform 20G comprises: data storage memory 20G1; flash memory (firmware storage) 20G2; a programmable micro-processor 20G3; a general purpose I/O (GPIO) interface 20G4; a GPS transceiver circuit/chip with matched antenna structure 20G5; and the system bus 201 which interfaces these components together and provides the necessary addressing, data and control signal pathways supported within the system 20.



[0351] As shown in FIG. 8B, the wireless communication subsystem 20H comprises: an RF-GSM modem transceiver 20H1; a T/X amplifier 20H2 interfaced with the RF-GSM modem transceiver 20H1; and a WIFI and Bluetooth wireless interfaces 20H3.

[0352] As shown in FIG. 8B, the GPS-tracked and remotely-controllable anti-fire (AF) chemical liquid spray control subsystem 20F comprises: anti-fire chemical liquid supply sensor(s) 20F1 installed in or on the anti-fire chemical liquid supply tank 20B to produce an electrical signal indicative of the volume or percentage of the AF liquid supply tank containing AF chemical liquid at any instant in time, and providing such signals to the AF liquid spraying system control interface 20F4; a power supply and controls 20F2 interfaced with the liquid pump spray subsystem 20C, and also the AF liquid spraying system control interface 20F4; manually-operated spray pump controls interface 20F3, interfaced with the AF liquid spraying system control interface 20F4; and the AF liquid spraying system control interface 20F4 interfaced with the micro-computing subsystem 20G, via the system bus 201. The flash memory storage 20G2 contains microcode that represents a control program that runs on the microprocessor 20G3 and realizes the various GPS-specified AF chemical liquid spray control, monitoring, data logging and management functions supported by the system 20.

[0353] In the preferred embodiment, the environmentally-clean anti-fire (AF) chemical liquid is the fire inhibiting biochemical compositions described and taught herein with respect to FIGS. 6A through 7A. When so treated, combustible products will prevent flames from spreading, and confine fire to the ignition source which can be readily extinguished, or go out by itself. In the presence of a flame, the chemical molecules in both dry and wet coatings, formed with the biochemical liquid of the present invention, and inhibiting fire by one or more pathways including interfering with the free radicals (H<sup>+</sup>, OH<sup>-</sup>, O) involved in the free-radical chemical reactions within the combustion phase of a fire, and breaking free-radical chemical reactions and extinguishing the fire's flames.

#### Specification of GPS-Tracked Manned or Autonomous Vehicle for Spraying Anti-Fire (AF) Liquid on Building and Ground Surfaces

[0354] FIG. 9A shows a mobile GPS-tracked manned or autonomous vehicle anti-fire (AF) liquid spray vehicle system 30 for spraying environmentally-clean anti-fire (AF) chemical liquid on exterior building surfaces and ground surfaces in accordance with the principles of the present invention. As shown, the vehicle system 30 is supported on a set of wheels 30A driven by a propulsion drive subsystem 30 and navigated by GPS-guided navigation subsystem 301, and carrying an integrated supply tank 30B with either rechargeable-battery-operated electric-motor driven spray pump, or gasoline/diesel or propane operated motor-driven spray pump, 30C, for deployment on private and public property parcels having building structures, for spraying the same with environmentally-clean anti-fire (AF) liquid using a spray nozzle assembly 30D connected to the spray pump 30C by way of a flexible hose 30E.

[0355] FIG. 9B shows the GPS-tracked mobile anti-fire liquid spraying system 30 of FIG. 7A as comprising a number of subcomponents, namely: a GPS-tracked and remotely-monitored AF chemical liquid spray control sub-

system 30F; a micro-computing platform or subsystem 30G interfaced with the GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 30F by way of a system bus 30I; a wireless communication subsystem 30H interfaced to the micro-computing platform 30G via the system bus 30I; and a vehicular propulsion and navigation subsystem 30I employing a propulsion subsystem 30I1 and AI-driven or manually-driven navigation subsystem 30I2.

[0356] As configured in the illustrative embodiment, the GPS-tracked mobile anti-fire liquid spraying system 30 enables and supports (i) the remote monitoring of the spraying of anti-fire (AF) chemical liquid from the system 30 when located at specific GPS-indexed location coordinates, and (ii) the logging of all such GPS-indexed spray application operations, and recording the data transactions thereof within a local database maintained within the micro-computing platform 30G, as well as in the remote network database 9C1 maintained at the data center 8 of the system network 1.

[0357] As shown in FIG. 9B, the micro-computing platform 30G comprises: data storage memory 30G1; flash memory (firmware storage) 30G2; a programmable microprocessor 30G3; a general purpose I/O (GPIO) interface 30G4; a GPS transceiver circuit/chip with matched antenna structure 30G5; and the system bus 30I which interfaces these components together and provides the necessary addressing, data and control signal pathways supported within the system 30. As such, the micro-computing platform 30G is suitably configured to support and run a local control program 30G2-X on microprocessor 30G3 and memory architecture 30G1, 30G2 which is required and supported by the enterprise-level mobile application 12 and the suite of services supported by the system network 1 of the present invention.

[0358] As shown in FIG. 9B, the wireless communication subsystem 30H comprises: an RF-GSM modem transceiver 30H1; a T/X amplifier 30H2 interfaced with the RF-GSM modem transceiver 30H1; and a WIFI interface and a Bluetooth wireless interface 30H3 for interfacing with WIFI and Bluetooth data communication networks, respectively, in a manner known in the communication and computer networking art.

[0359] As shown in FIG. 9B, the GPS-tracked and remotely-controllable anti-fire (AF) chemical liquid spray control subsystem 30F comprises: anti-fire chemical liquid supply sensor(s) 30F1 installed in or on the anti-fire chemical liquid supply tank 30B to produce an electrical signal indicative of the volume or percentage of the AF liquid supply tank containing AF chemical liquid at any instant in time, and providing such signals to the AF liquid spraying system control interface 30F4; a power supply and controls 30F2 interfaced with the liquid pump spray subsystem 30C, and also the AF liquid spraying system control interface 30F4; manually-operated spray pump controls interface 30F3, interfaced with the AF liquid spraying system control interface 30F4; and the AF liquid spraying system control interface 30F4 interfaced with the micro-computing subsystem 30G, via the system bus 30I. The flash memory storage 30G2 contains microcode for a control program that runs on the microprocessor 20G3 and realizes the various GPS-specified AF chemical liquid spray control, monitoring, data logging and management functions supported by the system 30.



[0360] Notably, because the chemical components of wildfire inhibiting biochemical solution of the present invention completely dissolve in water, without crystal formation in solution, it's possible to spray the biochemical liquid using atomization and/or misting spray techniques so that very fine liquid droplets of micron dimensions can be formed and projected over long throw distances—during spraying operations. This pure-liquid property of the fire inhibiting biochemical composition (i) allows its active fire inhibiting chemistry (e.g. potassium mineral salts) to efficiently cling onto combustible surfaces of natural fuels distributed widely across ground surfaces in the rapidly expanding WUI region, and (ii) promotes surface infusion of the potassium mineral salts within the microstructure of the sprayed surfaces during atomization spraying and quick drying operations. This promotes the formation of ultra-thin potassium salt crystal coatings that offer improved duration of fire protection of TPC-based potassium mineral salts contained in the wildfire inhibitor, when exposed to moisture and/or high levels of relative humidity.

[0361] Deposition of potassium mineral salt crystal coatings within the molecular surface structure surfaces being treated with the biochemical solution of the present invention using atomization-based spraying techniques, preferably at elevated spraying temperatures under the arid hot Sun, the wildfire inhibiting potassium salt coating, once dried, can be made either insensitive or less sensitive to water exposure, a property which will improve the wildfire inhibitor's duration of fire protection in the presence of rain and ambient moisture levels. Also, it is believed that better surface deposition of the biochemical composition of the present invention can be achieved by reducing the size of the spray or misting droplets of the wildfire inhibitor as small as possible using, for example, atomization-based spraying/misting techniques applied at elevated spraying temperatures. Such techniques will promote water molecules to be rapidly evaporated during spray application, and promote deposition and bonding of potassium mineral salts within surface molecules of the sprayed surface substrate, as they are deposited onto the organic fuel surfaces to be protected against the threat of ignition by wildfire. Such insights and practices inform and support optimized methods of wildfire inhibitor deposition in outdoor environments.

Using GPS-Tracking, Mapping and Recording Techniques to Know where Clean-Chemistry Wild Fire Breaks and Zones where Formed by Whom, and when

[0362] Using the cloud-based wildfire defense network's integrated GPS-tracking, mapping and recording techniques, as illustrated in FIGS. 8 through 32B, fire jurisdictions can plan and implement clean-chemistry wildfire breaks and zones (e.g. around telephone poles) to proactively protect property and life from raging wildfires—by effectively inhibiting specific regions of combustible fuel from ignition, along the path towards targeted property and life to be protected from the incidence of wildfire. Proactive wildfire protection according to the principles of the present invention is simple. Wherever combustible ground cover is sprayed/misted with the fire inhibiting biochemical composition of the present invention, illustrated in FIGS. 6A through 7A, the free radical chain reactions driving the combustible phase of wildfire will be interrupted, taking the energy out of a raging wildfire, reducing the production of smoke, and protecting property that has been treated in advance of a wildfire incidence.

[0363] In hot dry climates, conditioned by hot dry prevailing winds, the relative humidity will be expectedly low, and in the absence of rain, the all-natural (clear) wild fire inhibiting sprayed over wild fire break and zone regions, will last for durations into weeks and months in many situations. However, whenever rain occurs, the Network will know and advise fire departments and homeowner alike that clean-chemistry wildfire breaks and zones need to be maintained by an additional spraying of wildfire inhibiting biochemical liquid, while GPS-tracked, mapped and recorded for management purposes.

Specification of GPS-Tracked Autonomously-Driven Drone System Adapted for Spraying Anti-Fire (AF) Liquid on Buildings and Ground Surfaces

[0364] FIG. 10A shows a mobile GPS-tracked unmanned airborne system (UAS) or drone 40 adapted for misting and spraying environmentally-clean anti-fire (AF) chemical liquid of the present invention on exterior building surfaces and ground surfaces in accordance with the principles of the present invention.

[0365] As shown, the drone vehicle system 40 comprises: a lightweight airframe 40A0 supporting a propulsion subsystem 40I provided with a set of eight (8) electric-motor driven propellers 40A1-40A8, driven by electrical power supplied by a rechargeable battery module 409, and controlled and navigated by a GPS-guided navigation subsystem 40I2; an integrated supply tank 40B supported on the airframe 40A0, and connected to either rechargeable-battery-operated electric-motor driven spray pump, or gasoline/diesel or propane operated motor-driven spray pump, 40C, for deployment on private and public property parcels having building structures; a spray nozzle assembly 40D connected to the spray pump 40C by way of a flexible hose 40E, for misting and spraying the same with environmentally-clean anti-fire (AF) liquid under the control of GPS-specified coordinates defining its programmed flight path when operating to suppress or otherwise fight wild fires.

[0366] FIG. 10B shows the GPS-tracked anti-fire liquid spraying system 40 of FIG. 8A as comprising a number of subcomponents, namely: a GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 40F; a micro-computing platform or subsystem 40G interfaced with the GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 40F by way of a system bus 40I; a wireless communication subsystem 40H interfaced to the micro-computing platform 40G via the system bus 40I; and a vehicular propulsion and navigation subsystem 40I employing propulsion subsystem 40I1, and AI-driven or manually-driven navigation subsystem 40I2.

[0367] As configured in the illustrative embodiment, the GPS-tracked anti-fire liquid spraying system 40 enables and supports (i) the remote monitoring of the spraying of anti-fire (AF) chemical liquid from the system 40 when located at specific GPS-indexed location coordinates, and (ii) the logging of all such GPS-indexed spray application operations, and recording the data transactions thereof within a local database maintained within the micro-computing platform 40G, as well as in the remote network database 9C1 maintained at the data center 8 of the system network 1.

[0368] As shown in FIG. 10B, the micro-computing platform 40G comprises: data storage memory 40G1; flash memory (firmware storage) 40G2; a programmable micro-processor 40G3; a general purpose I/O (GPIO) interface



**40G4**; a GPS transceiver circuit/chip with matched antenna structure **40G5**; and the system bus **40I** which interfaces these components together and provides the necessary addressing, data and control signal pathways supported within the system **40**. As such, the micro-computing platform **40G** is suitably configured to support and run a local control program **40G2-X** on microprocessor **40G3** and memory architecture **40G1**, **40G2** which is required and supported by the enterprise-level mobile application **12** and the suite of services supported by the system network **1** of the present invention.

[0369] As shown in FIG. 10B, the wireless communication subsystem **30H** comprises: an RF-GSM modem transceiver **40H1**; a T/X amplifier **40H2** interfaced with the RF-GSM modem transceiver **40H1**; and a WIFI interface and a Bluetooth wireless interface **40H3** for interfacing with WIFI and Bluetooth data communication networks, respectively, in a manner known in the communication and computer networking art.

[0370] As shown in FIG. 10B, the GPS-tracked and remotely-controllable anti-fire (AF) chemical liquid spray control subsystem **40F** comprises: anti-fire chemical liquid supply sensor(s) **40F1** installed in or on the anti-fire chemical liquid supply tank **30B** to produce an electrical signal indicative of the volume or percentage of the AF liquid supply tank containing AF chemical liquid at any instant in time, and providing such signals to the AF liquid spraying system control interface **40F4**; a power supply and controls **40F2** interfaced with the liquid pump spray subsystem **40C**, and also the AF liquid spraying system control interface **40F4**; manually-operated spray pump controls interface **40F3**, interfaced with the AF liquid spraying system control interface **30F4**; and the AF liquid spraying system control interface **40F4** interfaced with the micro-computing subsystem **40G**, via the system bus **40I**. The flash memory storage **40G2** contains microcode for a control program that runs on the microprocessor **40G3** and realizes the various GPS-specified AF chemical liquid spray control, monitoring, data logging and management functions supported by the system **40**.

Specification of GPS-Tracked Aircraft (i.e. Helicopter) for Spraying Anti-Fire (AF) Liquid on Ground Surfaces

[0371] FIG. 11A shows a mobile GPS-tracked manned aircraft (i.e. helicopter) system **50** adapted for misting and spraying environmentally-clean anti-fire (AF) chemical liquid of the present invention on ground surfaces and over buildings in accordance with the principles of the present invention.

[0372] As shown, the aircraft system **50** comprises: a lightweight airframe **50A0** supporting a propulsion subsystem **50I** provided with a set of axially-mounted helicopter blades **50A1-50A2** and **50A5**, driven by combustion-engine and controlled and navigated by a GPS-guided navigation subsystem **50I2**; an integrated supply tank **50B** supported on the airframe **50A0**, and connected to a gasoline/diesel operated motor-driven spray pump, **50C**, for deployment on private and public property parcels having building structures; a spray nozzle assembly **50D** connected to the spray pump **50C** by way of a hose **50E**, for misting and/or spraying the same with environmentally-clean anti-fire (AF) liquid under the control of GPS-specified coordinates defining its programmed flight path when operating to suppress or otherwise fight wild fires.

[0373] FIG. 11B shows the GPS-tracked anti-fire liquid spraying system **50** of FIG. 9A as comprising a number of subcomponents, namely: a GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem **50F**; a micro-computing platform or subsystem **50G** interfaced with the GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem **50F** by way of a system bus **50I**; a wireless communication subsystem **50H** interfaced to the micro-computing platform **50G** via the system bus **50I**; and a vehicular propulsion and navigation subsystem **50I** employing propulsion subsystem **50I1**, and AI-driven or manually-driven navigation subsystem **50I2**.

[0374] As configured in the illustrative embodiment, the GPS-tracked anti-fire liquid spraying system **50** enables and supports (i) the remote monitoring of the spraying of anti-fire (AF) chemical liquid from the system **50** when located at specific GPS-indexed location coordinates, and (ii) the logging of all such GPS-indexed spray application operations, and recording the data transactions thereof within a local database maintained within the micro-computing platform **50G**, as well as in the remote network database **9C1** maintained at the data center **8** of the system network **1**.

[0375] As shown in FIG. 9B, the micro-computing platform **50G** comprises: data storage memory **50G1**; flash memory (firmware storage) **50G2**; a programmable microprocessor **50G3**; a general purpose I/O (GPIO) interface **50G4**; a GPS transceiver circuit/chip with matched antenna structure **50G5**; and the system bus **40I** which interfaces these components together and provides the necessary addressing, data and control signal pathways supported within the system **50**. As such, the micro-computing platform **50G** is suitably configured to support and run a local control program **50G2-X** on microprocessor **50G3** and memory architecture **50G1**, **40G2** which is required and supported by the enterprise-level mobile application **12** and the suite of services supported by the system network **1** of the present invention.

[0376] As shown in FIG. 11B, the wireless communication subsystem **50H** comprises: an RF-GSM modem transceiver **50H1**; a T/X amplifier **50H2** interfaced with the RF-GSM modem transceiver **50H1**; and a WIFI interface and a Bluetooth wireless interface **50H3** for interfacing with WIFI and Bluetooth data communication networks, respectively, in a manner known in the communication and computer networking art.

[0377] As shown in FIG. 11B, the GPS-tracked and remotely-controllable anti-fire (AF) chemical liquid spray control subsystem **50F** comprises: anti-fire chemical liquid supply sensor(s) **50F1** installed in or on the anti-fire chemical liquid supply tank **50B** to produce an electrical signal indicative of the volume or percentage of the AF liquid supply tank containing AF chemical liquid at any instant in time, and providing such signals to the AF liquid spraying system control interface **50F4**; a power supply and controls **50F2** interfaced with the liquid pump spray subsystem **50C**, and also the AF liquid spraying system control interface **50F4**; manually-operated spray pump controls interface **50F3**, interfaced with the AF liquid spraying system control interface **50F4**; and the AF liquid spraying system control interface **50F4** interfaced with the micro-computing subsystem **50G**, via the system bus **50I**. The flash memory storage **50G2** contains microcode for a control program that runs on the microprocessor **50G3** and realizes the various GPS-



specified AF chemical liquid spray control, monitoring, data logging and management functions supported by the system 50.

#### Specification of GPS-Tracked Autonomously-Driven Aircraft for Spraying Anti-Fire (AF) Liquid on Building and Ground Surfaces

[0378] FIG. 12A shows a mobile GPS-tracked manned all-terrain vehicle (ATV) system 60 adapted for misting and spraying environmentally-clean anti-fire (AF) chemical liquid of the present invention on ground surfaces in accordance with the principles of the present invention.

[0379] As shown, the aircraft system 60 comprises: a lightweight frame/chassis 60A0 supporting a propulsion subsystem 60I provided with a set of wheels 60A1-60A4, driven by combustion-engine, and controlled and navigated by a GPS-guided navigation subsystem 60I2; an integrated supply tank 60B supported on the frame 60A0, and connected to a gasoline/diesel operated motor-driven spray pump, 60C, for deployment on private and public property parcels; a spray nozzle assembly 60D connected to the spray pump 60C by way of a hose 60E, for misting and/or spraying the same with environmentally-clean anti-fire (AF) liquid under the control of GPS-specified coordinates defining its programmed flight path when operating to suppress or otherwise fight wild fires.

[0380] FIG. 12B shows the GPS-tracked anti-fire liquid spraying system 60 of FIG. 10A as comprising a number of subcomponents, namely: a GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 60F; a micro-computing platform or subsystem 60G interfaced with the GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 60F by way of a system bus 60I; a wireless communication subsystem 60H interfaced to the micro-computing platform 60G via the system bus 50I; and a vehicular propulsion and navigation subsystem 60I employing propulsion subsystem 60I1, and AI-driven or manually-driven navigation subsystem 60I2.

[0381] As configured in the illustrative embodiment, the GPS-tracked anti-fire liquid spraying system 60 enables and supports (i) the remote monitoring of the spraying of anti-fire (AF) chemical liquid from the system 60 when located at specific GPS-indexed location coordinates, and (ii) the logging of all such GPS-indexed spray application operations, and recording the data transactions thereof within a local database maintained within the micro-computing platform 60G, as well as in the remote network database 9C1 maintained at the data center 8 of the system network 1.

[0382] As shown in FIG. 12B, the micro-computing platform 60G comprises: data storage memory 60G1; flash memory (firmware storage) 60G2; a programmable micro-processor 60G3; a general purpose I/O (GPIO) interface 60G4; a GPS transceiver circuit/chip with matched antenna structure 60G5; and the system bus 60I which interfaces these components together and provides the necessary addressing, data and control signal pathways supported within the system 60. As such, the micro-computing platform 60G is suitably configured to support and run a local control program 60G2-X on microprocessor 60G3 and memory architecture 60G1, 60G2 which is required and supported by the enterprise-level mobile application 12 and the suite of services supported by the system network 1 of the present invention.

[0383] As shown in FIG. 12B, the wireless communication subsystem 50H comprises: an RF-GSM modem transceiver 60H1; a T/X amplifier 60H2 interfaced with the RF-GSM modem transceiver 60H1; and a WIFI interface and a Bluetooth wireless interface 60H3 for interfacing with WIFI and Bluetooth data communication networks, respectively, in a manner known in the communication and computer networking art.

[0384] As shown in FIG. 12B, the GPS-tracked and remotely-controllable anti-fire (AF) chemical liquid spray control subsystem 60F comprises: anti-fire chemical liquid supply sensor(s) 60F1 installed in or on the anti-fire chemical liquid supply tank 60B to produce an electrical signal indicative of the volume or percentage of the AF liquid supply tank containing AF chemical liquid at any instant in time, and providing such signals to the AF liquid spraying system control interface 60F4; a power supply and controls 60F2 interfaced with the liquid pump spray subsystem 60C, and also the AF liquid spraying system control interface 60F4; manually-operated spray pump controls interface 60F3, interfaced with the AF liquid spraying system control interface 60F4; and the AF liquid spraying system control interface 60F4 interfaced with the micro-computing subsystem 60G, via the system bus 60I. The flash memory storage 60G2 contains microcode for a control program that runs on the microprocessor 60G3 and realizes the various GPS-specified AF chemical liquid spray control, monitoring, data logging and management functions supported by the system 60.

#### Specification of GPS-Tracking Backpack-Mounted Atomizing Spray Cannon System for Spraying Environmentally-Clean Anti-Fire/Fire Inhibiting Liquid Biochemical Composition on the Surfaces of Buildings and Property Ground Surfaces

[0385] FIG. 13A shows a mobile GPS-tracked backpack-mounted atomizing spray cannon (ASC) system 70 adapted for misting and spraying environmentally-clean anti-fire (AF) chemical liquid on ground surfaces in accordance with the principles of the present invention.

[0386] As shown, the GPS-tracked spray cannon system 70 comprises: a lightweight frame/chassis 60A a GPS-guided navigation subsystem 60I2 for providing the user with navigation control during GPS-tracked and mapped spraying operations; an integrated supply tank 60B supported on the frame 60A0, and connected to a gasoline/diesel or battery-powered operated motor-driven spray pump, 60C, for deployment on private and public property parcels; a spray nozzle assembly 60D connected to the spray pump 60C by way of a hose 60E, for misting and/or spraying the same with environmentally-clean anti-fire (AF) liquid under the control of GPS-specified coordinates defining its programmed flight path when operating to suppress or otherwise fight wild fires.

[0387] FIG. 13B shows the GPS-tracked anti-fire liquid spraying system 70 of FIG. 13A as comprising a number of subcomponents, namely: a GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 60F; a micro-computing platform or subsystem 60G interfaced with the GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 60F by way of a system bus 60I; a wireless communication subsystem 60H interfaced to the micro-computing platform 60G via the system bus 50I; and a vehicular propulsion and navigation subsystem 60I



employing propulsion subsystem 60I1, and AI-driven or manually-driven navigation subsystem 60I2.

[0388] As configured in the illustrative embodiment, the GPS-tracked anti-fire liquid spraying system 70 enables and supports (i) the remote monitoring of the spraying of anti-fire (AF) chemical liquid from the system 70 when located at specific GPS-indexed location coordinates, and (ii) the logging of all such GPS-indexed spray application operations, and recording the data transactions thereof within a local database maintained within the micro-computing platform 60G, as well as in the remote network database 9C1 maintained at the data center 8 of the system network 1.

[0389] As shown in FIG. 13B, the micro-computing platform 60G comprises: data storage memory 60G1; flash memory (firmware storage) 60G2; a programmable micro-processor 60G3; a general purpose I/O (GPIO) interface 60G4; a GPS transceiver circuit/chip with matched antenna structure 60G5; and the system bus 60I which interfaces these components together and provides the necessary addressing, data and control signal pathways supported within the system 70. As such, the micro-computing platform 60G is suitably configured to support and run a local control program 60G2-X on microprocessor 60G3 and memory architecture 60G1, 60G2 which is required and supported by the enterprise-level mobile application 12 and the suite of services supported by the system network 1 of the present invention.

[0390] As shown in FIG. 13B, the wireless communication subsystem 50H comprises: an RF-GSM modem transceiver 60H1; a T/X amplifier 60H2 interfaced with the RF-GSM modem transceiver 60H1; and a WIFI interface and a Bluetooth wireless interface 60H3 for interfacing with WIFI and Bluetooth data communication networks, respectively, in a manner known in the communication and computer networking art.

[0391] As shown in FIG. 13B, the GPS-tracked and remotely-controllable anti-fire (AF) chemical liquid spray control subsystem 60F comprises: anti-fire chemical liquid supply sensor(s) 60F1 installed in or on the anti-fire chemical liquid supply tank 60B to produce an electrical signal indicative of the volume or percentage of the AF liquid supply tank containing AF chemical liquid at any instant in time, and providing such signals to the AF liquid spraying system control interface 60F4; a power supply and controls 60F2 interfaced with the liquid pump spray subsystem 60C, and also the AF liquid spraying system control interface 60F4; manually-operated spray pump controls interface 60F3, interfaced with the AF liquid spraying system control interface 60F4; and the AF liquid spraying system control interface 60F4 interfaced with the micro-computing subsystem 60G, via the system bus 60I. The flash memory storage 60G2 contains microcode for a control program that runs on the microprocessor 60G3 and realizes the various GPS-specified AF chemical liquid spray control, monitoring, data logging and management functions supported by the system network of the present invention.

Specification of GPS-Tracked Mobile Atomizing Spray Cannon System for Spraying Environmentally-Clean Anti-Fire Biochemical Liquid on Buildings and Ground Surfaces

[0392] FIG. 14A shows a mobile GPS-tracked backpack-mounted atomizing spray cannon (ASC) system 80 adapted for misting and spraying environmentally-clean anti-fire/

fire-inhibiting biochemical liquid on ground surfaces in accordance with the principles of the present invention.

[0393] As shown, the GPS-tracked system 80 comprises: a lightweight frame/chassis 60A provided with a set of wheels mounted on a trailer 60B2 that is pulled by tractor 60A2 driven by combustion-engine or electric battery-powered motor, that is controlled and navigated by a GPS-guided navigation subsystem 60I2; an integrated supply tank 60B supported on the frame 60A3, and connected to a gasoline/diesel operated motor-driven spray pump, 60C, for deployment on private and public property parcels; an atomizing spray nozzle assembly 60D comprising a ring a atomizing spray nozzles mounted about the aperture of a cannon-like air-blowing engine, powered by a turbine fan blower unit, and connected to the spray pump 60C by way of a hose 60E, for producing a forceful airstream enriched with atomized mist developed from a supply of the environmentally-clean fire inhibiting (i.e. anti-fire) biochemical liquid of the present invention, under the control of GPS-specified coordinates defining its programmed path or course when operating to suppress or otherwise fight wild fires.

[0394] FIG. 14B shows the GPS-tracked anti-fire liquid spraying system cannon 80 of FIG. 14A as comprising a number of subcomponents, namely: a GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 60F; a micro-computing platform or subsystem 60G interfaced with the GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 60F by way of a system bus 60I; a wireless communication subsystem 60H interfaced to the micro-computing platform 60G via the system bus 50I; and a vehicular propulsion and navigation subsystem 60I employing propulsion subsystem 60I1, and AI-driven or manually-driven navigation subsystem 60I2.

[0395] As configured in the illustrative embodiment, the GPS-tracked anti-fire liquid spraying system 80 enables and supports (i) the remote monitoring of the spraying of anti-fire (AF) chemical liquid from the system 80 when located at specific GPS-indexed location coordinates, and (ii) the logging of all such GPS-indexed spray application operations, and recording the data transactions thereof within a local database maintained within the micro-computing platform 60G, as well as in the remote network database 9C1 maintained at the data center 8 of the system network 1.

[0396] As shown in FIG. 14B, the micro-computing platform 60G comprises: data storage memory 60G1; flash memory (firmware storage) 60G2; a programmable micro-processor 60G3; a general purpose I/O (GPIO) interface 60G4; a GPS transceiver circuit/chip with matched antenna structure 60G5; and the system bus 60I which interfaces these components together and provides the necessary addressing, data and control signal pathways supported within the system 80. As such, the micro-computing platform 60G is suitably configured to support and run a local control program 60G2-X on microprocessor 60G3 and memory architecture 60G1, 60G2 which is required and supported by the enterprise-level mobile application 12 and the suite of services supported by the system network 1 of the present invention.

[0397] As shown in FIG. 14B, the wireless communication subsystem 50H comprises: an RF-GSM modem transceiver 60H1; a T/X amplifier 60H2 interfaced with the RF-GSM modem transceiver 60H1; and a WIFI interface and a Bluetooth wireless interface 60H3 for interfacing with



WIFI and Bluetooth data communication networks, respectively, in a manner known in the communication and computer networking art.

[0398] As shown in FIG. 14B, the GPS-tracked and remotely-controllable anti-fire (AF) chemical liquid spray control subsystem 60F comprises: anti-fire chemical liquid supply sensor(s) 60F1 installed in or on the anti-fire chemical liquid supply tank 60B to produce an electrical signal indicative of the volume or percentage of the AF liquid supply tank containing AF chemical liquid at any instant in time, and providing such signals to the AF liquid spraying system control interface 60F4; a power supply and controls 60F2 interfaced with the liquid pump spray subsystem 60C, and also the AF liquid spraying system control interface 60F4; manually-operated spray pump controls interface 60F3, interfaced with the AF liquid spraying system control interface 60F4; and the AF liquid spraying system control interface 60F4 interfaced with the micro-computing subsystem 60G, via the system bus 60I. The flash memory storage 60G2 contains microcode for a control program that runs on the microprocessor 60G3 and realizes the various GPS-specified AF chemical liquid spray control, monitoring, data logging and management functions supported by the system network of the present invention.

#### Specification of GPS-Tracking Mobile Atomizing Spray Cannon System for Spraying Environmentally-Clean Anti-Fire Biochemical Liquid on Buildings and Ground Surfaces

[0399] FIG. 15A shows a mobile GPS-tracked mobile atomizing spray cannon (ASC) system 90 capable of being towed along a course or desired pathway, and specially adapted for misting and spraying environmentally-clean fire inhibiting biochemical liquid composition of the present invention on ground and other property surfaces in accordance with the principles of the present invention.

[0400] As shown in FIG. 15A, the GPS-tracked cannon-type spraying system 90 comprises: a lightweight frame/chassis 60A0 supporting a propulsion subsystem 60I provided with a set of wheels 60A1-60A3, and tow bar 60A4; an integrated supply tank 60B supported on the vehicle towing the spray cannon system 90, and connected to a gasoline/diesel or electric-motor operated motor-driven spray pump, 60C, for deployment on private and public property parcels; an electric turbine fan 60I1 for producing forced air stream through the cylindrical cannon or barrel like structure as shown; an atomizing spray nozzle assembly 60D connected to the spray pump 60C by way of a hose 60E, for misting and/or spraying the same with environmentally-clean anti-fire (AF) biochemical liquid of the present invention under the control of GPS-specified coordinates defining its programmed path or course when operating to suppress or otherwise fight wild fires.

[0401] FIG. 15B shows the GPS-tracked anti-fire liquid spraying system 90 of FIG. 15A as comprising a number of subcomponents, namely: a GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 60F; a micro-computing platform or subsystem 60G interfaced with the GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 60F by way of a system bus 60I; a wireless communication subsystem 60H interfaced to the micro-computing platform 60G via the system bus 50I;

and a vehicular propulsion and navigation subsystem 60I, and AI-driven or manually-driven navigation subsystem 60I2.

[0402] As configured in the illustrative embodiment, the GPS-tracked anti-fire liquid spraying system 90 enables and supports (i) the remote monitoring of the spraying of anti-fire (AF) chemical liquid from the system 90 when located at specific GPS-indexed location coordinates, and (ii) the logging of all such GPS-indexed spray application operations, and recording the data transactions thereof within a local database maintained within the micro-computing platform 60G, as well as in the remote network database 9C1 maintained at the data center 8 of the system network 1.

[0403] As shown in FIG. 15B, the micro-computing platform 60G comprises: data storage memory 60G1; flash memory (firmware storage) 60G2; a programmable micro-processor 60G3; a general purpose I/O (GPIO) interface 60G4; a GPS transceiver circuit/chip with matched antenna structure 60G5; and the system bus 60I which interfaces these components together and provides the necessary addressing, data and control signal pathways supported within the system 90. As such, the micro-computing platform 60G is suitably configured to support and run a local control program 60G2-X on microprocessor 60G3 and memory architecture 60G1, 60G2 which is required and supported by the enterprise-level mobile application 12 and the suite of services supported by the system network 1 of the present invention.

[0404] As shown in FIG. 15B, the wireless communication subsystem 50H comprises: an RF-GSM modem transceiver 60H1; a T/X amplifier 60H2 interfaced with the RF-GSM modem transceiver 60H1; and a WIFI interface and a Bluetooth wireless interface 60H3 for interfacing with WIFI and Bluetooth data communication networks, respectively, in a manner known in the communication and computer networking art.

[0405] As shown in FIG. 15B, the GPS-tracked and remotely-controllable anti-fire (AF) chemical liquid spray control subsystem 60F comprises: anti-fire chemical liquid supply sensor(s) 60F1 installed in or on the anti-fire chemical liquid supply tank 60B to produce an electrical signal indicative of the volume or percentage of the AF liquid supply tank containing AF biochemical liquid at any instant in time, and providing such signals to the AF liquid spraying system control interface 60F4; a power supply and controls 60F2 interfaced with the liquid pump spray subsystem 60C, and also the AF liquid spraying system control interface 60F4; manually-operated spray pump controls interface 60F3, interfaced with the AF liquid spraying system control interface 60F4; and the AF liquid spraying system control interface 60F4 interfaced with the micro-computing subsystem 60G, via the system bus 60I. The flash memory storage 60G2 contains microcode for a control program that runs on the microprocessor 60G3 and realizes the various GPS-specified AF chemical liquid spray control, monitoring, data logging and management functions supported by the system network of the present invention.

#### Specification of GPS-Tracking Back-Pack Atomizing-Spray Fire Extinguishing System for Spraying Environmentally-Clean Fire-Inhibiting Biochemical Liquid Compositions on Active Fires, and Also on Surfaces of Buildings and Ground Surfaces to be Protectively Protected Against Fire

[0406] FIG. 16A shows a mobile GPS-tracked backpack-mounted atomizing fire extinguishing system 100 adapted



for spraying clouds of environmentally-clean anti-fire (AF) biochemical liquid mist onto fire outbreaks (e.g. all Classes of fire A, B, C and D) wherever they may exit, to quickly extinguish the same in accordance with the principles of the present invention. The system can also be used to apply clean fire protective coatings as well using atomizing sprays of clean biochemical liquid compositions of the present invention, disclosed herein.

[0407] As shown, the backpack-mounted fire extinguishing system **100** comprises: a liquid storage tank **401** containing 5 gallons of environmentally-clean water-based free-radical chemical-reaction interrupting liquid of the present invention, charged with 100 [PSIG] pressure from a small pressurized air or CO<sub>2</sub> tank **402** integrated with the housing **403**. The hand-activated gun-style misting head (i.e. spray misting gun) **404** is provided with a stainless-steel misting nozzle **45** that is connected to two flexible hoses **406A** and **406B**. Hose **406A** is connected to the water tank **41** and hose **406B** is connected to the pressurized air tank **402**. The hand-held gun-style misting head **404** with misting nozzle **405** is manually activated by the user depressing a finger-activated trigger **406** to discharge clean-chemistry water-based chemical-reaction interrupting mist clouds **407** from the nozzle **405** onto a fire for quick suppression and extinguishment. The portable system can be either back-mounted, or carried in one hand while the other hand is used to hold and operate the spray-misting gun **404**. Fire Inhibitor chemical liquid **410** has the required free-radical chemical reaction interrupting chemistry of the present invention, such that the chemical molecules in chemical liquid will interfere with the free radicals generated during the combustion phase of a fire, and interrupt these free-radical chemical reactions within the combustion phase, to suppress and extinguish the fire. Specifically, the biochemical liquid has the required metal ions to interrupt free-radical chemical reaction interrupting chemistry of the present invention, such that chemical molecules in the chemical liquid, when transformed into a clean-chemistry-water-based mist, provides a countless supply of water-based micro-droplets, each containing dissolved ions (i.e. electrically-charged atoms or molecules) supplying free-electrons that pair with and stabilize the free-radicals (H<sup>+</sup>, OH<sup>-</sup>, O) before any other molecules in the combustion phase can do so to sustain the chemical-reactions (i.e. free-electrons that reduce and stabilize the free-radicals before rapidly-oxidizing molecules within the combustion phase of the fire to sustain the chemical-reactions), and thereby quickly suppressing and extinguishing the fire.

[0408] The superior performance of system **400** over conventional portable water mist systems can be attributed to the fact that the micro-droplets of the clean-chemistry water mist **407** will vaporize when absorbing the radiant heat energy of the hot fire, rapidly expanding into a vapor, cooling down the fire, and displacing oxygen. Also the chemical molecules in the micro-droplets will interfere with the free radicals (H<sup>+</sup>, OH<sup>-</sup>, O) and interrupt these free-radical chemical reactions within the combustion phase of a fire, and extinguishing the fire.

[0409] FIG. 16B shows the GPS-tracked mobile fire extinguishing system **100** of FIG. 16A as comprising a number of subcomponents, namely: a GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem **60F**; a micro-computing platform or subsystem **60G** interfaced with the GPS-tracked and remotely-monitored AF chemical

liquid spray control subsystem **60F** by way of a system bus **60I**; a wireless communication subsystem **60H** interfaced to the micro-computing platform **60G** via the system bus **50I**.

[0410] As configured in the illustrative embodiment, the GPS-tracked anti-fire liquid spraying system **100** enables and supports (i) the remote monitoring of the spraying of anti-fire (AF) biochemical liquid from the system **100** when located at specific GPS-indexed location coordinates, and (ii) the logging of all such GPS-indexed spray application operations, and recording the data transactions thereof within a local database maintained within the micro-computing platform **60G**, as well as in the remote network database **9C1** maintained at the data center **8** of the system network **1**.

[0411] As shown in FIG. 16B, the micro-computing platform **60G** comprises: data storage memory **60G1**; flash memory (firmware storage) **60G2**; a programmable micro-processor **60G3**; a general purpose I/O (GPIO) interface **60G4**; a GPS transceiver circuit/chip with matched antenna structure **60G5**; and the system bus **60I** which interfaces these components together and provides the necessary addressing, data and control signal pathways supported within the system **100**. As such, the micro-computing platform **60G** is suitably configured to support and run a local control program **60G2-X** on microprocessor **60G3** and memory architecture **60G1**, **60G2** which is required and supported by the enterprise-level mobile application **12** and the suite of services supported by the system network **1** of the present invention.

[0412] As shown in FIG. 16B, the wireless communication subsystem **50H** comprises: an RF-GSM modem transceiver **60H1**; a T/X amplifier **60H2** interfaced with the RF-GSM modem transceiver **60H1**; and a WIFI interface and a Bluetooth wireless interface **60H3** for interfacing with WIFI and Bluetooth data communication networks, respectively, in a manner known in the communication and computer networking art.

[0413] As shown in FIG. 16B, the GPS-tracked and remotely-controllable anti-fire (AF) chemical liquid spray control subsystem **60F** comprises: anti-fire chemical liquid supply sensor(s) **60F1** installed in or on the anti-fire chemical liquid supply tank **60B** to produce an electrical signal indicative of the volume or percentage of the AF liquid supply tank containing AF biochemical liquid of the present invention at any instant in time, and providing such signals to the AF liquid spraying system control interface **60F4**; a power supply and controls **60F2** interfaced with the liquid pump spray subsystem **60C** controlling the mixing of gas source **402** with biochemical liquid source **410**, and also the AF liquid spraying system control interface **60F4**; manually-operated trigger **404** controlled spray pump controls interface **60F3**, is interfaced with the AF liquid spraying system control interface **60F4**; and the AF liquid spraying system control interface **60F4** is interfaced with the micro-computing subsystem **60G**, via the system bus **60I**. The flash memory storage **60G2** contains microcode for a control program that runs on the microprocessor **60G3** and realizes the various GPS-specified AF biochemical liquid spray control, monitoring, data logging and management functions supported by the system network of the present invention.



Specification of GPS-Tracking Mobile Remotely-Controllable Atomizing Spray Cannon System for Spraying Environmentally-Clean Anti-Fire (AF) Liquid on Buildings and Ground Surfaces

[0414] FIG. 17A shows a mobile GPS-tracked backpack-mounted atomizing spray cannon (ASC) system 110 adapted for misting and spraying environmentally-clean anti-fire (AF) biochemical liquid on ground surfaces in accordance with the principles of the present invention.

[0415] As shown in FIG. 17A, the GPS-tracked mobile spraying cannon system 110 comprises: a lightweight frame/chassis 60A0 supporting a propulsion subsystem 60I provided with a set of wheels 60A1-60A4, driven by combustion-engine, and controlled and navigated by a GPS-guided navigation subsystem 60I2; an integrated supply tank 60B supported on the frame 60A0, and connected to a gasoline/diesel operated motor-driven spray pump, 60C, for deployment on private and public property parcels; a spray nozzle assembly 60D connected to the spray pump 60C by way of a hose 60E, for misting and/or spraying the same with environmentally-clean anti-fire (AF) liquid under the control of GPS-specified coordinates defining its programmed path when operating to suppress or otherwise fight wild fires.

[0416] FIG. 17B shows the GPS-tracked anti-fire liquid spraying system 110 of FIG. 17A as comprising a number of subcomponents, namely: a GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 60F; a micro-computing platform or subsystem 60G interfaced with the GPS-tracked and remotely-monitored AF chemical liquid spray control subsystem 60F by way of a system bus 60I; a wireless communication subsystem 60H interfaced to the micro-computing platform 60G via the system bus 50I; and AI-driven or manually-driven navigation subsystem 60I2.

[0417] As configured in the illustrative embodiment, the GPS-tracked anti-fire liquid spraying cannon system 110 enables and supports (i) the remote monitoring of the spraying of anti-fire (AF) chemical liquid from the system 110 when located at specific GPS-indexed location coordinates, and (ii) the logging of all such GPS-indexed spray application operations, and recording the data transactions thereof within a local database maintained within the micro-computing platform 60G, as well as in the remote network database 9C1 maintained at the data center 8 of the system network 1.

[0418] As shown in FIG. 17B, the micro-computing platform 60G comprises: data storage memory 60G1; flash memory (firmware storage) 60G2; a programmable microprocessor 60G3; a general purpose I/O (GPIO) interface 60G4; a GPS transceiver circuit/chip with matched antenna structure 60G5; and the system bus 60I which interfaces these components together and provides the necessary addressing, data and control signal pathways supported within the system 110. As such, the micro-computing platform 60G is suitably configured to support and run a local control program 60G2-X on microprocessor 60G3 and memory architecture 60G1, 60G2 which is required and supported by the enterprise-level mobile application 12 and the suite of services supported by the system network 1 of the present invention.

[0419] As shown in FIG. 17B, the wireless communication subsystem 50H comprises: an RF-GSM modem transceiver 60H1; a T/X amplifier 60H2 interfaced with the RF-GSM modem transceiver 60H1; and a WIFI interface

and a Bluetooth wireless interface 60H3 for interfacing with WIFI and Bluetooth data communication networks, respectively, in a manner known in the communication and computer networking art.

[0420] As shown in FIG. 17B, the GPS-tracked and remotely-controllable anti-fire (AF) chemical liquid spray control subsystem 60F comprises: anti-fire chemical liquid supply sensor(s) 60F1 installed in or on the anti-fire chemical liquid supply tank 60B to produce an electrical signal indicative of the volume or percentage of the AF liquid supply tank containing AF chemical liquid at any instant in time, and providing such signals to the AF liquid spraying system control interface 60F4; a power supply and controls 60F2 interfaced with the liquid pump spray subsystem 60C, and also the AF liquid spraying system control interface 60F4; manually-operated spray pump controls interface 60F3, interfaced with the AF liquid spraying system control interface 60F4; and the AF liquid spraying system control interface 60F4 interfaced with the micro-computing subsystem 60G, via the system bus 60I. The flash memory storage 60G2 contains microcode for a control program that runs on the microprocessor 60G3 and realizes the various GPS-specified AF chemical liquid spray control, monitoring, data logging and management functions supported by the system network of the present invention.

Specification of an Exemplary Network Database Schema for Supporting the System Network of the Present Invention and GPS-Specified Operations Involving the Spraying of Anti-Fire (AF) Liquid on GPS-Specified Ground, Property and Building Surfaces in Regions at Risk Prior to and During the Outbreak of Wild Fires

[0421] FIG. 18 shows an exemplary schema for the network database (RDBMS) 9C1 supported by the system network of the present invention, showing the primary enterprise level objects supported in the database tables created in the network database 9C using the schema, and the relationships that are specified or indicated. This exemplary database schema is for supporting the system network of the present invention and GPS-specified operations involving the spraying of anti-fire (AF) liquid on GPS-specified ground, property and building surfaces in regions at risk prior to and during the outbreak of wild fires.

[0422] As shown in FIG. 18, the exemplary database schema for the system network 1 includes a number of high-level enterprise objects such as, for example: Users, with properties including User ID, Residence, Age, User Class (e.g. Wild Fire Management Administrator, Wild Fire Spray Applicator, Real Property Owner, Home Owner, Business Owner, Property Owner, Resident, etc.), and Pets; Real Property, with properties including Ownership/Lease, Location, Buildings, GPS Addresses, County, State; Vehicles, with properties such as Model, Year, Brand, Registered Owner; Water Crafts, with properties Model, ID # etc.; Anti-Fire Chemical Liquid Supplies, with properties Manufacturer, Location, Quantity, Date Delivered; Anti-Fire (AF) Liquid Spraying Aircraft Systems, with properties Manufacturer, Model, ID #; Anti-Fire Liquid Spraying Ground Systems, including Manufacturer, Model, ID #; Portable Anti-Fire Liquid Spraying Systems; Anti-Fire (AF) Chemical Liquid Spray Application Orders, including Location, ID #; Anti-Fire Chemical Liquid Spray Application Reports, with properties such as State, County, GPS Addresses; and Weather Data, with properties State, County, and GPS Addresses.



Specification of Exemplary Graphical User Interfaces Supported on the Mobile Application Deployed on System Network of the Present Invention, for the Purpose of Delivering the Various Services Supported on the System Network

**[0423]** FIG. 19 illustrates an exemplary wire-frame model of a graphical user interface (GUI) 13 of the mobile application 120 for use by registered users (e.g. property parcel owners, contractors and/or agents, and other stakeholders on the system network) to request and receive services supported by the system network of the present invention. As shown in this exemplary GUI screen 13, supports a number of pull-down menus under the titles: messages 13A, where the user can view messages sent via messaging services supported by the application; maps 13B, where wild fires have been identified and mapped, tracked and ranked in terms of risk to the user and associated property; and tasks 13C, where AF liquid spray tasks have been scheduled, have been completed, or are in progress, by the user.

**[0424]** FIG. 19A shows an exemplary graphical user interface supported by the mobile application 12 showing a user updating the registration profile as a task on the system network. The GUI screen is accessed and delivered to LCD screen of the mobile computing device 11 when the user selects the Tasks menu to display a menu of commands, and then selects the Update command from the command menu. During this service, the user can update various information items relating to the user profile, such as, name and address, contact information (e.g. email and SMS number), property parcel linked to one's profile, and GPS-tracked spray system deployed or assigned to the user and/or property parcel(s).

**[0425]** FIG. 19B shows an exemplary graphical user interface supported by the mobile application 12 showing a user receiving a message "notice of request to wild-fire spray protect a property parcel" (via email, SMS messaging and/or push-notifications) issued from the command center 19 to spray GPS-specified private property parcel(s) with clean anti-fire (AF) chemical liquid and registered GPS-tracked spray equipment.

**[0426]** FIG. 19C shows an exemplary graphical user interface supported by the mobile application 12 showing a user receiving a notice of order (via email, SMS messaging and/or push-notifications) to wild-fire spray-protect GPS-specified public property parcel(s) with clean anti-fire (AF) liquid to create and maintain a GPS-specified public fire-break (e.g. Firebreak No. 120).

**[0427]** FIG. 19D shows an exemplary graphical user interface supported by the mobile application showing a user requesting a refill of clean anti-fire (AF) chemical liquid for supply to GPS-specified spray equipment registered on the system network. The user selects the Tasks menu to display a set of commands, and then selects the Refill command from the displayed command menu. The user confirms the refill order and when ready selects the Send Request command from the display screen, sending the command to the command center 19 and related data center 8 for processing and fulfillment. All operations are logged and tracked in the system network database 9C1 shown in FIG. 4A.

**[0428]** In the illustrative embodiment, the mobile application 12 on mobile computing device 11 supports many functions to provide many services: (i) sends automatic notifications from the command center 19 to home/business owners with the mobile application 12, instructing them to spray their real property and home/building at certain times

with anti-fire (AF) liquid contained in the tanks of GPS-tracked AF liquid spraying systems 20, 30, 40, 40, 50 and 60; (ii) automatically monitors consumption of sprayed AF-liquid and generate auto-replenish order (via its onboard GSM-circuits) so as to achieve compliance with the home/neighborhood spray defense program, and report AF chemical liquid levels in each home-owner tank; and (iii) shows status of wild fire risk in the region, and actions to be taken before wild fire outbreak.

**[0429]** FIG. 20 shows an exemplary graphical user interface 13' supported by the mobile application 12 configured for use by command center administrators to issue wild-fire protection orders, plan wild-fire protection tasks, generate wild-fire and protection reports, and send and receive messages to users on the system network, to carry out a wild fire suppression and management program in the region where the system network is deployed. As shown, GUI screen 13' supports a number of pull-down menus under the titles: Messages 13A', where project administrator and spray technicians can view messages sent via messaging services supported by the application; Maps 13B', where wild fires have been identified, tracked, and ranked in terms of risk to certain regions at a given moment in time; Planning 13C', wherein plans have been made to fight wild fires using the methods described in FIGS. 24 through 32B, status of specific plans, which one are in progress; and Reports 13D', where reports are issued to the mobile application 12 running on mobile client systems 11 in operable communication with the web, application and database servers 9A, 9B and 9C at the data center 8, supported by the system network 1.

**[0430]** FIG. 20A shows an exemplary graphical user interface supported by the mobile application configured for use by command center administrators to issue wild-fire protection orders using the system network of the present invention. As shown, the user selects the Planning menu and displays a set of planning commands, and then selects the Property command, where the user is then giving to choice to select one or more parcels of property in a given region, and then select an Action (e.g. Wild Fire Spray Protect). The user selects the property parcel(s), and then the required Action (i.e. Wild Fire Spray Protect), and Order is set up for the command center action. When the command center selects execute from the menu, the system network issues the order and sends notice of orders to all property parcel owners or agents to oversee the immediate spraying of the GPS-specified property parcels with clean anti-fire (AF) chemical liquid supply to the property owners or agents as the case may be.

**[0431]** FIG. 20B shows an exemplary graphical user interface supported by the mobile application 12 configured for use by command center administrators to issue wild-fire protection orders involving the creation and maintenance of a clean AF-based chemical firebreak, as illustrated in FIG. 25, for example, using the methods of the present invention described herein. As shown, the administrator selects the Planning menu, and displays a menu of Planning commands, from which the user selects Firebreaks. In the case example shown in FIG. 20B, the administrator issues an Order to apply or rather practice the dual-region clean AF chemical firebreak method illustrated in FIG. 25, at GPS-specified coordinates GPS LAT-X/LONG-Y using AF chemical liquid misting and spraying airborne operations. As shown the order will specify the deployment of specific GPS-tracked



AF spray vehicle systems, and identify them by system ID #. The order may also identify or request users (e.g. pilots) assigned to the AF chemical firebreak project/task.

[0432] FIG. 20C shows an exemplary graphical user interface supported by mobile application 12 configured for use by command center administrators to order the creation and/or maintenance of a GPS-specified clean AF-based chemical firebreak on one or more public/private property parcels. As shown, the administrator selects the Planning menu, and displays a menu of Planning commands, from which the user selects Firebreaks. In the case example shown in FIG. 13C, the administrator issues an Order to practice the Wild Fire Spray Protect Method alongside one or more parcels of public property, which may be a long strip of land/brush alongside or near a highway. The method may be the AF chemical firebreak method as illustrated in the FIG. 25 and described in FIGS. 26A, 26B and 26C, at GPS-specified coordinates GPS LAT-X/LONG-Y using ground-based AF chemical liquid spraying operations. As shown, the order will specify the deployment of specific GPS-tracked AF spray vehicle systems, and identify them by system ID #. The order may also identify or request users (e.g. drivers) assigned to the AF chemical firebreak project/task. Alternatively, the method disclosed in FIGS. 28A through 28C can be used to construct the clean biochemistry fire break shown in FIG. 27, and the method disclosed in FIGS. 30A, 30B and 30C can be used to construct the clean chemistry wildfire break shown in FIG. 29.

[0433] FIG. 20D shows an exemplary graphical user interface for mobile application configured used by command center administrators to receive messages from users including property owners and contractors, requesting refills for clean anti-fire (AF) chemical liquid for GPS-specified spray system equipment.

[0434] FIG. 21 shows an exemplary fire hazard severity zone (FHSZ) map generated by the CAL FIRE® System in state responsibility areas of the State of California. Such maps can be used by the system network 1 to inform the strategic application of environmentally-clean anti-fire (AF) liquid spray using the system network of the present invention. Such maps also can be displayed on the mobile application 12 to provide greater awareness of risks created by wild fires in a specific region, at certain moments in time.

Specification of an Exemplary Anti-Fire (AF) Spray Protection Map Generated by the System Network of the Present Invention

[0435] FIG. 22 shows an exemplary GPS-specified anti-fire (AF) biochemical liquid spray protection map generated by the system network 1, showing properties, houses and buildings that were sprayed, and not-sprayed, with state/county-issued anti-fire liquid as of report date, 15 Dec. 2017. The system network will periodically update these AF chemical liquid spray protection maps (e.g. every 5 minutes or less) for display to users and neighbors to see whose property/land parcels and homes/building have been spray protected with anti-fire (AF) biochemical liquid of the present invention, and whose parcels and home/buildings have not been AF-spray protected against wild fires, so that they can or may volunteer to lend a helping hand in spray protecting their neighbors properties as time and anti-fire chemical supplies allow, to provide a stronger defense against one or more wild fires raging towards their neighborhood.

[0436] In accordance with the principles of the present invention, the application servers 9B supported by the system network 1 will automatically generate anti-fire (AF) chemical liquid spray-protection task reports, as illustrated in FIG. 23, based on the analysis of spray-protection maps as shown in FIG. 22, and based on many other kinds of intelligence collected by the system, and analyzed by human analysts, as well as artificial intelligence (AI) expert systems. Based on such automated intelligence efforts, the application servers 9B will generate periodically, and as needed, AF chemical liquid (AFCL) Spray Command Program files containing GPS/Time-Frame-indexed commands and instructions that are wirelessly transmitted to assigned GPS-tracked anti-fire (AF) chemical liquid spraying systems 30, 40, 50, 60, 70, 80, 90, and 110 so that the operators of such GPS-tracked biochemical liquid spraying systems will know when and where to mist and/or spray AF biochemical liquid over and one certain GPS-specified properties, in their effort to defend against the threat of wild fires.

[0437] The AFCL Spray Command Program files, containing GPS-indexed commands and instructions, generated by the application servers 9B are transmitted over the system network 1 to the numerous deployed GPS-tracked AF liquid spraying systems 30, 40, 50, 60, 70, 80, 90, and 110 so as to orchestrate and choreograph the spray application of clean anti-fire (AF) chemical liquid over GPS-specified properties, before and during the presence of wild fires, so as to implement an orchestrated strategic and collective defense against wild fires that break out for various reasons, threatening states, counties, towns, neighborhoods homes, business, and human and animal life.

[0438] In some embodiments, the application servers 9B will generate and issue AFCL Spray Command Program files that are transmitted to specific GPS-tracked AF liquid spraying systems 30, 40, 50, 60, 70, 80, 90, and 110 and containing automated instructions (i.e. commands) on when and where (i.e. in terms of time frame and GPS-specified coordinates) the GPS-tracked AF liquid spraying systems should automatically apply, via spraying operations, clean AF biochemical liquid on GPS-specified property during their course of movement over land. During such spraying operations, the system network 1 will automatically meter, dispense and log how much clean AF chemical liquid has been sprayed over and on certain GPS-specified properties. Real-time wind-speed measurements can be made and used to compensate for spraying operations in real-time, as may be required under certain weather conditions.

[0439] In other embodiments, the application servers 9B will generate and issue AFCL Spray Command Program files that are transmitted to other GPS-tracked AF liquid spraying systems 30, 40, 50, 60, 70, 80, 90, and 110 providing automated instructions (i.e. commands) on when and where the GPS-tracked AF liquid spraying systems should spray-apply clean AF chemical liquid on GPS-specified property during course of movement over land, but allowing the human operator to override such spraying instructions, and compensate and ensure greater accuracy, using human operator skill and judgment during spraying operations. While such spraying operations, the system will automatically meter, log and record all dispensed AF biochemical liquid sprayed over and over certain GPS-specified properties under the supervision and control of the human operator.



Specification of an Exemplary Anti-Fire Spray Protection Task Report Generated by the System of the Present Invention

[0440] FIG. 23 shows an exemplary GPS-specified anti-fire spray protection task report generated by the system network 1 for state/county xxx on 15 Dec. 2017, indicating which properties on what streets, in what town, county, state, requires the reapplication of AF chemical liquid spray treatment in view of factors such as weather (e.g. rainfall, sunlight) and passage of time since last spray application. Such task reports will be transmitted by the command center 19 to registered users, along with an SMS and/or email message to attend to the AF spray task, so the requested user will promptly spray protect their land parcels and home with clean AF chemical liquid, as conditions require or suggest, using the mobile/portable GPS-tracked AF liquid spraying system 20 assigned to the property owner, and deployed over the system network 1.

[0441] As contracted AF-spray operators, and home owners alike, protect properties and homes using the GPS-tracked AF liquid spraying systems (20, 30, 40, 50, 60 70, 80, 90, and 110) the system network 1 automatically receives GSM or other RF-based signals transmitted from the GPS-tracked anti-fire (AF) chemical liquid spraying systems, indicating that certain amounts of AF chemical liquid has been dispensed and sprayed from the system onto GPS-specified property. Notably, the amounts of AF chemical liquid dispensed and sprayed from the system over and onto GPS-specified property should closely match the amounts requested in the task report transmitted to the user, to achieve the AF spray protection task directed by AI-driven management processes supported by the wild fire suppression system network of the present invention.

Specification of New and Improved Wild Fire Suppression Methods in Accordance with Principles of the Present Invention

[0442] Having described the various GPS-tracked anti-fire (AF) chemical liquid spraying systems of the illustrative embodiments 20, 30, 40, 50 60, 70, 80, 90, and 110 shown in the Figure Drawings, and the various functions supported by the mobile application 12 supported by the data center 8 of the system network 1, it is appropriate at this juncture to now described the various new and improved wild fire suppression methods in accordance with principles of the present invention, each involving GPS-guided spray application of clean anti-fire (AF) chemical liquid having a chemistry that works to break a wild fire by interfering with the free-radicals produced during the combustion phase of a ranging wild fire. The benefits and advantages provided by such new and improved methods will become apparent hereinafter.

Specification of a Method of Suppressing a Wild Fire Raging Across a Region of Land in the Direction of the Prevailing Winds

[0443] FIG. 24 shows a plan view of a wild fire 70 emerging from a forest region 71A and approaching a neighboring town 72 surrounded by other forest regions 71B, 71B and 71C, and moving in the direction determined by prevailing winds, indicated by a pair of bold arrows. This example closely resembles the pathway of many wild fires recently destroying countless acres of land (i.e. real property) in the State of California in 2017.

[0444] FIG. 25 illustrates the various steps involved in carrying out the method of suppressing a wild fire raging across a region of land. Specifically, the method involves forming a multi-stage anti-fire chemical fire-break system illustrated in FIG. 25 using the remotely-managed GPS-controlled application of both anti-fire (AF) liquid mist streams and AF chemical liquid spray streams from ground and air based GPS-tracked anti-fire (AF) liquid spray vehicles, as illustrated for example in FIGS. 8A through 17B.

[0445] As illustrated in FIG. 25, the method generally involves: (a) applying, prior to the wild fire reaching the specified target region of land 74, a low-density anti-fire (AF) liquid mist stream in advance of the wild fire 75 so as to form a fire stall region 76, while providing a non-treated region 77 of sufficient size between the front of the wild fire 75 approaching the target region of land 73 and the fire stall region 76; and (b) applying a high-density anti-fire (AF) liquid spray stream in advance of the wild fire 75 to form a fire break region 74 beyond and contiguous with the fire stall region 76, and also continuous with the target region 73 to be protected from the wild fire.

[0446] As illustrated in FIG. 25, the fire stall region 76 is formed before the wild fire reaches the fire stall region 76. The fire stall region 76 operates to reduce the free-radical chemical reactions raging in the wild fire 75. This fire stall region 76 helps to reduce the destructive energy of the wild fire by the time the wild fire reaches the fire break region 74, and enabling the fire break region 74 to operate and significantly break the free radical chemical reactions in the wild fire 75 when the wild fire reaches the fire break region 74. This helps to suppress the wild fire 75 and protect the target region of land 73.

[0447] FIGS. 26A and 26B describe the method of suppressing a wild fire raging towards a target region of land 73 (and beyond) in a direction determined by prevailing winds and other environmental and weather factors, as illustrated in FIG. 25. Typically, the system used to practice this method of the present invention will employ a centralized GPS-indexed real-property/land database system 7 shown in FIG. 4 containing GPS-indexed maps of all land regions under management and fire-protection, developed using methods, equipment and services known in the GPS mapping art. Such GPS-indexed maps will contain the GPS coordinates for the vertices of each and every parcel in any given state, county and town in the country in which system is deployed. As shown in FIG. 4A, this central GPS-indexed real property database 7 will be operably connected to the TCP/IP infrastructure 10 of the Internet, and accessible by system network 1 of the present invention.

[0448] As indicated at Block A in FIG. 26A, prior to the wild fire reaching the specified target region of land, a GPS-tracked AF spray vehicle 50 as shown for example in FIG. 11A applies a low-density anti-fire (AF) liquid mist 80 in advance of the wild fire so as to form a fire stall region 76 while providing a non-treated region 77 of sufficient size between the front of the wild fire approaching the target region of land 73 and the fire stall region 76. The fire stall region 76 is formed by a first GPS-guided aircraft system flying over the fire stall region during multiple passes and applying the low-density AF chemical liquid mist 80 over the fire stall region 76. The non-treated region 77 is defined by a first set of GPS coordinates  $\{GPS_1(x,y)\}$  and, the fire stall region 76 is defined by a second set of GPS coordinates



$\{\text{GPS}_2(x,y)\}$ . Each of these regions are mapped out using global positioning system (GPS) methods, the GPS-indexed land database system 7, drone-type aircraft systems 40 as shown in FIG. 10A, and space-based land-imaging satellites 14 having multi-spectral imaging capabilities, and operably connected to the infrastructure of the Internet. When used alone and/or together, these systems are capable of capturing real-time intelligence on the location and spread of a particular wild fire, its direction of propagation, intensity and other attributes. This captured data is provided to application servers in the data center 8 which, in turn, generate GPS coordinates determining the planned pathways of the GPS-traced AF chemical liquid spraying/misting aircraft systems, to provide the anti-fire protection over the GPS-indexed fire stall region 76 and GPS-specified non-treated region 75, as described in greater detail below.

[0449] As indicated at Block B in FIG. 26A, a second GPS-tracked AF spray vehicle 50 as shown in FIG. 11A, or other suitable spraying vehicle deployed on the system network, applies a high-density anti-fire (AF) liquid spray 81 over the land in advance of the wild fire to form a GPS-specified fire break region 74 beyond and contiguous with the GPS-specified fire stall region 76. The fire break region 74 is formed by the second GPS-guided aircraft flying over the fire break region 74 during multiple passes and applying the high-density AF chemical liquid spray 81 over the fire break region 74. The fire break region 74 is defined by a third set of GPS coordinates  $\{\text{GPS}_3(x,y)\}$  mapped out using global positioning system (GPS) methods, the GPS-indexed land database system 7, drone-type aircraft systems as shown in FIG. 8A, and/or space-based land-imaging satellites 14 having multi-spectral imaging capabilities, and operably connected to the infrastructure of the Internet. When used alone and/or together, these systems are capable of capturing real-time intelligence on the location and spread of a particular wild fire, its direction of propagation, intensity and other attributes. This captured data is provided to application servers in the data center 8 which, in turn, generate GPS coordinates determining the planned pathways of the GPS-traced AF chemical liquid spraying/misting aircraft systems, to provide the anti-fire protection over GPS-specified fire break region 74, as described in greater detail below.

[0450] As indicated at Block C in FIG. 26B, the fire stall region 76 is formed before the wild fire 75 reaches the fire stall region 76, and operates to reduce the free-radical chemical reactions raging in the wild fire so as to reduce the destructive energy of the wild fire by the time the wild fire 75 reaches the fire break region 74, and enabling the fire break region 74 to operate and significantly break the free radical chemical reactions in the wild fire 75 when the wild fire reaches the fire break region 74, and thereby suppress the wild fire 75 and protect the target region of land 73 and beyond.

Specification of a Method of Reducing the Risks of Damage to Private Property Due to Wild Fires by Managed Application of Anti-Fire (AF) Liquid Spray

[0451] FIG. 27 illustrates a method of reducing the risks of damage to private property due to wild fires by managed application of anti-fire (AF) liquid spray. FIGS. 28A, 28B and 28C illustrates a method of reducing the risks of damage to private property due to wild fires by managed application of anti-fire (AF) liquid spray. Typically, this method is

carried out using the system network of FIG. 4A and any one or more of the GPS-tracked anti-fire (AF) liquid spray vehicle systems 14A through 14D represented in FIG. 4A and illustrated in FIGS. 8A through 17B.

[0452] As indicated at Block A in FIG. 28A, the system registers each GPS-specified parcel of private real property in a specified County and State, which may or may not have buildings constructed thereon, and identifying the owner and tenants, as well as all pets, vehicles and watercrafts associated with the registered parcel of private property. Typically, the system will request the address of the property parcel, and will automatically determine its GPS coordinates that specify the vertices of the parcel using databases, and data processing methods, equipment and services, known in the GPS mapping art.

[0453] As indicated at Block B in FIG. 28A, the system collects intelligence relating to the County, risks of wild fires in the surrounding region, and historical data maintained in a network database, and generating GPS-specified anti-fire (AF) spray protection maps and task reports for execution.

[0454] As indicated at Block C in FIG. 28A, an AF chemical liquid spraying system is provided to a GPS-specified location for spraying one or more registered parcels of private property with AF chemical liquid spray.

[0455] As indicated at Block D in FIG. 28A, a supply of AF chemical liquid spray is provided to the GPS-specified location of the AF chemical liquid spraying system.

[0456] As indicated at Block E in FIG. 28A, the AF chemical liquid spraying system is provided with the supply of AF chemical liquid,

[0457] As indicated at Block F in FIG. 28B, based on the GPS-specified anti-fire (AF) spray protection maps and task reports, the system issues orders to the private property owner, or its contractor, to apply AF chemical liquid spray on the private property using the AF chemical liquid spraying system.

[0458] As indicated at Block G in FIG. 28B, the private property owner executes the order and applies AF chemical liquid spray on the private property using the AF chemical liquid spraying system, and the system remotely monitors the consumption and application of AF chemical liquid at the private property on a given time and date, and automatically records the transaction in the network database 9C prior to the arrival and presence of wild fire in the region.

[0459] As indicated at Block H in FIG. 28B, the system updated the records in the network database associated with each application of AF chemical liquid spray on a GPS-specified parcel of private property.

[0460] As indicated at Block I in FIG. 28B, the system scheduled the next application of AF chemical liquid spray on the GPS-specified parcel of private property, factoring weather conditions and the passage of time.

[0461] As indicated at Block J in FIG. 28B, the system issues another order to the GPS-specified parcel of private property to re-apply AF chemical liquid spray on the private property to maintain active wild fire protection.

[0462] As indicated at Block K in FIG. 28C, the property owner executes (i.e. carries out) the order to reapply AF chemical liquid spray on the parcel of private property using the AF chemical liquid spraying system, and the system remotely monitors the application of AF chemical liquid at the private property on a given time and date, and records this transaction in the network database 9C.



[0463] As indicated at Block L in FIG. 28C, the system updates records on AF chemical liquid spray application in the network database 9C associated with reapplication of AF chemical liquid on the parcel of private property.

[0464] As indicated at Block M in FIG. 28C, the system schedules the next application of AF chemical liquid spray on the parcel of private property, factoring weather conditions and the passage of time.

Specification of a Method of Reducing the Risks of Damage to Public Property Due to Wild Fires, by Managed Spray Application of Fire Inhibiting Biochemical Liquid to Ground Cover and Building Surfaces Prior to the Arrival of Wild Fires

[0465] FIG. 29 illustrates a method of reducing the risks of damage to public property due to wild fires, by managed spray application of AF chemical liquid to ground cover and building surfaces prior to the arrival of wild fires. FIGS. 23A, 23B and 23C illustrate a method of reducing the risks of damage to public property due to wild fires by managed application of anti-fire (AF) liquid spray. Typically, this method is carried out using the system network of FIG. 4A and any one or more of the GPS-tracked anti-fire (AF) liquid spray vehicle systems 14A through 14D represented in FIG. 4A and shown in FIGS. 8A through 17B.

[0466] As indicated at Block A in FIG. 30A, each GPS-specified parcel of public real property in a specified County and State is registered with the system. Such parcels of property may or may not have buildings constructed thereon. As part of registration with the system network 1, supported by the network database 9C, it is necessary to identify the owner and tenants, as well as all pets, vehicles and watercrafts associated with the registered parcel of public property. Typically, the system will request the address of the property parcel, and will automatically determine its GPS coordinates that specify the vertices of the parcel using databases, and data processing methods, equipment and services, known in the GPS mapping art.

[0467] As indicated at Block B in FIG. 30A, the system collects various kinds of intelligence relating to the County, risks of wild fires in the surrounding region, and historical weather and related data maintained in a network database 9C, and generates GPS-specified anti-fire (AF) spray protection maps and task reports for review and execution, along with GPS-specified spray plans (e.g. flight plans) for GPS-tracked anti-fire (AF) liquid spray vehicle systems 30 and 60, and GPS-specified spray plans.

[0468] As indicated at Block C in FIG. 30A an AF chemical liquid spraying system is provided to a GPS-specified location for spraying one or more registered parcels of public property with AF chemical liquid spray.

[0469] As indicated at Block D in FIG. 30A, a supply of AF chemical liquid spray is provided to the registered location of the AF chemical liquid spraying system.

[0470] As indicated at Block E in FIG. 30A, the AF chemical liquid spraying system is filled with the provided supply of AF chemical liquid.

[0471] As indicated at Block F in FIG. 30, based on the anti-fire (AF) spray protection maps and task reports, the system issues orders to the public property owner, or its contractor, to apply AF chemical liquid spray on the public property using the AF chemical liquid spraying system 60.

[0472] As indicated at Block G in FIG. 30B, the public property owner executes the order and applies AF chemical

liquid spray on the public property using the AF chemical liquid spraying system, and the system remotely monitors the consumption and application of AF chemical liquid at the public property on a given time and date, and automatically records the transaction in the network database prior to the presence of wild fire in the region.

[0473] As indicated at Block H in FIG. 30B, the system updates records in the network database 9C associated with each application of AF chemical liquid spray on a GPS-specified parcel of public property.

[0474] As indicated at Block I in FIG. 30B, the system schedules the next application of AF chemical liquid spray on the GPS-specified parcel of public property, factoring weather conditions and the passage of time.

[0475] As indicated at Block J in FIG. 30B, the system issues another order to the GPS-specified parcels of public property to re-apply AF chemical liquid spray on the public property to maintain active fire protection.

[0476] As indicated at Block K in FIG. 30C, the property owner executes the order to reapply AF chemical liquid spray on the GPS-specified parcels of public property using the AF chemical liquid spraying system, and the system remotely monitors the application of AF chemical liquid at the public property on a given time and date, and records this transaction in the network database 9C.

[0477] As indicated at Block L in FIG. 30C, the system updates records on AF chemical liquid spray application in the network database 9C associated with reapplication of AF chemical liquid on the GPS-specified parcels of public property.

[0478] As indicated at Block M in FIG. 30C, the system schedules the next application of AF chemical liquid spray on the GPS-specified parcels of public property, factoring weather conditions and the passage of time.

Specification of a Method of Remotely Managing the Application of Anti-Fire (AF) Liquid Spray to Ground Cover and Buildings so as to Reduce the Risks of Damage Due to Wild Fires

[0479] FIG. 31 is a graphical illustration showing a method of remotely managing the application of anti-fire (AF) liquid spray to ground cover and buildings so as to reduce the risks of damage due to wild fires. FIGS. 32A and 32B describes the high level steps carried out by the method in FIG. 24 to reduce the risks of damage due to wild fires. Typically, this method is carried out using the system network of FIG. 4A and any one or more of the GPS-tracked anti-fire (AF) biochemical liquid spray vehicle systems 14A-14D represented in FIG. 4A and shown in FIGS. 8A, through 17B.

[0480] As indicated at Block A in FIG. 32A, the system registers each GPS-specified parcel of real property in a specified County and State, which may or may not have buildings constructed thereon, and identifying the owner and tenants, as well as all pets, vehicles and water crafts associated with the registered parcel of real property. Typically, the system will request the address of the property parcel, and will automatically determine (or estimate) its GPS coordinates that specify the vertices of the parcels using databases, and data processing methods, equipment and services, known in the GPS mapping art. The GPS address of each parcel will be stored in the centralized GPS-indexed land database system 7 shown in FIG. 4



[0481] As indicated at Block B in FIG. 32A, the system collects intelligence relating to the County, risks of wild fires in the surrounding region, and historical data maintained in a network database, and generates GPS-specified anti-fire (AF) spray protection maps and task reports for execution.

[0482] As indicated at Block C in FIG. 32A, an AF chemical liquid spraying system is provided to a GPS-specified location for spraying the GPS-specified parcels of real property with AF chemical liquid spray.

[0483] As indicated at Block D in FIG. 32A, a supply of AF chemical liquid spray is provided to the GPS-specified location of the AF chemical liquid spraying system.

[0484] As indicated at Block E in FIG. 32A, the AF chemical liquid spraying system is filled with the provided supply of AF chemical liquid.

[0485] As indicated at Block F in FIG. 32B, prior to the arrival of a wild fire to the region, and based on the anti-fire (AF) spray protection maps generated by the system, the system issues a request to property owners, or their registered contractors, to apply AF chemical liquid spray on GPS-specified properties using deployed AF chemical liquid spraying systems.

[0486] As indicated at Block G in FIG. 32B, in response to the issued request, the property owner or contractor thereof applies AF chemical liquid spray on the real property using the AF chemical liquid spraying system, and the system remotely monitors the consumption and application of the AF biochemical liquid on the property on a given date, and automatically records the transaction in the network database.

[0487] As indicated at Block H in FIG. 32B, the system updates records in the network database associated with each application of AF chemical liquid spray on one or more GPS-specified parcels of real property.

[0488] In the illustrative embodiment, the fire inhibiting biochemical liquid of the present invention is used when practicing the present invention. A liquid dye of a preferred color can be added to biochemical liquid to help visually track where AF chemical liquid has been sprayed during the method of wild fire suppression. However, in some applications, it may be desired to maintain the AF biochemical liquid in a clear state, and not employ a colorant.

Method of and Apparatus for Applying Fire and Smoke Inhibiting Slurry Compositions on Ground Surfaces Before the Incidence of Wild-Fires, and Also Thereafter, Upon Smoldering Ambers and Ashes to Reduce Smoke and Suppress Fire Re-Ignition

[0489] FIGS. 33A, 33B and 33C show the clean fire and smoke inhibiting slurry spray application vehicle 500 carrying a high-capacity (e.g. 3000 gallon) stainless steel mixing tank 93 with an integrated agitator mechanism (e.g. motor driven mixing paddles) 94, and a hydraulic pumping apparatus and spray nozzle 101 for mixing and spraying the environmentally-clean aqueous-based clean fire and smoke inhibiting slurry 102 (i) on ground surfaces to create CFIC-based fire breaks (105) around regions to be protected from wildfires as illustrated in FIGS. 30 and 31, (ii) to cover smoldering ambers and ash after the present of wildfires to reduce toxic waste water runoff and smoke production as shown in FIG. 40, and (iii) on burning fires destroying buildings as well as outdoor combustion material as shown in FIG. 41.

[0490] FIG. 34 shows the clean fire and smoke inhibiting slurry spray application vehicle 500 comprising: a mobile slurry mixing and spray vehicle chassis 91 having a propulsion and transport subsystem 92, with a vehicle chassis supporting a high-capacity (e.g. 3000 gallon) stainless steel mixing tank 93, with an integrated agitator mechanism (e.g. motor driven mixing paddles) 94, and having a filling chute 93A through which slurry ingredients (e.g. thermally processed wood fibers, cellulose fibers, wetting agents, tacking agents 96, and a supply of clean fire inhibiting biochemical liquid 97 of the present invention as taught herein; a water pumping subsystem 99 for pumping water 98 from an external source into the mixing tank 93 to blend with the chemicals and fiber material 96 and CFIC material 97, and produce an environmentally-clean fire and smoke inhibiting mixture 102; a hydraulic pumping apparatus and spray nozzle 101, for mixing and spraying the clean aqueous-based clean fire and smoke inhibiting slurry mixture 102 (i) on ground surfaces to create CFIC-based fire breaks around regions to be protected from wildfires, (ii) over smoldering ambers and ash after the present of wildfires to reduce toxic waste water runoff and smoke production, and (iii) on active burning fires in buildings and/or burning land and brush. As shown, the vehicle system 500 includes A GPS receiver and controls 100 for controlling apparatus specified by 91, 92, 93, 94, 98, and 101.

[0491] The system 500 also includes a second CFIC liquid tank 112 for storing a secondary CFIC liquid 113, and supplying an air-less spray system 111 for spraying CFIC liquid 113 using a spray nozzle applicator 111A. The spray applicator 112 can be mounted on the vehicle 90, alongside or in tandem with primary slurry spray nozzle 101A, or it can be via connected to a reel of hose for application of CFIC liquid 113 to the surface of the slurry coating 102 after it has been applied to the ground surface. Preferably, biochemical liquid spray 113 will be provided with a colored dye to assist in spray application over the fire and smoke inhibiting slurry 102. By providing a vehicle 90 with two tanks, one tank 93 containing the slurry mixture 102, and the other tank 112 containing a CFIC liquid 113, the system 90 has an added capacity to suppress fire and smoke created by wildfires, and other sources of fire.

[0492] FIG. 35 describes the method of applying fire and smoke inhibiting slurry compositions of the present invention on ground surfaces before the incidence of wild-fires, and also thereafter, upon smoldering ambers and ashes to reduce smoke and suppress fire re-ignition.

[0493] As indicated at Block A in FIG. 35, the first of the method involves measuring and staking out area using GPS coordinates to ensure proper application rates.

[0494] As indicated at Block B in FIG. 35, the processed wood fibers, cellulose fiber, wetting agents, tackling agents 96, and clean fire inhibiting biochemicals (CFIC) 97 are blended with a supply of water 98 to make up a fire and smoke inhibiting slurry composition 102.

[0495] In the illustrative embodiment, the processed wood fibers, cellulose fiber, wetting agents, tackling agents 96 can be provided in a number of different ways and formulations. For example, one can use Hydro-Blanket® Bonded Fiber Matrix (BFM) from Profile Products, which combines Profile Product's Thermally Refined® wood fiber and multi-dimensional pacifiers for greater water-holding capacity. This BFM anchors intimately to the soil through proprietary cross-linked, hydro-colloidal pacifiers and activators and is



completely biodegradable and non-toxic. When Hydro-Blanket® Bonded Fiber Matrix is blended and mixed with CFIC 97, and water 98, the slurry compositing 102 sprays on as mulch, but dries to form a breathable blanket that bonds more completely with the soil. Thermally Refined® wood fiber starts with 100% recycled wood chips which are thermally processes to create fine, long and highly absorbent fibers, engineered fibers are the source for Profile's superior: yield and coverage; water-holding capacity; growth establishment; wet-bond strength; and erosion control performance. Profile Products offers other brands of wood, cellulose, wood-cellulose blended hydraulically-applied mulches which are preblended with one or more performance enhancing additions.

**[0496]** Because paper does not hold as much moisture, and does not prevent erosion nearly as well as thermally refined wood fiber mulch, many states and provinces have prohibited the use of paper mulch. Large-scale independent testing has shown that paper mulch is only 25% effective at preventing erosion, whereas wood fiber mulch with no performance enhancing additives is 45% effective at preventing erosion. ASTM standard testing methods also indicate that wood fiber mulches are superior to paper at promoting vegetation establishment. In addition, where steeper or longer slopes exist, and where greater erosion protection is required (greater than 50% effective), there are advanced technologies, beyond basic paper and wood fiber mulches, that are indicated to ensure erosion prevention and vegetation establishment.

**[0497]** Examples of preblended mulch materials from Profile Products which may be used to practice the manufacture of the fire and smoke inhibiting slurry mixtures of the present invention 102, include the following wood-based and paper-based mulches described below. The Base Hydraulic Mulch Loading Chart shown in FIG. 36 can be used to estimate how much Profile® brand mulch fiber products (e.g. packaged in 50 lb. bales) will be required to make a fire and smoke inhibiting slurry 102 of the present invention for use on particular incline ground surfaces, of particular slope lengths, over particular surface areas (e.g. in acres). The Hydraulic Loading Chart shown in FIG. 36 for Profile® mulch fiber products provides the required hydraulic loading for specified application rates required by specific Profile® brand mulch fiber materials used on particular slopes, and provided for three specific application rates, namely 1500 lb./acre, 2000 lb./acre, and 2500 lb./acre.

#### Wood Fiber Mulch

**[0498]** Materials: 100% wood fiber, made from thermally processed (within a pressurized vessel) wood fiber heated to a temperature greater than 380 degrees Fahrenheit (193 degrees Celsius) for 15 minutes at a pressure greater than 80 psi (552 kPa) and dark green marker dye.

Moisture Content: 12%+/-3%

**[0499]** Water-Holding Capacity: 1,100% minimum  
Approved Large-Scale Erosion Control Effectiveness: 45% minimum.

When comparing the four base paper and wood mulches listed below, the key items to note are the differences in the maximum slope inclinations, slope lengths and the erosion prevention capabilities.

#### Cellulose (Paper) Fiber Mulch

**[0500]** Maximum slope inclination: 4:1  
Appl. rate on maximum slope: 1,500-2,000 pounds/acre  
Maximum slope length\*: 18 feet  
Functional longevity: up to 3 months  
Erosion control effectiveness: 25%  
Cellulose (Paper) Fiber Mulch with Tackifier  
Maximum slope inclination: 4:1  
Appl. rate on maximum slope: 1,500-2,000 pounds/acre  
Maximum slope length\*: 20 feet  
Functional longevity: up to 3 months  
Erosion control effectiveness: 30%

#### Wood Fiber Mulch

**[0501]** Maximum slope inclination: 2:1  
Appl. rate on maximum slope: 3,000 pounds/acre  
Maximum slope length\*: 28 feet  
Functional longevity: up to 3 months  
Erosion control effectiveness: 45%  
Wood Fiber Mulch with Tackifier  
Maximum slope inclination: 2:1  
Appl. rate on maximum slope: 3,000 pounds/acre  
Maximum slope length\*: 30 feet  
Functional longevity: up to 3 months  
Erosion control effectiveness: 50%  
\*Maximum slope length is based on a 4H:1V slope. For applications on steeper slopes, the maximum slope length may need to be reduced based on actual site conditions. If greater than 50% erosion prevention effectiveness is desired, then the technologies should be specified and not the four base mulch products listed above.

#### Stabilized Mulch Matrix (SMM)

**[0502]** Maximum slope inclination: 2:1  
Appl. rate on maximum slope: 3,500 pounds/acre  
Maximum slope length\*\*: 50 feet  
Minimum cure time: 24 hours  
Functional longevity: 3 to 6 months  
Erosion control effectiveness: 90%

#### Bonded Fiber Matrix (BFM)

**[0503]** Maximum slope inclination: 1:1  
Appl. rate on maximum slope: 4,000 pounds/acre  
Maximum slope length\*\*: 75 feet  
Minimum cure time: 24 hours  
Functional longevity: 6 to 12 months  
Erosion control effectiveness: 95%

#### Engineered Fiber Matrix™ (EFM)

**[0504]** Maximum slope inclination: >2:1  
Appl. rate on maximum slope: 3,500 pounds/acre  
Maximum slope length\*\*: 50 feet  
Minimum cure time: 24-48 hours  
Functional longevity: Up to 12 months  
Erosion control effectiveness: >95%

#### High Performance-Flexible Growth Medium™ (HP-FGM™)

**[0505]** Maximum slope inclination: >1:1  
Appl. rate on maximum slope: 4,500 pounds/acre  
Maximum slope length\*\*: 100 feet  
Minimum cure time: 2 hours\*



Functional longevity: 12 to 18 months  
Erosion control effectiveness: 99.9%

Extended-Term Flexible Growth Medium (ET-FGM)

**[0506]** Maximum slope inclination: >1:1  
Appl. rate on maximum slope: 4,500 pounds/acre  
Maximum slope length\*\*: 125 feet  
Minimum cure time: 2 hours\*  
Functional longevity: 18 to 24 months  
Erosion control effectiveness: 99.95%

**[0507]** Profile Product's HP-FGM and ET-FGM mulches have very short cure times, and therefore, fire and smoke inhibiting slurry mixtures, employing these mulches, can be applied onto wet soils and during a light rainfall. Maximum slope length is based on a 3H:1V slope. For applications on steeper slopes, the maximum slope length may need to be reduced based on actual site conditions.

**[0508]** In applications where the fire and smoke inhibiting slurry **102** is applied onto smoldering ashes and ambers of houses destroyed by wildfires, there slope will be generally zero. However, alongside roads and embankments, where wildfires may travel, following specified application rates for specified ground slopes should be followed for optimal performance and results.

**[0509]** In the illustrative embodiments, the CFIC liquid component **97**, added to the fire and smoke inhibiting slurry mixture **102**, will be realized using biochemical clean anti-fire inhibiting biochemical liquid compositions specified in FIGS. 6A1 through 6C2, and described in detail above.

**[0510]** When blending the fire inhibiting biochemical liquid composition **97** with Profile's hydraulic mulch fiber products in the mixing tank **93**, the following mixture ratio should be used for biochemical liquid **97**: about 1 gallon of biochemical liquid composition per 10 gallons of water added to the mixing tank **93** during the blending and mixing of the fire and smoke inhibiting slurry **102**. So, as shown in FIG. 30, when mixing 2800 gallons of water to 1450 lbs. of mulch fiber (29x50 lb Profile® mulch fiber bales) to make a batch of fire and smoke inhibiting slurry **102**, at least 280 gallons of biochemical liquid **97** will be added to the mixing tank **93** and mixed well with the 2800 gallons water and 1450 lbs. of mulch fiber, preferably from Profile Products, LLC of Buffalo Grove, Ill., when using the 1500 lb./acre application rate.

**[0511]** However, additional amounts of biochemical liquid **97** can be added to the 2800 gallons of water so as to increase the amount of fire inhibiting biochemical liquid that infuses into the surface of the mulch fibers when being mixed within the mixing tank **93** during the blending and mixing steps of the process. Notably, a large percentage of the water in the mixing tank **93** will function as a hydraulic carrier fluid when spraying biochemical liquid infused mulch fibers in the slurry mixture to the ground surface being coated during spray applications, and thereafter, this water will quickly dry off when curing under the hot Sun, leaving behind infused fire inhibiting biochemicals (e.g. potassium citrate salt crystal structures) embodied within the mulch fibers to provide during proactive fire protection.

**[0512]** As indicated at Block C in FIG. 35, the blended fire and smoke inhibiting slurry mixture is mixed in the mixing tank **93** on the mobile vehicle **500** supporting hydraulic spray equipment **101**.

**[0513]** As indicated at Block D in FIG. 35, the mixed fire and smoke inhibiting slurry mixture **102** is then hydraulically

sprayed on the specific ground surface using hydraulic spray equipment **101** supported on the mobile spray vehicle **500**. The slurry spray process can be guided by GPS coordinates of the staked out ground surface regions, using GPS receiver and controls **100**.

**[0514]** As indicated at Block E in FIG. 35, a secondary biochemical liquid **113** is sprayed over the fire and smoke inhibiting slurry coating **102** after it has been hydraulically sprayed onto the ground. Once the slurry coating **102** has dried, and adheres to the ground surface, it will provide erosion control, as well as fire protection and smoke reduction in the presence of a wildfire in accordance with the scope and spirit of the present invention.

**[0515]** FIG. 37 shows a neighborhood of houses surrounded by a high-risk wildfire region. As shown, a wild-fire break region **105A** is sprayed on the ground surface region all around a neighborhood of houses, using the clean fire and smoke inhibiting slurry composition of the present invention **102** hydraulically sprayed onto the ground surface.

**[0516]** FIG. 38 shows a highway surrounded by high-risk wildfire regions on both sides of the highway. As shown, the wild-fire break regions **105A** on both sides of the highway are sprayed using the clean fire and smoke inhibiting slurry composition **102** hydraulically sprayed from the vehicle **500** onto the ground surface. Spray operators can stand on top of the platform above the mixing tank **93** and use the mounted spray gun to coat the ground surface with the wet slurry mixture **102**. Fire inhibiting biochemical liquid of the present invention **113** can then be sprayed upon the surface of the slurry coating **102** on the ground, if and as desired by the application at hand. By applying the clean fire and smoke inhibiting slurry composition **102** over a smoldering fire, followed with an biochemical spray coating, this double coating functions like a blanket for chemically breaking the combustion phase of a traveling wildfire and reducing smoke, and the need for water reduced to prevent reignition to neighboring areas.

**[0517]** FIG. 40 shows a house that just burned to the ground after a wildfire passed through an unprotected neighborhood. As shown, the clean fire and smoke inhibiting slurry composition **102** is sprayed over the glowing ambers and fire ash to suppress and prevent re-ignition of the fire, and reduce the production of smoke and creation of toxic water runoff during post fire management operations. Spray operators can stand on top of the platform above the mixing tank **93** and use the mounted spray gun to coat the ground surface with the wet slurry mixture **102**. The biochemical liquid **113** can then be sprayed upon the surface of the slurry coating **102** on hot glowing ambers and ashes. By applying the clean fire and smoke inhibiting slurry composition **102** over a smoldering fire, followed with a biochemical spray coating, this double coating functions like a blanket for chemically breaking the combustion phase of a traveling wildfire and reducing smoke and the need for water to prevent reignition to neighboring areas.

**[0518]** FIG. 41 shows a house or building that is burning due to a fire within the building. As shown, the wet fire and smoke inhibiting slurry composition of the present invention **102** is hydraulically sprayed on and over the fire in effort to suppress the fire and reduce the production of smoke. In some applications, this method may be effective in fire and smoke suppression using a minimal amount of water.



Specification of the Automated Wildfire Ember Detection and Suppression System/Module of Present Invention

[0519] FIG. 42A shows a wildfire ember detection module 604A mounted on the top of each building 300. Each wildfire ember detection module 604A is configured in the wireless wildfire ember detection and notification network 600, for (i) receiving wildfire alerts and messages from neighboring modules 604A, (ii) sensing and processing IR thermal images for automated detection of wildfires and wildfire embers in the field of views (FOVs) of the module, (iii) sending and recording the CO<sub>2</sub> levels in the ambient air, (iv) measuring and recording the relative humidity (%) in the ambient air, (v) measuring and recording the temperature of the ambient air, and measuring and recording other parameters relating to the ambient environment which may be helpful in automated detection of wildfires and wildfire ember storms, so the anti-fire misting systems installed on property can be timely triggered to protect the building and property when a wildfire storm rages across the property. The advantage of being part of this network is that each module 4A can scout for wildfires and alert other modules in the network in terms of GPS coordinates so that the specific properties can timely prepare for any such wildfire outbreaks in the vicinity.

Specification of the Wireless GPS-Tracked Wildfire Ember Detection and Notification Network Employing the Wildfire Ember Detection and Suppression Systems of the Present Invention

[0520] FIG. 42A shows the wireless GPS-tracked Wildfire ember detection and notification network 604 employing with the Wildfire ember detection and suppression systems 4A. Each wireless GPS-tracked wildfire ember detection module 4A, deployed in the wireless Wildfire ember detection and notification network 604 comprises: a fire-protective housing cover 4A1; and various sensors and signal and data processing and storage components 4A2 through 4A19, shown in schematic block diagram of FIG. 42B.

[0521] As shown in FIG. 42B, the sensors and signal and data processing and storage components arranged and configured about a microprocessor 4A20 and flash memory (i.e. control subsystem) 4A21 include: one or more passive infra-red (PIR) thermal-imaging sensors 4A2 connected together with suitable IR optics to project IR signal reception field of view (FOV) before the IR receiving array; multiple pyrometric sensors 4A3 for detecting the spectral radiation of burning, organic substances such as wood, natural gas, gasoline and various plastics; a GPS antenna 4A4; a GPS signal receiver 4A5; voltage regulator 4A6; an Xbee antenna 4A7; an Xbee radio transceiver 4A8; a voltage regulator 4A9; an external power connector 4A10; a charge controller 4A11; a battery 4A12; thermistors 4A13; a power switch 4A14; a voltage regulator 4A15; external and internal temperature sensors 4A16; power and status indicator LEDs 4A17; programming ports 4A18; a digital/video camera 4A19; and other environment sensors adapted for collecting and assessing building intelligence, in accordance with the spirit of the present invention. Alternatively, the wildfire detection module 4A and wireless wildfire intelligence network 4 can be realized using the technical disclosure of U.S. Pat. No. 8,907,799, incorporated herein by reference.

[0522] In the illustrative embodiment, the wildfire ember detection system 604A supports a computing platform,

network-connectivity (i.e. IP Address), and is provided with native application software installed on the system as client application software designed to communicate over the system network and cooperate with application server software running on the application servers of the system network, thereby fully enabling the functions and services supported by the system, as described above. In the illustrative embodiment, a wireless mess network is implemented using conventional IEEE 802.15.4-based networking technologies to interconnect these wireless subsystems into subnetworks and connect these subnetworks to the internet infrastructure of the system of the present invention.

[0523] Preferably, the optical bandwidth of the IR sensing arrays 4A2 used in the thermal sensors will be adequate to perform 360 degrees thermal-activity analysis operations, and automated detection of wildfire and wildfire embers. Specifically, thermal sensing in the range of the sensor can be similar to the array sensors installed in forward-looking infrared (FLIR) cameras, as well as those of other thermal imaging cameras, use detection of infrared radiation, typically emitted from a heat source (thermal radiation) such as fire, to create an image assembled for video output and other image processing operations to generate signals for use in early fire detection and elimination system of the present invention.

[0524] Pixel processing algorithms known to those skilled in the art will be used to automatically process captured and buffered pixels from different color channels and automatically determine the presence of fire, wildfire and flying embers within the field of view (FOV) of the wildfire ember detection module 604A. Reference can be made to “Automatic Fire Pixel Detection Using Image Processing: A Comparative Analysis of Rule-based and Machine Learning Methods” by Tom Loulouse et al, 2015, University of Corsica, France; and “Fast Detection of Deflagrations Using Image Processing” by Thomas Schroeder et al, Helmut Schmidt University, Hamburg, Germany, 2014.

[0525] The pyroelectric detectors 4A3 detect the typical spectral radiation of burning, organic substances such as wood, natural gas, gasoline and various plastics. To distinguish a flame from the sun or other intense light source such as light emissions from arc welding, and thus exclude a false alarm, the following independent criteria are considered: a typical flame has a flicker frequency of (1 . . . 5) Hz; a hydrocarbon flame produces the combustion gases carbon monoxide (CO) and carbon dioxide (CO<sub>2</sub>); and in addition, burning produces water which can also be detected in the infrared range. Each pyroelectric detector 4A3 is an infrared sensitive optoelectronic component specifically used for detecting electromagnetic radiation in a wavelength range from (2 to 14)  $\mu\text{m}$ . A receiver chip of a pyroelectric infrared detector consists of single-crystalline lithium tantalite. On the upper electrode of the crystal, an absorbing layer (black layer) is applied. When this layer interacts with infrared radiation, the pyroelectric layer heats up and surface charge arises. If the radiation is switched off, a charge of the opposite polarity originates. However, the charge is very low. Before the finite internal resistance of the crystal can equalize the charges, extremely low-noise and low leakage current field-effect transistors (JFET) or operational amplifier (Pomp) convert the charges into a signal voltage.

[0526] In general, most streams of digital intelligence captured by the wireless network 604 will be time and data stamped, as well as GPS-indexed by a local GPS receiver



within the sensing module, so that the time and source of origin of each data package is recorded within the system database. The GPS referencing system supporting the system transmits GPS signals from satellites to the Earth's surface, and local GPS receivers located on each networked device or machine on the system network receive the GPS signals and compute locally GPS coordinates indicating the location of the networked device within the GPS referencing system.

**[0527]** When practicing the wireless network of the present invention, any low power wireless networking protocol of sufficient bandwidth can be used. In one illustrative embodiment, a Zigbee® wireless network would be deployed inside the wood-framed or mass timber building under construction, so as to build a wireless internetwork of a set of wireless PIR thermal-imaging fire outbreak detection systems deployed as a wireless subnetwork deployed within the building under construction. While Zigbee® technology, using the IEEE 802.15.1 standard, is illustrated in this schematic drawing, it is understood that any variety of wireless networking protocols including Zigbee®, WIFI and other wireless protocols can be used to practice various aspects of the present invention. Notably, Zigbee® offers low-power, redundancy and low cost which will be preferred in many, but certainly not all applications of the present invention. In connection therewith, it is understood that those skilled in the art will know how to make use of various conventional networking technologies to interconnect the various wireless subsystems and systems of the present invention, with the internet infrastructure employed by the system of the present invention.

The Automated Hybrid Clean Wildfire Inhibitor Misting and Sprinkler System of the Present Invention, Controlled by the Wireless Automated Wildfire Ember Detection and Notification Network

**[0528]** As disclosed in Applicant's prior US patent applications, when treating combustible organic materials so they will not burn in the presence of a wildfire, it will be helpful in many instances to spray clean anti-fire chemical liquid over the target surfaces so that the droplets are relatively large and an adequate coating of anti-fire chemical dries over the treated surface. This way, when the chemically treated organic material is exposed to fire, the treated surface has adequate chemicals to break the free-radical chain reactions of the fire and thereby quickly suppress and/or extinguish the fire.

**[0529]** However, during wildfire storms, producing burning wildfire embers flying through dried heated air, driven by strong prevailing winds, it has been discovered that clean aqueous-based anti-fire biochemical liquid of the present invention will perform as a more effective fire suppressant if provided to the burning fire in the form of a mist cloud, so that it can work on a wildfire and its embers, as described in the wildfire ember suppression process described herein.

**[0530]** While most mist producing apparatus disclosed herein operates on the principle of transmitting an anti-fire chemical liquid through a misting nozzle under low, medium or high hydraulic pressure, it is understood that when spraying anti-fire chemical liquids over the surfaces of organic material during fire-protection treating operations, then spray-type nozzles will be often used as provided on the mobile spraying apparatus 5 shown herein. Using spray-type nozzles, it is possible to quickly deposit and form sufficient

coatings of anti-fire chemical material on the treated surfaces, because spray-type nozzles produce liquid drops substantially larger in size than microscopic droplets formed by misting nozzles during misting operations, illustrated herein.

**[0531]** FIGS. 42C and 42D shows automated hybrid clean wildfire inhibitor misting system of the present invention 600, providing both an anti-fire chemical misting system for suppressing wildfire embers impacting a building as shown in FIG. 42C and a lawn and ground anti-fire chemical liquid misting system impacting the lawn and ground around the building as shown in FIG. 42C, both automatically controlled by an automated wildfire ember detection and notification network shown in FIGS. 42A and 42B. All of these system components are integrated into the system network shown in FIG. 4A.

**[0532]** FIG. 42A shows a piping manifold 6G, a network of piping, and a set of misting nozzles 6H used to supply and produce anti-fire chemical misting droplets from the automated hybrid clean wildfire misting system 6 shown in FIGS. 13A and 13B.

**[0533]** As shown in FIG. 13A, automated multi-mode hybrid clean wildfire inhibitor misting system 600 comprises: an dual-mode anti-fire lawn and ground misting system 6A shown in FIG. 13B for either misting water from a main water supply, or misting environmentally-clean anti-fire chemical liquid of the present invention over lawns (e.g. dried out grass) and ground surfaces covered with organic material; a wildfire ember misting controller 6B (e.g. programmable microcontroller supported by a memory architecture) for controlling the various modes of the system 6; lithium battery pack and controller 6C for supplying electrical power to the electronic components in the system 6 including the DC or AC electric motor of hydraulic (e.g. diaphragm-type) liquid pumping system 6F; a photovoltaic solar cell panel 6D for recharging the lithium-ion battery back 6C while collecting sunlight with the PV solar panel 6D as solar conditions allow; a supply tank containing an adequate supply (e.g. 100 gallons) of a liquid anti-fire chemical liquid realizable using anti-fire biochemical liquid of the present invention; a liquid spray misting pump system 6F (e.g. self-priming DC or AC electrical-motor powered diaphragm liquid pump) for hydraulically pumping the anti-fire chemical liquid 6E from its supply tank (e.g. 50-100 gallons) to a plurality of misting nozzles 6H mounted all around a building being protected, and connected through adequate heat-resistant piping (e.g. 1/8", 1/4" or 1/2" metal tubing, or high-heat resistant plastic tubing such as PET) extending over relatively short distances under adequate hydraulic pressure, to support sufficient flow rates of anti-fire chemical liquid during a wildfire ember storm, determined in a manner well known in the fluid hydraulic arts; a piping manifold 6G and piping network including a set of misting nozzles 6H for producing clean anti-fire (AF) chemical mist; a GPRS/GSM transceiver 6I with suitable antennas 6J, connected to the controller 6B, and adapted for transmitting and receiving digital data packets using GPRS and GSM communication protocols, over the system network 1, to support a suite of digital communication services and protocols specified herein; a suite of communication services and protocols 6L (e.g. email, SMS alert, PUSH protocol, XML, PDMS, and CALL alert) supported by GSM, for sending and receiving messages; and at least one electronic Wildfire ember detection module 4A, with 360



degrees of sensing and associated field of views (FOVs), and in wireless communication with the wireless wildfire ember detection and notification network 4 of the present invention.

[0534] As shown in FIG. 42D, the lawn misting system 6A comprises: a water supply 6Q connected to a network of underground piping 6R; misting-type sprinklers 6O (e.g. misting nozzles) connected to the underground piping 6R; misting-type rotors 6P connected to the piping 6R; valves 6N connected to the underground piping 6R, the local water supply 6Q, and the liquid pumping system 6F, which is operably connected to the supply of clean wildfire inhibitor liquid 6E using piping; and a timer/controller 6M connected to the controllable valves 6N, and controlled by the wildfire ember misting controller 6B, which is managed by the automated wildfire ember detection and notification network 604, shown in FIG. 13A.

[0535] The dual-mode lawn misting system 6A shown in FIG. 43D has two modes of operation. During its first mode of operation, when no wildfire storm is detected, the lawn misting system 6A automatically mists the lawn with water supplied from the local water supply 6Q. During its second mode, when a wildfire storm is detected, the lawn misting system 6A automatically mists the lawn with an environmentally anti-fire (AF) liquid 6E supplied from a local supply of anti-fire (AF) liquid pumped from a pumping system 6F.

[0536] In the preferred embodiment the hybrid wildfire misting system 600 also has at least two modes operation: (i) a manual mode where a building/home owner or manager can manually activate and operate the anti-fire chemical liquid misting system 600 to protect either the building 17 and/or the lawn and ground surfaces around the building 17, as desired or required, based on intelligence in the possession of the human operator or manager; and (ii) an automated mode where the wildfire ember misting controller 6B, in cooperation with the local electronic wildfire and ember detection module 604A and associated wireless wildfire detection network 4, automatically activate and operate the anti-fire chemical liquid misting system 600 to protect both the building 17 and/or the lawn and ground surfaces around the building 17, as required, based on intelligence automatically collected by the wireless wildfire detection and notification network.

[0537] Preferably, modules 6I, 6K, 6B, 6C, 6E and 6F shown in FIG. 42C will be mounted and safely protected in the wildfire-protected shed or closet structure 50, disclosed in great technical detail in Applicant's published U.S. patent application Ser. No. 15/925,796, incorporated herein by reference. In the manual mode, a touch-screen or touch-type control panel associated with the controller 6B is used by the operator to simply operate the system 6 in its manual mode, or automatically arm the system 600 to operate in its automated, artificial intelligence (AI) mode of operation.

[0538] The system 600 will be remotely controllable by the building manager/home-owner using a mobile computing system 11 running the mobile application 12, as shown and described in FIG. 43. Suitable graphical user interfaces (GUIs) will be supported on the mobile application 12 to enable the user to monitor and control the system 600 locally, or from a remote location, in real-time, provided the wireless communication infrastructure is not disrupted by a wildfire. In the case of active wildfires, the wildfire detection and notification network 4 should be accessible by a remote

user provided with the mobile application 12. As the system will continuously collect, record and monitor intelligence about specific regions of land and any wildfires detected in such regions, and advise any specific home/building owner of the status of any specific building before, during and after a wildfire.

[0539] The system 600 will include and supported automated mechanisms for remotely monitoring and reporting the amount of anti-fire chemical liquid 6E available and remaining for use in supporting anti-fire misting operations, during an automatically detected wildfire ember storm. Preferably, adequate reserves of anti-fire chemical liquid 6E will be stored on each property before any given wildfire strike, to support several hours of wildfire ember suppression misting operations, which is typically expected during a wildfire storm before passes through and consumes the organic material that is desperately seeks to fuel its combustion process.

[0540] As shown in FIG. 42G, the sprinkler-type head(s) shown in FIGS. 42F1 and 42F2, mounted on home building rooftops and driven by automated pumps, automatically create and maintain a clean-biochemistry fire break all around the home's property area, to maintain a proactive fire defense coverage against an advancing wildfire, so as to help reduce risk of destruction of property and life by wildfire. Preferably, wildfire fire defense system is provided with a mode of operation in the system 600 is locked and loaded, with a full supply of clean fire inhibiting liquid biochemical composition of the present invention stored in a storage tank, and connected to a battery-powered pump, connected to the sprinkler and misting heads installed on and around the property, and on the home building roof structure, equipped and ready to automatically spray the predetermined amount of fire inhibiting liquid biochemical from the sprinklers 60 and misting nozzles 64 over all the property surfaces, so as to create a proactive zone of fire protection from potassium citrate salt crystalline structures formed and dried on all combustible property surfaces, and effective against hot embers and fire from an wildfire advancing toward the proactively protected property.

[0541] Typically, the locked and loaded home wildfire defense system will be manually triggered by the owners several hours and just before the owners are required to evacuate their homes and property for safety reasons, by authorities such as the local fire chief and deputies. Alternatively, the wildfire home defense system can also be remotely triggered using a mobile smartphone, if required, with the property owners not home to manually triggering the spraying defense mode of the system.

[0542] To provide adequate protection against flying wildfire embers combusting in a low humidity environment, the misting nozzles 64 will be mounted about the building 17 so as to provide adequate coverage over all air-inlet vents provided on the specific building being equipped with the wildfire misting/sprinkler system of the present invention, as well as on wood and other organic surfaces that might be vulnerable to hot wildfire embers during a wildfire ember storm, as illustrated in FIG. 42A. The liquid spraying pattern of each misting nozzle 6H and sprinkler 60 being used in the misting system 600 will be considered and exploited to provide the adequate misting protection required by the wildfire protection application at hand. Computer software tools may be developed and distributed to installers to assist



in the design and installation of a hybrid wildfire misting system in accordance with the principles of the present invention.

**[0543]** In the illustrative embodiment, the clean anti-fire biochemical liquid to be used for wildfire ember misting operations disclosed herein. It is expected that service-oriented businesses will support the rapid design, installation and installation of the automated wildfire detection and misting suppression systems of the present invention, as well as the supplying and replenishing of clean anti-fire chemical liquid on each GPS-indexed property. It is expected that this can occur with the efficiency currently provided by conventional liquid propane supply companies around the country. Because of the reduced risk of loss of wood-framed or other buildings to wildfire, which the systems and method of the present invention will provide, while advancing the best practices for home and building property protection against wildfires, it is expected that fire insurance companies will embrace the best practices represented by the present invention, for reason of the great benefits such inventions will provide, predicted by Benjamin Franklin's time-honored principle of fire protection: "An ounce of prevention is worth a pound of cure."

**[0544]** When encountering the cloud of anti-fire liquid droplets, combustible wildfire embers will be suppressed or readily extinguished. The chemical molecules in the droplets formed with fire inhibiting biochemical liquid of the present invention will interfere with the free radicals ( $H^+$ ,  $OH^-$ ,  $O$ ) involved in the free-radical chemical reactions within the combustion phase of a fire, or wildfire embers, breaking these free-radical chemical reactions and extinguishing the fire's flames. Also, the droplets will vaporize when absorbing the radiant heat energy of the hot wildfire ember(s), rapidly expanding into a vapor, cooling down the embers, and displaying oxygen, causing the combustion phase of the embers to be suppressed if not extinguished.

Best Practices on when to Spray and Respray the Fire Inhibiting Biochemical Compositions of the Present Invention on Dry Native Vegetation, Decks, Fences, Patio Covers and Vertical Framing for Best Early Fire Elimination Results

**[0545]** Below are some best practices on when to spray and respray the fire inhibiting biochemical compositions of the present invention on dry native vegetation, decks, fences, patio covers and vertical framing, to get the best early fire elimination results.

**[0546]** When vegetation is bright green, it will not support easy ember ignition and fire advance the way it will when vegetation starts to dry out and turn brown and gold in color. When proactively spraying the biochemical compositions of the present invention, its best to spray using spray equipment during dry conditions when lumber and vegetation are bone dry. Also, after wood or dry vegetation has been sprayed with the biochemical compositions, one needs to consider any reduction in early proactive fire defense that may have resulted to sprayed fire-protective salt crystal structure coatings that may have been exposed to heavy water sprays from rain, garden sprinklers and/or a hose. After any type of heavy watering or rain to treated surfaces, it is advised to respray to surfaces to be re-protected and insure that the best proactive fire defense has been provided to such surfaces. When property owners or contractors are proactive spraying the biochemical compositions for wildfire defense, best results can be obtained using the GPS-tracking back-pack atomizing spray cannon system shown in FIGS. 13A

through 13C. When spraying framing lumber on a construction site or elsewhere, the best results can be achieved using an airless paint sprayer provided with a #25 tip. In both applications, for best results, vegetation and lumber should be dry, with less than 19% moisture content.

Bio-Degradable Fire Extinguishing Liquid Concentrate (LC) Compositions of the Present Invention for Producing Fire Extinguishing Sprays, Mists and Clouds of Fine Vapor Engineered for Use in Actively Fighting Fires Involving Class A/B Fuels, as Well as Proactively Inhibiting Fires to Defend Against Fire Ignition, Flame Spread, and Flash Over

**[0547]** Referring now to FIGS. 44 through 50, technical details will now be provided teaching how practice the bio-degradable fire extinguishing liquid concentrate (LC) compositions of the present invention for producing fire extinguishing sprays, mists and clouds of fine vapor engineered for use in actively fighting fires involving Class A/B fuels, as well as proactively inhibiting fires to defend against fire ignition, flame spread, and flash over.

Specification of Fire Extinguishing Liquid Concentrate Compositions of the Present Invention for Mixing with Water to Produce Streams of Fire Extinguishing Water, Mist and Vapor Clouds for Use in Actively Extinguishing Fire Involving Class A Fuel and/or Class B Fuel

**[0548]** As will be described herein, the fire extinguishing biochemical liquid compositions of the present invention can be premixed and bottled/containerized at full strength for final usage and application, as illustrated for use in applications shown in FIGS. 7A, 8A, 10A, 11A, 12A, 13A, and 16A. The dry chemical components of the fire inhibiting compositions can be premixed and packaged in a container, for subsequent mixing with water to produce final liquid compositions for extinguishing fire and suppressing combustible vapors. The biochemical liquid compositions can be made into a liquid concentrate, and then bottled/containerized, and transported to an intermediate, or end user location, for mixing with a supply of clean water in correct proportions, to produce fire extinguishing liquid compositions in a batch mode, with proper chemical constituent proportions, as described herein, as illustrated for use in applications shown in FIGS. 7A, 8A, 10A, 11A, 12A, 13A, and 16A. Alternatively, these biochemical liquid compositions can be made into a liquid concentrate, and then bottled/containerized, and transported to end user location, for mixing with a supply of clean water in correct proportions using a hydraulic inductor device, to produce fire inhibiting liquid compositions in an in-line proportioning/mixing mode, with proper chemical constituent proportions, as described herein, as illustrated for use in applications shown in FIGS. 14A, 14B, 15A, 17A, 42A, 42C, 46, and 57.

**[0549]** FIG. 44 shows the primary components of a first environmentally-clean aqueous-based fire extinguishing biochemical liquid concentrate (i.e. fire extinguishing biochemical additive) of the present invention comprising: (i) major amounts of a fire inhibiting agent realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate (TPC), for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (ii) minor amounts of triethyl citrate (TEC) as a low-surface tension surfactant and dispersing agent, and (iii) major amounts of water functioning as a solvent, carrier and dispersant, to form the fire extinguishing biochemical liquid concentrate (LC) designed to be added to and mixed in-line



with a pressurized supply of water in pre-specified proportions so as to produce an environmentally-clean fire extinguishing aqueous liquid for spraying onto an actively combusting fire involving Class A fuel and/or Class B fuel.

**[0550]** FIG. 45 illustrates the primary components of a second environmentally-clean aqueous-based fire extinguishing biochemical liquid concentrate (i.e. fire extinguishing additive) of the present invention comprising: (i) major amounts of a fire inhibiting agent realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate (TPC), for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (ii) minor amounts of triethyl citrate (TEC) as a low-surface tension surfactant and dispersing agent; (iii) minor amounts of citric acid as a buffering agent; and (iv) major amounts of water functioning as a solvent, carrier and dispersant, to form the fire extinguishing biochemical liquid concentrate (LC) designed to be added to and mixed in-line with a pressurized supply of water in pre-specified proportions so as to produce an environmentally-clean fire extinguishing aqueous liquid for spraying onto an actively combusting fire involving Class A fuel and/or Class B fuel.

Specification of Environmentally-Clean Aqueous-Based Fire Extinguishing Bio-Chemical Liquid Compositions and Formulations, and Methods of Making the Same in Accordance with the Principles of the Present Invention

**[0551]** Another object of the present invention is to provide new and improved environmentally-clean aqueous-based fire extinguishing biochemical solutions (i.e. liquid compositions and liquid concentrate for proportioning and mixing with prespecified supplies of water) for producing biochemical liquid products that demonstrate very good immediate extinguishing effects when applied to extinguish a burning or smoldering fire.

**[0552]** In general, the novel biochemical fire extinguishing liquid compositions of the present invention comprise: (a) a dissolving agent in the form of a quantity of water, for dissolving salts and dispersing metal ions dissolved in water; (b) a fire extinguishing agent in the form of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid, for providing metal ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (c) a dispersing agent in the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), such as triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal ions when the fire extinguishing liquid composition is applied to a surface to be protected against fire; and (d) optional additives to the resulting liquid solution, to provide coloring and stability.

**[0553]** As will be described herein, the fire extinguishing biochemical liquid compositions of the present invention can be premixed and bottled/containerized at full strength for final usage and direct spray application, as illustrated using the apparatus shown in FIGS. 43, 44, 47, 51, and other conventional spray equipment used by fire departments around the world.

**[0554]** Alternatively, these biochemical liquid compositions can be formulated into a liquid concentrate (LC), and then bottled/containerized, and transported to end user location, for mixing with a supply of clean water in correct prespecified proportions using a hydraulic inductor device, to produce fire inhibiting liquid compositions in an in-line proportioning/mixing mode, with proper chemical constitu-

ent proportions maintained, as described herein, as illustrated for use in applications shown in FIGS. 14A, 14B, 15A, 17A, 42A, and 42C.

**[0555]** In general, useful alkali metal salts of nonpolymeric saturated carboxylic acids for inclusion in the compositions of the present invention preferably comprise: alkali metal salts of oxalic acid; alkali metal salts of gluconic acid; alkali metal salts of citric acid; and also alkali metal salts of tartaric acid. Alkali metal salts of citric acid are particularly preferred, as will be further explained hereinafter.

**[0556]** Notably, while the efficacy of the alkali metal salts increases in the order of lithium, sodium, potassium, cesium and rubidium, the salts of sodium and salts of potassium are preferred for cost of manufacturing reasons. Potassium carboxylates are very particularly preferred, but tripotassium citrate monohydrate (TPC) is the preferred alkali metal salt for use in formulating the environmentally-clean fire extinguishing biochemical compositions of the present invention.

**[0557]** While it is understood that other alkali metal salts are available to practice the biochemical compositions of the present invention, it should be noted that the selection of tripotassium citrate as the preferred alkali metal salt, includes the follow considerations: (i) the atomic ratio of carbon to potassium (the metal) in the utilized alkali metal salt (i.e. tripotassium citrate); (ii) that tripotassium citrate is relatively stable at transport and operating temperatures; (iii) tripotassium citrate is expected to be fully dissociated to citrate and potassium when dissolved in water, and that the dissociation constant is not relevant for the potassium ions, while citric acid/citrate has three ionizable carboxylic acid groups, for which pKa values of 3.13, 4.76 and 6.4 at 25° C. are reliably reported the European Chemicals Agency (ECHA) handbook; and (iv) that tripotassium citrate produces low carbon dioxide levels when dissolved in water.

**[0558]** Preferably, the water soluble dispersing agent should have a melting point at least 32 F (0 C) or lower in temperature, and be soluble in water. Triethyl citrate (TEC) is a preferred dispersing agent when used in combination with tripotassium citrate (TPC) having excellent compatibility given that both chemical compounds are derived from citric acid. Triethyl citrate (TEC) also functions as a surfactant, reducing the surface tension of the final liquid solution to improve wettability or sticking to Class A fuel surfaces, and promote thin film formation on Class B fuels.

**[0559]** The fire extinguishing liquid biochemical compositions of the present invention are produced and prepared by mixing the components in specified amounts with pre-specified quantities of water to produce the fire extinguishing liquid compositions. The order of mixing is discretionary. However, it is advantageous to produce aqueous preparations by first mixing the components other than water, into a quantity of water.

**[0560]** The fire extinguishing liquid compositions of the present invention have a very good immediate fire extinguishing effect, and a good inhibiting effect after application to reduce reignition of fire once extinguished. This mixing of the constituent biochemical compounds can take place before or during their use. It is preferred, however, that an aqueous preparation is made and kept ready for fire extinguishing use. Preferably, the fire extinguishing liquid solution is made as a liquid concentrate (e.g. 1% by volume) to be added to 99% water by volume, using conventional in-line venturi-based fluid proportioning/mixing devices (e.g. eductor device) available from numerous manufactur-



ers around the globe. However, other concentrations such as 3%, 6% and other concentration percentages can be prepared as required or desired by the applications at hand. When prepared in the form of a fire extinguishing liquid concentrate (LC), the liquid solution can be mixed with the correct proportions of pressured water at a fire pumping engine during a fire outbreak, to produce streams of water with the biochemical additive of the present invention, so as to significantly enhance the fire extinguishing properties of water, while reducing the quantity of water required to extinguish any given fire involving Class A and/or B type fuels.

**[0561]** Using such biochemical fire extinguishing additives of the present invention, it is possible to more quickly extinguish fires using water as a fire extinguishing agent, while reducing water damage to property during fire extinguishing operations. Also, while extinguishing Class A and/or B fires are preferred fire applications for the biochemical additives of the present invention, it is noted that the fire extinguishing liquid compositions of the present invention are also useful as fire extinguishing agents when fighting fires involving Class D fuels as well, and even Class C (electrical) fires if and when the situation requires.

#### Specification of Preferred Embodiment of the Aqueous-Based Biochemical Fire Extinguishing Liquid Concentrate (LC) Compositions of the Present Invention

**[0562]** In the preferred embodiment of the fire extinguishing biochemical liquid concentrate (LC) composition of the present invention, the components are realized as follows: (a) the dissolving agent is realized in the form of a quantity of water, for dissolving salts and dispersing metal ions dissolved in the water; (b) the fire extinguishing agent is realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate, for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; and (c) a dispersing agent realized in the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), specifically triethyl citrate, an ester of citric acid, for dispersing the metal potassium ions when the fire extinguishing liquid composition is applied to surfaces to extinguish active fire and/or protected against fire ignition and flame spread, while (i) water molecules in the water stream are evaporated by the heat of the fire to cool off fuel surfaces, and (ii) metal potassium ions interact with free radical hydrocarbon vapors in the combustion phase of the fire to interrupt such chemical reactions and help extinguish the fire using less quantities of water.

**[0563]** Once the preferred formulation specified above is prepared in the form of a biochemical liquid concentrate (LC) composition (e.g. 1% liquid concentrate by volume to be proportioned and mixed with 99% water by volume, 3% liquid concentrate by volume to be proportioned and mixed with 97% water by volume, or 6% liquid concentrate by volume to be proportioned and mixed with 94% water by volume), then this fire extinguishing liquid concentrate (LC) is then stored in a container, bottle or tote (i.e. package) suitable for the end user application in mind. Then, the filled container of liquid concentrate should be sealed with appropriate sealing technology and immediately labeled with a specification of (i) its biochemical components, with weight percent measures where appropriate, and the date and time of manufacture, printed and recorded in accordance with

good quality control (QC) practices well known in the art. Where necessary or desired, barcode symbols and/or barcode/Rfid identification tags and labels can be produced and applied to the sealed package to efficiently track each barcoded package containing a specified quantity of clean fire extinguishing biochemical composition concentrate. All product and QC information should be recorded in globally accessible network database, for use in tracking the movement of the package as it moves along the supply chain from its source of manufacture, toward its end use at a GPS specified location.

#### Selecting Tripotassium Citrate (TCP) as a Preferred Fire Extinguishing Agent for Use in the Fire Exhibiting Liquid Biochemical Compositions of the Present Invention

**[0564]** In the preferred embodiments of the present invention, tripotassium citrate (TPC) is selected as active fire extinguishing chemical component in fire extinguishing biochemical composition of the present invention. In dry form, TPC is known as tripotassium citrate monohydrate ( $C_6H_5K_3O_7 \cdot H_2O$ ) which is the common tribasic potassium salt of citric acid, also known as potassium citrate. It is produced by complete neutralization of citric acid with a high purity potassium source, and subsequent crystallization. Tripotassium citrate occurs as transparent crystals or a white, granular powder. As discussed above, it is an odorless substance with a cooling, salty taste. It is slightly deliquescent when exposed to moist air, freely soluble in water and almost insoluble in ethanol (96%).

**[0565]** Tripotassium citrate is a non-toxic, slightly alkaline salt with low reactivity. It is chemically stable if stored at ambient temperatures. In its monohydrate form, TPC is very hygroscopic and must be protected from exposure to humidity. Care should be taken not to expose tripotassium citrate monohydrate to high pressure during transport and storage as this may result in caking. Tripotassium citrate monohydrate is considered "GRAS" (Generally Recognized As Safe) by the United States Food and Drug Administration without restriction as to the quantity of use within good manufacturing practice. CAS Registry Number: [6100-05-6]. E-Number: E332.

**[0566]** Tripotassium citrate monohydrate (TPC) is a non-toxic, slightly alkaline salt with low reactivity. It is a hygroscopic and deliquescent material. It is chemically stable if stored at ambient temperatures. In its monohydrate form, it is very hygroscopic and must be protected from exposure to humidity. Its properties are:

- [0567]** Monohydrate
- [0568]** White granular powder
- [0569]** Cooling, salty taste profile, less bitter compared to other potassium salts
- [0570]** Odorless
- [0571]** Very soluble in water
- [0572]** Potassium content of 36%
- [0573]** Slightly alkaline salt with low reactivity
- [0574]** Hygroscopic
- [0575]** Chemically and microbiologically stable
- [0576]** Fully biodegradable
- [0577]** Allergen and GMO free

**[0578]** Jungbunzlauer (JBL), a leading Swiss manufacturer of biochemicals, manufactures and distributes TPC for food-grade, healthcare, pharmaceutical and over the counter (OTC) applications around the world. As disclosed in JBL's product documents, TPC is an organic mineral salt which is



so safe to use around children and adults alike. Food scientists worldwide have added TPC to (i) baby/infant formula powder to improve the taste profile, (ii) pharmaceuticals/OTC products as a potassium source, and (iii) soft drinks as a soluble buffering salt for sodium-free pH control in beverages, improving stability of beverages during processing, heat treatment and storage.

Selecting Triethyl Citrate (TEC) as a Preferred Dispersing Agent with Surface Tension Reducing and Surfactant Properties for Use in the Fire Extinguishing Biochemical Liquid Compositions of the Present Invention

**[0579]** In the preferred illustrative embodiments of the present invention, the dispersing agent and surfactant used in the fire extinguishing biochemical compositions of the present invention is realized as a food-grade additive component, namely, triethyl citrate (TEC) which functions as a dispersing agent with strong surface tension reducing properties and surfactant properties as well. Triethyl citrate belongs to the family of tricarboxylic acids (TCAs) and derivatives, organic compounds containing three carboxylic acid groups (or salt/ester derivatives thereof).

**[0580]** In the aqueous-based fire extinguishing liquid composition concentrate, the dispersing agent functions as temporary dispersing agent for dispersing the metal ions dissolved and disassociated in aqueous solution. As water molecules evaporate from a spray coating of the biochemical composition, typically spray/atomized applied to a surface being attacked by or to be proactively protected from fire, the dispersing agent (i.e. TEC) allows and promotes the formation of thin metallic (e.g. potassium) salt crystal films on surfaces of Class A and Class B fuels. Also, the dispersing agent promotes durability against water and ambient moisture once dried. Thus, spraying smoldering ashes of an extinguished fire, using pressurized water treated with the fire extinguishing additive of the present invention, should significantly help to prevent reignition of fire, while significantly reducing smoke production, and mitigate further damage to the environment.

**[0581]** In the preferred embodiment, a relatively minor quantity of triethyl citrate (TEC) liquid is blended with a major quantity of TCP powder in specific quantities by weight and dissolved in a major quantity of water to produce a clear, completely-dissolved liquid biochemical formulation consisting of food-grade biochemicals mixed with water and having highly effective fire extinguishing properties, as proven by testing. The resulting aqueous biochemical solution remains stable without the formation of solids at expected operating temperatures (e.g. 34 F to 120 F).

**[0582]** Jungbunzlauer (JBL) also manufactures and distributes its CITROFOL® A1 branded bio-based citrate esters for food-grade, healthcare, pharmaceutical and over the counter (OTC) applications around the world. CITROFOL® A1 triethyl citrate (TEC) esters have an excellent toxicological and eco-toxicological profile, and provide good versatility and compatibility with the tripotassium citrate (TPC) component of the biochemical compositions of the present invention. CITROFOL® A1 branded citrate esters are particularly characterized by highly efficient solvation, low migration and non-VOC (volatile organic compound) attributes. As an ester of citric acid, triethyl citrate is a colorless, odorless liquid which historically has found use as a food additive (E number E1505) to stabilize foams, especially as a whipping aid for egg whites.

**[0583]** Broadly described, the fire extinguishing biochemical liquid concentrate solution of the present invention, once proportioned and mixed with proper quantities of water, consist of an aqueous dispersion medium such as water which carries dissolved metal salt cations that (i) interfere with the free radical chemical reactions with the combustion vapor phase of the active fire, and (ii) eventually form thin metal salt crystalline films on the surface substrates being protected from the ignition of fire, as the case may be. The aqueous dispersion medium may be an organic solvent, although the preferred option is water when practicing the present invention, because of its abundance on the planet earth. After the application of liquid spray onto the combustible surface to extinguish an active fire, or to protect against fire ignition, flame spread and smoke development, the aqueous dispersion medium evaporates in the presence of heat, causing the metal salt (i.e. potassium salt) cations to interfere with the free radical chemical reactions within the vapor combustion phase of the fire.

**[0584]** While offering some surface tension reducing effects, the main function of the dispersing agent in the biochemical liquid composition is to ensure a relatively uniform and optimal formation of the salt crystalline films on combustible surfaces to be protected, as well as desired uniform dispersion of metal (i.e. potassium) salt cations within the sprayed droplets, misting, add/or vapor clouds produced from the nose of the fire nozzle used to extinguish an actively burning fire, or fuel being sprayed with the fire extinguishing liquid compensation.

**[0585]** The fact that CITROFOL® A1 triethyl citrate (TEC) esters are bio-based, odorless, biodegradable and label-free, represents a great advantage over most other dispersing agents, and fully satisfies the toxicological and environmental safety requirements desired when practicing the biochemical compositions of the present invention.

**[0586]** In the preferred embodiments of the present invention, the use of CITROFOL® A1 triethyl citrate (TEC) esters with tripotassium citrate monohydrate (TPC) dissolved in water as a dispersion solvent, produce fire extinguishing biochemical liquid formulations that demonstrate excellent fire extinguishing properties, and once applied to active fires and Class A fuel, very good fire inhibiting properties. The chemical and colloidal nature of potassium salt ions (which are mineral salt dispersions) present in TPC dissolved in water, is highly compatible with the CITROFOL® A1 triethyl citrate (TEC) ester used as the dispersing agent in the preferred embodiments of the present invention. Also, CITROFOL® A1 triethyl citrate esters are REACH registered and are safe, if not ideal, for use in environmentally sensitive products such as fire extinguishing agents which must not adversely impact human, animal and plant life, ecological systems, or the natural environment.

Specification of Preferred Formulations for the Fire Extinguishing Biochemical Liquid Compositions of Matter According to the Present Invention

#### Example #1: Liquid-Based Fire Extinguishing Biochemical Composition

**[0587]** FIG. 44 illustrates the primary components of a first environmentally-clean aqueous-based fire extinguishing liquid biochemical composition of the present invention consisting of tripotassium citrate (TPC) and triethyl citrate



(TEC) formulated as a dispersing agent and surfactant, with water functioning as a solvent, carrier and dispersant in the biochemical composition.

**[0588]** Example 1: Schematically illustrated in FIG. 44: A fire-extinguishing biochemical composition was produced by stirring the components into water. The composition comprising: 0.05 pounds by weight of triethyl citrate as coalescing agent, (20.3 milliliters by volume); 5.2 pounds by weight of tripotassium citrate (64 fluid ounces by volume); and 4.4 pounds by weight of water (64 fluid ounces by volume), to produce a resultant solution of total weight of 9.61 pounds having 128 ounces or 1 gallon of volume.

Preferred Weights Percentages of the Components of the Fire Extinguishing Biochemical Liquid Formulation of the Present Invention

**[0589]** In the biochemical compositions of the present invention, the ratio of the ester of citrate (e.g. triethyl citrate) to the alkali metal salt of a nonpolymeric carboxylic acid (e.g. tripotassium citrate) may be major amount between 1:100: to 1:1000 and is typically in the range from 1:1 to 1:100, preferably in the range from 1:2 to 1:50, more preferably in the range from 1:4 to 1:25 and most preferably in the range from 1:8 to 1:15.

**[0590]** A preferred biochemical liquid composition according to the present invention comprises: a major amount from 1% to 65% by weight, preferably from 20% to 50% by weight and more preferably from 30% to 55% by weight, of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid (e.g. tripotassium citrate monohydrate or TPC); and minor amount from 0.08% to 5% by weight, preferably from 0.5% to 2% by weight and more preferably from 0.1% to 1.0% by weight, of triethyl citrate (an ester of citrate acid); wherein the sum by % weight of the components (a) and (b) should not exceed 100% by weight.

**[0591]** In a preferred embodiment, the fire extinguishing composition further comprises water. The water content is present in a major amount and is typically not less than 30% by weight, preferably not less than 40% by weight, more preferably not less than 50% by weight and most preferably not less than 60% by weight and preferably not more than 60% by weight and more preferably not more than 70% by weight, all based on the fire extinguishing biochemical composition.

**[0592]** The viscosity of the aqueous preparation is preferably at least 5 [mPas] (millipascal-seconds, in SI units, defined as the internal friction of a liquid to the application of pressure or shearing stress determined using a rotary viscometer), and preferably not more than 50 [mPas], or 50 centipois) [cps], for most applications. Preferably, the pH of the aqueous solution is in the range of 6.0 to 8.0.

Methods of Blending, Making and Producing the Biochemical Liquid Formulations

**[0593]** The fire extinguishing liquid chemical compositions illustrated above are reproducible by mixing the components described above. The order of mixing is discretionary. However, it is advantageous to produce aqueous preparations by first mixing the components other than water, into the quantity of water.

Conversion of the Fire Extinguishing Liquid Composition Make-Up Formulation into X % Liquid Concentrate (LC) Formulation

**[0594]** A highly-effective fire extinguishing solution is produced by adding the chemical additive of the present invention to a pressurized water stream being supplied to a fire hose having a spray gun nozzle at its terminal end. Such mixing can be achieved using an in-line venturi-type proportioning/mixing device (e.g. eductor) between a 200 PSI water pumping engine and the spray gun nozzle being directed at an active fire. Examples of suitable proportioning/mixing devices to practice the present invention are made and sold by LEADER Group S.A.S. of France. Specifically, the LEADER MIX 200-1000 V2 automatic proportioning/mixing system is capable proportioning all types of liquid and foam concentrates from 0.03% to 6.0% over a large range of flow rates from 200 to 1050 liter/min. (43 to 230 gallons/min), over a wide operating pressure range from 5 to 16 bar (72 to 232 PSI). When used in combination with the apparatus specified in FIG. 46,

**[0595]** To perform an X % Liquid Concentrate (LC) conversion of the mass and volume measures of the components of the fire extinguishing liquid composition (make-up working solution), it is necessary to first list all mass measures in the formulation, including the amount of water to be mixed with the components to make up the working solution formulation. Then, a total volume is selected for the amount of liquid concentrate to be formulated in a standard container (e.g. 30 gallons), and a X % is selected for the liquid concentrate formulation (e.g. 6%). Based on such selections, 94 gallons of water will be added to each 6 gallons of liquid concentrate suctioned (e.g. educated) from the 30 gallon liquid concentrate container, and  $30/6 \times 94 = 564$  gallons of working liquid solution will be produced from the 30 gallons of liquid concentrate, for a 6% proportioning/mixing. Thus, using the original formulation for the fire extinguishing liquid composition of the present invention, one calculates how much mass (lbs.) of each chemical ingredient/component (e.g. TPC, TEC and water) must be dissolved in 30 gallons of water to make a 6% liquid concentrate for the fire extinguishing liquid composition of the present invention.

Specification of Portable Apparatus for Proportioning and Mixing Fire Extinguishing Liquid Concentrate of the Present Invention with Pressurized Streams of Water, and then with Air within an Aerating Nozzle to Generate Fire Extinguishing Spray, Mist or Vapor Clouds for Use in Actively Extinguishing Fire Involving Class A Fuel and/or Class B Fuel

**[0596]** FIG. 46 shows a mobile and/or portable fire extinguishing system for mixing and proportioning the fire extinguishing liquid concentrate of present invention with pressurized water to produce a fire extinguishing water solution for use in fighting active fires involving Class A and/or Class B fuels.

**[0597]** As shown in FIG. 46, the system comprises: (i) a venturi-based fluid eductor-type mixing/proportioning device operably connected to (i) a pressurized supply of water output at 200+ PSI pressure from a hydraulic pumping engine connected to a supply of water and pressurized by a hydraulic pump system driven by gasoline, diesel or electric engine; (ii) a supply of fire extinguishing liquid concentrate (LC) of the present invention contained within a 20+ gallon container; and (iii) one or more aerating/atomizing-type fire hose spray nozzles manually-actuable for producing a manually-adjustable water stream containing a proportioned quantity of fire extinguishing additive for every propor-



tioned quantity of water, and comprising fine water droplets, mist and/or vapor having dimensions in the range of about 1500 microns to about 50 microns, as required for rapidly extinguishing an particular fire involving Class A and/or Class B fuels.

**[0598]** FIG. 47 shows a conventional in-line type venturi-based proportioning/mixing (i.e. eductor) device for proportioning and mixing the fire extinguishing concentrate (e.g. additive) of the present invention, into a pressurized water stream flowing into the eductor device, while spraying a pressurized stream of water from an atomizing-type or spray-type nozzle assembly connected to a length of fire hose, as schematically illustrated in FIG. 46.

**[0599]** FIG. 48 shows a portable spray cart containing a supply of fire extinguishing liquid concentrate additive in a tank, supported on a set of wheels, and equipped with an in-line venturi-based proportioning/mixing device (i.e. eductor) device for drawing liquid concentrate into a pressurized water stream, as shown in FIG. 46, and being (i) operably connected to a length of fire hose terminated with an adjustable aerating/atomizing-type spray nozzle, and (ii) operably connected to a pressurized water pumping engine as illustrated in FIG. 46. As shown, the output of the proportioning/mixing device is to mix a proportioned quantity (i.e. %) of fire extinguishing liquid concentrate with a pressurized supply of water flowing through the eductor device, along the length of fire hose to the adjustable spray nozzle, spraying an active fire involving a Class A and/or B fuel.

**[0600]** FIG. 49 shows a portable triple tote spray trailer designed to be pulled and driven by a pressurized water pumping firetruck, and having a trailer platform supporting three liquid concentrate totes, each containing 200 gallons of fire extinguishing liquid concentrate additive of the present invention. As shown, the device is operably connected to an in-line venturi-based proportioning/mixing (i.e. eductor) device as shown in FIG. 46, and also to an adjustable spray nozzle gun assembly mounted for spraying operations, and also being operably connected to the pressurized water pumping engine aboard the water pumping firetruck, as illustrated in FIG. 46, so as to mix a proportioned quantity (i.e. 1%, 3% or 6%) of fire extinguishing liquid concentrate with a pressurized supply of water flowing through the venturi-based proportioning/mixing device, to the adjustable spray nozzle, while the spraying pressurized water from the spray gun nozzle, during an actively combustible fire.

Specification of Fixed/Stationary Apparatus for Proportioning and Mixing Fire Extinguishing Liquid Concentrate of the Present Invention with Pressurized Streams of Water, and then with Air within an Aerating Nozzle to Generate Fire Extinguishing Spray, Mist or Vapor Clouds for Use in Actively Extinguishing Fire Involving Class A Fuel and/or Class B Fuel

**[0601]** FIG. 50 shows a stationary and/or fixed fire extinguishing system for mixing and proportioning the fire extinguishing liquid concentrate of present invention with pressurized water to produce a fire extinguishing enhanced or treated water solution for use in fighting active fires involving Class A and/or Class B fuels.

**[0602]** As shown in FIG. 50, the system comprises: (i) a venturi-based fluid mixing/proportioning (i.e. eductor) device operably connected to (i) a pressurized supply of water output at 200+ PSI pressure from a hydraulic pumping engine connected to a supply of water and pressurized by a

hydraulic pump system driven by an electric, propane or other engine; (ii) a supply of fire extinguishing liquid concentrate (LC) of the present invention contained within a 20+ gallon container; and (iii) one or more aerating-type spray nozzles, typically triggered by electronic-controlled sensors, IR cameras and/or controllers, for automatically producing a cloud of water mist or vapor comprising fine water microdroplets in the range of about 500 microns to about 50 microns, with proportioned quantities of fire extinguishing biochemical additives, as required for extinguishing an particular fire involving Class A and/or Class B fuels, with improved fire extinguishing efficacy and efficiency using reduced quantities of water to minimize water damage to property during a fire outbreak, and production of smoke which contributes to environmental pollution.

Overview of the Use of Firefighting Foam Concentrates and Aerating/Aspirating Nozzles in the Generation of Finished Firefighting Foam Materials for Use in Extinguishing Fires

**[0603]** Foam is a great tool for extinguishing multiple types of fires, and does so in a shorter timeframe, making more efficient use of water, and providing more firefighting capability from the same volume of water, while resulting in more cost-effective and simpler operations. Foam is made by first mixing foam concentrate with water to create a foam solution, using a foam proportioner/mixer commercially available on the market. Once the solution is made, it must then be combined with air, and the mixture must be agitated to create finished foam materials consisting of billions of bubbles.

**[0604]** There are three commonly used methods of agitation. The most common method is using pumping foam solution out of a fog nozzle, which allows agitation to occur when the product hits its target. The second method involves using a compressed air foam system (CAFS) in which air is injected into the foam solution at the discharge, and when agitation occurs as the mixture rubs against the inside of the hose. The third method is to use an aspirating foam nozzle, which creates agitation and forms the foam bubbles within the nozzle.

**[0605]** Aspirating nozzles have several key qualities that make them an optimal choice when using producing firefighting foam. They are a low-energy system, meaning that the only energy available to produce bubbles in the nozzle comes from the water pump. They're easy to make, but there's also a wide variety of aspirating nozzles on the market. Some are fixed tubes with no adjustment. Others are adjustable usually by changing the stream pattern. Each nozzle manufacturer also makes clip-on aspirating foam producing nozzles, which attach to the bumper of a fog nozzle when needed. Today, manufacturers produce a wide variety of foam types and volumes, and can be used with both Class A and B concentrates. The foam forming nozzles can be designed to draw air into either the front or the back of the nozzle, using the Venturi effect. As the foam solution passes through the center of the nozzle, a low level of pressure is created, which allows the air to enter the nozzle. The more air that is drawn into the nozzle, the more energy is consumed (i.e. resulting in a pressure drop), and causing a reduction in the "reach" of the foam stream—measured in the distance the finished foam can travel towards a target.

**[0606]** The "expansion ratio" of an aerating/aspirating nozzle determines the difference between the volume of foam solution pumped into the nozzle, and the volume of



finished foam bubbles exiting the foam producing nozzle. For example, if one gallon of foam solution enters the nozzle and 50 gallons of foam bubbles exit the nozzle, the expansion ratio is 50/1. Expansion ratios are broken into three categories: low, medium and high. Low expansion starts at 1/1 and goes up to 20/1. Medium expansion starts at 21/1 and goes up to 200/1. High expansion begins at 201/1 and can go as high as 1,000/1.

**[0607]** Low-expansion nozzles are typically fixed tubes with no adjustment, and produce a wet foam, which is appropriate for many tactical applications, such as fire attack and mop-up. These nozzles typically operate at 80 to 100 psi nozzle pressure with a Class A foam percentage of 0.5%.

**[0608]** Medium-expansion nozzles are typically adjustable, allowing variations in foam volume and foam consistency. Tactical applications include wildland firebreak (i.e. fireline) work, and mop-up and overhaul of vehicle fires. The operating pressure for these medium-expansion nozzles is typically 60 psi. This lower pressure is necessary because as the bubble size increases, the foam bubbles become more fragile. Higher pressure will cause the bubbles to break, reducing the effective production of the nozzle. Larger foam bubbles also require more structure, which comes from an increase in foam percentage, usually 0.5% to 0.7%.

**[0609]** High-expansion nozzles produce a large volume of dry foam due to the large volume of air taken in and low water content. As the foam bubbles become even bigger, the nozzle pressures drop to around 40 psi and the foam percentage must be increased to the range of 0.7% to 1.0%, which is the maximum percentage for Class A foams. High-expansion foams are typically used on compartment fires or to fill void spaces, such as in large aircraft hangers.

**[0610]** As the expansion ratio increases, the foam stream reach of the nozzle will decrease. Again, this is due to the entrainment of more air consuming more energy and the lower nozzle pressure that's needed to prevent destruction of the bubbles in the nozzle.

**[0611]** The medium-expansion nozzles are the most effective because they are the most versatile. These nozzles can be used for everything from initial attack, to mop up and overhaul, to wildland fire breaks and pretreatment of fuels. As a general rule, every foam-capable apparatus, including compressed-air-foam-system (CAFS) equipped rigs, should be equipped with an adjustable, medium-expansion nozzle. Bio-Degradable Fire Extinguishing Foam Concentrate (FC) Compositions of the Present Invention for Producing Finished Fire Extinguishing Foam Material for Use in Actively Fighting Fires Involving Class A/B Fuels, as Well as Proactively Inhibiting Fires to Defend Against Fire Ignition, Flame Spread, and Flash Over

**[0612]** Referring now to FIGS. 51 through 57, technical details will now be provided teaching how to practice the bio-degradable fire extinguishing foam concentrate (FC) compositions of the present invention for producing finished fire extinguishing foam material for use in actively fighting fires involving Class A/B fuels, as well as proactively inhibiting fires to defend against fire ignition, flame spread, and flash over.

Specification of Fire Extinguishing Foam Concentrate Compositions of the Present Invention for Mixing with Pressurized Streams of Water and Air to Generate Fire Extinguishing Foam for Use in Actively Extinguishing Fire Involving Class A Fuel and/or Class B Fuel

**[0613]** FIG. 51 shows the primary components of a first environmentally-clean aqueous-based fire extinguishing

biochemical foam concentrate (i.e. fire extinguishing additive) of the present invention.

**[0614]** As shown in FIG. 51, the foam concentrate composition comprises: (i) major amounts of a fire inhibiting agent realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate (TPC), for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (ii) minor amounts of triethyl citrate (TEC) as a low-surface tension surfactant agent; (iii) major amounts of hydrolyzed protein isolate (HPI) or protein hydrolysates, preferably 85-90% purity or higher, protein derived from plant sources such as soy, whey, soy whey, or protein derived from such animal parts, and functioning as a foaming agent, as disclosed and taught in U.S. Pat. Nos. 2,361,057; 4,424,133; and 5,824,238, each incorporated herein by reference; (iv) major amounts of water functioning as a solvent, carrier and dispersant, to form the fire extinguishing biochemical foam concentrate (FC) designed to be added to and mixed with a pressurized supply of water in pre-specified proportions so as to produce an environmentally-clean aqueous fire extinguishing foam material of high stability, suitable for spraying onto an actively combusting fire involving Class A fuel and/or Class B fuel; and (v) other additives such as preservative, colorants, and inhibitors desired or required to inhibit processes such as fermentation, and the like, from occurring the liquid solution during storage in its container.

**[0615]** FIG. 52 shows a table illustrating the performance characteristics and chemical components associated with the bio-degradable Class AB firefighting foam concentrate of the present invention specified in FIG. 51. As described, the biodegradable fire extinguishing foam concentrate of the present invention is designed for producing environmentally-safe, biodegradable firefighting foam for use in extinguishing fires involving Class A & B fuels, as well as inhibiting fires involving Class A and B fuels, after application.

**[0616]** As shown in FIG. 52, the performance characteristics of the finished foam material include, for example: high foam stability; and excellent fire inhibiting and extinguishing capabilities. The foam concentrate composition is formulated using generally safe food-grade chemicals, namely: food-grade foaming agents; food-grade fluorine-free low-surface tension surfactants and dispersants; food-grade and environmentally-safe fire inhibiting agents and extinguishing agents; and 100% free of phosphates and ammonia compounds.

Specification of Environmentally-Clean Aqueous-Based Fire Extinguishing Bio-Chemical Foam Concentrate Formulations, and Methods of Producing the Same in Accordance with the Principles of the Present Invention

**[0617]** Another object of the present invention is to provide new and improved environmentally-clean aqueous-based fire extinguishing biochemical foam concentrate (FC) composition designed and engineered for proportioning and mixing with pressurized supplies/quantities of water, and then aerated/aspirated within an aerating/aspirating spray foam forming nozzle so as to generate a stream of finished fire extinguishing biochemical foam that demonstrates (i) excellent immediate fire extinguishing effects when applied to an active fire involving Class A and/or Class B fuels, and



(ii) excellent fire inhibiting effects after a fire has been extinguished, to prevent fire reignition.

**[0618]** In general, the novel biochemical fire extinguishing foam concentrate compositions of the present invention comprise: (a) a dissolving agent in the form of a quantity of water, for dissolving salts and dispersing metal ions dissolved in water; (b) a fire extinguishing agent in the form of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid, for providing metal ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (c) a foaming agent consisting of hydrolyzed protein isolate (HPI) material, such as plant-based protein (e.g. soy, whey or soy whey) or animal-based protein, having fine particle size (e.g. 600 to 100 microns) and being at least 80% pure (i.e. free of fat, starch, and sugar molecules) and dissolvable in water, with other surfactants and fire extinguishing components, for setting up the finished structure of foam during the aeration process, as disclosed and taught in U.S. Pat. Nos. 2,361,057; 4,424,133; and 5,824,238, incorporated herein by reference; (d) a dispersing agent in the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), such as triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal ions when the finished fire extinguishing foam is applied to a surface to be protected against fire; and (e) optional additives to the resulting foam concentrate solution, to provide coloring, preserve shelf-life of the contained product, and inhibit fermentation processes within the resulting foam concentrate solution while stored in its container awaiting use.

**[0619]** As will be described herein, the fire extinguishing biochemical foam concentrate of the present invention can be premixed and bottled/containerized at its prescribed concentrated strength (e.g. 1%, 3% or 6%) for transport to end-users (e.g. fire departments) who store the foam concentrate containers aboard the pumping fire engine truck, having onboard portable or fixed fire extinguishing foam generation equipment, as illustrated in FIGS. 53, 54, 55, 56 and 57. When there is a need to extinguish a Class A and/or B fuel fire using fire extinguishing foam, the liquid foam concentrate (FC) stored aboard a fire engine pumper truck is mixed with a pressurized supply of clean water (e.g. 200 PSI) in correct proportions using a Venturi-based proportioning/mixing system as shown in FIG. 54, to produce a foam mixture that is provided to an aerating foam generating nozzle, that generates a finished fire extinguishing foam with excellent fire extinguishing properties, as described herein.

**[0620]** In general, useful alkali metal salts of nonpolymeric saturated carboxylic acids for inclusion in the foam concentrate (FC) compositions of the present invention preferably comprise: alkali metal salts of oxalic acid; alkali metal salts of gluconic acid; alkali metal salts of citric acid; and also alkali metal salts of tartaric acid. Alkali metal salts of citric acid are particularly preferred, as will be further explained hereinafter.

**[0621]** Notably, while the efficacy of the alkali metal salts increases in the order of lithium, sodium, potassium, cesium and rubidium, the salts of sodium and salts of potassium are preferred for cost of manufacturing reasons. Potassium carboxylates are very particularly preferred, but tripotassium citrate monohydrate (TPC) is the preferred alkali metal salt for use in formulating the fire extinguishing foam concentrate (FC) compositions of the present invention.

**[0622]** While it is understood that other alkali metal salts are available to practice the biochemical foam compositions of the present invention, it should be noted that the selection of tripotassium citrate as the preferred alkali metal salt, includes the following considerations: (i) the atomic ratio of carbon to potassium (the metal) in the utilized alkali metal salt (i.e. tripotassium citrate); (ii) that tripotassium citrate is relatively stable at transport and operating temperatures; (iii) tripotassium citrate is expected to be fully dissociated to citrate and potassium when dissolved in water, and that the dissociation constant is not relevant for the potassium ions, while citric acid/citrate has three ionizable carboxylic acid groups, for which pKa values of 3.13, 4.76 and 6.4 at 25° Celsius are reliably reported the European Chemicals Agency (ECHA) handbook; and (iv) that tripotassium citrate produces low carbon dioxide levels when dissolved in water.

**[0623]** Preferably, the water soluble dispersing agent should have a melting point at least 32 F (0 C) or lower in temperature, and be soluble in water. Triethyl citrate (TEC) is a preferred dispersing agent when used in combination with tripotassium citrate (TPC) having excellent compatibility given that both chemical compounds are derived from citric acid. Triethyl citrate (TEC) also functions as a surfactant, reducing the surface tension of the final proportioned and mixed foam liquid solution to be injected to an aerating/aspirating foam forming nozzle or gun, and to improve wettability or sticking of resulting foam to Class A fuel surfaces, and promote thin film liquid formation on Class B fuels after application of finished foam.

**[0624]** The fire extinguishing biochemical foam concentrates (FC) of the present invention are produced and prepared by mixing the components in specified amounts with water to produce the fire extinguishing foam concentrate (LC) liquid material. The order of mixing is discretionary. It is advantageous to produce aqueous preparations by first mixing the components other than water, into a predetermined quantity of water.

**[0625]** The fire extinguishing foam concentrates of the present invention have a good immediate fire extinguishing effect, and a good inhibiting effect after application to reduce reignition of fire once extinguished. This mixing of the constituent biochemical compounds takes place before use while making the foam concentrate (FC) liquid material. For example, it is preferred that the fire extinguishing foam concentrate be made as a liquid concentrate (e.g. 1%, 3% or 6% by volume) to be added to 99%, 97% or 94% water by volume, respectively, using conventional in-line venturi-based fluid proportioning/mixing devices (e.g. eductor device) shown in FIG. 54 and available from numerous manufacturers around the globe, including the LEADER GROUP S.A. S, with headquarters in France. When prepared in the form of a fire extinguishing foam concentrate (FC), the viscous liquid concentrate solution can be later mixed with the correct proportions of pressured water at a fire pumping engine, or other location having a fire outbreak, to produce streams of liquid foam containing the biochemical additives of the present invention, which are then aerated within a foam-type spray nozzle so as to generate streams of finished fire extinguishing foam, capable of extinguishing Class A and/or B fires, while significantly reducing the quantity of water required to do so. While extinguishing Class A and/or B fires are preferred applications for the fire extinguishing foam compositions of the present invention, it is noted that the fire extinguishing foam compositions are



also useful as a fire extinguishing agent for fighting fires involving Class D fuel as well.

#### Specification of Preferred Embodiment of the Aqueous-Based Biochemical Fire Extinguishing Foam Concentrates of the Present Invention

**[0626]** In the preferred embodiment of the fire extinguishing biochemical foam concentrate of the present invention, the components are realized as follows: (a) the dissolving agent is realized in the form of a quantity of water, for dissolving salts and dispersing metal ions dissolved in the water; (b) the fire extinguishing agent is realized in the form of an alkali metal salt of a nonpolymeric saturated carboxylic acid, specifically, tripotassium citrate, for providing metal (potassium) ions dispersed in the water when the at least one alkali metal salt is dissolved in the water; (c) a foaming agent consisting of hydrolyzed protein isolate (HPI) material, such as soy, whey or soy whey protein isolate (SPI) or animal protein isolate having a fine particle size (e.g. 500 to 50 microns) and being at least 80% purity (i.e. at least 80% free of fat, starch, and sugar molecules) and dissolvable in water, and other surfactant and fire extinguishing components, for setting up the structure of foam during the aeration process; (d) a dispersing agent in the form of an organic compound containing three carboxylic acid groups (or salt/ester derivatives thereof), such as triethyl citrate, an ester of citric acid, for dispersing and coalescing the metal ions when the finished fire extinguishing foam is applied to a surface to be protected against fire, while water molecules are being evaporated by the heat of the fire, to cool off fuel surfaces, and metal potassium ions interact with free radical hydrocarbon vapors in the combustion phase of the fire, so as to interrupt these free-radical chemical reactions and help extinguish the fire using significantly less water; and (e) optional additives to the resulting foam concentrate solution that provide coloring, stability to components, and inhibiting fermentation processes involving the hydrolyzed protein isolate (HPI) added to the foam concentrate.

**[0627]** The preferred formulation specified above is prepared as an X % foam concentrate (FC) composition (e.g. 1% liquid concentrate by volume to be proportioned and mixed with 99% water by volume; 3% liquid concentrate by volume to be proportioned and mixed with 97% water by volume; or 6% liquid concentrate by volume to be proportioned and mixed with 94% water by volume). Then this fire extinguishing foam concentrate (FC) is stored in a container, bottle or tote (i.e. package) suitable for the end user application in mind. The filled container should be sealed with appropriate sealing technology and immediately labeled with a specification of (i) its biochemical components, with weight percent measures where appropriate, and the date and time of manufacture, printed and recorded in accordance with good quality control (QC) practices well known in the art. Where necessary or desired, barcode symbols and/or barcode/RFID identification tags and labels can be produced and applied to the sealed package to efficiently track each barcoded package containing a specified quantity of clean fire extinguishing biochemical composition concentrate. All product and QC information should be recorded in globally accessible network database, for use in tracking the movement of the package as it moves along the supply chain from its source of manufacture, toward its end use at a GPS specified location.

#### Selecting Tripotassium Citrate (TCP) as a Preferred Fire Extinguishing Agent for Use in the Fire Exhibiting Foam Biochemical Foam Concentrate Compositions of the Present Invention

**[0628]** In the preferred embodiments of the present invention, tripotassium citrate (TPC) is selected as active fire extinguishing chemical component in fire extinguishing biochemical foam concentrates of the present invention. In dry form, TPC is known as tripotassium citrate monohydrate ( $C_6H_5K_3O_7 \cdot H_2O$ ) which is the common tribasic potassium salt of citric acid, also known as potassium citrate. It is produced by complete neutralization of citric acid with a high purity potassium source, and subsequent crystallization. Tripotassium citrate occurs as transparent crystals or a white, granular powder. It is an odorless substance with a cooling, salty taste. It is slightly deliquescent when exposed to moist air, freely soluble in water and almost insoluble in ethanol (96%).

**[0629]** Tripotassium citrate is a non-toxic, slightly alkaline salt with low reactivity. It is chemically stable if stored at ambient temperatures. In its monohydrate form, TPC is very hygroscopic and must be protected from exposure to humidity. Tripotassium citrate monohydrate is considered "GRAS" (Generally Recognized As Safe) by the United States Food and Drug Administration without restriction as to the quantity of use within good manufacturing practice. CAS Registry Number: [6100-05-6]. E-Number: E332.

#### Selecting Triethyl Citrate (TEC) as a Preferred Dispersing Agent with Surface Tension Reducing and Surfactant Properties for Use in the Fire Exhibiting Biochemical Foam Concentrates of the Present Invention

**[0630]** In the preferred illustrative embodiments of the present invention, the dispersing agent and surfactant used in the fire extinguishing biochemical foam concentrates of the present invention is realized as a food-grade additive component, namely, triethyl citrate (TEC) which functions as a dispersing agent with surface tension reducing properties and surfactant properties as well. Triethyl citrate belongs to the family of tricarboxylic acids (TCAs) and derivatives, organic compounds containing three carboxylic acid groups (or salt/ester derivatives thereof).

**[0631]** In the aqueous-based fire extinguishing foam concentrate, the dispersing agent and surfactant functions to disperse the metal ions dissolved and disassociated in aqueous solution. As foam concentrate expands when air is injected into the structure of dissolved solution of hydrolyzed protein molecules, surfactant molecules and water, the potassium ions are able to uniformly disperse throughout the structure of the finished foam generated during aeration and expansion of the finished foam material produced from the aerating/aspirating foam nozzle.

**[0632]** A relatively minor quantity of triethyl citrate (TEC) liquid is blended with a major quantity of TCP powder and foaming agent (e.g. soy protein molecules) in specific quantities by weight and dissolved in a major quantity of water to produce a translucent completely-dissolved biochemical foam concentration consisting of food-grade biochemicals mixed with water and having highly effective fire extinguishing properties, as proven by testing. The resulting aqueous foam solution remains stable without the formation of solids at expected operating temperatures (e.g. 34 F to 120 F).

**[0633]** Jungbunzlauer (JBL) manufactures and distributes its CITROFOL® A1 branded bio-based citrate esters for



food-grade, healthcare, pharmaceutical and over the counter (OTC) applications around the world. CITROFOL® A1 triethyl citrate (TEC) esters have an excellent toxicological and eco-toxicological profile, and provide good versatility and compatibility with the tripotassium citrate (TPC) component of the biochemical compositions of the present invention. CITROFOL® A1 branded citrate esters are particularly characterized by highly efficient solvation, low migration and non-VOC (volatile organic compound) attributes. As an ester of citric acid, triethyl citrate is a colorless, odorless liquid which historically has found use as a food additive (E number E1505) to stabilize foams, especially as a whipping aid for egg whites.

**[0634]** The primary function of the dispersing agent and surfactant (i.e. TEC) in the biochemical foam composition is to ensure relatively uniform dispersal of dissolved potassium ions in solution, while significantly reducing surface tension of the liquid foam solution containing dissolved hydrolyzed protein molecules and surfactant (i.e. TEC), so that the finished fire extinguishing foam is produced from the aerating foam nozzle with the desired stability and wetness/dryness.

**[0635]** The fact that CITROFOL® A1 triethyl citrate (TEC) esters are bio-based, odorless, biodegradable and label-free, represents a great advantage over most other surfactants (e.g. fluoro-surfactants used in conventional Fluoroprotein foam concentrates adversely impacting the environment), and fully satisfies the toxicological and environmental safety requirements desired when practicing the biochemical concentrate of the present invention.

**[0636]** The chemical and colloidal nature of potassium salt ions (which are mineral salt dispersions) present in TPC dissolved in water with dissolved hydrolyzed protein isolate, is highly compatible with the CITROFOL® A1 triethyl citrate (TEC) ester used as the dispersing agent and surfactant in the preferred embodiments of the present invention. Also, CITROFOL® A1 triethyl citrate esters are REACH registered and are safe, if not ideal, for use in environmentally sensitive products such as fire extinguishing agents which must not adversely impact human, animal and plant life, ecological systems, or the natural environment.

Specification of Preferred Formulations for Fire Extinguishing Biochemical Foam Concentrates According to the Present Invention

#### Example #1: Liquid-Based Fire Extinguishing Biochemical Foam Concentrate

**[0637]** FIG. 51 illustrates the primary components of a first environmentally-clean aqueous-based fire extinguishing biochemical foam concentrate of the present invention consisting of tripotassium citrate (TPC), triethyl citrate (TEC) formulated as a dispersing agent and surfactant, and hydrolyzed protein isolate (HPI) of at least 80% purity, with water functioning as a solvent, carrier and dispersant in the biochemical liquid foam composition.

**[0638]** Example 1: Schematically illustrated in FIG. 51: A fire-extinguishing biochemical foam concentrate was produced by stirring the components into water. The foam concentrate composition comprising: 0.4 pounds by weight of triethyl citrate as dispersing agent/surfactant, (20.3 milliliters by volume); 2.6 pounds by weight of tripotassium citrate (64 fluid ounces by volume); 2.6 pounds of hydrolyzed protein isolate (HPI) of at least 80% purity or greater;

and 4.0 pounds by weight of water (64 fluid ounces by volume), to produce a resultant foam concentrate solution of total weight of 9.60 pounds having 128 ounces or 1 gallon of volume.

#### Preferred Weights Percentages of the Components of the Fire Extinguishing Biochemical Foam Concentrate Formulation of the Present Invention

**[0639]** In the biochemical foam concentrates of the present invention, the ratio of the ester of citrate (e.g. triethyl citrate) to the alkali metal salt of a nonpolymeric carboxylic acid (e.g. tripotassium citrate) may be major amount between 1:100: to 1:1000 and is typically in the range from 1:1 to 1:100, preferably in the range from 1:2 to 1:50, more preferably in the range from 1:4 to 1:25 and most preferably in the range from 1:8 to 1:15. Also, the ratio of the alkali metal salt of a nonpolymeric carboxylic acid (e.g. tripotassium citrate) to the hydrolyzed protein isolate (e.g. HPI of 80% or greater purity) may be major amount between 1:100: to 1:1000 and is typically in the range from 1:1 to 1:100, preferably in the range from 1:2 to 1:50, more preferably in the range from 1:2 to 1:30 and most preferably in the range from 1:2 to 1:4.

**[0640]** A preferred biochemical foam concentrate according to the present invention comprises: a major amount from 1% to 65% by weight, preferably from 20% to 50% by weight and more preferably from 30% to 55% by weight, of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid (e.g. tripotassium citrate monohydrate or TPC); a major amount from 1% to 65% by weight, preferably from 20% to 50% by weight and more preferably from 30% to 55% by weight, of at least one hydrolyzed protein isolate (e.g. soy, whey or soy whey protein isolate of 80% purity or greater; and minor amount from 0.08% to 5% by weight, preferably from 0.5% to 2% by weight and more preferably from 0.1% to 1.0% by weight, of triethyl citrate (an ester of citrate acid) as dispersing agent and surfactant; wherein the sum by % weight of the components (a) and (b) should not exceed 100% by weight.

**[0641]** In a preferred embodiment, the fire extinguishing foam concentrate further comprises water. The water content is present in a major amount and is typically not less than 30% by weight, preferably not less than 40% by weight, more preferably not less than 50% by weight and most preferably not less than 60% by weight and preferably not more than 60% by weight and more preferably not more than 70% by weight, all based on the fire extinguishing biochemical foam concentrate composition.

**[0642]** The viscosity of the aqueous preparation is preferably at least 4 [mPas] (millipascal-seconds, in SI units, defined as the internal friction of a liquid to the application of pressure or shearing stress determined using a rotary viscometer), and preferably not more than 50 [mPas], or 50 centipois [cps], for most applications. Preferably, the pH of the aqueous concentrate solution is in the range of 6.0 to 8.0.

#### Methods of Blending, Making and Producing the Biochemical Foam Concentrate Formulation

**[0643]** The fire extinguishing foam concentrate illustrated above is reproducible by mixing the components described above. The order of mixing is discretionary. However, it is



advantageous to produce aqueous preparations by first mixing the components other than water, into a predetermined quantity of water.

**[0644]** When seeking to produce a highly-effective fire extinguishing foam material, the foam concentrate is added and mixed with a proportioned quantity of pressurized water flowing through a venturi-driven proportioning/mixing system (e.g. as shown in FIG. 54), and thereafter, downstream, the mixed foam solution is forced under pressure through a length of fire hose terminated with an aerating/aspirating foam forming spray nozzle that generates a stream of finished fire extinguishing foam material. Typically, it is convenient to use an automatic venturi-type proportioning/mixing device installed in-line between (i) a hydraulic water pumping engine with an output water pressure of about 200 psi, and (ii) an aerating/aspirating foam forming spray nozzle, as shown in FIGS. 53 and 54, to generate streams of finished fire extinguishing foam which is then directed towards an active fire involving Class A and/or B fuel. Examples of automated fluid proportioning/mixing devices that can be used in generating fire extinguishing foam according to the present invention are made and sold by many manufacturers around the world, including the LEADER Group S.A.S. of France, which makes and sells the LEADER MIX 200-1000 V2 automatic proportioning/mixing system. This in-line system is capable proportioning all types of liquid and foam concentrates from 0.03% to 6.0% over a large range of flow rates from 200 to 1050 liter/min. (43 to 230 gallons/min), over a wide operating pressure range from 5 to 16 bar (72 to 232 PSI).

**[0645]** To formulate an X % foam concentrate (FC), all of the mass measures in the foam concentrate formulation should be listed in a table format, including the amount of water to be mixed with the components to make up the foam concentrate formulation. For a 6% foam concentration, 94 gallons of water will be added to each 6 gallons of foam concentrate suctioned by the venturi-type proportioning/mixing system shown in FIG. 54. So, for 30 gallons of liquid concentrate,  $30/6 \times 94 = 564$  gallons of working foam concentrate solution will be produced from the 30 gallons of foam concentrate, using 6% proportioning/mixing settings. Using ordinary experimentation and testing, the mass quantities measured in [lbs.] are determined or otherwise estimated for each chemical ingredient/component (e.g. TPC, TEC, and HPI) to be dissolved in 30 gallons of water to make a 6% liquid concentrate for the fire extinguishing foam composition of the present invention. Similar methods can be practiced for determining different X % foam concentrates of the present invention, such as 1% foam concentrates, 2% foam concentrates, 3% foam concentrates, and so on, as required by the application at hand.

Specification of Portable Apparatus for Proportioning and Mixing Fire Extinguishing Foam Concentrate of the Present Invention with Pressurized Streams of Water, and then with Air within an Aerating Nozzle to Generate Fire Extinguishing Foam for Use in Actively Extinguishing Fire Involving Class A Fuel and/or Class B Fuel

**[0646]** FIG. 53 shows a mobile and/or portable fire extinguishing system for continuously mixing and proportioning the fire extinguishing foam concentrate of present invention with pressurized water, and then injecting air into the foam liquid solution to produce a finished fire extinguishing foam material for use in fighting active fires involving Class A and/or Class B fuels.

**[0647]** As shown in FIG. 53, the system comprises: (i) a venturi-based fluid mixing/proportioning device/system operably connected to a pressurized supply of water output at 200+ PSI pressure from a hydraulic pumping engine connected to a supply of water and pressurized by a hydraulic pump system driven by gasoline, diesel or electric engine; (ii) a supply of fire extinguishing liquid concentrate (LC) of the present invention contained within a 20+ gallon container and in fluid communication with the proportioning/mixing device via its suction/drawing tube; and (iii) one or more aerating/aspirating foam forming nozzles manually-actuable for producing finished foam material, for use in extinguishing any particular fire involving Class A and/or Class B fuels.

**[0648]** FIG. 54 shows a conventional in-line automated proportioning/mixing systems (e.g. eductor) for automatically proportioning and mixing a fire extinguishing foam concentrate (e.g. additive) with a pressurized water stream while spraying finished fire extinguishing foam generated from an aerating (i.e. aspirating) foam nozzle connected thereto, as schematically illustrated in FIG. 53.

**[0649]** FIG. 55 shows a portable spray cart, containing a supply of fire extinguishing foam concentrate, with an in-line proportioning/mixing system of FIG. 54, connected to a length of fire hose and an adjustable aerating-type foam spray nozzle, and operably connectable to a pressurized water pumping engine as illustrated in FIG. 53, to mix a proportioned quantity (i.e. %) of fire extinguishing foam concentrate with a pressurized supply of water flowing through the proportioning/mixing device, along the length of fire hose to the adjustable aerating-type foam spray nozzle, generating a finished foam for application to an active fire.

**[0650]** FIG. 56 shows a portable triple tote spray trailer designed to be pulled and driven by a pressurized water pumping firetruck, and having a trailer platform supporting three foam concentrate totes, each containing 200-500 gallons of fire extinguishing liquid concentrate additive of the present invention, and being operably connected to an in-line proportioning/mixing device as shown in FIG. 54, and also to an adjustable aerating/aspirating foam forming nozzle gun assembly mounted for spraying operations, and also being operably connectable to the pressurized water pumping engine aboard the water pumping firetruck, as illustrated in FIG. 53.

**[0651]** Using this system, a proportioned quantity (i.e. 1%, 3% or 6%) of fire extinguishing foam concentrate (FC) is mixed with a pressurized supply of water flowing through the automated venturi-based proportioning/mixing device that continuously produces, as output, a mixed foam liquid/solution to the input port of an aerating-type foam spray nozzle, flexibly mounted on platform, or hand-supportable, so as to generate a finished fire extinguishing foam from the foam gun nozzle for application to Class A and/or B fuel sources proactively, or during an active fire.

Specification of Fixed/Stationary Apparatus for Proportioning and Mixing Fire Extinguishing Foam Concentrate of the Present Invention with Pressurized Streams of Water, and then with Air within an Aerating Nozzle to Generate Fire Extinguishing Foam for Use in Actively Extinguishing Fire Involving Class A Fuel and/or Class B Fuel

**[0652]** FIG. 57 shows a stationary and/or fixed fire extinguishing system for mixing and proportioning the fire extinguishing foam concentrate of present invention with pressurized water, and then injecting air into the foam liquid



solution to produce a finished fire extinguishing foam for use in fighting active fires involving Class A and/or Class B fuels.

**[0653]** As shown in FIG. 57, the system comprises: (i) a venturi-based mixing/proportioning device operably connected to a pressurized supply of water output (at 200 psi) from a hydraulic pumping engine connected to a supply of water and pressurized by a hydraulic pump system driven by an electric, propane or other engine; (ii) a supply of fire extinguishing foam concentrate (FC) of the present invention contained within a 20+ gallon container; and (iii) one or more aerating/aspirating foam forming nozzles, manually triggered by a lever or automatically triggered by electronic-controlled sensors, IR cameras and/or controllers, for automatically producing fire extinguishing foam for extinguishing an particular fire involving Class A and/or Class B fuels, with improved fire extinguishing efficacy and efficiency using reduced quantities of water to minimize water damage to property during a fire outbreak.

**[0654]** The system shown in FIG. 57 can be realized in many different ways, using many different means, to meet the fire inhibiting and/or extinguishing requirements arising in the aviation, industrial, marine and offshore, and oil, gas and petrochemical industries.

Applications for the Fire Fighting Liquid Concentrate (LC) Compositions and Foam Concentrate (FC) Compositions of the Present Invention

**[0655]** The fire fighting (e.g. inhibiting and extinguishing) liquid concentrates and foam concentrate compositions of the present invention have diverse applications beyond inhibiting and extinguishing Class A and B fires in the wildfire and wildland fire defense industry including, for example, preventing and fighting against fires arising in the aviation, industrial, marine and offshore, and oil, gas and petrochemical industries.

**[0656]** Aviation applications include inhibiting and extinguishing fires in aircraft hangers, airports, heliports, maintenance bays, and engine test facilities.

**[0657]** Industrial application include inhibiting and extinguishing fires in chemical plants, hazardous material spills, blending operations, power plants, waste treatment, pumping stations, truck loading racks, and warehouses.

Marine and offshore application include inhibiting and extinguishing fires in engine and pump rooms, cargo holds, helidecks, offshore platforms, floating production storage and offloading units (FPSOs), floating offshore stations (FOSs), jetties, dry docks, and onshore-offshore storage.

**[0658]** Oil, gas and petrochemical applications include inhibiting and extinguishing fires in refineries, tank farms, storage tanks and dikes, petrochemical facilities, liquid natural gas (LNG) terminals, and pipelines.

Modifications to the Present Invention which Readily Come to Mind

**[0659]** The illustrative embodiments disclose the formulation, application and use of environmentally-clean fire inhibiting liquid concentrates, fire extinguishing liquid concentrates, and fire extinguishing foam concentrates. Such fire inhibiting compositions, methods and apparatus are disclosed and taught herein for use in proactively coating the surfaces of wood, lumber, and timber, and other combustible matter, wherever wild fires may travel. Such fire extinguishing liquid concentrates, methods and apparatus are disclosed and taught herein for use in actively producing fire extin-

guishing mists, sprays, clouds and streams of water with additives that more effectively extinguish fire involving Class A and/or B fuels, using significantly less water. Also, fire extinguishing foam concentrates, methods and apparatus are disclosed and taught herein for use in actively producing fire extinguishing foams that more effectively extinguish fire involving Class A and/or B fuels, using significantly less water.

**[0660]** These and other variations and modifications will come to mind in view of the present invention disclosure.

**[0661]** While several modifications to the illustrative embodiments have been described above, it is understood that various other modifications to the illustrative embodiment of the present invention will readily occur to persons with ordinary skill in the art. All such modifications and variations are deemed to be within the scope and spirit of the present invention as defined by the accompanying Claims to Invention.

1. A fire extinguishing liquid foam concentrate to be mixed with a proportioned quantity of water, and then mixed with air within an aerating/aspirating foam forming nozzle to generate finished fire extinguishing foam material, said fire extinguishing liquid foam concentrate comprising:

- a dispersing agent in the form of a quantity of water, for dispersing metal ions dissolved in said quantity of water;
- a fire inhibiting agent in the form of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid, for providing metal ions dispersed in the water when the at least one alkali metal salt is dissolved in said quantity of water;
- a foaming agent including hydrolyzed protein isolate (HPI) material dissolved in said quantity of water; and
- a dispersing agent in the form of an organic compound containing three carboxylic acid groups, or salt/ester derivatives thereof, for dispersing the metal ions in said quantity of water, and lowering the surface tension of the liquid solution formed by said fire inhibiting agent, said foaming agent and said dispersing agent dissolved in said quantity of water, to enable the forming of a fire extinguishing foam material when said liquid solution is mixed with air within an aerating/aspirating foam forming nozzle.

2. The fire extinguishing liquid foam concentrate according to claim 1, wherein the alkali metal salt is a sodium or potassium salt.

3. The fire extinguishing liquid foam concentrate according to claim 1, wherein the alkali metal salt is tripotassium citrate.

4. The fire extinguishing liquid foam concentrate according to claim 1, wherein said coalescing agent is triethyl citrate, an ester of citric acid.

5. A method of fighting a wildfire comprising the steps of applying the fire extinguishing foam material produced in claim 1 to the surfaces to be proactively protected from a wildfire.

6. A method of fighting a fire comprising the steps of applying the fire extinguishing foam material produced in claim 1 to surfaces ignited or consumed by fire to be extinguished by said fire extinguishing foam material.

7. An aqueous-based fire extinguishing liquid concentrate for mixing with a prespecified quantity of water to produce a fire extinguishing liquid solution that produces good immediate extinguishing effects when applied to extinguish



a burning or smoldering fire, and very good long-term fire inhibiting effects when being proactively applied to protect combustible surfaces against the threat of fire, said aqueous-based fire extinguishing liquid concentrate comprises:

- a dispersing agent realized in the form of a quantity of water, for dispersing metal ions dissolved in water;
- a fire inhibiting agent in the form of at least one alkali metal salt of a nonpolymeric saturated carboxylic acid, for providing metal ions dispersed in the water when the at least one alkali metal salt is dissolved in said quantity of water; and
- a dispersing agent in the form of an organic compound containing three carboxylic acid groups Or salt/ester derivatives thereof, such as triethyl citrate, an ester of citric acid, for dispersing the metal ions in said quantity of water, and lowering the surface tension of the liquid solution formed by said fire inhibiting agent, and said dispersing agent dissolved in said quantity of water, and forming of a fire extinguishing liquid solution that produces good immediate extinguishing effects when applied to extinguish a burning or smoldering fire, and very good long-term fire inhibiting effects when being proactively applied to protect combustible surfaces against the threat of fire.

8. The aqueous-based fire extinguishing liquid concentrate of claim 7, wherein said alkali metal salts of nonpolymeric saturated carboxylic acids for inclusion in the composition comprises: alkali metal salts of oxalic acid; alkali

metal salts of gluconic acid; alkali metal salts of citric acid; and also alkali metal salts of tartaric acid.

9. The aqueous-based fire extinguishing liquid concentrate of claim 7, wherein said alkali metal salts of nonpolymeric saturated carboxylic acids comprise potassium carboxylates

10. The aqueous-based fire extinguishing liquid concentrate of claim 7, wherein said alkali metal salts of nonpolymeric saturated carboxylic acids comprise tripotassium citrate monohydrate (TPC).

11. The aqueous-based fire extinguishing liquid concentrate according to claim 7, wherein the alkali metal salt is a sodium or potassium salt.

12. The aqueous-based fire extinguishing liquid concentrate according to claim 7, wherein the alkali metal salt is tripotassium citrate.

13. The aqueous-based fire extinguishing liquid concentrate according to claim 7, wherein said coalescing agent is triethyl citrate, an ester of citric acid.

14. A method of fighting a fire comprising the steps of applying fire extinguishing foam material produced in claim 7 to the surfaces to be proactively protected from a wildfire.

15. A method of fighting a fire comprising the steps of applying fire extinguishing foam material produced in claim 7 to surfaces ignited or consumed by fire to be extinguished by said fire extinguishing foam material.

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